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TECHNICAL SPECIFICATION

Mission Critical (MC) services;
Mission Critical Data (MCData) Application Server (AS)
Protocol conformance specification for
server-to-client interface;
Part 1: Test structure, configurations,
conformance requirement and test purposes

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee TETRA and Critical Communications Evolution (TCCE).

The present document is based on 3GPP TS 36.579-9 [i.8] produced by the 3rd Generation Partnership Project (3GPP).

The present document is part 1 of a multi-part deliverable covering the Mission Critical (MC) services; Mission Critical Data (MCData) Application Server (AS) Protocol conformance specification for server-to-client interface, as identified below:

Part 1: "Test structure, configurations, conformance requirement and test purposes";

Part 2: "Test Applicability and Implementation Conformance Statement (ICS) pro forma specification".

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Introduction

The ETSI Technical Specification used as the reference document for the present document is part 9 of a multi-part deliverable covering conformance test specification for Mission Critical Services over LTE consisting of:

- ETSI TS 136 579-1 [1]: "LTE; Mission Critical (MC) services over LTE; Part 1: Common test environment (3GPP TS 36.579-1)";
- ETSI TS 136 579-2 [i.4]: "LTE; Mission Critical (MC) services over LTE; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-2)";
- ETSI TS 136 579-3 [2]: "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3)";
- ETSI TS 136 579-4 [3]: "LTE; Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification (3GPP TS 36.579-4)";
- ETSI TS 136 579-5 [i.2]: "LTE; Mission Critical (MC) services over LTE; Part 5: Abstract test suite (ATS) (3GPP TS 36.579-5)";
- ETSI TS 136 579-6 [i.5]: "LTE; Mission Critical (MC) services over LTE; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-6)";
- ETSI TS 136 579-7 [4]: "LTE; Mission Critical (MC) services over LTE; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-7)";
- 3GPP TS 36.579-8 [31]: "Mission Critical (MC) services over LTE; Part 8: Mission Critical Video (MCVideo) Server Application conformance specification";
- **3GPP TS 36.579-9 [i.8]: "Mission Critical (MC) services over LTE; Part 9: Mission Critical Data (MCData) Server Application conformance specification".**

1 Scope

The present document specifies the protocol conformance testing for testing a MCData Server for compliance to the Mission Critical Data (MCData) over LTE protocol requirements defined by 3GPP. The present document addresses only MCData Server-Client communication scenarios. It does not cover e.g. MCData Server-EPS, MCData Server-SIP Core, etc. scenarios which involve interfaces which implementation may widely vary.

In particular the present specification contains:

- the overall test structure;
- the test configurations;
- the conformance requirement and reference to the core specifications;
- the test purposes; and
- a brief description of the test procedure, the specific test requirements and short message exchange table.

The present document is valid for MCData Servers implemented according to 3GPP releases starting from Release 14 up to the Release indicated on the cover page of the present document.

The following information relevant to testing specified in the present document could be found in accompanying specifications:

- default setting of the test parameters ETSI TS 136 579-1 [1] (3GPP TS 36.579-1);
- Implementation Conformance Statement (ICS) ETSI TS 136 579-4 [3] (3GPP TS 36.579-4) and Implementation extra Information for Testing (IXIT) ETSI TS 136 579-5 [i.2] (3GPP TS 36.579-5);
- the applicability of each test case ETSI TS 136 579-4 [3] (3GPP TS 36.579-4).

The present document does not specify the protocol conformance testing for the EPS (LTE) bearers which carry the MCData data sent or received by the MCData Server. The specification of such testing is out of the scope of RAN5.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 136 579-1: "LTE; Mission Critical (MC) services over LTE; Part 1: Common test environment (3GPP TS 36.579-1)".

[2] ETSI TS 136 579-3: "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3)".

[3] ETSI TS 136 579-4: "LTE; Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification (3GPP TS 36.579-4)".

[4] ETSI TS 136 579-7: "LTE; Mission Critical (MC) services over LTE; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-7)".

[5] ETSI TS 124 379: "LTE; Mission Critical Push To Talk (MCPTT) call control; Protocol specification (3GPP TS 24.379)".

[6] ETSI TS 124 481: "LTE; Mission Critical Services (MCS) group management; Protocol specification (3GPP TS 24.481)".

[7] ETSI TS 124 482: "LTE; Mission Critical Services (MCS) identity management; Protocol specification (3GPP TS 24.482)".

[8] ETSI TS 124 484: "LTE; Mission Critical Services (MCS) configuration management; Protocol specification (3GPP TS 24.484)".

[9] ETSI TS 136 509: "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Special conformance testing functions for User Equipment (UE) (3GPP TS 36.509)".

[10] ETSI TS 136 508: "LTE; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common test environments for User Equipment (UE) conformance testing (3GPP TS 36.508)".

[11] ETSI TS 133 310: "Universal Mobile Telecommunications System (UMTS); LTE; 5G; Network Domain Security (NDS); Authentication Framework (AF) (3GPP TS 33.310)".

[12] ETSI TS 133 180: "LTE; Security of the Mission Critical (MC) service (3GPP TS 33.180)".

[13] ETSI TS 124 282: "LTE; Mission Critical Data (MCData) signalling control; Protocol specification (3GPP TS 24.282)".

[14] IETF RFC 3326 (December 2002): "The Reason Header Field for the Session Initiation Protocol (SIP)".

[15] IETF RFC 4145 (September 2005): "TCP-Based Media Transport in the Session Description Protocol (SDP)".

[16] IETF RFC 3903 (October 2004): "Session Initiation Protocol (SIP) Extension for Event State Publication".

[17] IETF RFC 5366 (October 2008): "Conference Establishment Using Request-Contained Lists in the Session Initiation Protocol (SIP)".

[18] IETF RFC 2046 (November 1996): "Multipurpose Internet Mail Extensions (MIME) Part Two: Media Types".

[19] IETF RFC 5621 (September 2009): "Message Body Handling in the Session Initiation Protocol (SIP)".

[20] IETF RFC 7134 (March 2014): "The Management Policy of the Resource Priority Header (RPH) Registry Changed to "IETF Review"".

[21] IETF RFC 3261 (June 2002): "SIP: Session Initiation Protocol".

[22] IETF RFC 5031 (January 2008): "A Uniform Resource Name (URN) for Emergency and Other Well-Known Services".

[23] IETF RFC 4028 (April 2005): "Session Timers in the Session Initiation Protocol (SIP)".

[24] IETF RFC 4975 (September 2007): "The Message Session Relay Protocol (MSRP)".

[25] IETF RFC 5547 (May 2009): "A Session Description Protocol (SDP) Offer/Answer Mechanism to Enable File Transfer".

[26] IETF RFC 5322 (October 2008): "Internet Message Format".

[27] IETF RFC 2854 (June 2020): "The 'text/html' Media Type".

[28] W3C® Recommendation 24 December 1999: "HTML 4.01 Specification".

[29] IETF RFC 5245 (April 2010): "Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal for Offer/Answer Protocols".

[30] IETF RFC 6665 (July 2012): "SIP-Specific Event Notification".

[31] 3GPP TS 36.579-8: "Mission Critical (MC) services over LTE; Part 8: Mission Critical Video (MCVideo) Server Application conformance specification".

[32] ETSI TS 124 281: "LTE; Mission Critical Video (MCVideo) signalling control; Protocol specification (3GPP TS 24.281)".

[33] ETSI TS 126 114: "Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP Multimedia Subsystem (IMS); Multimedia Telephony; Media handling and interaction (3GPP TS 26.114)".

[34] IETF RFC 4867 (April 2007): "RTP Payload Format and File Storage Format for the Adaptive Multi-Rate (AMR) and Adaptive Multi-Rate Wideband (AMR-WB) Audio Codecs".

[35] IETF RFC 6335 (August 2011): "Internet Assigned Numbers Authority (IANA) Procedures for the Management of the Service Name and Transport Protocol Port Number Registry".

[36] IETF RFC 3556 (July 2003): "Session Description Protocol (SDP) Bandwidth Modifiers for RTP Control Protocol (RTCP) Bandwidth".

[37] ETSI TS 124 380: "LTE; Mission Critical Push To Talk (MCPTT) media plane control; Protocol specification (3GPP TS 24.380)".

[38] IETF RFC 4412 (February 2006): "Communications Resource Priority for the Session Initiation Protocol (SIP)".

[39] IETF RFC 8101 (March 2017): "IANA Registration of New Session Initiation Protocol (SIP) Resource-Priority Namespace for Mission Critical Push To Talk service".

[40] ETSI TS 124 582: "LTE; Mission Critical Data (MCData) media plane control; Protocol specification (3GPP TS 24.582)".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI TR 121 905: "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; Vocabulary for 3GPP Specifications (3GPP TR 21.905)".

[i.2] ETSI TS 136 579-5: "LTE; Mission Critical (MC) services over LTE; Part 5: Abstract test suite (ATS) (3GPP TS 36.579-5)".

[i.3] ETSI TS 124 483: "LTE; Mission Critical Services (MCS) Management Object (MO) (3GPP TS 24.483)".

[i.4] ETSI TS 136 579-2: "LTE; Mission Critical (MC) services over LTE; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-2)".

[i.5] ETSI TS 136 579-6: "LTE; Mission Critical (MC) services over LTE; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification (3GPP TS 36.579-6)".

[i.6] ETSI TS 122 282: "LTE; 5G; Mission Critical (MC) data (3GPP TS 22.282)".

[i.7] ETSI TS 123 282: "LTE; Functional architecture and information flows to support Mission Critical Data (MCData); Stage 2 (3GPP TS 23.282)".

[i.8] 3GPP TS 36.579-9: "Mission Critical (MC) services over LTE; Part 9: Mission Critical Data (MCData) Server Application conformance specification".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TR 121 905 [i.1] (3GPP TR 21.905) and the following apply.

NOTE: A term defined in the present document takes precedence over the definition of the same term, if any, in ETSI TR 121 905 [i.1] (3GPP TR 21.905).

For the purpose of the present document, the following terms given in ETSI TS 123 282 [i.7] (3GPP TS 23.282) apply:

- Auto-receive
- Conversation identifier
- Data stream
- FD disposition
- MCData client
- MCData group
- MCData group affiliation
- MCData group communication
- MCData group de-affiliation
- MCData ID
- MCData server
- MCData service
- MCData UE
- MCData user
- Reception control
- Reply identifier
- SDS data
- SDS disposition
- Standalone communication
- Transaction identifier

- Transmission control

For the purpose of the present document, the following terms given in ETSI TS 122 282 [i.6] (3GPP TS 22.282) apply:

- Conversation
- Conversation ID
- MCData Conversation Hang Time
- MCData System

For the purpose of the present document, the following terms given in ETSI TS 133 180 [12] (3GPP TS 33.180) apply:

- Client Server Key (CSK)
- Private Call Key (PCK)
- Signalling Protection Key (SPK)
- XML Protection Key (XPK)

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TR 121 905 [i.1] (3GPP TR 21.905) and the following apply:

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in ETSI TR 121 905 [i.1] (3GPP TR 21.905).

CSK	Client-Server Key
FD	File Distribution
GKTP	Group Key Transport Payload
ICE	Interactive Connectivity Establishment
ICS	Implementation Conformance Statement
IUT	Implementation Under Test
IXIT	Implementation eXtra Information for Testing
MC	Mission Critical
MCData group ID	MCData group Identity
MCData	Mission Critical Data
MCPTT	Mission Critical Push To Talk
MCS	Mission Critical Service
MCVideo	Mission Critical Video
MIME	Multipurpose Internet Mail Extensions
MSRP	Message Sending Relay Protocol
OIDC	OpenID Connect
PIDF	Presence Information Data Format
PLMN	Public Land Mobile Network
RTCP	Real Time Control Protocol
RTP	Real-time Transport Protocol
SDP	Session Description Protocol
SDS	Short Data Service
SIP	Session Initiation Protocol
SPK	Signalling Protection Key
SS	System Simulator
SSRC	Synchronization SouRCe
TP	Test Purpose
TP	Transmission Point

UE	User Equipment
URI	Uniform Resource Identifier
XCAP	eXtensible Markup Language (XML) Access Configuration Protocol
XDMC	XML Document Management Client
XPK	XML Protection Key

4 General

4.1 Test methodology

4.1.1 Testing of optional functions and procedures

Any function or procedure which is optional, may be subject to a conformance test if it is implemented in the MCData Server.

A declaration by the MCData Server supplier (to use the Implementation Conformance Statement (ICS) proforma specified in ETSI TS 136 579-4 [3] (3GPP TS 36.579-4)) is used to determine whether an optional function/procedure has been implemented.

4.1.2 Test interfaces and facilities

Detailed descriptions of the MCData test interfaces and special facilities for testing are provided in ETSI TS 136 509 [9] (3GPP TS 36.509).

4.2 Implicit testing

For some 3GPP MCData protocol features conformance is not verified explicitly in the present document. This does not imply that correct functioning of these features is not essential, but that these are implicitly tested to a sufficient degree in tests which are not explicitly dedicated to test the feature.

4.3 Repetition of tests

As a general rule, the test cases specified in the present document are highly reproducible and do not need to be repeated unless otherwise stated.

4.4 Handling of differences between conformance requirements in different releases of cores specifications

The conformance requirements which determine the scope of each test case are explicitly copy-pasted from relevant core specifications in the especially dedicated for this clause of each test with the title 'Conformance requirements'.

NOTE: When in the copy/pasted text there are references to other specifications the reference numbers will not match the reference numbers used in the present document. This approach has been taken in order to allow easy copy and then search for conformance requirements in those specifications.

When differences between conformance requirements in different releases of the cores specifications have impact on the Pre-test conditions, Test procedure sequence or/and the Specific message contents, the Conformance requirements related to different releases are specified separately with clear indication of the Release of the spec from which they were copied.

When there is no Release indicated for a conformance requirement text, this should be understood either as the Conformance requirements in the latest version of the spec with release = the TC Applicability release (which can be found in ETSI TS 136 579-4 [3] (3GPP TS 36.579-4), Table 4-1: Applicability of tests and additional information for testing, column 'Release'), or, as the Conformance requirements in the latest version of the spec of the release when the feature was introduced to the core specs.

4.5 Reference conditions

The reference environments used by all signalling and protocol tests is specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1). Where a test requires an environment that is different, this will be specified in the test itself.

For all test cases in the present document unless specified otherwise the condition MCDATA applies for all message contents.

4.6 Generic setup procedures

A set of basic generic procedures for MCData Client-Server communication are described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1). These procedures will be used in numerous test cases throughout the present document.

5 MCData Server - MCData Client Configuration

5.1 MCDATA Server - MCDATA Client / Configuration / Authentication / User Authorisation / UE Configuration / User Profile / Key generation

5.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCData client) activates an MCData application and requests MCData
initialisation }
    then { IUT (MCData Server) provides the initial UE configuration and performs MCData User
Authentication and provides id_token, access_token and refresh token to the successfully
authenticated user}
            }
```

(2)

```
with { IUT (MCData Server) having authenticated the SS-UE1 (MCData client) }
ensure that {
  when { the SS-UE1 (MCData Client) initiates key management authorization }
    then { IUT (MCData Server) provides identity management key material }
            }
```

(3)

```
with { IUT (MCData Server) having provided identity management key material }
ensure that {
  when { the SS-UE1 (MCData client) requests user service authorization }
    then { IUT (MCData Server) responds to the SS-UE1 (MCData Client) with SIP 200 (OK) messages }
            }
```

(4)

```
with { IUT (MCData Server) having provided service authorization }
ensure that {
  when { the SS-UE1 (MCData client) requests configuration management authorization}
    then { IUT (MCData Server) responds to the SIP SUBSCRIBE message with a SIP 200 (OK) message and
sends a SIP NOTIFY message containing the XCAP-URI of the documents and sends the MCData UE
Configuration Document and MCData User Profile Configuration Document MCData Service Configuration
Document via HTTP 200 (OK) messages in response to HTTP GET requests }
            }
```

(5)

```
with { IUT (MCData Server) having provided user configuration data }
ensure that {
  when { the SS-UE1 (MCData client) requests group management authorization }
    then { IUT (MCData Server) responds to the SS-UE1 (MCData Client) with SIP 200 (OK) messages and
sends the XCAP-URI of the Group documents via a SIP NOTIFY message and sends the Group Document
MCData UE Configuration document' via a HTTP 200 (OK) message in response to a HTTP GET request and
sends the group key transport payloads (GKTP) document via a SIP NOTIFY message }
           }
```

(6)

```
with { IUT (MCData Server) having provided all required configuration data }
ensure that {
  when { the SS-UE1 (MCData client) requests to refresh its service settings }
    then { IUT (MCData Server) responds to the SS-UE1 (MCData Client) with a SIP 200 (OK) message }
           }
```

5.1.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 482 [7] (3GPP TS 24.482) clauses 6.3.1, A.2.1.3, A.2.3, ETSI TS 124 484 [8] (3GPP TS 24.484) clauses 4.3.1, 4.3.4, 4.4, 6.3.13.2.3, 6.3.3.2.3, 6.3.3.2.1, 4.1.4, 10.2.1, 10.2.1A, 10.3.1, 10.3.1A, 10.4.1, 6.2.3, ETSI TS 124 481 [6] (3GPP TS 24.481) clauses 5.5, 5.2A, 6.2.4, 6.2.2, 6.3.3.2.3, 6.3.3.2.1, 6.3.13.2.2, 7.2.12.1, ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 4.1, 7.1, 7.3.1, 7.3.1A, 7.3.2, 7.3.3, 7.3.5, 7.3.6.1, 7.3.6.2, 7.3.7, ETSI TS 133 180 [12] (3GPP TS 33.180) clauses 5.1.3.1, 5.1.3.2.1, 5.1.3.2.2, 5.1.3.2.3, 5.2.1, 5.2.2, 5.8.1, 5.8.2. Unless otherwise stated these are Rel-14 requirements.

[ETSI TS 124 482 [7] (3GPP TS 24.482), clause 6.3.1]

"*Upon receipt of an OIDC Authentication Request message as specified in the OpenID Connect 1.0 [6] and IETF RFC 6749 [5] via a secure TLS tunnel between the identity management client and the authorisation endpoint of the IdM server, the IdM server:*

1) shall validate the received OIDC Authentication Request message as specified in the OpenID Connect 1.0 [6] and IETF RFC 6749 [5];

2) shall generate an HTTP 200 (OK) response according to IETF RFC 2616 [4] including form data to prompt the MC service user for their username and password credentials; and

NOTE 1: The username will be the MC service user's MC ID.

3) shall send the HTTP 200 (OK) response towards the IdM client.

Upon receipt of an HTTP POST request method from the IdM client containing the MC service user's username and password, the IdM server authenticates the MC service user and:

NOTE 2: Other methods of authentication can be used by the MC service provider and are not defined by the OIDC specifications. 3GPP TS 33.180 [17] has defined username and password as a mandatory authentication method to be supported for MC services, hence a procedure to realize that method is included here.

1) shall generate an OIDC Authentication Response message as specified in OpenID Connect 1.0 [6] and IETF RFC 6749 [5] with the following clarifications:

a) shall generate an HTTP 302 (FOUND) response according to IETF RFC 2616 [4]; and

b) shall include the required parameters including the authorization_code as specified in 3GPP TS 33.180 [17] in the query component of the redirection URI contained in the Location header field of the HTTP FOUND request method using the "application/x-www-form-urlencoded" format as specified in W3C.REC-html401-19991224 [7]; and

2) *shall send the HTTP 302 (FOUND) response towards the IdM client.*

Upon receipt of an OIDC Token Request message via a secure TLS tunnel established between the identity management client and the token endpoint of the IdM server, the IdM server:

1) *shall validate the OIDC Token Request message and if valid shall generate an OIDC Token Response message as specified in OpenID Connect 1.0 [6] and IETF RFC 6749 [5] with the following clarifications:*

   a) *shall generate an HTTP 200 (OK) response according to IETF RFC 2616 [4];*

   b) *shall based on the received MC ID obtained from the received user authentication credentials, determine the MC service ID of the MC service user;*

   c) *shall include an id_token, access_token and refresh_token and MC service ID as specified in 3GPP TS 33.180 [17]; and*

   d) *shall include the other required parameters as specified in OpenID Connect 1.0 [6] and IETF RFC 6749 [5]; and*

2) *shall send the HTTP 200 (OK) response towards the IdM client."*

[ETSI TS 124 482 [7] (3GPP TS 24.482), clause A.2.1.3]

"The HTTP client in the network entity is configured with the following parameters:

1) *a home HTTP proxy FQDN; and*

2) *a home HTTP proxy port.*

The HTTP client in the network entity shall send and receive all HTTP messages via the home HTTP proxy.

The HTTP client in the network entity shall insert an X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.109 [15] in the HTTP request and shall set X-3GPP-Asserted-Identity header field to the identity of the HTTP client in the network entity. The identity of the HTTP client in the network entity can be a public service identity, an MC service group ID, or an MC service ID."

[ETSI TS 124 482 [7] (3GPP TS 24.482), clause A.2.3]

"The HTTP server shall support the server role of IETF RFC 2616 [4].

Upon reception of an HTTP request:

1) *if the received HTTP request does not contain an Authorization header field with the "Bearer" authentication scheme and a bearer access token as specified in IETF RFC 6750 [14] and the received HTTP request does not contain an X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.109 [15], the HTTP server shall reject the request with HTTP 403 (Forbidden) response;*

2) *if the received HTTP request contains an Authorization header field with the "Bearer" authentication scheme and a bearer access token as specified in IETF RFC 6750 [14];*

   a) *the HTTP server shall validate the bearer access token as specified in IETF RFC 6750 [14]; and*

   b) *the HTTP server shall consider the MC service ID derived from the bearer access token as the identity of the sender of the HTTP request; and*

3) *if the received HTTP request does not contain an Authorization header field with the "Bearer" authentication scheme and a bearer access token as specified in IETF RFC 6750 [14] and the received HTTP request contains an X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.109 [15], the HTTP server shall consider the URI in the X-3GPP-Asserted-Identity header field as the identity of the sender of the HTTP request."*

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.3.1]

"The MCS server obtains the appropriate MCS service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCS from the configuration management server.

The MCS server obtains the MCS service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCS from the configuration management server.

The MCS server subscribes to the MCS service configuration document for each mission critical organisation that is provisioned that is supported by the MCS server using the procedure specified in subclause 6.3.13.2.3. How the MCS server is provisioned with the identities of the mission critical organisations is out of scope of the present document.

If the MCS service configuration document has been updated since the current version stored at the MCS server, then the MCS server will receive a SIP NOTIFY request containing an HTTPS URI of the MCS service configuration document. Retrieval by the MCS server, using the notified HTTPS URI, of the MCS service configuration document is performed as specified in subclause 6.3.3.2.3.

NOTE: The MCS server can be notified of changes to the MCS service management configuration document at any time while operating the MCS."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.3.4]

"The format of the MCData service configuration document downloaded to the MCData server is defined in subclause 10.4."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.4]

"The following applies to the configuration management server used for online configuration.

The configuration management server needs to convert the MCS UE initial configuration document received from an MCS administrator into an appropriate format for configuration of the MCS UE initial configuration MO.

If the MCS UE initial configuration MO contains a <default-user-profile> element that identifies an MCS user profile configuration document, the configuration management server needs to convert the identified MCS user profile configuration document received from an MCS administrator into an appropriate format for configuration of the MCS user profile configuration MO.

Once an MCS User Profile configuration document has been created or updated by the MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to store MCS user profile configuration document as the user profile in the MC service user database.

In order to download the MCS user profile configuration document to an MCS UE or to support an MCS UE updating the MCS user profile configuration document, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to obtain the MCS user profile from the MC service user database.

In order to be notified of changes to an MCS user profile configuration document(s) that have been subscribed to by an MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to be notified of changes to the MCS user profile stored in the MC service user database.

In order to delete the MCS user profile when requested by an MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to delete the MCS user profile from the MC service user database.

NOTE: The configuration management server and group management server functionality for offline configuration is out of scope of the present document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.13.2.3]

"In order to subscribe to an MCS service configuration document, an MCS server shall send an initial SIP SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [22] and IETF RFC 5875 [11]. In the initial SIP SUBSCRIBE request, MCS server:

a) shall set the Request URI to a SIP URI containing:

1) the base URI being equal to the public service identity of the CMS configured in the MCS server; and

2) the "auid" parameter set to the application usage identifying th MCS service configuration document;

b) shall include a P-Asserted-Identity header field containing the public service identity of the MCS server;

c) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [22]), in a P-Asserted-Service header field according to IETF RFC 6050 [23]; and

d) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field.

Upon receiving a SIP NOTIFY request associated with a subscription created as result of the sent initial SIP SUBSCRIBE request, the MCS server shall handle the SIP NOTIFY request according to IETF RFC 5875 [11].

In order to re-subscribe to notification of changes to an MCS service configuration document, an MCS server shall send a SIP re-SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [22] and IETF RFC 5875 [11]. In the SIP re-SUBSCRIBE request, MCS server:

a) shall set the Request URI to a SIP URI containing:

1) the base URI being equal to the public service identity of the CMS configured in the MCS server; and

2) the "auid" parameter set to the application usage identifying an MCS service configuration document; and

b) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.3.2.3]

"In order to retrieve a configuration management document via the CSC-5 reference point, an MCS Server shall perform the procedures in subclause 6.3.3.2.1 specified for GC. The MCS server shall set the Request-URI of the HTTP GET request to identify the XML document based on configuration and include the "auid" as per the appropriate application usage."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.3.2.1]

"In order to retrieve a configuration management document, a GC shall send an HTTP GET request with the Request URI that references the document to be updated to the network according to procedures specified in IETF RFC 4825 [14] "Retrieve a Document"."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.1.4]

"An MCData service administrator can, using an MCData UE configure the:

- *MCData UE configuration document;*
- *MCData user profile configuration document;*
- *MCData service configuration document; and*
- *MCData related group configuration data in the MCS group document.*

The format of the MCData UE configuration document is defined in subclause 10.2.

The format of the MCData user profile configuration document is defined in subclause 10.3.

The format of the MCData service configuration document is defined in subclause 10.4."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.2.1]

"The MCData UE configuration document is specified in this subclause. The MCData UE configuration document content is based on requirements of Annex A.2 of 3GPP TS 23.282 [24], in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData UE configuration document in the MCData service is described in 3GPP TS 24.282 [25] and 3GPP TS 24.582 [26]. The schema definition is provided in subclause 10.2.2.3. An MCData UE configuration document may apply to all MCData UEs of a mission critical organization or apply to specific MCData UEs identified in the <mcdata-UE-id> element. If there is no <mcdata-UE-id> element in the MCData UE configuration document, then by default the MCData UE configuration document applies to all MCData UEs of the mission critical organization that are not specifically identified in the <mcdata-UE-id> element of another MCData UE configuration document of the mission critical organization. Each MCData UE of a mission critical organization is configured with an MCData UE configuration document that is identified by the instance ID of the MCData UE.

MCData UE configuration documents of a MCData user are contained as "XDM collections" in the user's directory of the users tree, in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. In this case, the term "user" in the XCAP sense refers to the MCData ID, as the user has been already authenticated. A MCData UE configuration document corresponding to a specific MCData UE the MCData user has used to authenticate and is authorised to use the MCData service with is placed in the user directory of the MCData user.

The MCData UE configuration document acts as a template for the CMS to generate UE configuration documents that are downloaded to specific MCData UEs. The MCData UE configuration document that acts as a template is referred to as a "master MCData UE configuration document". The master MCData UE configuration document is stored in the users tree of that MCData system administrator. The master MCData UE configuration document does not directly apply to a specific MCData UE, but instead acts as template that the CMS uses to populate the MCData UE configuration documents of MCData UEs identified by elements of the <MCData-UE-id> element. For MCData UE configuration documents that correspond to a specific MCData UE, the name of the MCData UE configuration document is created from a value defined by the corresponding element that identifies the MCData UE within the <MCData-UE-id> element. For a master MCData UE configuration documents that does not contain a <MCData-UE-id> element, the name of the MCData UE configuration document stored in the user directory is "DEFAULT-MCData-UE.xml"."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.2.1A]

"The MCData UE configuration document is accessed using the same XCAP URI, regardless of whether the MCData UE has a specific MCData UE configuration document configured or the master MCData UE configuration document applies. The CMS shall generate the UE's MCData UE configuration document from the master MCData UE configuration document if the MCData administrator did not provision a specific MCData UE configuration document. In this generated document, the <MCData-UE-id> element shall be set to the MCData client's UE ID. The UE's MCData UE configuration document shall always be stored with the filename corresponding to the UE's MCData UE ID under the user's directory of the users tree.

The XCAP URI used by the MCData client to access the UE's MCData UE configuration document shall be:

- *CMSXCAPROOTURI/org.3gpp.mcdata.ue-config/users/sip:MCDATAID/MCSUEID."*

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.3.1]

"The MCData user profile configuration document is specified in this subclause. The MCData user profile configuration document content is based on requirements of Annex A.3 of 3GPP TS 23.282 [24], and structure and procedures of OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData user profile in the MCData service is described in 3GPP TS 24.282 [25]. The schema definition is provided in subclause 10.3.2.

MCData user profile documents are "XDM collections" in the user's directory of the users tree, in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. In this case, the term "user" in the XCAP sense refers to the MCData ID, as the user has been already authenticated.

The name of the MCData user profile document matches the value of the <ProfileName> element in the MCData user profile document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.3.1A]

"The XCAP URI used by the MCData client to access the MCData user's MCData user profile documents shall be:

CMSXCAPROOTURI/org.3gpp.mcdata.user-profile/users/sip:MCDATAID/mcdata-user-profile-INDEX.xml

Where INDEX is the index of the MCData user profile as defined in subclause 10.3.2.8."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.4.1]

"The MCData service configuration document is specified in this subclause. The MCData service configuration document content is based on requirements of Annex A.5 of 3GPP TS 23.282 [24], and structure and procedures of OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData service configuration in the MCData service is described in 3GPP TS 24.282[25] and 3GPP TS 24.582 [26]. The schema definition is provided in subclause 10.4.2.3. Each mission critical organization is configured with an MCData service configuration document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.2.3]

"The MCS server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.482 [6]."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 5.5]

"To be compliant with the procedures in the present document, an MCData server shall support role of an MCS server as specified in subclause 5.2A."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 5.2A]

"To be compliant with the procedures in the present document, an MCS server:

- *shall support the role of XCAP client as specified in IETF RFC 4825 [22];*
- *shall support the role of XDMC as specified in OMA OMA-TS-XDM_Group-V1_1_1 [3];*
- *shall support the procedure in subclause 6.2.4;*
- *shall support the procedure in subclause 6.3.3.2.3; and*
- *shall support the procedure in subclause 6.3.13.2.2."*

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.2.4]

"The MCS server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.482 [10].

The MCS server shall perform the procedures in subclause 6.2.2 specified for GC."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.2.2]

"In order to address an existing group document defining a group ID known by GC, the GC shall set the Request-URI of an HTTP request to a XCAP URI identifying a group document addressed by a group ID as described in subclause 7.2.10.2, where the group ID is set to the group ID known by GC and where the XCAP root URI is the XCAP root URI configured in the GC."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.3.2.3]

"In order to retrieve a group document, an MCS server shall perform the procedures in subclause 6.3.3.2.1 specified for a GC."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.3.2.1]

"In order to retrieve a group document, a GC shall send an HTTP GET request with the Request URI that references the document to be retrieved to the network according to procedures specified in IETF RFC 4825 [22] "Fetch a Document"."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.13.2.2]

"In order to subscribe to notification of changes of:

a) one or more MCS group documents of MCS groups identified by MCS group IDs;

an MCS server shall send an initial SIP SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [12] and IETF RFC 5875 [13]. In the initial SIP SUBSCRIBE request, MCS server:

a) shall include an application/resource-lists+xml MIME body. In the application/resource-lists+xml MIME body, the MCS server shall include one <entry> element for each document or element to be subscribed to, such that the "uri" attribute of the <entry> element:

1) contains a relative path reference:

A) with the base URI being equal to the XCAP root URI configured in the MCS server; and

B) identifying a group document addressed by a group ID as described in subclause 7.2.10.2 where the group ID is set to the MCS group ID;

b) shall set the Request-URI to the configured public service identity for performing subscription proxy function of the GMS;

c) shall include a P-Asserted-Identity header field containing the public service identity of the MCS server;

e) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [12]), in a P-Asserted-Service header field according to IETF RFC 6050 [14]; and

f) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field.

Upon receiving a SIP NOTIFY request associated with a subscription created as result of the sent initial SIP SUBSCRIBE request, the MCS server shall handle the SIP NOTIFY request according to IETF RFC 5875 [13].

In order to re-subscribe to notification of changes of a modified list of:

a) one or more MCS group documents of MCS groups identified by MCS group IDs,

the MCS server shall send a SIP re-SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [12] and IETF RFC 5875 [13]. In the SIP re-SUBSCRIBE request, MCS server:

a) shall include an application/resource-lists+xml MIME body. In the application/resource-lists+xml MIME body, the MCS server shall include one <entry> element for each document or element to be subscribed to, such that the "uri" attribute of the <entry> element:

1) contains a relative path reference:

A) with the base URI being equal to the XCAP root URI configured in the MCS server; and

B) identifying a group document addressed by a group ID as described in subclause 7.2.10.2 where the group ID is set to the MCS group ID; and

b) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 7.2.12.1]

"Authorization policies are described in the OMA OMA-TS-XDM_Group-V1_1_1 [3] "Authorization policies".

An authenticated identity is authorized to read the <group> element of an MCS group document.

An authenticated identity is authorized to read a <list-service> element of the <group> element of the MCS group document and any descendant elements of the <list-service> element:

a) except a <list> element of the <list-service> element; and

b) except descendant elements of the <list> element of the <list-service> element;

if the authenticated identity is a member of an MCS group defined by the <list-service> element.

An authenticated identity is authorized to read a <list> element of a <list-service> element of the <group> element of an MCS group document and any descendant elements of the <list> element if:

a) the authenticated identity is a member of MCS group defined by the <list-service> element; and

b) the MCS group document contains a <rule> element:

1) with the <conditions> element evaluating to true for the authenticated identity or without the <conditions> element; and

2) with the <actions> element containing the <on-network-allow-getting-member-list> element.

An authenticated identity is authorized to read an <entry> element of a <list> element of a <list-service> element of the <group> element of the MCS group document and any descendant elements of the <entry> element if:

a) the "uri" attribute of the <entry> element contains the authenticated identity;

b) the "uri" attribute of the <mcvideo-mcvideo-id> element of the <entry> element contains the authenticated identity; or

c) the "uri" attribute of the <mcdata-mcdata-id> element of the <entry> element contains the authenticated identity.

An authenticated identity is authorized to read an MCS group document if the authenticated identity is listed in the authorized MCS server list specified in subclause 6.2.5.1.

An authenticated identity is authorized to create, read, modify and delete an MCS group document if the MCS group document is stored:

a) in users tree of the XUI equal to the authenticated identity; or

b) in users tree of the another XUI, subject to MCS service provider policy. The MCS service provider policy can specified e.g. using an appropriate access permissions document specified in OMA OMA-TS-XDM_Core-V2_1 [2], stored in the user's tree of the other XUI;

and unless precluded by other statement of this subclause.

None is authorized to create, delete, and modify a <on-network-temporary> element of a <list-service> element of the <group> element of an existing MCS group document.

None is authorized to delete a group document containing a <on-network-regrouped> element of a <list-service> element of the <group> element of an MCS group document.

An authenticated identity is authorized to create and delete a <on-network-regrouped> element in the <list-service> element of the <group> element of an existing MCS group document if the authenticated identity is in authorized GMS list specified in subclause 6.2.5.1.

An authenticated identity is authorized to subscribe to notification of changes of an MCS group document if:

1) the authenticated identity is a member of an MCS group defined by a <list-service> element of the <group> element of the MCS group document;

2) the authenticated identity is listed in the authorized MCS server list specified in subclause 6.2.5.1; or

3) the authenticated identity is listed in the authorized GMS list specified in subclause 6.2.5.1 and the subscription is on behalf of a constituent MCS group of the MCS group defined by the MCS group document."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 4.1]

"…

The present document provides the signalling control protocol enhancements to support the MCData architectural procedures specified in 3GPP TS 23.282 [2].

For on-network communications, the present document makes use of the existing IMS procedures specified in 3GPP TS 24.229 [5].

…

The MCData procedures provided by the present document refer to:

- *the media plane procedures defined in 3GPP TS 24.582 [15];*
- *the group management procedures defined in 3GPP TS 24.481 [11];*
- *the identity management procedures defined in 3GPP TS 24.482 [24]; and*
- *the security procedures defined in 3GPP TS 33.180 [26].*

The MCData procedures provided by the present document access the configuration parameters provided by 3GPP TS 24.483 [42] and 3GPP TS 24.484 [12].

The following procedures are provided within this document:

- *common procedures are specified in clause 6;*
- *procedures for registration in the IM CN subsystem and service authorisation are specified in clause 7;*
- *procedures for affiliation are specified in clause 8;*

…

The MCData UE primarily obtains access to the MCData service via E-UTRAN, using the procedures defined in 3GPP TS 24.301 [43]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 7.1]

"This clause describes the procedures for SIP registration and MCData service authorization for the MCData client and the MCData service. The MCData UE can use SIP REGISTER or SIP PUBLISH for MCData service settings to perform service authorization for MCData. The decision which method to use is based on implementation and on availability of an access-token received as outcome of the user authentication procedure as described in 3GPP TS 24.482 [24].

If another MC service client (e.g. MCPTT, MCVideo) is operating at the same time on the same MC UE as the MCData client, then the MCData client shares the same SIP registration as the other MC service clients. The SIP REGISTER procedures in this clause are combined with the SIP REGISTER procedures for the other operating MC service clients to create a single SIP REGISTER request. If other MC service clients are already operating when the MCData client registers then a re-registration is performed containing the parameters for the other operating MC services.

Although the access-token can be the same for the MCData service as for other MC services when performing service authorization for MCData along with other MC services using SIP REGISTER multipart MIME bodies for each MC service are included in the SIP REGISTER request. The MCData server can therefore receive multipart MIME bodies in the SIP REGISTER request. Multiple contact addresses (one per MC service client) can be included in a SIP REGISTER request provided they all contain the same IP address and port number (see 3GPP TS 24.229 [5] for further details of including multiple contact addresses in a single SIP REGISTER request).

If the MCData client logs off from the MCData service but other MC service clients are to remain registered the MC UE performs a re-registration as specified in 3GPP TS 24.229 [5] without the supported g.3gpp.mcdata media feature tags and the g.3gpp.icsi-ref media feature tags containing the values of the supported MCData service ICSIs in the Contact header field of the SIP REGISTER request but with the parameters for the remaining operating MC service clients."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 7.3.1]

"The MCData server obtains information that it needs to implement service authorization specific requirements from:

a) any received third-party SIP REGISTER request (e.g. including information contained in the body of the third-party SIP REGISTER request) as specified in 3GPP TS 24.229 [5]. The body will carry the SIP REGISTER request as sent by the MCData client and may contain information needed for service authorization; or

b) any received SIP PUBLISH request for MCData server settings containing the g.3gpp.mcdata.sds media feature tag along with the "require" and "explicit" header field parameters. The body of the SIP PUBLISH request will contain information needed for service authorization."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 7.3.1A]

"When the MCData server receives a SIP REGISTER request sent from the MCData client contained within a message/sip MIME body of a received third-party SIP REGISTER request or a SIP PUBLISH request, it first determines whether XML MIME bodies included in the request are integrity protected. If XML MIME bodies are integrity protected the MCData server validates the signature of each of the XML MIME bodies. If the integrity protection check(s) pass or the XML MIME bodies are not integrity protected, the MCData server then determines whether the content in specific XML elements is confidentiality protected. If XML content is confidentiality protected, the MCData server decrypts the protected content.

Upon receiving:

- *a SIP REGISTER request containing an application/vnd.3gpp.mcdata-info+xml MIME body within a message/sip MIME body of the SIP REGISTER request sent from the MCData client; or*
- *a SIP PUBLISH request containing an application/vnd.3gpp.mcdata-info+xml MIME body and an application/poc-settings+xml MIME body;*

the MCData server:

1) *shall determine if integrity protection has been applied to XML MIME bodies in the SIP request by following the procedures in subclause 6.5.3.4.1 for each XML MIME body;*
2) *if integrity protection has been applied, shall use the keying data described in subclause 6.5.3.2 and the procedures described in subclause 6.5.3.4.2 to verify the integrity of each of the XML MIME bodies; and*
3) *if all integrity protection checks succeed, shall continue with the remaining steps of this subclause.*

Upon receiving:

- *a SIP REGISTER request containing an application/vnd.3gpp.mcdata-info+xml MIME body with an <mcdata-access-token> element and an <mcdata-client-id> element within a message/sip MIME body of the SIP REGISTER request sent from the MCData client; or*
- *a SIP PUBLISH request containing an application/vnd.3gpp.mcdata-info+xml MIME body with an <mcdata-access-token> element and an <mcdata-client-id> element, and an application/poc-settings+xml MIME body;*

the MCData server:

1) *shall determine if confidentiality protection has been applied to the <mcdata-access-token> element and the <mcdata-client-id> element in the application/vnd.3gpp.mcdata-info+xml MIME body, by following the procedures in subclause 6.5.2.4.1;*
2) *if confidentiality protection has been applied to the <mcdata-access-token> element and <mcdata-client-id> element:*
    a) *shall use the keying information received in the MIKEY-SAKKE I_MESSAGE as specified in 3GPP TS 33.180 [26], along with the procedures described in subclause 6.5.2.4.2 to:*
        i) *decrypt the received access token in the <mcdata-access-token> element in the application/vnd.3gpp.mcdata-info+xml MIME body; and*
        ii) *decrypt the received MCData client ID in the <mcdata-client-id> element in the application/vnd.3gpp.mcdata-info+xml MIME body;*
    b) *if the decryption procedure succeeds, shall identify the MCData ID and the MCData client ID from the decrypted values; and*
    c) *if the decryption procedure fails, shall determine that confidentiality protection has not been successful;*
3) *if confidentiality protection has been applied to only one of the <mcdata-access-token> element or the <mcdata-client-id> element:*
    a) *shall determine that confidentiality protection has not been successful;*
4) *if confidentiality protection has not been applied:*
    a) *shall identify the MCData ID from <mcdata-access-token> element received in the application/vnd.3gpp.mcdata-info+xml MIME body; and*
    b) *shall identify the MCData client ID from the <mcdata-client-id> element received in the application/vnd.3gpp.mcdata-info+xml MIME body.*

Upon receiving a SIP PUBLISH request containing an application/vnd.3gpp.mcdata-info+xml MIME body with an <mcdata-request-uri> element, an <mcdata-client-id> element, and an application/poc-settings+xml MIME body, the MCData server:

1) shall determine if confidentiality protection has been applied to the <mcdata-request-uri> element and the <mcdata-client-id> element in the application/vnd.3gpp.mcdata-info+xml MIME body by following the procedures in subclause 6.5.2.4.1;

2) if confidentiality protection has been applied to the <mcdata-request-uri> element and the <mcdata-client-id> element:

a) shall use the keying information described in subclause 6.5.2.2 along with the procedures described in subclause 6.5.2.4.2 to:

i) decrypt the received encrypted MCData ID in the <mcdata-request-uri> element in the application/vnd.3gpp.mcdata-info+xml MIME body; and

ii) decrypt the received encrypted MCData client ID in the <mcdata-client-id> element in the application/vnd.3gpp.mcdata-info+xml MIME body;

b) if all decryption procedures succeed, shall identify the MCData ID and MCData client ID from the decrypted values; and

c) if the decryption procedure fails, shall determine that confidentiality protection has not been successful;

3) if confidentiality protection has been applied to only one of the <mcdata-request-uri> element or <mcdata-client-id> element:

a) shall determine that confidentiality protection has not been successful;

4) if confidentiality protection has not been applied:

a) shall identify the MCData ID from the contents of the <mcdata-request-uri> element in the application/vnd.3gpp.mcdata-info+xml MIME body; and

b) shall identify the MCData client ID from the <mcdata-client-id> element received in the application/vnd.3gpp.mcdata-info+xml MIME body."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 7.3.2]

"The MCData server shall support obtaining service authorization specific information from the SIP REGISTER request sent from the MCData client and included in the body of a third-party SIP REGISTER request.

NOTE 1: 3GPP TS 24.229 [5] defines how based on initial filter criteria the SIP REGISTER request sent from the UE is included in the body of the third-party SIP REGISTER request.

Upon receiving a third party SIP REGISTER request with a message/sip MIME body containing the SIP REGISTER request sent from the MCData client containing an application/vnd.3gpp.mcdata-info+xml MIME body with an <mcdata-access-token> element and an <mcdata-client-id> element within a message/sip MIME body of the SIP REGISTER request sent from the MCData client, the MCData server:

1) shall identify the IMS public user identity from the third-party SIP REGISTER request;

2) shall identify the MCData ID from the SIP REGISTER request sent from the MCData client and included in the message/sip MIME body of the third-party SIP REGISTER request by following the procedures in subclause 7.3.1A;

2A) shall check if the number of maximum simultaneous authorizations supported for the MCData user as specified in the <max-simultaneous-authorizations> element of the <anyExt> element contained in the <OnNetwork> element of the MCData service configuration document (see the service configuration document in 3GPP TS 24.484 [12]) has been reached. If reached, the MCData server shall not continue with the rest of the steps in this clause;

3) shall perform service authorization for the identified MCData ID as described in 3GPP TS 33.180 [26];

4) if service authorization was successful, shall bind the MCData ID and the MCData client ID to the IMS public user identity;

4A) if service authorization was successful and if the service authorization request was from an MCData user who is previously MCData service authorized on another MCData client (as determined by a comparison of the received MCData client ID with the MCData client ID of existing bindings), keep the current bindings and create a new binding between the MCData ID, the MCData client ID and the IMS public user identity;

NOTE 2: The MCData server will store the binding MCData ID, MCData client ID, IMS public user identity and an identifier addressing the MCData server in an external database.

5) if a Resource-Share header field with the value "supported" is contained in the "message/sip" MIME body of the third-party REGISTER request, shall bind the MCData ID and the MCData client ID to the identity of the MCData UE identified by the "+g.3gpp.registration-token" header field parameter in the Contact header field of the incoming third-party REGISTER request; and

6) if more than one binding exists for the MCData ID, shall include in the SIP 200 (OK) response an application/vnd.3gpp.mcdata-info+xml MIME body as specified in annex D.1 with a <multiple-devices-ind> element set to the value "true"."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 7.3.3]

"The MCData server shall support obtaining service authorization specific information from a SIP PUBLISH request for MCData server settings.

Upon receiving a SIP PUBLISH request containing:

1) an Event header field set to the "poc-settings" value;

2) an application/poc-settings+xml MIME body; and

3) an application/vnd.3gpp.mcdata-info+xml MIME body containing an <mcdata-access-token> element and an <mcdata-client-id> element;

the MCData server:

1) shall identify the IMS public user identity from the P-Asserted-Identity header field;

2) shall perform the procedures in subclause 7.3.1A;

…

4) shall perform service authorization for the identified MCData ID as described in 3GPP TS 33.180 [26];

5) if service authorization was successful:

a) shall bind the MCData ID and MCData client ID to the IMS public user identity;

b) if the service authorization request was from an MCData user who is previously MCData service authorized on another MCData client (as determined by a comparison of the received MCData client ID with the MCData client ID of existing bindings), keep the current bindings and create a new binding between the MCData ID, MCData client ID and the IMS public user identity; and

c) if a Resource-Share header field with the value "supported" was included in the "message/sip" MIME body of the third-party REGISTER request, shall bind the MCData ID and MCData client ID to the identity of the MCData UE identified by the "+g.3gpp.registration-token" header field parameter in the Contact header field of the third-party REGISTER request that contained this IMS public user identity;

NOTE 1: The MCData server will store the binding MCData ID, MCData client ID, IMS public user identity and an identifier addressing the MCData server in an external database.

…

7) shall process the SIP PUBLISH request according to rules and procedures of IETF RFC 3903 [34] and if processing of the SIP request was not successful, do not continue with the rest of the steps;

8) shall cache the received MCData service settings until the MCData service settings expiration timer expires;

9) shall send a SIP 200 (OK) response according 3GPP TS 24.229 [5] with:

a) if more than one binding exists for the MCData ID, an application/vnd.3gpp.mcdata-info+xml MIME body as specified in annex D.1 with a <multiple-devices-ind> element set to the value "true";

10) shall download the MCData user profile from the MCData user database as defined in 3GPP TS 29.283 [37] if not already stored at the MCData server and use the <selected-user-profile-index> element of the poc-settings event package if included to identify the active MCData user profile for the MCData client;

NOTE 2: If the <selected-user-profile-index> element of the poc-settings event package is included then only that MCData user profile is needed to be downloaded from the MCData user database.

11) if there is no <selected-user-profile-index> element included in the poc-settings event package then if multiple MCData user profiles are stored at the MCData server or downloaded for the MCData user from the MCData user database, shall determine the pre-selected MCData user profile to be used as the active MCData user profile by identifying the MCData user profile (see the MCData user profile document in 3GPP TS 24.484 [12]) in the collection of MCData user profiles that contains a <Pre-selected-indication> element; and

NOTE 3: If only one MCData user profile is stored at the MCData server or only one MCData user profile is downloaded from the MCData user database, then by default this MCData user profile is the pre-selected MCData user profile.

12) if an <ImplicitAffiliations> element is contained in the <OnNetwork> element of the MCData user profile document with one or more <entry> elements containing an MCData group ID (see the MCData user profile document in 3GPP TS 24.484 [12]) for the served MCData ID, shall perform implicit affiliation as specified in subclause 8.3.2.15."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 7.3.5]

"Upon receiving a SIP PUBLISH request containing:

1) an Event header field set to the "poc-settings" value; and

2) an Expires header field set to 0;

the MCData server:

1) shall identify the IMS public user identity from the P-Asserted-Identity header field;

2) shall process the SIP PUBLISH request according to rules and procedures of IETF RFC 3903 [34] and if processing of the SIP request was successful, continue with the rest of the steps;

3) shall remove the MCData service settings;

4) shall remove the binding between the MCData ID and public user identity; and

5) shall send a SIP 200 (OK) response according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 7.3.6.1]

"Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the participating MCData function of the served MCData user;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the<mcdata-request-uri> element which identifies an MCData ID served by the MCData server;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata,sds" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7]; and

3) the Event header field of the SIP SUBSCRIBE request contains the 'poc-settings' event type.

the MCData server:

1) *shall identify the served MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;*

2) *if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the participating MCData function serving the MCData user, shall identify the originating MCData ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;*

3) *if the originating MCData ID is different than the served MCData ID, shall send a 403 (Forbidden) response and shall not continue with the rest of the steps; and*

4) *shall generate a 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [5], IETF RFC 6665 [36] and IETF RFC 4354 [35].*

For the duration of the subscription, the MCData server shall notify subscriber about changes of the MCData service settings of the subscribed MCData user, as described in subclause 7.3.6.2."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 7.3.6.2]

"In order to notify the subscriber about changes of the MCData service settings of the subscribed MCData client of the subscribed MCData user, the MCData server:

1) *shall generate an application/poc-settings+xml MIME body as defined in 3GPP TS 24.379 [10] containing:*

   a) *the <selected-user-profile-index> element identifying the active MCData user profile; and*

2) *send a SIP NOTIFY request according to 3GPP TS 24.229 [5], IETF RFC 6665 [36] and IETF RFC 4354 [35] with the constructed application/poc-settings+xml MIME body."*

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 7.3.7]

"If confidentiality protection is enabled as specified in subclause 6.5.2.3.1, and if the participating MCData function received a Client Server Key (CSK) within a SIP REGISTER request for service authorisation or SIP PUBLISH request for service authorisation, the participating MCData function may decide to update the CSK. In this case, the participating MCData function shall perform a key download procedure for the CSK. The participating MCData function:

1) *shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];*

2) *shall set the Request-URI to the URI received in the To header field in the third-party SIP REGISTER request;*

3) *shall include an Accept-Contact header field with the g.3gpp.icsi-ref media-feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata" along with parameters "require" and "explicit" according to IETF RFC 3841 [8];*

4) *shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcdata";*

5) *shall include an application/mikey MIME body containing the algorithm to be used to protect the MCData signalling fields, the CSK-ID and the CSK encrypted within a MIKEY message to the MC client as specified in subclause 9.2.1 of 3GPP TS 33.180 [26] in the body of the SIP MESSAGE request; and*

6) *shall send the SIP MESSAGE request towards the MCData client according to 3GPP TS 24.229 [5]."*

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.1.3.1]

"…

MCX User Service Authorization is the function that validates whether or not a MCX user has the authority to access certain MCX services. In order to gain access to MCX services, the MCX client in the UE presents an access token (acquired during user authentication as described in subclause 5.1.2) to each service of interest (i.e. Key Management, MCX server, Configuration Management, Group Management, etc.). If the access token is valid, then the user is granted the use of that service. Figure 5.1.3.1-1 shows the flow for user authorization which covers key management authorization, MCX user service authorization, configuration management authorization, and group management authorization.

NOTE: All HTTP traffic between the UE and HTTP proxy, and all HTTP traffic between the UE and KMS (if not going through the HTTP proxy) is protected using HTTPS.

For key management authorization, the KM client in the UE presents an access token to the KMS over HTTP. The access token shall be scoped for key management services as defined in annex B.4.2.2. The KMS validates the access token and if successful, provides one or more sets of user specific key material back to the UE KM client based on the MC service ID(s) present in the access token (MCPTT ID, MCVideo ID and/or MCData ID). User specific key material includes identity based key information for media and signalling protection…. This key management authorisation may be repeated for each KM service the user is authorised to use (MCPTT, MCVideo, MCData). In order to secure the transfer of user specific key material from the KMS to the KM client when using the TrK and InK, the KM client includes the TrK-ID and the InK-ID in the key management authorization request.

…

For MCData user service authorization, the MCData client in the UE presents an access token to the MCData server over SIP. The access token shall be scoped for MCData services as defined in annex B.4.2.2. The MCData server validates the access token and if successful, authorizes the user for full MCData services and sends an acknowledgement back to the MCData client. The MCData server then maps and maintains the IMPU to MCData ID association. The MCData ID to IMPU association shall only be known to the application layer. The SIP message used to convey the access token from the MCData client to the MCData server may be either a SIP REGISTER or SIP PUBLISH message.

The UE can now perform configuration management authorization and download the user profile for the service(s) (MCPTT, MCVideo, MCData). Following the flow described in subclause 10.1.4.3 of 3GPP TS 23.280 [36] "MC service user obtains the MC service user profile(s) from the network", the Configuration Management (CM) client in the UE sends an access token in the user profile query to the Configuration Management server over HTTP. The access token shall be scoped for configuration management services as defined in annex B.4.2.2. The CM server receives the request and validates the access token, and if valid, the CM server uses the identity from the access token (MCPTT ID, MCVideo ID, MCData ID) to obtain the user profile from the MCX user database. The CM server then sends the user profile back to the CM client over HTTP. This configuration management authorisation may be repeated for each CM service the user is authorised to use (MCPTT, MCVideo, MCData).

Upon receiving each user profile, the Group Management (GM) client in the UE can now perform group management authorization. The GM client obtains the user's group membership information from the user profile, and following the flow shown in clause 10.1.5.2 of 3GPP TS 23.280 [36] "Retrieve group configurations at the group management client", the Group Management (GM) client in the UE sends an access token in the Get group configuration request to the host GM server of the group membership over HTTP. The access token shall be scoped for group management services as defined in annex B.4.2.2. The GM server validates the access token, and if valid, completes the flow. As part of group management authorization, group key information is provided as per subclause 5.7 of the present document. This group management authorisation may be repeated for each GM service the user is authorised to use (MCPTT, MCVideo, MCData).

…"

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.1.3.2.1]

"Depending on implementation, MCX user service authorization may be performed by sending the access token to the MCX server over the SIP-1 and SIP-2 reference points using either a SIP REGISTER message or a SIP PUBLISH message. Clause 5.1.3.2.2 describes how to use the SIP REGISTER message to transport the access token to the MCX server and clause 5.1.3.2.3 describes how to use the SIP PUBLISH message to transport the access token to the MCX server.

…

If an access token is available before SIP registration, or if the UE becomes de-registered and a SIP re-registration is required, the SIP REGISTER message may include the access token without requiring the user to re-authenticate.

The access token may be sent over SIP to the MCX server to re-bind an IMPU and MC service ID (MCPTT ID, MCVideo ID or MCData ID) if either have changed (e.g. IMPU is different due to SIP deregistration/SIP re-registration, or user logs out and another user logs onto the same UE)."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.1.3.2.2]

"The use of a SIP REGISTER message to provide the access token to the MCX server is shown in figure 5.1.3.2.2-1. The inclusion of an access token in any particular SIP REGISTER message is optional.

…

Step 5 of figure 5.1.3.2.2-1 shows the access token message passed to the SIP core in a SIP REGISTER. Upon successful SIP authentication, the SIP core forwards the access token to the MCX server in the third part registration request message (Step 9).

In Steps 9 through 11, the MCX server receives the third part registration request message, validates the access token, binds the IMPU and MC service ID (MCPTT ID, MCVideo ID or MCData ID) if the access token is valid, and responds to the 3rd party registration message."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.1.3.2.3]

"The use of a SIP PUBLISH message to provide the access token to the MCX server is shown in figure 5.1.3.2.3-1. The inclusion of an access token in any particular SIP PUBLISH message is optional.

…

As shown in Step 1 of figure 5.1.3.2.3-1, the SIP PUBLISH message carries the access token through the SIP core to the MCX server.

In Steps 2 and 3, the MCX server receives the SIP PUBLISH message, validates the access token, binds the IMPU and MC service ID (MCPTT ID, MCVideo ID or MCData ID) if the access token is valid, and responds to the SIP PUBLISH message."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.2.1]

"…

The primary purpose of these procedures is to allow MCX entities to communicate with each other using end-to-end security. End-to-end security provides assurance to MCX users that no unauthorized access to communications is taking place within the MCX network. End-to-end communication security may be applied to media when operating on-network and media, floor control, transmission control, and media control when operating off-network.

A security domain is managed by a Key Management Server (KMS). The KMS is a component of the Common Services Core within the MCX system architecture. For any end-point to use or access end-to-end secure communications, it needs to be provisioned with key material associated to its identity by the KMS. Through the use of the KMS, MC administrators are able to manage the use of, and access to, secure communications within the MCX network.

…"

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.2.2]

"The security mechanism described in this clause allows a key, K, to be distributed from an initiating party to a receiving party. It provides confidentiality of the key, and integrity and authenticity of the payload. It is used within a number of different security procedures in this specification.

The key, K, is distributed encrypted specifically to the receiving entity and signed by the initiating entity. Prior to call commencement, both MCX UEs shall be provisioned by the KMS with time-limited key material associated with the MCX entity's URI. The key is distributed with a 32-bit Key Identifier (K-ID). This payload is a MIKEY-SAKKE I_MESSAGE, as defined in IETF RFC 6509 [11], which ensures the confidentiality of the key, plus integrity and authenticity of the payload.

The key is encrypted to the user identity (UID) associated to the receiving MCX entity using the security domain parameters provided in the public values in the certificate received from the KMS. The UID used to encrypt the data is derived from the receiving entity's URI (e.g. sip:user.002@mcptt.example.org) and a time-related parameter (e.g. the current year and month). The terminating entity's URI is added to the recipient field (IDRr) of the message.

The payload includes the encrypted key and the key identifier (K-ID). The key is unique within the MC domain. On creating the key, the initiator generates a 32-bit key identifier (K-ID). The 4 most significant bits of the K-ID shall indicate the purpose of the key, the other 28-bits shall be randomly generated. The key identifier (K-ID) is stored in the CSB-ID field of the MIKEY I_MESSAGE.

The payload is signed using (the KMS-provisioned key associated to) the identity of the initiating entity. The UID used to sign the data is derived from the initiating entity's URI (e.g. sip:user.001@mcptt.example.org) and a time-related parameter (e.g. the current year and month). The initiating entity's URI is added to the initiator field (IDRi) of the message.

NOTE: This solution is for the end-to-end protection of keys and does not protect the identities transmitted. Identities may be masked by transmitting the UID within the MIKEY ID fields as described in Annex E.7.

The security processes are summarized in figure 5.2.2-1."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.8.1]

"The 'key download' procedure is used to send keys from the MCX server to the MC client. It is used to distribute Multicast Signalling Keys (MuSiKs) to the MC clients, and it is used to update both the CSKs and MuSiKs.

Within the 'key download' procedure, keys (CSK or MuSiKs) are encrypted specifically to the MC user and signed using an identity representing the MC Server. Prior to group key distribution, each MC client shall be provisioned by the KMS with time-limited key material associated with the MC User as described clause 5.3. The MC Server shall also be provisioned by the MC KMS with key material for an identity which is authorised to act as an MCX Server.

The key (CSK or MuSiK) is distributed from the MCX Server to a MC client using the security mechanism described in clause 5.2.2, transported over the SIP bearer. End-point diversity is not required as end-points do not encrypt data, hence the extension in clause 5.2.3 is not applied. Additional parameters may be included as defined in clause 5.2.4. The SAKKE-to-self extension may be included as defined in clause 5.2.5. Identity hiding may be supported as defined in clause 5.2.6.

The initiating entity shall be the initiating MCX Server and the receiving entity shall be the terminating MC user. The initiating entity URI shall be the FQDN of the MCX Server (e.g. MDSI of the MC Domain) and the receiving entity URI shall be the MC Service ID of the terminating user. The distributed key, K, shall be the CSK or MuSiK and the key identifier K-ID shall be the CSK-ID or MuSiK-ID (respectively).

As a result of this 'key download' mechanism, the MC clients receive a new signalling key, CSK or MuSiK, identified by the 4 most significant bits of the key ID.

The MCData Service server may use the Key Download procedure to indicate or modify the algorithm used to protect the MCData signalling fields (i.e. MCData signaling parameters, Data signaling payload and End to end security parameters) by including a 'signalling algorithm' parameter. The 'signalling algorithm' parameter is described in clause 8.5.4.1. The available algorithms shall be as defined in clause 8.5.4.2."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.8.2]

"The procedure for key download is described in figure 5.8.2-1:

…

0. The MCX UE has been provisioned by a KMS with key material associated with the MC user. The MC UE has also registered with an MCX Server. As a consequence of this registration, the MC UE is subscribed to key download notifications from the MCX Server.

1. The MCX Server sends a key download message (SIP NOTIFY or SIP MESSAGE) to the MC UE. The MC UE extracts the signalling key from the key download message.

2. Upon successful extraction of the signalling key, the MC UE returns a key download success message (200 OK response) to the MCX Server. Upon receipt of a notification of success, the MCX Server is able to begin to use the key for protection of signalling traffic."

5.1.3 Test description

5.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client)

- For the underlying "transport bearer" over which the SS-UE1 (MCData client) and the MCData Server will communicate Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server)
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Distribution Function, the MCData User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server

Preamble:

- The IUT (MCData Server) is connected to PLMN1.
- The IUT (MCData Server) is connected to the SS-UE1 (MCData client) as defined in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Figure 4.2.4, where term "MCPTT" is replaced with "MCData".

5.1.3.2 Test procedure sequence

Table 5.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
0A	The SS-UE1 (MCData client) sends an HTTP GET (initial UE configuration) request to retrieve the initial UE configuration from the Server.	-->	HTTP GET	-	-
0B	Check: Does the IUT (MCData Server) respond by sending an HTTP 200 (OK) including the initial UE configuration document?	<--	HTTP 200 (OK)	1	P
0C	The SS-UE1 (MCData client) establishes a secure TLS tunnel as specified by ETSI TS 133 310 [11] (3GPP TS 33.310) to the authorisation endpoint of the IdM server as specified in ETSI TS 133 180 [12] (3GPP TS 33.180) using the configured URL of the authorisation endpoint of the IdM server as specified in the "<x>/OnNetwork/AppServerInfo/IDMSAuthEndpoint" leaf node, Table 5.5.8.1-1, ETSI TS 136 579-1 [1] (3GPP TS 36.579-1).	-	-	-	-
1	The SS-UE1 (MCData client) sends an OpenID Connect Authentication Request using HTTP POST	-->	HTTP POST (Authorization)	-	-
2	Check: Does the IUT (MCData Server) respond by sending an HTTP 200 (OK) including the HTML form requesting username and password?	<--	HTTP 200 (OK)	1	P
3	The SS-UE1 (MCData client) sends an HTTP POST Request message to the SS containing user name and password	-->	HTTP POST	-	-
4	Check: Does the IUT (MCData Server) send an HTTP 302 (Found) as the OpenID Connect Authentication Response?	<--	HTTP 302 (Found)	1	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
5	The SS-UE1 (MCData client) establishes a secure TLS tunnel as specified by ETSI TS 133 310 [11] (3GPP TS 33.310) to the token endpoint of the IdM server as specified in ETSI TS 133 180 [12] (3GPP TS 33.180) using the configured URL of the token endpoint of the IdM server as specified in the "/<x>/OnNetwork/AppServerInfo/IDMSTokenEndpoint" leaf node, Table 5.5.8.1-1, ETSI TS 136 579-1 [1] (3GPP TS 36.579-1)	-	-	-	-
6	The SS-UE1 (MCData client) sends an HTTP POST Request message to the IUT (MCData Server) over the TLS connection established to the IdM token endpoint (OIDC Token Request message) passing the authorization code sent in step 4.	-->	HTTP POST	-	-
7	Check: Does the IUT (MCData Server) send an HTTP 200 (OK) providing id_token, access_token and refresh token?	<--	HTTP 200 (OK)	1	P
7A	The SS-UE1 (MCData client) establishes a secure TLS tunnel as specified by ETSI TS 133 310 [11] (3GPP TS 33.310) to the HTTP Proxy as specified in ETSI TS 133 180 [12] (3GPP TS 33.180) using the configured URL of the HTTP Proxy as specified in the "/<x>/OnNetwork/AppServerInfo/HTTPproxy" leaf node, Table 5.5.8.1-1, ETSI TS 136 579-1 [1] (3GPP TS 36.579-1)	-	-	-	-
8	The SS-UE1 (MCData client) sends an HTTP POST message presenting an access token sent in step 7 for Key Management Initialisation.	-->	HTTP POST	-	-
9	Check: Does the IUT (MCData Server) respond by sending identity-specific key information?	<--	HTTP 200 (OK)	2	P
10	The SS-UE1 (MCData client) sends an HTTP POST message presenting an access token for Key Material Request	-->	HTTP POST	-	-
11	Check: Does the IUT (MCData Server) respond by sending identity-specific key information?	<--	HTTP 200 (OK)	2	P
12	The SS-UE1 (MCData client) sends a SIP REGISTER request for service authorisation	-->	SIP REGISTER	-	-
13	Check: Does the IUT (MCData Server) respond by sending an SIP 200 (OK) message?	<--	SIP 200 (OK)	3	P
14	The SS-UE1 (MCData client) sends a SIP SUBSCRIBE - subscription to multiple documents simultaneously - containing the access token and a resource list mime body containing a list of the following documents: MCData UE Configuration document, MCData User Profile Configuration Document, and the MCData Service configuration document. The base URI of each list entry is set to the CMS XCAP-ROOT-URI	-->	SIP SUBSCRIBE	-	-
15	Check: Does the IUT (MCData Server) respond by sending an SIP 200 (OK) message?	<--	SIP 200 (OK)	4	P
16	Check: Does the IUT (MCData Server) send a SIP NOTIFY message containing the XCAP-URI of the documents?	<--	SIP NOTIFY	4	P
17	The SS-UE1 (MCData client) responds with a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
18	The SS-UE1 (MCData client) sends an HTTP GET Request message that contains the access token and the XCAP-URI of the MCData UE Configuration document	-->	HTTP GET	-	-
19	Check: Does the IUT (MCData Server) send the HTTP 200 (OK) message including the MCData UE Configuration Document?	<--	HTTP 200 (OK)	4	P
20	The SS-UE1 (MCData client) sends an HTTP GET Request message that contains the access token and the XCAP-URI of the MCData User Profile Configuration Document	-->	HTTP GET	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
21	Check: Does the IUT (MCData Server) send the HTTP 200 (OK) message including the MCData User Profile Configuration Document?	<--	HTTP 200 (OK)	4	P
22	The SS-UE1 (MCData client) sends an HTTP GET Request message that contains the access token and the XCAP-URI of the MCData Service Configuration Document	-->	HTTP GET	-	-
23	Check: Does the IUT (MCData Server) send the HTTP 200 (OK) message including the MCData Service Configuration Document?	<--	HTTP 200 (OK)	4	P
24	The SS-UE1 (MCData client) sends a SIP SUBSCRIBE containing the access token and a resource list mime body and a list of the Groups to be obtained. The base URI of each list entry is set to the GMS XCAP-ROOT-URI, and the MCData group ID identifies a group document	-->	SIP SUBSCRIBE	-	-
25	Check: Does the IUT (MCData Server) respond with a HTTP 200 (OK) message	<--	SIP 200 (OK)	5	P
26	Check: Does the IUT (MCData Server) send a SIP NOTIFY message to the UE that contains the XCAP-URI of the Group documents?	<--	SIP NOTIFY	5	P
27	The SS-UE1 (MCData client) sends a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
28	The SS-UE1 (MCData client) sends an HTTP GET Request message that contains the access token and the XCAP-URI of the Group Configuration document	-->	HTTP GET	-	-
29	Check: Does the IUT (MCData Server) send an HTTP 200 (OK) message including the Group Document 'MCData UE Configuration document'?	<--	HTTP 200 (OK)	5	P
29A	Check: Does the IUT (MCData Server) send a SIP NOTIFY message to the UE that contains the group key transport payloads (GKTP) document.	<--	SIP NOTIFY	5	P
29B	The SS-UE1 (MCData client) sends a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
30	The SS-UE1 (MCData client) sends a SIP PUBLISH request for update of PoC-settings. NOTE: The PoC-settings document contains the user profile index of the selected user profile.	-->	SIP PUBLISH	-	-
31	Check: Does the IUT (MCData Server) send a SIP 200 (OK)?	<--	SIP 200 (OK)	6	P
32-33	Void	--	-	-	-

5.1.3.3 Specific message contents

Table 5.1.3.3-1: HTTP POST (Step 1, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition AUTH

Table 5.1.3.3-2: HTTP POST (Step 3, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition USERAUTH

Table 5.1.3.3-3: HTTP POST (Step 6, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition TOKEN

Table 5.1.3.3-4: HTTP POST (Step 8, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition KMSINIT.

Table 5.1.3.3-5: HTTP POST (Step 10, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition KMSKEY.

Table 5.1.3.3-5A: HTTP 200 (OK) (Step 0B, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition UEINITIALCONFIG

Table 5.1.3.3-6: HTTP 200 (OK) (Step 2, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type				
media-type	"text/html"		IETF RFC 2854 [27]	
Message-body				
HTML form	<!DOCTYPE html> <html> <body> <form action="/idms/userauth" method="post"> Username: <input type="text" name="user"> Password: <input type="password" name="password"><button type="submit">Login</button> </form> </body> </html>	"/idms/userauth" given by tsc_MCX_IdMS_userauth_UriPath is the URI to be used by the UE as request URI in the HTTP POST request for user authentication	HTML 4.01 Specification [28]	

Table 5.1.3.3-7: HTTP 200 (OK) (Step 7, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition TOKEN

Table 5.1.3.3-8: HTTP 200 (OK) (Step 9, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition KMSINIT.

Table 5.1.3.3-9: HTTP 200 (OK) (Step 11, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition KMSKEY.

Table 5.1.3.3-10: HTTP 200 (OK) (Step 19, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, conditions MCDATA and UECONFIG.

Table 5.1.3.3-11: HTTP 200 (OK) (Step 21, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, conditions MCDATA and UEUSERPROF.

Table 5.1.3.3-12: HTTP 200 (OK) (Step 23, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, conditions MCDATA and UESERVCONFIG.

Table 5.1.3.3-13: HTTP 200 (OK) (Step 29, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition GROUPCONFIG.

Table 5.1.3.3-14: HTTP 302 (Found) (Step 4, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.8-1, condition AUTH.

Table 5.1.3.3-14A: HTTP GET (Step 0A, Table 5.1.3.2-1)

Derivation Path: Table 5.5.4.2-1, condition UEINITIALCONFIG

Table 5.1.3.3-15: HTTP GET (Step 18, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.2-1, conditions MCDATA and UECONFIG.

Table 5.1.3.3-16: HTTP GET (Step 20, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.2-1, conditions MCDATA and UEUSERPROF.

Table 5.1.3.3-17: HTTP GET (Step 22, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.2-1, conditions MCDATA and UESERVCONFIG.

Table 5.1.3.3-18: HTTP GET (Step 28, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.2-1, condition GROUPCONFIG

Table 5.1.3.3-19: SIP REGISTER (Step 12, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.13-1, condition SIP_REGISTER_INITIAL, CONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCData_PublicServiceId_B	SIP URI of the home domain name		

Table 5.1.3.3-20: SIP SUBSCRIBE (Step 14, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition CONFIG

Table 5.1.3.3-20A: SIP SUBSCRIBE (Step 24, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 5.1.3.3-20B			

Table 5.1.3.3-20B: Resource-lists in SIP SUBSCRIBE (Table 5.1.3.3-20A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1, condition GROUPCONFIG

Table 5.1.3.3-21: SIP NOTIFY (Step 16, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition CONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact				
addr-spec				
user-info and host	tsc_MCData_PublicServiceId_B			

Table 5.1.3.3-21A: SIP NOTIFY (Step 26, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact				
addr-spec				
user-info and host	tsc_MCData_PublicServiceId_B			

Table 5.1.3.3-21B: SIP NOTIFY (Step 29A, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition GROUPCONFIG				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact				
addr-spec				
user-info and host	tsc_MCData_PublicServiceId_B			
Message-body				
xcap-diff document	xcap-diff document as described in Table 5.1.3.3-21C			

Table 5.1.3.3-21C: Xcap-Diff Document (Table 5.1.3.3-21B)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.12-2, condition GROUPKEY

Table 5.1.3.3-22: SIP PUBLISH (Step 30, Table 5.1.3.2-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.11-1, conditions MCDATA and POC-SETTINGS-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCData_PublicServiceId_B	The public service identity identifying the originating participating MCData function serving the MCData user		

Table 5.1.3.3-23: Void

Table 5.1.3.3-24: SIP 200 (OK) (Steps 13, Table 5.1.3.2-1))

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 condition REGISTER-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**

Table 5.1.3.3-25: SIP 200 (OK) (Steps 15, 25, Table 5.1.3.2-1))

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 condition SUBSCRIBE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Contact				
addr-spec				
user-info and host	tsc_MCData_PublicServiceId_B			

Table 5.1.3.3-26: SIP 200 (OK) (Step 31, Table 5.1.3.2-1))

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 condition PUBLISH-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**

5.2 MCDATA Server - MCDATA Client / Configuration / Group Creation / Group Regroup Creation / Group Regroup Teardown

5.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 and supports new MCData terminated group }
ensure that {
  when { the UE (MCData client) initiates creation of the terminated group by sending a HTTP PUT
request }
    then { IUT (MCData Server) responds to the UE (MCData Client) with a HTTP 201 (Created) }
            }
```

(2)

```
with { IUT (MCData Server) having establish terminated group }
ensure that {
  when { the SS-UE1 (MCData Client) initiates creation of the terminated temporary group by sending
a HTTP POST request }
    then { IUT (MCData Server) responds with a HTTP 200 (OK) containing the GMOP group-regroup-
creation-response }
            }
```

(3)

```
with { IUT (MCData Server) having establish a temporary group }
ensure that {
  when { the UE (MCData client) initiates tear down of the temporary group by sending a HTTP DELETE
request }
    then { IUT (MCData Server) responds to the UE (MCData Client) with HTTP 200 (OK) message }
            }
```

5.2.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 484 [8] (3GPP TS 24.484) clauses 4.3.1, 4.3.4, 4.4, 6.3.13.2.3, 6.3.3.2.3, 6.3.3.2.1, 4.1.4, 10.2.1, 10.2.1A, 10.3.1, 10.3.1A, 10.4.1, 6.2.3, ETSI TS 124 481 [6] (3GPP TS 24.481) clauses 5.5, 5.2A, 6.2.4, 6.2.2, 6.3.2.2.2, 6.3.2.2.1, 6.3.3.2.3, 6.3.3.2.1, 6.3.13.2.2, 6.3.14.2, 6.3.15.2, 6.3.2.3, 6.3.14.3.1, 6.3.15.3.1, ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 6.3.3, 6.3.4, 6.3.5, 8.3.3.3, 8.3.3.4, 8.3.3.5, ETSI TS 133 180 [12] (3GPP TS 33.180) clauses 5.7.1, 5.7.3, 5.8.1, 7.3.2, 7.3.3.2. Unless otherwise stated these are Rel-14 requirements.

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.3.1]

"The MCS server obtains the appropriate MCS service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCS from the configuration management server.

The MCS server obtains the MCS service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCS from the configuration management server.

The MCS server subscribes to the MCS service configuration document for each mission critical organisation that is provisioned that is supported by the MCS server using the procedure specified in subclause 6.3.13.2.3. How the MCS server is provisioned with the identities of the mission critical organisations is out of scope of the present document.

If the MCS service configuration document has been updated since the current version stored at the MCS server, then the MCS server will receive a SIP NOTIFY request containing an HTTPS URI of the MCS service configuration document. Retrieval by the MCS server, using the notified HTTPS URI, of the MCS service configuration document is performed as specified in subclause 6.3.3.2.3.

NOTE: The MCS server can be notified of changes to the MCS service management configuration document at any time while operating the MCS."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.3.4]

"The format of the MCData service configuration document downloaded to the MCData server is defined in subclause 10.4."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.4]

"The following applies to the configuration management server used for online configuration.

The configuration management server needs to convert the MCS UE initial configuration document received from an MCS administrator into an appropriate format for configuration of the MCS UE initial configuration MO.

If the MCS UE initial configuration MO contains a <default-user-profile> element that identifies an MCS user profile configuration document, the configuration management server needs to convert the identified MCS user profile configuration document received from an MCS administrator into an appropriate format for configuration of the MCS user profile configuration MO.

Once an MCS User Profile configuration document has been created or updated by the MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to store MCS user profile configuration document as the user profile in the MC service user database.

In order to download the MCS user profile configuration document to an MCS UE or to support an MCS UE updating the MCS user profile configuration document, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to obtain the MCS user profile from the MC service user database.

In order to be notified of changes to an MCS user profile configuration document(s) that have been subscribed to by an MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to be notified of changes to the MCS user profile stored in the MC service user database.

In order to delete the MCS user profile when requested by an MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to delete the MCS user profile from the MC service user database.

NOTE: The configuration management server and group management server functionality for offline configuration is out of scope of the present document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.13.2.3]

"In order to subscribe to an MCS service configuration document, an MCS server shall send an initial SIP SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [22] and IETF RFC 5875 [11]. In the initial SIP SUBSCRIBE request, MCS server:

a) shall set the Request URI to a SIP URI containing:

1) the base URI being equal to the public service identity of the CMS configured in the MCS server; and

2) the "auid" parameter set to the application usage identifying th MCS service configuration document;

b) shall include a P-Asserted-Identity header field containing the public service identity of the MCS server;

c) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [22]), in a P-Asserted-Service header field according to IETF RFC 6050 [23]; and

d) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field.

Upon receiving a SIP NOTIFY request associated with a subscription created as result of the sent initial SIP SUBSCRIBE request, the MCS server shall handle the SIP NOTIFY request according to IETF RFC 5875 [11].

In order to re-subscribe to notification of changes to an MCS service configuration document, an MCS server shall send a SIP re-SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [22] and IETF RFC 5875 [11]. In the SIP re-SUBSCRIBE request, MCS server:

a) shall set the Request URI to a SIP URI containing:

1) the base URI being equal to the public service identity of the CMS configured in the MCS server; and

2) the "auid" parameter set to the application usage identifying an MCS service configuration document; and

b) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.3.2.3]

"In order to retrieve a configuration management document via the CSC-5 reference point, an MCS Server shall perform the procedures in subclause 6.3.3.2.1 specified for GC. The MCS server shall set the Request-URI of the HTTP GET request to identify the XML document based on configuration and include the "auid" as per the appropriate application usage."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.3.2.1]

"In order to retrieve a configuration management document, a GC shall send an HTTP GET request with the Request URI that references the document to be updated to the network according to procedures specified in IETF RFC 4825 [14] "Retrieve a Document"."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.1.4]

"An MCData service administrator can, using an MCData UE configure the:

- *MCData UE configuration document;*
- *MCData user profile configuration document;*
- *MCData service configuration document; and*
- *MCData related group configuration data in the MCS group document.*

The format of the MCData UE configuration document is defined in subclause 10.2.

The format of the MCData user profile configuration document is defined in subclause 10.3.

The format of the MCData service configuration document is defined in subclause 10.4."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.2.1]

"The MCData UE configuration document is specified in this subclause. The MCData UE configuration document content is based on requirements of Annex A.2 of 3GPP TS 23.282 [24], in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData UE configuration document in the MCData service is described in 3GPP TS 24.282 [25] and 3GPP TS 24.582 [26]. The schema definition is provided in subclause 10.2.2.3. An MCData UE configuration document may apply to all MCData UEs of a mission critical organization or apply to specific MCData UEs identified in the <mcdata-UE-id> element. If there is no <mcdata-UE-id> element in the MCData UE configuration document, then by default the MCData UE configuration document applies to all MCData UEs of the mission critical organization that are not specifically identified in the <mcdata-UE-id> element of another MCData UE configuration document of the mission critical organization. Each MCData UE of a mission critical organization is configured with an MCData UE configuration document that is identified by the instance ID of the MCData UE.

MCData UE configuration documents of a MCData user are contained as "XDM collections" in the user's directory of the users tree, in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. In this case, the term "user" in the XCAP sense refers to the MCData ID, as the user has been already authenticated. A MCData UE configuration document corresponding to a specific MCData UE the MCData user has used to authenticate and is authorised to use the MCData service with is placed in the user directory of the MCData user.

The MCData UE configuration document acts as a template for the CMS to generate UE configuration documents that are downloaded to specific MCData UEs. The MCData UE configuration document that acts as a template is referred to as a "master MCData UE configuration document". The master MCData UE configuration document is stored in the users tree of that MCData system administrator. The master MCData UE configuration document does not directly apply to a specific MCData UE, but instead acts as template that the CMS uses to populate the MCData UE configuration documents of MCData UEs identified by elements of the <MCData-UE-id> element. For MCData UE configuration documents that correspond to a specific MCData UE, the name of the MCData UE configuration document is created from a value defined by the corresponding element that identifies the MCData UE within the <MCData-UE-id> element. For a master MCData UE configuration documents that does not contain a <MCData-UE-id> element, the name of the MCData UE configuration document stored in the user directory is "DEFAULT-MCData-UE.xml"."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.2.1A]

"The MCData UE configuration document is accessed using the same XCAP URI, regardless of whether the MCData UE has a specific MCData UE configuration document configured or the master MCData UE configuration document applies. The CMS shall generate the UE's MCData UE configuration document from the master MCData UE configuration document if the MCData administrator did not provision a specific MCData UE configuration document. In this generated document, the <MCData-UE-id> element shall be set to the MCData client's UE ID. The UE's MCData UE configuration document shall always be stored with the filename corresponding to the UE's MCData UE ID under the user's directory of the users tree.

The XCAP URI used by the MCData client to access the UE's MCData UE configuration document shall be:

- *CMSXCAPROOTURI/org.3gpp.mcdata.ue-config/users/sip:MCDATAID/MCSUEID."*

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.3.1]

"The MCData user profile configuration document is specified in this subclause. The MCData user profile configuration document content is based on requirements of Annex A.3 of 3GPP TS 23.282 [24], and structure and procedures of OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData user profile in the MCData service is described in 3GPP TS 24.282 [25]. The schema definition is provided in subclause 10.3.2.

MCData user profile documents are "XDM collections" in the user's directory of the users tree, in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. In this case, the term "user" in the XCAP sense refers to the MCData ID, as the user has been already authenticated.

The name of the MCData user profile document matches the value of the <ProfileName> element in the MCData user profile document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.3.1A]

"The XCAP URI used by the MCData client to access the MCData user's MCData user profile documents shall be:

CMSXCAPROOTURI/org.3gpp.mcdata.user-profile/users/sip:MCDATAID/mcdata-user-profile-INDEX.xml

Where INDEX is the index of the MCData user profile as defined in subclause 10.3.2.8."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.4.1]

"The MCData service configuration document is specified in this subclause. The MCData service configuration document content is based on requirements of Annex A.5 of 3GPP TS 23.282 [24], and structure and procedures of OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData service configuration in the MCData service is described in 3GPP TS 24.282[25] and 3GPP TS 24.582 [26]. The schema definition is provided in subclause 10.4.2.3. Each mission critical organization is configured with an MCData service configuration document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.2.3]

"The MCS server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.482 [6]."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 5.5]

"To be compliant with the procedures in the present document, an MCData server shall support role of an MCS server as specified in subclause 5.2A."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 5.2A]

"To be compliant with the procedures in the present document, an MCS server:

- *shall support the role of XCAP client as specified in IETF RFC 4825 [22];*
- *shall support the role of XDMC as specified in OMA OMA-TS-XDM_Group-V1_1_1 [3];*
- *shall support the procedure in subclause 6.2.4;*
- *shall support the procedure in subclause 6.3.3.2.3; and*
- *shall support the procedure in subclause 6.3.13.2.2."*

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.2.4]

"The MCS server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.482 [10].

The MCS server shall perform the procedures in subclause 6.2.2 specified for GC."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.2.2]

"In order to address an existing group document defining a group ID known by GC, the GC shall set the Request-URI of an HTTP request to a XCAP URI identifying a group document addressed by a group ID as described in subclause 7.2.10.2, where the group ID is set to the group ID known by GC and where the XCAP root URI is the XCAP root URI configured in the GC."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.2.2.2]

"In order to create a group document, a GMC shall perform the procedures in subclause 6.3.2.2.1 specified for GC.

NOTE: When the GMC has not received the assigned MCS group ID from the GMS, the GMC can set empty value to the "uri" attribute of the <list-service> element of the <group> element of the XML document specified in subclause 7.2.1.

…"

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.2.2.1]

"In order to create a group document, a GC shall create an XML document of the application usage specified in subclause 7.2.1 and shall send the XML document to the network according to procedures specified in IETF RFC 4825 [22] "Create or Replace a Document". The GC shall set the Request-URI of the HTTP PUT request to an XCAP URI in users tree where the XUI is set to a group creation XUI configuration parameter."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.3.2.3]

"In order to retrieve a group document, an MCS server shall perform the procedures in subclause 6.3.3.2.1 specified for a GC."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.3.2.1]

"In order to retrieve a group document, a GC shall send an HTTP GET request with the Request URI that references the document to be retrieved to the network according to procedures specified in IETF RFC 4825 [22] "Fetch a Document"."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.13.2.2]

"In order to subscribe to notification of changes of:

a) one or more MCS group documents of MCS groups identified by MCS group IDs;

an MCS server shall send an initial SIP SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [12] and IETF RFC 5875 [13]. In the initial SIP SUBSCRIBE request, MCS server:

a) shall include an application/resource-lists+xml MIME body. In the application/resource-lists+xml MIME body, the MCS server shall include one <entry> element for each document or element to be subscribed to, such that the "uri" attribute of the <entry> element:

1) contains a relative path reference:

A) with the base URI being equal to the XCAP root URI configured in the MCS server; and

B) identifying a group document addressed by a group ID as described in subclause 7.2.10.2 where the group ID is set to the MCS group ID;

b) shall set the Request-URI to the configured public service identity for performing subscription proxy function of the GMS;

c) shall include a P-Asserted-Identity header field containing the public service identity of the MCS server;

e) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [12]), in a P-Asserted-Service header field according to IETF RFC 6050 [14]; and

f) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field.

Upon receiving a SIP NOTIFY request associated with a subscription created as result of the sent initial SIP SUBSCRIBE request, the MCS server shall handle the SIP NOTIFY request according to IETF RFC 5875 [13].

In order to re-subscribe to notification of changes of a modified list of:

a) one or more MCS group documents of MCS groups identified by MCS group IDs,

the MCS server shall send a SIP re-SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [12] and IETF RFC 5875 [13]. In the SIP re-SUBSCRIBE request, MCS server:

a) shall include an application/resource-lists+xml MIME body. In the application/resource-lists+xml MIME body, the MCS server shall include one <entry> element for each document or element to be subscribed to, such that the "uri" attribute of the <entry> element:

1) contains a relative path reference:

A) with the base URI being equal to the XCAP root URI configured in the MCS server; and

B) identifying a group document addressed by a group ID as described in subclause 7.2.10.2 where the group ID is set to the MCS group ID; and

b) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.14.2]

"In order to form a temporary MCS group, a GMC shall send a HTTP POST request according to procedures specified in IETF RFC 2616 [21] and subclause 6.2.3. In the HTTP POST request, the GMC:

a) shall set the Request-URI to an XCAP URI:

1) in users tree where the XUI is set to a group creation XUI configuration parameter; and

2) with the document selector identifying the temporary MCS group to be created; and

b) shall include an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup creation specified in subclause 7.3.4.3, with a <group> element containing a group document for an MCS group. In the group document, the GMC shall include the <on-network-temporary> element according to subclause 7.2. In the <on-network-temporary> element, the GMC shall include <constituent-MCPTT-group-IDs> element according to subclause 7.2. In the <constituent-MCPTT-group-IDs> element, the GMC shall include one <constituent-MCPTT-group-ID> element according to subclause 7.2 for each MCS group to be combined.

Upon reception of an HTTP 2xx response to the sent HTTP POST request, the GMC shall consider the temporary MCS group formation as successful.

Upon reception of an HTTP 409 (Conflict) response with at least one <alt-value> element in the <uniqueness-failure> error element, the GMC may repeat procedures of the present subclause and identify the temporary MCS group being formed with an MCS Group ID indicated in an <alt-value> element."

[ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.2.3]

"Upon reception of an HTTP PUT request where the Request-URI of the HTTP PUT request identifies an XML document of the application usage specified in subclause 7.2

then the GMS:

a) shall determine the identity of the sender of the received HTTP PUT request as specified in subclause 6.2.5;

b) if the identity of the sender of the received HTTP PUT request is not authorized to initiate group creation, shall respond with a HTTP 403 (Forbidden) response to the HTTP PUT request and shall not continue with rest of the steps;

c) if value of the "uri" attribute of the <list-service> element of the <group> element of the XML document specified in subclause 7.2.1 of the received HTTP PUT request does not conform to local policy, shall respond with an HTTP 409 (Conflict) response to the HTTP PUT request. The <uniqueness-failure> error element shall identify the error condition. The GMS shall include at least one <alt-value> element in the <uniqueness-failure> error element, whereby each <alt-value> element contains an MCS Group ID acceptable for the GMS. The GMS shall not continue with rest of the steps; and

d) shall support receiving an XML document of the application usage specified in subclause 7.2.1 according to procedures specified in IETF RFC 4825 [22] "PUT Handling" where the Request-URI of the HTTP PUT request identifies an XML document of the application usage specified in subclause 7.2."

[ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.14.3.1]

"Upon reception of an HTTP POST request:

a) with a Request-URI with an XCAP URI identifying a non-existing group document; and

b) with an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup creation specified in subclause 7.3.4.3;

then the GMS:

a) shall determine the identity of the sender of the received HTTP POST request as specified in subclause 6.2.5;

b…

d) for each MCS group ID of an MCS group to be combined indicated in content of a <constituent-MCPTT-group-ID> element of the <constituent-MCPTT-group-IDs> element of the <on-network-temporary> element of the group document of the <group> element of the GMOP document requesting group regroup creation specified in subclause 7.3.4.3 of the received HTTP POST request:

1) shall send a HTTP POST request according to procedures specified in IETF RFC 2616 [21] and subclause 6.2.5. In the HTTP POST request, the GMS:

A) shall set the Request-URI to an XCAP URI:

i) with the document selector identifying a group document addressed by a group ID as specified in subclause 7.2.10.2, where the group ID is set to the MCS group ID of the MCS group to be combined;

ii) with the node selector identifying a <on-network-regrouped> element of the constituent MCS group such that the <on-network-regrouped> element has the "temporary-MCPTT-group-ID" attribute set to the content of the "uri" attribute of the <list-service> element of the group document included in the received HTTP POST request;

iii) if the MCS group ID of the MCS group to be combined is owned by the MCS provider of the GMS, with the XCAP root URI of the MCS provider of the GMS; and

iv) if the MCS group ID of the MCS group to be combined is owned by an MCS provider other than the MCS provider of the GMS, with XCAP root URI derived using the group ID routing database as specified in subclause 6.2.5.2 and the MCS group ID of the MCS group to be combined;

B) shall set the X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.482 [10] to a public service identity of the GMS; and

C) shall include an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document requesting group regroup check specified in subclause 7.3.4.4 with a <on-network-regrouped> element. In the <on-network-regrouped> element, the GMS:

i) shall set the "temporary-MCPTT-group-ID" attribute to the content of the "uri" attribute of the <list-service> element of the group document included in the received HTTP POST request;

ii) shall set the "temporary-MCPTT-group-requestor" attribute to the identity of the sender of the received HTTP POST request;

iii) shall include the <constituent-MCPTT-group-IDs> element set to content of the <constituent-MCPTT-group-IDs> element of <on-network-temporary> element of the <list-service> element of the group document of the MCS group included in the received HTTP POST request;

iv) if an <on-network-group-priority> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <on-network-group-priority> element set to content of the <on-network-group-priority> element of the <list-service> element of the group document included in the received HTTP POST request;

v) if a <protect-media> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <protect-media> element set to content of the <protect-media> element of the <list-service> element of the group document included in the received HTTP POST request;

vi) if a <protect-floor-control-signalling> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <protect-floor-control-signalling> element set to content of the <protect-floor-control-signalling> element of the <list-service> element of the group document included in the received HTTP POST request; and

vii) if a <require-multicast-floor-control-signalling> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <require-multicast-floor-control-signalling> element.

…

Upon reception of HTTP 2xx responses to all sent HTTP POST requests, the GMS:

a) for each MCS group ID of an MCS group to be combined indicated in content of a <constituent-MCPTT-group-ID> element of the <constituent-MCPTT-group-IDs> element of the <on-network-temporary> element of the <group> element of the GMOP document requesting group regroup creation specified in subclause 7.3.4.3 of the received HTTP POST request:

1) shall send an HTTP POST request according to procedures specified in IETF RFC 2616 [21] and subclause 6.2.5. In the HTTP POST request, the GMS:

A) shall set the Request-URI to an XCAP URI:

i) with the document selector identifying a group document addressed by a group ID as specified in subclause 7.2.10.2, where the group ID is set to the MCS group ID of the MCS group to be combined;

ii) with the node selector identifying a <on-network-regrouped> element of the constituent MCS group such that the <on-network-regrouped> element has the "temporary-MCPTT-group-ID" attribute set to the content of the "uri" attribute of the <list-service> element of the group document included in the received HTTP POST request;

iii) if the MCS group ID of the MCS group to be combined is owned by the MCS provider of the GMS, with the XCAP root URI of the MCS provider of the GMS; and

iv) if the MCS group ID of the MCS group to be combined is owned by an MCS provider other than the MCS provider of the GMS, with XCAP root URI derived using the group ID routing database as specified in subclause 6.2.5.2 and the MCS group ID of the MCS group to be combined;

B) shall set the X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.482 [10] to a public service identity of the GMS; and

C) shall include an application/vnd.3gpp.GMOP+xml MIME body with a GMOP document requesting group regroup notification specified in subclause 7.3.4.5. In the GMOP document requesting group regroup notification, the GMS:

i) shall include a <on-network-regrouped> element. In the <on-network-regrouped> element, the GMS:

- *shall set the "temporary-MCPTT-group-ID" attribute to the content of the "uri" attribute of the <list-service> element of the group document included in the received HTTP POST request;*
- *shall set the "temporary-MCPTT-group-requestor" attribute to the identity of the sender of the received HTTP POST request;*
- *shall include the <constituent-MCPTT-group-IDs> element set to content of the <constituent-MCPTT-group-IDs> element of <on-network-temporary> element of the <list-service> element of the group document of the MCS group included in the received HTTP POST request;*

- *if an <on-network-group-priority> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <on-network-group-priority> element set to content of the <on-network-group-priority> element of the <list-service> element of the group document included in the received HTTP POST request;*

- *if a <protect-media> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <protect-media> element set to content of the <protect-media> element of the <list-service> element of the group document included in the received HTTP POST request;*

- *if a <protect-floor-control-signalling> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <protect-floor-control-signalling> element set to content of the <protect-floor-control-signalling> element of the <list-service> element of the group document included in the received HTTP POST request; and*

- *if a <require-multicast-floor-control-signalling> element is included in the <list-service> element of the group document included in the received HTTP POST request, shall include the <require-multicast-floor-control-signalling> element.*

NOTE: GMK is not included in the GMOP document requesting group regroup notification as GMK is provided only using SIP.

Upon reception of HTTP 2xx responses to all sent HTTP POST requests, the GMS shall create the group document of the temporary MCS group at the location specified by the Request-URI of the received HTTP POST request and shall send an HTTP 2xx response to the received HTTP request. In the HTTP 2xx response, the GMS shall include an application/vnd.3gpp.GMOP+xml MIME body containing a GMOP document with group regroup creation response specified in subclause 7.3.4.6 with a <temporary-group-document-ETag> element set to the current value of the entity tag for the created group document of the temporary MCS group."

[ETSI TS 124 481 [6] (3GPP TS 24.481) clause 6.3.15.3.1]

"Upon reception of an HTTP DELETE request with Request-URI with an XCAP URI identifying a group document of a temporary MCS group, the GMS:

a) for each constituent MCS group indicated in the group document indicated by Request-URI:

1) shall send an HTTP DELETE request. In the HTTP DELETE request, the GMS:

A) shall set the Request-URI to an XCAP URI:

i) with the document selector identifying a group document addressed by a group ID as specified in subclause 7.2.10.2, where the group ID is set to the MCS group ID of the constituent MCS group;

ii) with the node selector identifying a <on-network-regrouped> element of the constituent MCS group, such that the "temporary-MCPTT-group-ID" attribute of the <on-network-regrouped> element contains the temporary MCS group ID of the temporary MCS group;

iii) if the MCS group ID of the constituent MCS group is owned by the MCS provider of the GMS, with the XCAP root URI of the MCS provider of the GMS; and

iv) if the MCS group ID of the constituent MCS group is owned by an MCS provider other than the MCS provider of the GMS, with XCAP root URI derived using the group ID routing database as specified in subclause 6.2.5.2 and the MCS group ID of the constituent MCS group; and

B) shall set the X-3GPP-Asserted-Identity header field as specified in 3GPP TS 24.482 [10] to a public service identity of the GMS.

Upon reception of an HTTP response to all sent HTTP DELETE requests, the GMS shall remove the group document of the temporary MCS group and shall send an HTTP 2xx response to the received HTTP request."

[ETSI TS 124 481 [6] (3GPP TS 24.481), clause 6.3.15.2]

"In order to tear down a temporary MCS group, the GMC shall send an HTTP DELETE request with Request-URI with an XCAP URI identifying a group document of the temporary MCS group according to procedures specified in IETF RFC 4825 [22] "Delete an Element"."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 6.3.3]

"This subclause describes how an MCData server accesses a group document from a group management server.

Upon receipt of a SIP request:

1) if the MCData server is not yet subscribed to the group document for the group identity in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP request, the MCData server shall subscribe to the "xcap-diff" event-package for the group document of this group identity as specified in 3GPP TS 24.481 [11];

NOTE: As a group document can potentially have a large content, the MCData server can subscribe to the group document indicating support of content-indirection as defined in IETF RFC 4483 [13], by following the procedures in 3GPP TS 24.481 [11].

2) upon receipt of a SIP 404 (Not Found) response as a result of attempting to subscribe to the "xcap-diff" event-package for the group document of the group identity in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP request as specified in 3GPP TS 24.481 [11], the MCData server shall send the SIP 404 (Not Found) response with the warning text set to "113 group document does not exist" in a Warning header field as specified in subclause 4.9. Otherwise, continue with the rest of the steps; and

3) upon receipt of any other SIP 4xx, SIP 5xx or SIP 6xx response as a result of attempting to subscribe to the "xcap-diff" event-package for the group document of the group identity in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP INVITE request as specified in 3GPP TS 24.481 [11], the MCData server shall send the SIP final response with the warning text set to "114 unable to retrieve group document" in a Warning header field as specified in subclause 4.4 and shall not continue with the rest of the steps;"

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 6.3.4]

"The MCData server shall only send MCData messages to affiliated group members.

The MCData server determines whether a user is affiliated to a group by following the procedures in subclause 6.3.5.

The MCData server performs the affiliation check in subclause 6.3.5 on each entry contained in the <list> element of the group document."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 6.3.5]

"The MCData server shall determine that the MCData user, with MCData User ID, is affiliated to the MCData group, with MCData Group ID, at the MCData client, with MCData client ID, if the elements, as described in subclause 8.3.3.2, exist with their expected values, as below:

1. an MCData group information entry with MCData group ID same as the MCData group ID under consideration;

2. in the MCData group information entry found in 1, an MCData user information entry with the MCData ID same as the MCData ID under consideration;

3. in the MCData user information entry found in 2, an MCData client information entry with MCData Client ID same as the MCData client ID under consideration; and

4. in the MCData user information entry found in 2, an expiration time, which has not expired."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.3.3]

"Upon receiving a SIP PUBLISH request such that:

1) Request-URI of the SIP PUBLISH request contains the public service identity of the controlling MCData function associated with the served MCData group;

2) the SIP PUBLISH request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the <mcdata-request-uri> element and the <mcdata-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7];

4) the Event header field of the SIP PUBLISH request contains the "presence" event type; and

5) SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-group affiliation information constructed according to subclause 8.4.1;

then the MCData server:

1) shall identify the served MCData group ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;

2) shall identify the handled MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;

…

6) shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to 3GPP TS 24.229 [5], IETF RFC 3903 [34]. In the SIP 200 (OK) response, the MCData server:

a) shall set the Expires header field according to IETF RFC 3903 [34], to the selected expiration time;

7) if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served MCData group ID, shall not continue with the rest of the steps;

8) if the handled MCData ID is different from the MCData ID in the "id" attribute of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request, shall not continue with the rest of the steps;

9) shall consider an MCData group information entry such that:

a) the MCData group information entry is in the list of MCData group information entries described in subclause 8.3.3.2; and

b) the MCData group ID of the MCData group information entry is equal to the served MCData group ID;

as the served MCData group information entry;

10) if the selected expiration time is zero:

a) shall remove the MCData user information entry such that:

i) the MCData user information entry is in the list of the MCData user information entries of the served MCData group information entry; and

ii) the MCData user information entry has the MCData ID set to the served MCData ID;

11) if the selected expiration time is not zero:

a) shall consider an MCData user information entry such that:

i) the MCData user information entry is in the list of the MCData user information entries of the served MCData group information entry; and

ii) the MCData ID of the MCData user information entry is equal to the handled MCData ID;

as the served MCData user information entry;

b) if the MCData user information entry does not exist:

i) shall insert an MCData user information entry with the MCData ID set to the handled MCData ID into the list of the MCData user information entries of the served MCData group information entry; and

ii) shall consider the inserted MCData user information entry as the served MCData user information entry; and

c) shall set the following information in the served MCData user information entry:

i) set the MCData client ID list according to the "client" attributes of the <affiliation> elements of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

ii) set the expiration time according to the selected expiration time;

12) shall identify the handled p-id in the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

13) shall perform the procedures specified in subclause 8.3.3.5 for the served MCData group ID."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.3.4]

"NOTE: Usage of one SIP SUBSCRIBE request to subscribe for notification about change of affiliation state of *several MCData users served by the same MCData server is not supported in this version of the specification.*

Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains the public service identity of the controlling MCData function associated with the served MCData group;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the<mcdata-request-uri> element and the <mcdata-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7];

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type; and

5) the SIP SUBSCRIBE request contains an application/simple-filter+xml MIME body indicating per-user restrictions of presence event package notification information according to subclause 8.4.2 indicating the same MCData ID as in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

then the MCData server:

1) shall identify the served MCData group ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

2) shall identify the handled MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

3) if the Expires header field of the SIP SUBSCRIBE request is not included or has nonzero value lower than 4294967295, shall send a SIP 423 (Interval Too Brief) response to the SIP SUBSCRIBE request, where the SIP 423 (Interval Too Brief) response contains a Min-Expires header field set to 4294967295, and shall not continue with the rest of the steps;

4) if an MCData group for the served MCData group ID does not exist in the group management server according to 3GPP TS 24.481 [11], shall reject the SIP SUBSCRIBE request with SIP 403 (Forbidden) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [5], IETF RFC 3903 [34] and IETF RFC 3856 [39] and skip the rest of the steps;

5) if the handled MCData ID is not a member of the MCData group identified by the served MCData group ID, shall reject the SIP SUBSCRIBE request with SIP 403 (Forbidden) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [5], IETF RFC 3903 [34] and IETF RFC 3856 [39] and skip the rest of the steps; and

6) shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [5], IETF RFC 6665 [36].

For the duration of the subscription, the MCData server shall notify subscriber about changes of the information of the served MCData ID, as described in subclause 8.3.3.5."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.3.5]

"In order to notify the subscriber identified by the handled MCData ID about changes of the affiliation status of the served MCData group ID, the MCData server:

1) shall consider an MCData group information entry such that:

a) the MCData group information entry is in the list of MCData group information entries described in subclause 8.3.3.2; and

b) the MCData group ID of the MCData group information entry is equal to the served MCData group ID;

2) shall consider an MCData user information entry such:

a) the MCData user information entry is in the list of the MCData user information entries of the served MCData group information entry; and

b) the MCData ID of the MCData user information entry is equal to the handled MCData ID;

as the served MCData user information entry;

3) shall generate an application/pidf+xml MIME body indicating per-group affiliation information according to subclause 8.4.1 and the served list of the served MCData user information entry of the MCData group information entry with following clarifications:

a) the MCData server shall include the "expires" attribute in the <affiliation> element; and

b) if this procedures is invoked by procedure in subclause 8.3.3.3 where the handled p-id was identified, the MCData server shall set the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id value; and

4) send a SIP NOTIFY request according to 3GPP TS 24.229 [5], and IETF RFC 6665 [36] for the subscription created in subclause 8.3.3.4. In the SIP NOTIFY request, the MCData server shall include the generated application/pidf+xml MIME body indicating per-group affiliation information."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.7.1]

"To create the group's security association, a Group Master Key (GMK) and associated identifier (GMK-ID) is distributed to MCX UEs by a Group Management Server (GMS). The GMK is distributed encrypted specifically to a user and signed using an identity representing the Group Management Server. Prior to group key distribution, each MCX UE within the group shall be provisioned by the MCX Key Management Server (KMS) with time-limited key material associated with the MCX user as described in clause 5.3. The Group Management Server shall also be provisioned by the MCX KMS with key material for the GMS's identity (the GMS Server URI).

The GMK is distributed from the GMS to a MCX client using the security mechanism described in clause 5.2.2, transported over the SIP bearer. For GMKs, end-point diversity is required and hence the extension in clause 5.2.3 is applied. Additional parameters may be included as defined in clause 5.2.4. The SAKKE-to-self extension may be included as defined in clause 5.2.5. Identity hiding may be supported as defined in clause 5.2.6. The receiving MCX client and any MCX Server through which the SIP INVITE is routed, may respond with a KMS Redirect Response (KRR) as described in clause 5.2.8.

GMKs may be managed individually per Group ID or the same GMK may be assigned to multiple MC Group IDs (using the MIKEY general extension payload defined in Clause E.6). This means that each specified MC Group ID in the MIKEY general extension payload shall use this GMK for group communications. Assigned MC Group IDs may include any combination of MCPTT Group IDs, MCData Group IDs or MCVideo Group IDs. Assigning the same GMK to multiple Group IDs does not prevent individual key management at a later time or vice versa.

An MC client may have multiple active GMKs associated with a Group ID. When this occurs, the MC client shall use the active GMK with the most recent Activation Time (as defined in Clause E.6.4) when encrypting group media.

The initiating entity shall be the initiating GMS. The initiating entity URI shall be the URI of the GMS (e.g. sip: gp.manager@mcptt.example.org). The receiving entity shall be the terminating MCX user. The receiving entity URI shall be the MCX service ID of the terminating user. The distributed key, K, shall be the GMK, the key identifier K-ID shall be the GMK-ID and the end-point-specific key identifier, UK-ID shall be the GUK-ID.

Clause E.3 provides MIKEY message structure for GMK distribution."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.7.3]

"In some situations, the membership of a group may be modified whereby an MC user may be added to or removed from an MCX group. Users are alerted to these changes by a user profile update from the CMS for which they are subscribed. The updated user configuration profile indicates the group ID to which the group membership change is associated.

When users are added to a new or existing group they may be implicitly affiliated to that group in which case the user is automatically subscribed to group configuration updates from the GMS. The user shall be authorised for group management services to the GMS before the GMS provides the associated group management records and the GMK. Once the user is authorised, the GMS sends the group management record as well as the GMK to the UE. The user may join in on the group communication immediately after receiving the group update and GMK.

When the user configuration record indicates the user has been added to a new or existing group but is required to explicitly affiliate to the group, the user shall be authorised for group management services to the GMS followed by a subscription to group updates from the GMS. The user shall be authorised for group management services and the subscription shall be validated before the GMS provides group management records and the GMK. Once the user is authorised and the subscription processed by the GMS, the GMS sends the group management record and the GMK to the UE. The user may then join in on the group communication immediately after receiving the group update and GMK.

When a user is removed from a group, the UE receives a user profile update from the CMS indicating the user is no longer a member of the specified group ID(s). Upon receiving the user profile update, ending of any group communication(s) associated with that group, and if the GMK associated with the group ID is not associated with another group that the user remains a member, the UE shall immediately and securely delete the GMK associated with that group ID. If the group ID is associated to more than one service (i.e. MCPTT, MCData and/or MCVideo) then upon the ending of any group communication(s) associated with that group ID, and if the GMKs associated with that group ID is not associated with another group that the user remains a member, the GMKs associated with that group ID shall be immediately and securely deleted.

When a user is removed from the group, the Group Management Server may choose to update the GMK associated with the group ID (depending on the security profile of the group)."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.8.1]

"The 'key download' procedure is used to send keys from the MCX server to the MC client. It is used to distribute Multicast Signalling Keys (MuSiKs) to the MC clients, and it is used to update both the CSKs and MuSiKs.

Within the 'key download' procedure, keys (CSK or MuSiKs) are encrypted specifically to the MC user and signed using an identity representing the MC Server. Prior to group key distribution, each MC client shall be provisioned by the KMS with time-limited key material associated with the MC User as described clause 5.3. The MC Server shall also be provisioned by the MC KMS with key material for an identity which is authorised to act as an MCX Server.

The key (CSK or MuSiK) is distributed from the MCX Server to a MC client using the security mechanism described in clause 5.2.2, transported over the SIP bearer. End-point diversity is not required as end-points do not encrypt data, hence the extension in clause 5.2.3 is not applied. Additional parameters may be included as defined in clause 5.2.4. The SAKKE-to-self extension may be included as defined in clause 5.2.5. Identity hiding may be supported as defined in clause 5.2.6.

The initiating entity shall be the initiating MCX Server and the receiving entity shall be the terminating MC user. The initiating entity URI shall be the FQDN of the MCX Server (e.g. MDSI of the MC Domain) and the receiving entity URI shall be the MC Service ID of the terminating user. The distributed key, K, shall be the CSK or MuSiK and the key identifier K-ID shall be the CSK-ID or MuSiK-ID (respectively).

As a result of this 'key download' mechanism, the MC clients receive a new signalling key, CSK or MuSiK, identified by the 4 most significant bits of the key ID.

The MCData Service server may use the Key Download procedure to indicate or modify the algorithm used to protect the MCData signalling fields (i.e. MCData signaling parameters, Data signaling payload and End to end security parameters) by including a 'signalling algorithm' parameter. The 'signalling algorithm' parameter is described in clause 8.5.4.1. The available algorithms shall be as defined in clause 8.5.4.2."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 7.3.2]

"The group creation procedure is described in clause 10.2.3 of 3GPP TS 23.280 [36] and applies to the MCPTT scenario of normal group creation by an MC administrator and user regrouping operations by an authorized user/dispatcher. To establish the security context for the group, the GMS follows the procedures in clause 5.7 to create a new GMK and GMK-ID.

The encapsulated GMK and GUK-ID is sent to group members by the GMS within a notification message (step 4 in clause 10.2.3 of 3GPP TS 23.280 [36]). The procedure is equivalent to that described in clause 5.7 of this specification."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 7.3.3.2]

"Group Regroup procedures for the MC system are described in clause 10.2.4.1 of 3GPP TS 23.280 [36]. To create the security context for the temporary group, the GMS follows the procedures in clause 5.7, creating a new GMK and GMK-ID for the temporary group.

An encapsulated GMK and GUK-ID is sent to the temporary group members by the GMS within a notification message (step 5 in clause 10.2.4.1 of 3GPP TS 23.280 [36]). The procedure is equivalent to that described in clause 5.7."

5.2.3 Test description

5.2.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client)
- For the underlying "transport bearer" over which the SS-UE1 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server)
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Distribution Function, the MCData User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCData Server) is the acting Group Management Server.

Preamble:

- The IUT (MCData Server) is connected to PLMN1

- The IUT (MCData Server) is connected to the SS-UE1 (MCData client) as defined in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Figure 4.2.4, where term "MCPTT" is replaced with "MCData".
- The SS-UE1 (MCData client) performs the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.
- The SS-UE1 (MCData client) is member of group A.

5.2.3.2 Test procedure sequence

Table 5.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server correctly perform procedure 'MCX Server Terminated Group Creation' as described in Annex A, Table A.2.4.3-1 to create group B?	-	-	1	P
2	Check: Does the MCData Server correctly perform procedure 'MCX Server Terminated Temporary Group Creation' as described in Annex A, Table A.2.5.3-1 to create temporary group T?	-	-	2	P
3	Check: Does the MCData Server correctly perform procedure 'MCX Server Originated notifications regarding temporary group creation or tear down' as described in Annex A, Table A.2.3.3-1 to update the group document of group A and retrieve the GMK for the temporary group T	-	-	2	P
4	Check: Does the MCData Server correctly perform procedure 'MCX Server Terminated Temporary Group Tear Down ' as described in Annex A, Table A.2.6.3-1 to delete temporary group T?	-	-	3	P
5	Check: Does the MCData Server correctly perform procedure 'MCX Server Originated notifications regarding temporary group creation or tear down' as described in Annex A, Table A.2.3.3-1 to update the group document and the GKTP document of group A?	-	-	3	P

5.2.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications:

- Subclauses referring to messages from UE relate to SS-UE1 (MCX Client).

5.3 MCDATA Server - MCDATA Client / Configuration / Group Affiliation / Implicit Affiliation / Remote change / De-affiliation / Home MCDATA system

5.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) having received the provisioned MCData Client's group information or a
pointer to the group information, indicating that the MCData Client is authorized to affiliate with
the specified group}
ensure that {
  when { the SS-UE1 (MCData client ) requests for own affiliation status and to subscribe to
affiliation status changes for the MCData User with SIP SUBSCRIBE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
          }
```

(2)

with { IUT (MCData Server) having received the provisioned MCData Client's group information or a pointer to the group information, indicating that the MCData Client is authorized to affiliate with the specified group }
ensure that {
  when { the SS-UE1 (MCData client) requests to de-affiliate user A from an MCData group with SIP PUBLISH }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message **and** sends SIP NOTIFY message to inform the change progress status **and** sends SIP NOTIFY message to inform the final status }
            }

(3)

with { IUT (MCData Server) having received the provisioned MCData Client's group information or a pointer to the group information, indicating that the MCData Client is authorized to affiliate with the specified group }
ensure that {
  when { the SS-UE1 (MCData client) requests to affiliate user A to an MCData group with SIP PUBLISH message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message **and** sends SIP NOTIFY message to inform the change progress status **and** sends SIP NOTIFY message to inform the final status }
            }

(4)

with { IUT (MCData Server) having received the provisioned MCData Client's group information or a pointer to the group information, indicating that the MCData Client is authorized to affiliate with the specified group }
ensure that {
  when { the SS-UE2 (MCData client) requests to de-affiliate user B from an MCData group with SIP PUBLISH message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message **and** sends SIP NOTIFY message to inform the change progress status **and** sends SIP NOTIFY message to inform the final status }
            }

(5)

with { IUT (MCData Server) having received the provisioned MCData Client's group information or a pointer to the group information, indicating that the MCData Client is allowed to make affiliation changes for another user with the specified group }
ensure that {
  when { the SS-UE1 (MCData client) requests to affiliate status changes of targeted user with SIP SUBSCRIBE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message **and** sends a SIP NOTIFY message }
            }

(6)

with { IUT (MCData Server) having received the provisioned MCData Client's group information or a pointer to the group information, indicating that the MCData Client is allowed to make affiliation changes for another user with the specified group }
ensure that {
  when { the SS-UE1 (MCData client) requests to affiliate targeted user B to an MCData group via mandatory mode with SIP PUBLISH message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message **and** sends SIP NOTIFY message to inform the change progress status **and** sends SIP NOTIFY message to inform the final status }
            }

(7)

with { IUT (MCData Server) having received the provisioned MCData Client's group information or a pointer to the group information, indicating that the MCData Client is allowed to make affiliation changes for another user with the specified group }
ensure that {
  when { the SS-UE1 (MCData client) requests to de-affiliate targeted user from an MCData group via mandatory mode with SIP PUBLISH message }
    then { IUT (MCData Server) responds to the SS-UE (MCData client) with a SIP 200 (OK) message **and** sends SIP NOTIFY message to inform the change progress status **and** sends SIP NOTIFY message to inform the final status }
            }

(8)

```
with { IUT (MCData Server) having received the provisioned MCData Client's group information or a
pointer to the group information, indicating that the MCData Client is allowed to make affiliation
changes for another user with the specified group }
ensure that {
  when { SS-UE1 (MCData Client) requests that a target user B be affiliated to an MCData group via
negotiated mode by sending a SIP MESSAGE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) }
            }
```

(9)

```
with { MCData Client already affiliated to the MCData group }
ensure that {
  when { the SS-UE1 (MCData Client) requests to affiliate targeted user to a group via negotiated
mode }
    then { IUT (MCData Server) informs user B about a remote request by sending a SIP MESSAGE
message to the SS-UE2 (MCData Client) }
            }
```

(10)

```
with { IUT (MCData Server) having received the provisioned MCData Client's group information or a
pointer to the group information, indicating that the MCData Client is authorized to affiliate with
the specified group }
ensure that {
  when { the SS-UE2 (MCData client) requests to affiliate user B to an MCData group with SIP PUBLISH
message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends SIP NOTIFY message to
inform the change progress status and sends SIP NOTIFY message to inform the final status }
            }
```

(11)

```
with { IUT (MCData Server) having received the provisioned MCData Client's group information or a
pointer to the group information, indicating that the MCData Client is allowed to make affiliation
changes for another user with the specified group }
ensure that {
  when { SS-UE1 (MCData Client) requests that a target user B be de-affiliated to an MCData group
via
negotiated mode by sending a SIP MESSAGE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) }
            }
```

(12)

```
with { MCData Client already de-affiliated from the MCData group }
ensure that {
  when { the SS-UE1 (MCData Client) requests to de-affiliate targeted user to a group via negotiated
mode }
    then { IUT (MCData Server) informs user B about a remote request by sending a SIP MESSAGE
message to the SS-UE2 (MCData Client) }
            }
```

(13)

```
with { IUT (MCData Server) having received the provisioned MCData Client's group information or a
pointer to the group information, indicating that the MCData Client is authorized to affiliate with
the specified group}
ensure that {
  when { the SS-UE1 (MCData client ) requests to de-subscribe from affiliation status changes for
the MCData User with SIP SUBSCRIBE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

5.3.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 484 [8] (3GPP TS 24.484) clauses 4.3.1, 4.3.4, 4.4, 6.3.13.2.3, 6.3.3.2.3, 6.3.3.2.1, 10.2.1, 10.2.1A, 10.3.1, 10.3.1A, 10.4.1, 6.2.3, ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 6.3.5, 8.3.2.3, 8.3.2.4, 8.3.2.5, 8.3.2.8, 8.2.3.9, 8.2.3.10, 8.3.2.11, 8.3.2.12, 8.3.2.13, 8.3.2.14, 8.3.2.15, 8.3.3.3, 8.3.3.4, 8.3.3.5, 8.3.3.7. Unless otherwise stated these are Rel-14 requirements.

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.3.1]

"The MCS server obtains the appropriate MCS service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCS from the configuration management server.

The MCS server obtains the MCS service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCS from the configuration management server.

The MCS server subscribes to the MCS service configuration document for each mission critical organisation that is provisioned that is supported by the MCS server using the procedure specified in subclause 6.3.13.2.3. How the MCS server is provisioned with the identities of the mission critical organisations is out of scope of the present document.

If the MCS service configuration document has been updated since the current version stored at the MCS server, then the MCS server will receive a SIP NOTIFY request containing an HTTPS URI of the MCS service configuration document. Retrieval by the MCS server, using the notified HTTPS URI, of the MCS service configuration document is performed as specified in subclause 6.3.3.2.3.

NOTE: The MCS server can be notified of changes to the MCS service management configuration document at any time while operating the MCS."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.3.4]

"The format of the MCData service configuration document downloaded to the MCData server is defined in subclause 10.4."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.4]

"The following applies to the configuration management server used for online configuration.

The configuration management server needs to convert the MCS UE initial configuration document received from an MCS administrator into an appropriate format for configuration of the MCS UE initial configuration MO.

If the MCS UE initial configuration MO contains a <default-user-profile> element that identifies an MCS user profile configuration document, the configuration management server needs to convert the identified MCS user profile configuration document received from an MCS administrator into an appropriate format for configuration of the MCS user profile configuration MO.

Once an MCS User Profile configuration document has been created or updated by the MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to store MCS user profile configuration document as the user profile in the MC service user database.

In order to download the MCS user profile configuration document to an MCS UE or to support an MCS UE updating the MCS user profile configuration document, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to obtain the MCS user profile from the MC service user database.

In order to be notified of changes to an MCS user profile configuration document(s) that have been subscribed to by an MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to be notified of changes to the MCS user profile stored in the MC service user database.

In order to delete the MCS user profile when requested by an MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to delete the MCS user profile from the MC service user database.

NOTE: The configuration management server and group management server functionality for offline configuration is out of scope of the present document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.13.2.3]

"In order to subscribe to an MCS service configuration document, an MCS server shall send an initial SIP SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [22] and IETF RFC 5875 [11]. In the initial SIP SUBSCRIBE request, MCS server:

a) shall set the Request URI to a SIP URI containing:

1) the base URI being equal to the public service identity of the CMS configured in the MCS server; and

2) the "auid" parameter set to the application usage identifying th MCS service configuration document;

b) shall include a P-Asserted-Identity header field containing the public service identity of the MCS server;

c) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [22]), in a P-Asserted-Service header field according to IETF RFC 6050 [23]; and

d) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field.

Upon receiving a SIP NOTIFY request associated with a subscription created as result of the sent initial SIP SUBSCRIBE request, the MCS server shall handle the SIP NOTIFY request according to IETF RFC 5875 [11].

In order to re-subscribe to notification of changes to an MCS service configuration document, an MCS server shall send a SIP re-SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [22] and IETF RFC 5875 [11]. In the SIP re-SUBSCRIBE request, MCS server:

a) shall set the Request URI to a SIP URI containing:

1) the base URI being equal to the public service identity of the CMS configured in the MCS server; and

2) the "auid" parameter set to the application usage identifying an MCS service configuration document; and

b) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.3.2.3]

"In order to retrieve a configuration management document via the CSC-5 reference point, an MCS Server shall perform the procedures in subclause 6.3.3.2.1 specified for GC. The MCS server shall set the Request-URI of the HTTP GET request to identify the XML document based on configuration and include the "auid" as per the appropriate application usage."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.3.2.1]

"In order to retrieve a configuration management document, a GC shall send an HTTP GET request with the Request URI that references the document to be updated to the network according to procedures specified in IETF RFC 4825 [14] "Retrieve a Document"."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.2.1]

"The MCData UE configuration document is specified in this subclause. The MCData UE configuration document content is based on requirements of Annex A.2 of 3GPP TS 23.282 [24], in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData UE configuration document in the MCData service is described in 3GPP TS 24.282 [25] and 3GPP TS 24.582 [26]. The schema definition is provided in subclause 10.2.2.3. An MCData UE configuration document may apply to all MCData UEs of a mission critical organization or apply to specific MCData UEs identified in the <mcdata-UE-id> element. If there is no <mcdata-UE-id> element in the MCData UE configuration document, then by default the MCData UE configuration document applies to all MCData UEs of the mission critical organization that are not specifically identified in the <mcdata-UE-id> element of another MCData UE configuration document of the mission critical organization. Each MCData UE of a mission critical organization is configured with an MCData UE configuration document that is identified by the instance ID of the MCData UE.

…. A MCData UE configuration document corresponding to a specific MCData UE the MCData user has used to authenticate and is authorised to use the MCData service with is placed in the user directory of the MCData user.

The MCData UE configuration document acts as a template for the CMS to generate UE configuration documents that are downloaded to specific MCData UEs. The MCData UE configuration document that acts as a template is referred to as a "master MCData UE configuration document". The master MCData UE configuration document is stored in the users tree of that MCData system administrator. The master MCData UE configuration document does not directly apply to a specific MCData UE, but instead acts as template that the CMS uses to populate the MCData UE configuration documents of MCData UEs identified by elements of the <MCData-UE-id> element. For MCData UE configuration documents that correspond to a specific MCData UE, the name of the MCData UE configuration document is created from a value defined by the corresponding element that identifies the MCData UE within the <MCData-UE-id> element. For a master MCData UE configuration documents that does not contain a <MCData-UE-id> element, the name of the MCData UE configuration document stored in the user directory is "DEFAULT-MCData-UE.xml"."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.2.1A]

"The MCData UE configuration document is accessed using the same XCAP URI, regardless of whether the MCData UE has a specific MCData UE configuration document configured or the master MCData UE configuration document applies. The CMS shall generate the UE's MCData UE configuration document from the master MCData UE configuration document if the MCData administrator did not provision a specific MCData UE configuration document. In this generated document, the <MCData-UE-id> element shall be set to the MCData client's UE ID. The UE's MCData UE configuration document shall always be stored with the filename corresponding to the UE's MCData UE ID under the user's directory of the users tree.

The XCAP URI used by the MCData client to access the UE's MCData UE configuration document shall be:

- *CMSXCAPROOTURI/org.3gpp.mcdata.ue-config/users/sip:MCDATAID/MCSUEID."*

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.3.1]

"The MCData user profile configuration document is specified in this subclause. The MCData user profile configuration document content is based on requirements of Annex A.3 of 3GPP TS 23.282 [24], and structure and procedures of OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData user profile in the MCData service is described in 3GPP TS 24.282 [25]. The schema definition is provided in subclause 10.3.2.

MCData user profile documents are "XDM collections" in the user's directory of the users tree, in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. In this case, the term "user" in the XCAP sense refers to the MCData ID, as the user has been already authenticated.

The name of the MCData user profile document matches the value of the <ProfileName> element in the MCData user profile document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.3.1A]

"The XCAP URI used by the MCData client to access the MCData user's MCData user profile documents shall be:

CMSXCAPROOTURI/org.3gpp.mcdata.user-profile/users/sip:MCDATAID/mcdata-user-profile-INDEX.xml

Where INDEX is the index of the MCData user profile as defined in subclause 10.3.2.8."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.4.1]

"The MCData service configuration document is specified in this subclause. The MCData service configuration document content is based on requirements of Annex A.5 of 3GPP TS 23.282 [24], and structure and procedures of OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData service configuration in the MCData service is described in 3GPP TS 24.282[25] and 3GPP TS 24.582 [26]. The schema definition is provided in subclause 10.4.2.3. Each mission critical organization is configured with an MCData service configuration document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.2.3]

"The MCS server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.482 [6]."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 6.3.5]

"The MCData server shall determine that the MCData user, with MCData User ID, is affiliated to the MCData group, with MCData Group ID, at the MCData client, with MCData client ID, if the elements, as described in subclause 8.3.3.2, exist with their expected values, as below:

1. *an MCData group information entry with MCData group ID same as the MCData group ID under consideration;*
2. *in the MCData group information entry found in 1, an MCData user information entry with the MCData ID same as the MCData ID under consideration;*
3. *in the MCData user information entry found in 2, an MCData client information entry with MCData Client ID same as the MCData client ID under consideration; and*
4. *in the MCData user information entry found in 2, an expiration time, which has not expired."*

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.3]

"Upon receiving a SIP PUBLISH request such that:

1) *Request-URI of the SIP PUBLISH request contains either the public service identity identifying the originating participating MCData function serving the MCData user, or the public service identity identifying the terminating participating MCData function serving the MCData user;*
2) *the SIP PUBLISH request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the<mcdata-request-uri> element which identifies an MCData ID served by the MCData server;*
3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7];*
4) *the Event header field of the SIP PUBLISH request contains the "presence" event type; and*
5) *SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-user affiliation information according to subclause 8.4.1;*

then the MCData server:

1) *shall identify the served MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;*
2) *if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the originating participating MCData function serving the MCData user, shall identify the originating MCData ID from public user identity in the P-Asserted-Identity header field of the SIP PUBLISH request;*
3) *if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the terminating participating MCData function serving the MCData user, shall identify the originating MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;*

…

6) *if the Expires header field of the SIP PUBLISH request has nonzero value, shall determine the candidate expiration interval to according to IETF RFC 3903 [34];*
7) *if the Expires header field of the SIP PUBLISH request has zero value, shall set the candidate expiration interval to zero;*
8) *shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to 3GPP TS 24.229 [5], IETF RFC 3903 [34]. In the SIP 200 (OK) response, the MCData server:*
    a) *shall set the Expires header field according to IETF RFC 3903 [34], to the candidate expiration time;*
9) *if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served MCData ID, shall not continue with the rest of the steps;*

10) shall identify the served MCData client ID in the "id" attribute of the <tuple> element of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request;

11) shall consider an MCData user information entry such that:

a) the MCData user information entry is in the list of MCData user information entries described in subclause 8.3.2.2; and

b) the MCData ID of the MCData user information entry is equal to the served MCData ID;

as the served MCData user information entry;

12) shall consider an MCData client information entry such that:

a) the MCData client information entry is in the list of MCData client information entries of the served MCData user information entry; and

b) the MCData client ID of the MCData client information entry is equal to the served MCData client ID;

as the served MCData client information entry;

13) shall consider a copy of the list of the MCData group information entries of the served MCData client information entry as the served list of the MCData group information entries;

14) if the candidate expiration interval is nonzero:

a) shall construct the candidate list of the MCData group information entries as follows:

i) for each MCData group ID which has an MCData group information entry in the served list of the MCData group information entries, such that the expiration time of the MCData group information entry has not expired yet, and which is indicated in a "group" attribute of an <affiliation> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

A) shall copy the MCData group information entry into a new MCData group information entry of the candidate list of the MCData group information entries;

B) if the affiliation status of the MCData group information entry is "deaffiliating" or "deaffiliated", shall set the affiliation status of the new MCData group information entry to the "affiliating" state and shall reset the affiliating p-id of the new MCData group information entry; and

C) shall set the expiration time of the new MCData group information entry to the current time increased with the candidate expiration interval;

ii) for each MCData group ID which has an MCData group information entry in the served list of the MCData group information entries, such that the expiration time of the MCData group information entry has not expired yet, and which is not indicated in any "group" attribute of the <affiliation> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

A) shall copy the MCData group information entry into a new MCData group information entry of the candidate list of the MCData group information entries; and

B) if the affiliation status of the MCData group information entry is "affiliated" or "affiliating":

- *shall set the affiliation status of the new MCData group information entry to the "de-affiliating" state; and*

- *shall set the expiration time of the new MCData group information entry to the current time increased with twice the value of timer F; and*

iii) for each MCData group ID:

A) which does not have an MCData group information entry in the served list of the MCData group information entries; or

B) which has an MCData group information entry in the served list of the MCData group information entries, such that the expiration time of the MCData group information entry has already expired;

and which is indicated in a "group" element of the <affiliation> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

A) shall add a new MCData group information entry in the candidate list of the MCData group information list for the MCData group ID;

B) shall set the affiliation status of the new MCData group information entry to the "affiliating" state;

C) shall set the expiration time of the new MCData group information entry to the current time increased with the candidate expiration interval; and

D) shall reset the affiliating p-id of the new MCData group information entry;

b) determine the candidate number of MCData group IDs as number of different MCData group IDs which have an MCData group information entry:

i) in the candidate list of the MCData group information entries; or

ii) in the list of the MCData group information entries of an MCData client information entry such that:

A) the MCData client information entry is in the list of the MCData client information entries of the served MCData user information entry; and

B) the MCData client ID of the MCData client information entry is not equal to the served MCData client ID;

with the affiliation status set to the "affiliating" state or the "affiliated" state and with the expiration time which has not expired yet; and

c) if the candidate number of MCData group IDs is bigger than N2 value of the served MCData ID, shall based on MCData service provider policy reduce the candidate MCData group IDs to that equal to N2;

NOTE: The MCData service provider policy can determine to remove an MCData group ID based on the order it appeared in the PUBLISH request or based on the importance or priority of the MCData group or some other policy to determine which MCData groups are preferred.

15) if the candidate expiration interval is zero, constructs the candidate list of the MCData group information entries as follows:

a) for each MCData group ID which has an entry in the served list of the MCData group information entries:

i) shall copy the MCData group entry of the served list of the MCData group information into a new MCData group information entry of the candidate list of the MCData group information entries;

ii) shall set the affiliation status of the new MCData group information entry to the "de-affiliating" state; and

iii) shall set the expiration time of the new MCData group information entry to the current time increased with twice the value of timer F;

16) shall replace the list of the MCData group information entries stored in the served MCData client information entry with the candidate list of the MCData group information entries;

17) shall perform the procedures specified in subclause 8.3.2.6 for the served MCData ID and each MCData group ID:

a) which does not have an MCData group information entry in the served list of the MCData group information entries and which has an MCData group information entry in the candidate list of the MCData group information entries with the affiliation status set to the "affiliating" state;

b) which has an MCData group information entry in the served list of the MCData group information entries with the expiration time already expired, and which has an MCData group information entry in the candidate list of the MCData group information entries with the affiliation status set to the "affiliating" state;

c) which has an MCData group information entry in the served list of the MCData group information entries with the affiliation status set to the "deaffiliating" state or the "deaffiliated" state and with the expiration time not expired yet, and which has an MCData group information entry in the candidate list of the MCData group information entries with the affiliation status set to the "affiliating" state; or

d) which has an MCData group information entry in the served list of the MCData group information entries with the affiliation status set to the "affiliated" state and with the expiration time not expired yet, and which has an MCData group information entry in the candidate list of the MCData group information entries with the affiliation status set to the "de-affiliating" state;

18) shall identify the handled p-id in the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

19) shall perform the procedures specified in subclause 8.3.2.5 for the served MCData ID."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.4]

"Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains either the public service identity identifying the originating participating MCData function serving the MCData user, or the public service identity identifying the terminating participating MCData function serving the MCData user;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the<mcdata-request-uri> element which identifies an MCData ID served by the MCData server;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7]; and

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type;

the MCData server:

1) shall identify the served MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

2) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the originating participating MCData function serving the MCData user, shall identify the originating MCData ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;

3) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the terminating participating MCData function serving the MCData user, shall identify the originating MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

4) if the originating MCData ID is different than the served MCData ID and the originating MCData ID is not authorized to modify affiliation status of the served MCData ID, shall send a 403 (Forbidden) response and shall not continue with the rest of the steps; and

5) shall generate a 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [5], IETF RFC 6665 [36].

For the duration of the subscription, the MCData server shall notify the subscriber about changes of the information of the served MCData ID, as described in subclause 8.3.2.5."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.5]

"In order to notify the subscriber about changes of the served MCData ID, the MCData server:

1) shall consider an MCData user information entry such that:

a) the MCData user information entry is in the list of MCData user information entries described in subclause 8.3.2.2; and

b) the MCData ID of the MCData user information entry is equal to the served MCData ID;

as the served MCData user information entry;

2) shall consider the list of the MCData client information entries of the served MCData user information entry as the served list of the MCData client information entries;

3) shall generate an application/pidf+xml MIME body indicating per-user affiliation information according to subclause 8.4.1 and the served list of the MCData client information entries with the following clarifications:

a) the MCData server shall not include information from an MCData group information entry with the expiration time already expired;

b) the MCData server shall not include information from an MCData group information entry with the affiliation status set to the "deaffiliated" state;

c) if the SIP SUBSCRIBE request creating the subscription of this notification contains an application/simple-filter+xml MIME body indicating per-client restrictions of presence event package notification information according to subclause 8.4.2, the MCData server shall restrict the application/pidf+xml MIME body according to the application/simple-filter+xml MIME body; and

d) if this procedures is invoked by procedure in subclause 8.3.2.3 where the handled p-id value was identified, the MCData server shall set the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id value; and

4) send a SIP NOTIFY request according to 3GPP TS 24.229 [5], and IETF RFC 6665 [36] for the subscription created in subclause 8.3.2.4. In the SIP NOTIFY request, the MCData server shall include the generated application/pidf+xml MIME body indicating per-user affiliation information."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.8]

"Upon receiving a SIP MESSAGE request such that:

1) Request-URI of the SIP MESSAGE request contains the public service identity identifying the terminating participating MCData function serving the MCData user;

2) the SIP MESSAGE request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the<mcdata-request-uri> element and the <mcdata-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7]; and

4) the SIP MESSAGE request contains an application/vnd.3gpp.mcdata-affiliation-command+xml MIME body;

then the MCData server:

1) shall identify the served MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request;

2) shall identify the originating MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request;

3) if the originating MCData ID is not authorized to send an affiliation status change request in negotiated mode to the served MCData ID, shall send a 403 (Forbidden) response and shall not continue with the rest of the steps;

4) shall set the Request-URI of the SIP MESSAGE request to the public user identity bound to the served MCData ID in the MCData server; and

5) *shall include an Accept-Contact header field with the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8];*

before forwarding the SIP MESSAGE request further."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.9]

"Upon receiving a SIP PUBLISH request such that:

1) *Request-URI of the SIP PUBLISH request contains the public service identity identifying the originating participating MCData function serving the MCData user;*

2) *the SIP PUBLISH request contains an application/vnd.3gpp.mcdata-info MIME body containing the<mcdata-request-uri> element which identifies an MCData ID not served by the MCData server;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7];*

4) *the Event header field of the SIP PUBLISH request contains the "presence" event type; and*

5) *SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-user affiliation information according to subclause 8.4.1;*

then the MCData server:

1) *shall identify the target MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info MIME body of the SIP PUBLISH request;*

2) *shall identify the originating MCData ID from public user identity in the P-Asserted-Identity header field of the SIP PUBLISH request;*

3) *shall generate a SIP PUBLISH request from the received SIP PUBLISH request. In the generated SIP PUBLISH request, the MCData server:*

    a) *shall set the Request-URI to the public service identity identifying the terminating participating MCData function serving the target MCData ID;*

    b) *shall include a P-Asserted-Identity header field containing the public service identity identifying the originating participating MCData function serving the MCData user;*

    c) *shall include an application/vnd.3gpp.mcdata-info+xml MIME body. In the application/vnd.3gpp.mcdata-info+xml MIME body, the MCData server:*

        A) *shall include the <mcdata-request-uri> element set to the target MCData ID; and*

        B) *shall include the <mcdata-calling-user-id> element set to the originating MCData ID; and*

    d) *shall include other signalling elements from the received SIP PUBLISH request; and*

4) *shall send the generated SIP PUBLISH request according to 3GPP TS 24.229 [5].*

The MCData server shall forward received SIP responses to the SIP PUBLISH request."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.10]

"Upon receiving a SIP SUBSCRIBE request such that:

1) *Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the originating participating MCData function serving the MCData user;*

2) *the SIP SUBCRIBE request contains an application/vnd.3gpp.mcdata-info MIME body containing the<mcdata-request-uri> element which identifies an MCData ID not served by MCData server;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7]; and*

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type;

then the MCData server:

1) shall identify the target MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info MIME body of the SIP SUBSCRIBE request;

2) shall identify the originating MCData ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;

3) shall generate a SIP SUBSCRIBE request from the received SIP SUBSCRIBE request. In the generated SIP SUBSCRIBE request, the MCData server:

a) shall set the Request-URI to the public service identity identifying the terminating participating MCData function serving the target MCData ID;

b) shall include a P-Asserted-Identity header field containing the public service identity identifying the originating participating MCData function serving the MCData user;

c) shall include an application/vnd.3gpp.mcdata-info+xml MIME body. In the application/vnd.3gpp.mcdata-info+xml MIME body, the MCData server:

A) shall include the <mcdata-request-uri> element set to the target MCData ID; and

B) shall include the <mcdata-calling-user-id> element set to the originating MCData ID; and

d) shall include other signalling elements from the received SIP SUBSCRIBE request; and

4) shall send the generated SIP SUBSCRIBE request according to 3GPP TS 24.229 [5].

The MCData server shall forward any received SIP responses to the SIP SUBSCRIBE request, any received SIP NOTIFY request and any received SIP responses to the SIP NOTIFY request."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.11]

"This subclause is referenced from other procedures.

If the participating MCData function needs to determine the affiliation status of an MCData user to an MCData group, the participating function:

1) shall find the user information entry in the list of MCData user information entries described in subclause 8.3.2.2 such that the MCData ID of the MCData user information entry is equal to the MCData ID of the originator of the received SIP request;

a) if the applicable MCData group information entry cannot be found, then the participating MCData function shall determine that the MCData user is not affiliated to the MCData group at the MCData client and the skip the rest of the steps;

2) shall find the MCData client information entry in the list of MCData client information entries of MCData user information entry found in step 1) in which the MCData client id matches the value of the <mcdata-client-id> element contained in the application/vnd.3gpp.mcdata-info+xml MIME body in the received SIP request;

a) if the applicable MCData client information entry cannot be found, then the participating MCData function shall determine that the MCData user is not affiliated to the MCData group at the MCData client and the skip the rest of the steps;

3) shall find the MCData group information entry in the list of MCData group information entries of MCData client information entry found in step 2 such that the MCData group identity matches the value of the identity of the targeted MCData group;

a) if the applicable MCData group information entry was found in step 3) and the affiliation status of the MCData group information entry is "affiliating" or "affiliated", shall determine that the MCData user at the MCData client to be affiliated to the targeted MCData group and skip the rest of the steps;

b) if the applicable MCData group information entry was found in step 3) and the affiliation status of the MCData group information entry is "deaffiliating" or "deaffiliated", shall determine that the MCData user at the MCData client to not be affiliated to the targeted MCData group and skip the rest of the steps; or

c) if the applicable MCData group information entry was not found in step 3), shall determine that the MCData user at the MCData client is not affiliated to the targeted MCData group."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.12]

"This subclause is referenced from other procedures.

Upon receiving a SIP request that requires implicit affiliation of the sending MCData client to an MCData group, the participating MCData function:

1) shall determine the served MCData client ID from the <mcdata-client-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the received SIP request;

2) shall determine the MCData group ID from the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the received SIP request;

3) shall determine the served MCData ID by using the public user identity in the P-Asserted-Identity header field of the SIP request;

NOTE 1: The MCData ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in subclause 7.3.

4) shall consider an MCData user information entry such that:

a) the MCData user information entry is in the list of MCData user information entries described in subclause 8.3.2.2; and

b) the MCData ID of the MCData user information entry is equal to the served MCData ID;

as the served MCData user information entry;

5) shall consider an MCData client information entry such that:

a) the MCData client information entry is in the list of MCData client information entries of the served MCData user information entry; and

b) the MCData client ID of the MCData client information entry is equal to the served MCData client ID;

as the served MCData client information entry;

6) shall consider a copy of the list of the MCData group information entries of the served MCData client information entry as the served list of the MCData group information entries;

7) shall construct the candidate list of the MCData group information entries as follows:

a) for each MCData group ID which has an MCData group information entry in the served list of the MCData group information entries shall copy the MCData group information entry into a new MCData group information entry of the candidate list of the MCData group information entries; and

b) if the determined MCData group ID does not have an MCData group information entry in the served list of the MCData group information entries or has an MCData group information entry in the served list of the MCData group information entries, such that the expiration time of the MCData group information entry has already expired:

i) shall add a new MCData group information entry in the candidate list of the MCData group information list for the determined MCData group ID;

ii) shall set the affiliation status of the new MCData group information entry to the "affiliating" state; and

iii) shall set the expiration time of the new MCData group information entry to the current time increased with the candidate expiration interval;

8) determine the candidate number of MCData group IDs as the number of different MCData group IDs which have an MCData group information entry:

a) in the candidate list of the MCData group information entries; or

b) in the list of the MCData group information entries of an MCData client information entry such that:

i) the MCData client information entry is in the list of the MCData client information entries of the served MCData user information entry; and

ii) the MCData client ID of the MCData client information entry is not equal to the served MCData client ID;

with the affiliation status set to the "affiliating" state or the "affiliated" state and with the expiration time which has not expired yet; and

9) if the candidate number of MCData group IDs is bigger than the N2 value of the served MCData ID, shall based on MCData service provider policy reduce the candidate MCData group IDs to that equal to N2;

10) if the determined MCData group ID cannot be added to the the candidate list of the MCData group information entries due to exceeding the MCData user's N2 limit, shall discard the candidate list of the MCData group information entries and skip the remaining steps of the current procedure; and

11) shall replace the list of the MCData group information entries stored in the served MCData client information entry with the candidate list of the MCData group information entries."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.13]

"This subclause is referenced from other procedures.

If the participating MCData function has received a SIP 2xx response from the controlling MCData function to a SIP request that had triggered performing the procedures of subclause 8.3.2.12, the participating MCData function:

1) shall set the affiliation status of the MCData group information entry added to the candidate list of the MCData group information entries by the procedures of subclause 8.3.2.12 to "affiliated"; and

2) shall perform the procedures specified in subclause 8.3.2.5 for the served MCData ID."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.14]

"This subclause is referenced from other procedures.

If the participating MCData function determines that a received SIP request that had triggered performing the procedures of subclause 8.3.2.12 needs to be rejected or if the participating MCData function receives a SIP 4xx, 5xx or 6xx response from the controlling MCData function for the received SIP request, the participating MCData function:

1) shall remove the MCData group ID entry added by the procedures of subclause 8.3.2.12 such that:

a) the MCData group information entry has the MCData group ID set to the MCData group ID of the MCData group targeted by the received SIP request;

b) the MCData group information entry is in the list of the MCData group information entries of an MCData client information entry containing the MCData client ID included in the received SIP request; and

c) the MCData client information entry is in the list of the MCData client information entries of the MCData user information entry containing the MCData ID of the sender of the received SIP request."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.2.15]

"This subclause is referenced from other procedures.

If the participating MCData function has successfully performed service authorization for the MCData ID identified in the service authorisation procedure as described in 3GPP TS 33.179 [56], the participating MCData function:

1) shall identify the MCData ID included in the SIP request received for service authorisation procedure as the served MCData ID;

2) shall identify the MCData client ID from the <mcdata-client-id> element contained in the application/vnd.3gpp.mcdata-info+xml MIME body included in the SIP request received for service authorisation as the served MCData client ID;

3) shall download the MCData user profile from the MCData user database as defined in 3GPP TS 29.283 [37] if not already stored at the participating MCData function;

4) if no <ImplicitAffiliations> element is contained in the <OnNetwork> element of the MCData user profile document (see the MCData user profile document in 3GPP TS 24.484 [12]) for the served MCData ID or the <ImplicitAffiliations> element contains no <entry> elements containing an MCData group ID, shall skip the remaining steps;

5) shall consider an MCData user information entry such that:

a) the MCData user information entry is in the list of MCData user information entries described in subclause 8.3.2.2; and

b) the MCData ID of the MCData user information entry is equal to the served MCData ID;

as the served MCData user information entry;

6) shall consider an MCData client information entry such that:

a) the MCData client information entry is in the list of MCData client information entries of the served MCData user information entry; and

b) the MCData client ID of the MCData client information entry is equal to the served MCData client ID;

as the served MCData client information entry;

7) shall consider a copy of the list of the MCData group information entries of the served MCData client information entry as the served list of the MCData group information entries;

8) shall construct the candidate list of the MCData group information entries as follows:

a) for each MCData group ID which has an MCData group information entry in the served list of the MCData group information entries shall copy the MCData group information entry into a new MCData group information entry of the candidate list of the MCData group information entries;

b) for each MCData group ID contained in an <entry> element of the <ImplicitAffiliations> element in the <OnNetwork> element of the MCData user profile document (see the MCData user profile document in 3GPP TS 24.484 [12]) for the served MCData ID that does not have an MCData group information entry in the served list of the MCData group information entries or has an MCData group information entry in the served list of the MCData group information entries such that the expiration time of the MCData group information entry has already expired:

i) shall add a new MCData group information entry in the candidate list of the MCData group information list for the MCData group ID;

ii) shall set the affiliation status of the new MCData group information entry to the "affiliating" state; and

iii) shall set the expiration time of the new MCData group information entry to the current time increased with the candidate expiration interval;

c) if in step b) above, no new MCData group information entries were added to the candidate list of the MCData group information list for the MCData group ID:

i) shall discard the candidate list; and

ii) shall skip the remaining steps;

9) determine the candidate number of MCData group IDs as the number of different MCData group IDs which have an MCData group information entry:

a) in the candidate list of the MCData group information entries; or

b) in the list of the MCData group information entries of an MCData client information entry such that:

i) the MCData client information entry is in the list of the MCData client information entries of the served MCData user information entry; and

ii) the MCData client ID of the MCData client information entry is not equal to the served MCData client ID;

with the affiliation status set to the "affiliating" state or the "affiliated" state and with the expiration time which has not expired yet; and

c) if the candidate number of MCData group IDs is bigger than the N2 value of the served MCData ID, shall based on MCData service provider policy reduce the candidate MCData group IDs to that equal to N2;

10) shall replace the list of the MCData group information entries stored in the served MCData client information entry with the candidate list of the MCData group information entries; and

11) for each MCData group ID contained in an <entry> element of the <ImplicitAffiliations> element in the <OnNetwork> element of the MCData user profile document (see the MCData user profile document in 3GPP TS 24.484 [12]) for the served MCData ID and which has an MCData group information entry in the candidate list of the MCData group information entries with an affiliation status of "affiliating", shall perform the procedures specified in subclause 8.3.2.6 for the served MCData ID and each MCData group ID.

NOTE 2: To learn of the MCData groups successfully affiliated to, the MCData client can subscribe to that information by the procedures specified in subclause 8.2.3."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.3.3]

"Upon receiving a SIP PUBLISH request such that:

1) Request-URI of the SIP PUBLISH request contains the public service identity of the controlling MCData function associated with the served MCData group;

2) the SIP PUBLISH request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the <mcdata-request-uri> element and the <mcdata-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7];

4) the Event header field of the SIP PUBLISH request contains the "presence" event type; and

5) SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-group affiliation information constructed according to subclause 8.4.1;

then the MCData server:

1) shall identify the served MCData group ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;

2) shall identify the handled MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;

…

6) shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to 3GPP TS 24.229 [5], IETF RFC 3903 [34]. In the SIP 200 (OK) response, the MCData server:

a) shall set the Expires header field according to IETF RFC 3903 [34], to the selected expiration time;

7) if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served MCData group ID, shall not continue with the rest of the steps;

8) if the handled MCData ID is different from the MCData ID in the "id" attribute of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request, shall not continue with the rest of the steps;

9) shall consider an MCData group information entry such that:

a) the MCData group information entry is in the list of MCData group information entries described in subclause 8.3.3.2; and

b) the MCData group ID of the MCData group information entry is equal to the served MCData group ID;

as the served MCData group information entry;

10) if the selected expiration time is zero:

a) shall remove the MCData user information entry such that:

i) the MCData user information entry is in the list of the MCData user information entries of the served MCData group information entry; and

ii) the MCData user information entry has the MCData ID set to the served MCData ID;

11) if the selected expiration time is not zero:

a) shall consider an MCData user information entry such that:

i) the MCData user information entry is in the list of the MCData user information entries of the served MCData group information entry; and

ii) the MCData ID of the MCData user information entry is equal to the handled MCData ID;

as the served MCData user information entry;

b) if the MCData user information entry does not exist:

i) shall insert an MCData user information entry with the MCData ID set to the handled MCData ID into the list of the MCData user information entries of the served MCData group information entry; and

ii) shall consider the inserted MCData user information entry as the served MCData user information entry; and

c) shall set the following information in the served MCData user information entry:

i) set the MCData client ID list according to the "client" attributes of the <affiliation> elements of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

ii) set the expiration time according to the selected expiration time;

12) shall identify the handled p-id in the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

13) shall perform the procedures specified in subclause 8.3.3.5 for the served MCData group ID."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.3.4]

"Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains the public service identity of the controlling MCData function associated with the served MCData group;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the<mcdata-request-uri> element and the <mcdata-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7];

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type; and

5) the SIP SUBSCRIBE request contains an application/simple-filter+xml MIME body indicating per-user restrictions of presence event package notification information according to subclause 8.4.2 indicating the same MCData ID as in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

then the MCData server:

1) shall identify the served MCData group ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

2) shall identify the handled MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

…

6) shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [5], IETF RFC 6665 [36].

For the duration of the subscription, the MCData server shall notify subscriber about changes of the information of the served MCData ID, as described in subclause 8.3.3.5."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.3.5]

"In order to notify the subscriber identified by the handled MCData ID about changes of the affiliation status of the served MCData group ID, the MCData server:

1) shall consider an MCData group information entry such that:

a) the MCData group information entry is in the list of MCData group information entries described in subclause 8.3.3.2; and

b) the MCData group ID of the MCData group information entry is equal to the served MCData group ID;

2) shall consider an MCData user information entry such:

a) the MCData user information entry is in the list of the MCData user information entries of the served MCData group information entry; and

b) the MCData ID of the MCData user information entry is equal to the handled MCData ID;

as the served MCData user information entry;

3) shall generate an application/pidf+xml MIME body indicating per-group affiliation information according to subclause 8.4.1 and the served list of the served MCData user information entry of the MCData group information entry with following clarifications:

a) the MCData server shall include the "expires" attribute in the <affiliation> element; and

b) if this procedures is invoked by procedure in subclause 8.3.3.3 where the handled p-id was identified, the MCData server shall set the <p-id> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id value; and

4) send a SIP NOTIFY request according to 3GPP TS 24.229 [5], and IETF RFC 6665 [36] for the subscription created in subclause 8.3.3.4. In the SIP NOTIFY request, the MCData server shall include the generated application/pidf+xml MIME body indicating per-group affiliation information."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.3.7]

"This subclause is referenced from other procedures.

Upon receiving a SIP request for an MCData group that the MCData user is not currently affiliated to and that requires the controlling MCData function to perform an implicit affiliation to, the controlling MCData function:

1) shall identify the served MCData group ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP request;

2) shall identify the handled MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP request;

3) shall consider an MCData group information entry such that:

a) the MCData group information entry is in the list of MCData group information entries described in subclause 8.3.3.2; and

b) the MCData group ID of the MCData group information entry is equal to the served MCData group ID;

as the served MCData group information entry;

4) shall consider an MCData user information entry such that:

a) the MCData user information entry is in the list of the MCData user information entries of the served MCData group information entry; and

b) the MCData ID of the MCData user information entry is equal to the handled MCData ID;

as the served MCData user information entry;

c) if the MCData user information entry does not exist:

i) shall insert an MCData user information entry with the MCData ID set to the handled MCData ID into the list of the MCData user information entries of the served MCData group information entry; and

ii) shall consider the inserted MCData user information entry as the served MCData user information entry; and

d) shall make the following modifications in the served MCData user information entry:

i) add the MCData client ID derived from the received SIP request to the MCData client ID list if not already present; and

ii) set the expiration time as determined by local policy;

5) shall perform the procedures specified in subclause 8.3.3.5 for the served MCData group ID."

5.3.3 Test description

5.3.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client)
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server)
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Distribution Function, the MCData User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCData Server) is connected to PLMN1
- The IUT (MCData Server) is connected to the SS-UE1 (MCData client) and the SS-UE2 (MCData client) as defined in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Figure 4.2.4, where term "MCPTT" is replaced with "MCData".
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

5.3.3.2 Test procedure sequence

Table 5.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation Status Determination and Subscription' as described in Annex A, Table A.4.11.3-1 to subscribe the SS-UE1 (MCData client) to own affiliation status changes for user A? (Note)	-	-	1	P
2	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to de-affiliate user A from group A using the SS-UE1 (MCData client)?	-	-	2	P
3	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to affiliate user A to group A using the SS-UE1 (MCData client)?	-	-	3	P
4	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation Status Determination and Subscription' as described in Annex A, Table A.4.11.3-1 to subscribe the SS-UE2 (MCData client) to own affiliation status changes for user B? (Note)	-	-	1	P
5	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to de-affiliate user B from group A using the SS-UE2 (MCData client)?	-	-	4	P
6	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation Status Determination and Subscription' as described in Annex A, Table A.4.11.3-1 to subscribe the SS-UE1 (MCData client) to affiliation status changes of targeted user B?	-	-	5	P
7	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to request from the SS-UE1 (MCData Client) to affiliate targeted user B to group A (mandatory mode)?	-	-	6	P
8	Check: Does the MCData Server correctly perform steps 3-6 of procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to inform user B in the SS-UE2 (MCData client) that is being remotely affiliated to group A?	-	-	6	P
9	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to request from the SS-UE1 (MCData Client) to de-affiliate targeted user B from group A (mandatory mode)?	-	-	7	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
10	Check: Does the MCData Server correctly perform steps 3-6 of procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to inform user B in the SS-UE2 (MCData Client) that is being remotely de-affiliated from group A?	-	-	7	P
11	Check: Does the MCData Server correctly perform procedure 'MCX Participating Terminated SIP MESSAGE' as described in Annex A, Table A.2.8.3-1 to request from the SS-UE1 (MCData Client) to affiliate targeted user B to group A via negotiated mode?	-	-	8	P
12	Check: Does the MCData Server correctly perform procedure 'MCX Participating Originated SIP MESSAGE' as described in Annex A, Table A.2.9.3-1 to inform user B in the SS-UE2 (MCData Client) about a remote request to affiliate to group A via negotiated mode?	-	-	9	P
13	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to affiliate user B to group A using the SS-UE2 (MCData client)?	-	-	10	P
14	Check: Does the MCData Server correctly perform steps 3-6 of procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to inform user A in the SS-UE1 (MCData Client), that user B is being affiliated to group A?	-	-	10	P
15	Check: Does the MCData Server correctly perform procedure 'MCX Participating Terminated SIP MESSAGE' as described in Annex A, Table A.2.8.3-1 to request from the SS-UE1 (MCData Client) to de-affiliate targeted user B to group A via negotiated mode?	-	-	11	P
16	Check: Does the MCData Server correctly perform procedure 'MCX Participating Originated SIP MESSAGE' as described in Annex A, Table A.2.9.3-1 to inform user B in the SS-UE2 (MCData Client) about a remote request to de-affiliate to group A via negotiated mode	-	-	11	P
17	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to de-affiliate user B to group A using the SS-UE2 (MCData client)?	-	-	4	P
18	Check: Does the MCData Server correctly perform steps 3-6 of procedure 'MCData Group Affiliation status change' as described in Annex A, Table A.4.12.3-1 to inform user A in the SS-UE1 (MCData Client), that user B is being de-affiliated from group A?	-	-	4	P
19	Check: Does the MCData Server correctly perform procedure 'MCData Group Affiliation Status Determination and Subscription' as described in Annex A, Table A.4.11.3-1 to de-subscribe the SS-UE1 (MCData Client) from affiliation status changes for targeted user B?	-	-	13	P
NOTE: Affiliation status change triggered by implicit affiliation.					

5.3.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications:

Table 5.3.3.3-1: SIP SUBSCRIBE from the SS-UE1 (Step 1, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.11.3-1)

Derivation Path: Annex A, Table A.4.11.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Annex A, Table A.2.7.4-2			
MIME body part		**SIMPLE-FILTER**		
MIME-part-body	Simple-Filter as described in Table 5.3.3.3-2			

Table 5.3.3.3-2: Simple-Filter in SIP SUBSCRIBE (Table 5.3.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.6-1, condition PER-CLIENT

Table 5.3.3.3-3: SIP SUBSCRIBE from the SS-UE2 (Step 4, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.11.3-1)

Derivation Path: Annex A, Table A.4.11.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)			
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-4			
MIME body part		**SIMPLE-FILTER**		
MIME-part-body	Simple-Filter as described in Table 5.3.3.3-2			

Table 5.3.3.3-4: MCData-Info in SIP SUBSCRIBE (Table 5.3.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	Encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-3A.				

Table 5.3.3.3-5: SIP SUBSCRIBE from the SS-UE1 (Step 6, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.11.3-1)

Derivation Path: Annex A, Table A.4.11.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-4			

**Table 5.3.3.3-6: SIP SUBSCRIBE from the SS-UE1
(Step 19, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.11.3-1)**

Derivation Path: Annex A, Table A.4.11.4-1, condition re-SUBSCRIBE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
delta-seconds	"0"			
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-4			

**Table 5.3.3.3-7: SIP PUBLISH from the SS-UE1
(Step 2, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.12.3-1)**

Derivation Path: Annex A, Table A.4.12.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
delta-seconds	"0"			

**Table 5.3.3.3-8: SIP PUBLISH from the SS-UE1
(Step 3, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.12.3-1)**

Derivation Path: Annex A, Table A.4.12.4-1

**Table 5.3.3.3-9: SIP PUBLISH from the SS-UE2
(Steps 5 and 17, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.12.3-1)**

Derivation Path: Annex A, Table A.4.12.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)			
Expires				
delta-seconds	"0"			
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-10			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-11			

Table 5.3.3.3-10: MCData-Info in SIP PUBLISH (Table 5.3.3.3-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	Encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-3A.				

Table 5.3.3.3-11: PIDF in SIP PUBLISH (Table 5.3.3.3-9)

Derivation Path: Annex A, Table A.4.12.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (note) with value set to px_MCData_ID_User_B			
tuple				
Id attribute	Encrypted URI (note) with value set to px_MCX_Client_B_ID			
status	not present			
..p-id	any allowed value	p-id shall be present acc. to ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.2.2		
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.13.3-1.				

Table 5.3.3.3-12: SIP PUBLISH from the SS-UE1 (Step 7, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.11.3-1)

Derivation Path: Annex A, Table A.4.12.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-13			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-14			

Table 5.3.3.3-13: MCData-Info in SIP PUBLISH (Table 5.3.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	Encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-3A.				

Table 5.3.3.3-14: PIDF in SIP PUBLISH (Table 5.3.3.3-12)

Derivation Path: Annex A, Table A.4.12.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (note) with value set to px_MCData_ID_User_B			
tuple				
Id attribute	Encrypted URI (note) with value set to px_MCX_Client_B_ID			
status	not present			
..p-id	any allowed value	p-id shall be present acc. to ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.2.2		
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.13.3-1.				

Table 5.3.3.3-15: SIP PUBLISH from the SS-UE1
(Step 9, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.12.3-1)

Derivation Path: Annex A, Table A.4.12.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
delta-seconds	"0"			
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-13			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-14			

Table 5.3.3.3-16: SIP PUBLISH from the SS-UE2
(Step 13, Table 5.3.3.2-1; Step 1, Annex A, Table A.4.12.3-1)

Derivation Path: Annex A, Table A.4.12.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From				
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)			
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-10			
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-11			

Table 5.3.3.3-17: SIP NOTIFY from the MCData Server
(Step 1, Table 5.3.3.2-1; Step 3, Annex A, Table A.4.11.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-18			

Table 5.3.3.3-18: PIDF in SIP NOTIFY (Table 5.3.3.3-17)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliated"			

Table 5.3.3.3-19: SIP NOTIFY from the MCData Server (Step 3, Table 5.3.3.2-1; Step 5, Annex A, Table A.4.12.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-20			

Table 5.3.3.3-20: PIDF in SIP NOTIFY (Table 5.3.3.3-19)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliated"			
..p-id	same value as received in corresponding SIP PUBLISH (Step 3)			

Table 5.3.3.3-21: SIP NOTIFY from the MCData Server (Step 3, Table 5.3.3.2-1; Step 3, Annex A, Table A.4.12.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-22			

Table 5.3.3.3-22: PIDF in SIP NOTIFY (Table 5.3.3.3-21)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliating"			
..p-id	same value as received in corresponding SIP PUBLISH (Step 3)			

Table 5.3.3.3-23: SIP NOTIFY from the MCData Server (Step 2, Table 5.3.3.2-1; Step 3, Annex A, Table A.4.12.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-24			

Table 5.3.3.3-24: PIDF in SIP NOTIFY (Table 5.3.3.3-23)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"deaffiliating"			
..p-id	same value as received in corresponding SIP PUBLISH (Step 2)			

Table 5.3.3.3-25: SIP NOTIFY from the MCData Server (Step 2, Table 5.3.3.2-1; Step 5, Annex A, Table A.4.12.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-26			

Table 5.3.3.3-26: PIDF in SIP NOTIFY (Table 5.3.3.3-25)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	not present			

Table 5.3.3.3-27: SIP NOTIFY from the MCData Server (Step 4, Table 5.3.3.2-1; Step 3, Annex A, Table A.4.11.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-28			

Table 5.3.3.3-28: PIDF in SIP NOTIFY (Table 5.3.3.3-27)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (note) with value set to px_MCData_ID_User_B			
tuple				
Id attribute	Encrypted URI (note) with value set to px_MCX_Client_B_ID			
status				
affiliation				
status	"affiliated"			
NOTE: Encrypted attribute as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.13.3-1.				

Table 5.3.3.3-29: SIP NOTIFY from the MCData Server (Steps 7, 8, 13, 14, Table 5.3.3.2-1; Step 5, Annex A, Table A.4.12.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-30			

Table 5.3.3.3-30: PIDF in SIP NOTIFY (Table 5.3.3.3-29)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (note) with value set to px_MCData_ID_User_B			
tuple				
Id attribute	Encrypted URI (note) with value set to px_MCX_Client_B_ID			
status				
affiliation				
status	"affiliated"			
..p-id	same value as received in corresponding SIP PUBLISH (Steps 7 and 13)			
NOTE: Encrypted attribute as described in Table 5.5.13.3-1.				

Table 5.3.3.3-31: SIP NOTIFY from the MCData Server (Steps 7, 8, 13, 14, Table 5.3.3.2-1; Step 3, Annex A, Table A.4.12.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-32			

Table 5.3.3.3-32: PIDF in SIP NOTIFY (Table 5.3.3.3-31)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (note) with value set to px_MCData_ID_User_B			
tuple				
Id attribute	Encrypted URI (note) with value set to px_MCX_Client_B_ID			
status				
affiliation				
status	"affiliating"			
..p-id	same value as received in corresponding SIP PUBLISH (Steps 7 and 13)			
NOTE: Encrypted attribute as described in Table 5.5.13.3-1.				

Table 5.3.3.3-33: SIP NOTIFY from the MCData Server (Steps 5, 9, 10, 17, 18, Table 5.3.3.2-1; Step 3, Annex A, Table A.4.12.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-34			

Table 5.3.3.3-34: PIDF in SIP NOTIFY (Table 5.3.3.3-33)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (note) with value set to px_MCData_ID_User_B			
tuple				
Id attribute	Encrypted URI (note) with value set to px_MCX_Client_B_ID			
status				
affiliation				
status	"deaffiliating"			
..p-id	same value as received in corresponding SIP PUBLISH (Steps 5, 9 and 17)			
NOTE: Encrypted attribute as described in Table 5.5.13.3-1.				

Table 5.3.3.3-35: SIP NOTIFY from the MCData Server (Steps 5, 6, 9, 10, 17, 18, Table 5.3.3.2-1; Step 5, Annex A, Table A.4.12.3-1; Step 3, Annex A, Table A.4.11.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-36			

Table 5.3.3.3-35: PIDF in SIP NOTIFY (Table 5.3.3.3-36)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
entity attribute	Encrypted URI (note) with value set to px_MCData_ID_User_B			
tuple				
Id attribute	Encrypted URI (note) with value set to px_MCX_Client_B_ID			
status	not present			
NOTE: Encrypted attribute as described in Table 5.5.13.3-1.				

Table 5.3.3.3-36: SIP NOTIFY from the MCData Server (Step 19, Table 5.3.3.2-1; Step 3, Annex A, Table A.4.11.3-1)

Derivation Path: Annex A, Table A.4.11.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [30]	
substate-value	"terminated"			
expires	not present			
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.3.3.3-35			

Table 5.3.3.3-37: SIP MESSAGE from the SS-UE1 (Step 11, Table 5.3.3.2-1; Step 1, Annex A, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, conditions MCDATA and AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-38			

Table 5.3.3.3-38: MCData-Info in SIP MESSAGE (Table 5.3.3.3-37)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	Encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-3A.				

Table 5.3.3.3-39: SIP MESSAGE from the MCData Server (Step 12, Table 5.3.3.2-1; Step 1, Annex A, Table A.2.17.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, conditions MCDATA and AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-40			

Table 5.3.3.3-40: MCData-Info in SIP MESSAGE (Table 5.3.3.3-39)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	Encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
mcdata-calling-user-id	not present			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A.				

Table 5.3.3.3-41: SIP MESSAGE from the SS-UE1
(Step 15, Table 5.3.3.2-1; Step 1, Annex A, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, conditions MCDATA and AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**Affiliation-Command**		
MIME-part-body	Affiliation-Command as described in Table 5.3.3.3-43			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-42			

Table 5.3.3.3-42: MCData-Info in SIP MESSAGE (Table 5.3.3.3-41)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	Encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-3A.				

Table 5.3.3.3-43: Affiliation-Command in SIP MESSAGE (Table 5.3.3.3-41)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.7-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
command-list				
affiliate	not present			
de-affiliate				
group[1]	px_MCData_Group_A_ID	MCData group name		

Table 5.3.3.3-44: SIP MESSAGE from the MCData Server
(Step 16, Table 5.3.3.2-1; Step 1, Annex A, Table A.2.17.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, conditions MCDATA and AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**Affiliation-Command**		
MIME-part-body	Affiliation-Command as described in Table 5.3.3.3-45			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.3.3.3-46			

Table 5.3.3.3-45: MCData-Info in SIP MESSAGE (Table 5.3.3.3-44)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	Encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
mcdata-calling-user-id	not present			
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A.				

Table 5.3.3.3-46: Affiliation-Command in SIP MESSAGE (Table 5.3.3.3-44)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.7-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
command-list				
affiliate	not present			
de-affiliate				
group[1]	px_MCData_Group_A_ID	MCData group name		

5.4 MCDATA Server - MCDATA Client / Configuration / Pre-established Session Establishment / Pre-established Session Modification / Pre-established Session Release

5.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) is able to create pre-established session }
ensure that {
  when { the SS-UE1 (MCData client ) requests creation of pre-established session with sending SIP
INVITE message }
    then { IUT (MCData Server) responds with optional SIP 100 Trying and mandatory 200 (OK) message
}
            }
```

(2)

```
with { IUT (MCData Server) is able to release pre-established session }
ensure that {
  when { the SS-UE1 (MCData client ) requests to terminate MCX session with sending SIP BYE message
}
    then { IUT (MCData Server) responds with 200 (OK) message }
            }
```

(3)

```
with { IUT (MCData Server) is able to release pre-established Originating session }
ensure that {
  when { the IUT (MCData Server) triggers release session }
    then { the IUT (MCData Server) requests release session sending SIP BYE message }
```

5.4.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 18.1, 18.3.2.2, 18.3.3.2.1, 18.3.2.2.2, 18.3.4.2.1, 18.3.4.2.2. Unless otherwise stated these are Rel-16 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 18.1]

"The MCData client may establish one or more pre-established sessions to the participating MCData function at any time after SIP registration and setting the service settings as defined in subclause 7.2.2 or subclause 7.2.3.

The MCData client may use the pre-established session for originating standalone SDS using media plane or SDS session after pre-established session establishment.

The participating MCData function may use the pre-established session for terminating standalone SDS using media plane or SDS session after pre-established session establishment.

The use of a pre-established session requires the use of resource sharing as specified in 3GPP TS 29.214 [49] and 3GPP TS 24.229 [5] by the participating MCData function. The participating MCData function use of resource sharing is defined in subclause 18.2."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 18.3.2.2]

"Upon receipt of a "SIP INVITE request for establishing a pre-established session" the participating MCData function:

1) shall check whether the public service identity is allocated and if it is not allocated, shall return a SIP 404 (Not Found) response and skip the rest of the steps;

…

3) shall determine whether resource sharing is supported (see subclause 18.2);

4) if resource sharing is supported by the SIP core, determine that there is a binding between the MCData ID of the MCData user establishing the pre-established session and the MCData UE identified by the "+g.3gpp.registration-token" header field parameter in the Contact header field of the third-party REGISTER request (see subclause 18.2) and that this UE identity matches the identity in the "+g.3gpp.registration-token" header field parameter in the Feature-Caps header field in the "SIP INVITE request for establishing a pre-established session";

…

8) shall allocate a URI to be used to identify the pre-established session;

9) shall generate a SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [5]; and

a) shall include a Contact header field containing the URI that identifies the pre-established session;

b) shall include the public service identity in the P-Asserted-Identity header field;

c) shall include a Supported header field containing the "norefersub" option tag;

d) shall if the SIP core supports resource sharing, include a Resource-Share header field answer as specified in 3GPP TS 24.229 [5] with:

A) the value "media-sharing";

B) an "origin" header field parameter set to "session-initiator";

C) a "timestamp" header field parameter; and

D) a "rules" header field parameter with one resource sharing rule per media stream in the same order the corresponding m-line appears in the SDP. Each resource sharing rule is constructed as follows:

- *a "new-sharing-key" part; and*
- *a "directionality" part indicating the direction of the pre-established media stream; and*

e) shall include an SDP answer as specified in 3GPP TS 24.229 [5] with the clarifications in subclause 18.3.1.2 and include ICE candidates in the SDP answer as per IETF RFC 5245 [50];

10) shall interact with the media plane as specified in 3GPP TS 24.582 [15];

11) shall send the SIP 200 (OK) response towards the MCData client according to the rules and procedures of the 3GPP TS 24.229 [5]; and

12) shall evaluate the ICE candidates according to IETF RFC 5245 [50].

NOTE: If ICE candidate evaluation results in candidate pairs other than the default candidate pair being selected a further offer answer exchange using the procedures in subclause 18.3.4 will be needed."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 18.3.3.2.1]

"Upon receiving a SIP BYE request from the MCData client within a pre-established session the participating MCData function:

1) shall check whether there is a MCData session using the pre-established session, and:

a) if there is an established MCData session then the participating MCData function shall remove the MCData client from the MCData session by performing the procedures as specified in subclause 13.2.2.2.3.1;

b) if there is a MCData session in the process of being established, then the participating MCData function:

i) shall send a SIP CANCEL request to cancel the MCData session in the process of being established as specified in 3GPP TS 24.229 [5]; and

ii) shall release the MCData session as specified in the subclause 13.2.2.2.3.1, if a SIP 200 (OK) response for the SIP INVITE request is received from the remote side; and

c) if there is no MCData session using the pre-established session, then the participating MCData function shall:

i)_ interact with the media plane as specified in 3GPP TS 24.582 [15] for disconnecting the media plane resources towards the MCData client; and

ii) shall generate and send a SIP 200 (OK) response to the SIP BYE request according to rules and procedures of 3GPP TS 24.229 [5].

Upon receiving a SIP 200 (OK) response to the SIP BYE request from the remote side, the participating MCData function:

1) shall interact with the media plane as specified in 3GPP TS 24.582 [15] for releasing media plane resources towards the remote side;

2) shall interact with the media plane as specified in 3GPP TS 24.582 [15] for releasing media plane resources towards the MCData client; and

3) shall send a SIP 200 (OK) response to the SIP BYE request to the MCData client."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 18.3.2.2.2]

"When a participating MCData function needs to release a pre-established session as created in subclause 8.2.2, the participating MCData function:

1) shall first release any participants of all MCData calls that are using the pre-established session. The participating MCData function shall remove the MCData client from the MCData session by performing the procedures as specified in subclause 13.2.2.2.3.1;

2) shall generate a SIP BYE request according to rules and procedures of 3GPP TS 24.229 [5];

3) shall set the Request-URI of the SIP BYE request to the URI that identifies the pre-established session;

4) shall send the SIP BYE request towards the MCData client within the SIP dialog of the pre-established session according to rules and procedures of the 3GPP TS 24.229 [5]; and

5) shall, upon receiving a SIP 200 (OK) response to the SIP BYE request interact with the media plane as specified in 3GPP TS 24.582 [15]."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 18.3.4.2.1]

"Upon receiving a SIP UPDATE request or a SIP re-INVITE request to modify an existing pre-established session without associated MCData session, the participating MCData function:

1) shall validate that the received SDP offer includes at least one media stream for which the media parameters and the MSRP URI is acceptable by the participating MCData function and if not reject the request with a SIP 488 (Not Acceptable Here) response. Otherwise, continue with the rest of the steps; and

2) shall generate a SIP 200 (OK) response as follows:

a) include an SDP answer according to 3GPP TS 24.229 [5] based on the received SDP offer with the clarifications given in the subclause 18.3.1.2, and include ICE candidates in the SDP answer as per IETF RFC 5245 [50], if required; and

b) include a Contact header field containing the URI that identifies the pre-established session and send a SIP 200 (OK) response according to the rules and procedures of 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 18.3.4.2.2]

"When the participating MCData function needs to modify the pre-established session outside of an MCData session, the participating MCData function:

1) shall generate a SIP UPDATE request or a SIP re-INVITE request according to 3GPP TS 24.229 [5];

2) shall include an SDP offer according to 3GPP TS 24.229 [5], and include ICE candidates in the SDP offer as per IETF RFC 5245 [50], if required; and

3) shall send the SIP request towards the MCData client according to the rules and procedures of 3GPP TS 24.229 [5].

On receipt of the SIP 200 (OK) response, the participating MCData function:

1) shall interact with the media plane as specified in 3GPP TS 24.582 [15], if there is change in media parameters or the MSRP URI in the received SDP answer, compared to those in the previously agreed SDP;

2) shall interact with the media plane as specified in 3GPP TS 24.582 [15], if there is a media stream, that is currently used in the pre-established session, is removed in the received SDP answer; and

3) shall interact with the media plane as specified in 3GPP TS 24.582 [15], if there is a media stream accepted in the received SDP answer, that is not currently used by the participant in the pre-established session.

NOTE: The participating MCData function keeps resources for previously agreed media stream, media parameters and the MSRP URI until it receives a SIP 200 (OK) response."

5.4.3 Test description

5.4.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client)
- For the underlying "transport bearer" over which the SS-UE1 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server)
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Distribution Function, the MCData User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCData Server) is the acting Participating Server.

Preamble:

- The IUT (MCData Server) is connected to PLMN1
- The IUT (MCData Server) is connected to the SS-UE1 (MCData client) as defined in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Figure 4.2.4, where term "MCPTT" is replaced with "MCData".
- The SS-UE1 (MCData client) performs the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

5.4.3.2 Test procedure sequence

Table 5.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData server correctly perform test procedure 'MCX Participating Terminated pre-established session establishment' as described in Annex A, Table A.2.13.3-1 to create a pre-established session?	-	-	1	P
2	Check: Does the MCData Server correctly perform test procedure MCX Participating Terminated call release' as described in Annex A, Table A.2.15.3-1 to release the pre-established session?	-	-	2	P
3	Check: Does the MCData server correctly perform test procedure 'MCX Participating Terminated pre-established session establishment' as described in Annex A, Table A.2.13.3-1 to create a pre-established session?	-	-	1	P
4	Check: Does the MCData Server correctly perform test procedure MCX Participating Originated call release' as described in Annex A, Table A.2.17.3-1 to release the pre-established session?	-	-	3	P

5.4.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 5.4.3.3-1: SIP INVITE from the SS-UE1
(Steps 1 and 3, Table 5.4.3.2-1; Step 1, Annex A, Table A.2.13.3-1)

Derivation Path: Annex A, Table A.2.13.4-1, condition MCDATA_SDS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	"1800"	The recommended initial value is 1 800 in IETF RFC 4028 [23].		
Max-Forwards				
value	"70"	The recommended initial value is 70 in IETF RFC 3261 [21].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.4.3.3-2			

Table 5.4.3.3-2: SDP for SIP INVITE (Table 5.4.3.3-1)

Derivation Path: Table 5.5.3.1.1-3, condition PRE_ESTABLISHED_SESSION, SDP_OFFER, SDS_SESSION, MCDATA_SDS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line		
Connection Data	not present	c= line		
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	port number assigned by the SS-UE	The transport port to which the media stream is sent		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Session attribute		a=line attribute="ice-lite"	IETF RFC 5245 [29]	
ice-lite				
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	not present			
host	IP address of the SS-UE			
port	same value as in the media line			
session id	assigned by the SS-UE			
transport	"tcp"			
URI-parameter	not present			
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for TCP/MSRP		
foundation	1234	arbitrarily selected		
component-id	1	according to IETF RFC 5245 [29] clause 4.1.1.1		
transport	"TCP/MSRP"			
priority	2130706431	IETF RFC 5245 [29] clause 4.2: 224 * 126 + 28 * 65535 + 256 - component id		
connection-address	IP address of the SS-UE (same IP address as in the c=line)	default candidate		
port	same port number as in the m= line			
cand-type	"host"			

Table 5.4.3.3-3: SIP 200 (OK) from the MCData Server (Steps 1 and 3, Table 5.4.3.2-1; Step 3, Annex A, Table A.2.13.3-1)

Derivation Path: Annex A, Table A.2.13.4-2, condition MCDATA_SDS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 5.4.3.3-2			

Table 5.4.3.3-4: SDP for SIP 200 (OK) (Table 5.4.3.3-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-3, condition PRE_ESTABLISHED_SESSION, SDP_ANSWER, SDS_SESSION, MCDATA_SDS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	any allowed value	"-" indicating the concept of user IDs not being supported		
sess-id	any allowed value	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	any allowed value			
Session Name	any allowed value	s= line		
Connection Data	not required if included in all media	c= line		
nettype	"IN"			
Addrtype	"IP4" or "IP6" depending on IP address"			
connection-address	IP address of the MCData Server			
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	any allowed value	The transport port to which the media stream is sent		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCData Server			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	any value if present			
host	any allowed value			
port	same value as in the media line if present			
session id	any allowed value if present			
transport	"tcp"			
URI-parameter	not present			
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for TCP/MSRP		
foundation	any value			
component-id	1	according to IETF RFC 5245 [29] clause 4.1.1.1		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-3, condition PRE_ESTABLISHED_SESSION, SDP_ANSWER, SDS_SESSION, MCDATA_SDS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
transport	"TCP/MSRP"			
priority	any value			
connection-address	same IP address as in media's c= line or in the session's c= line if the media does not have a c= line	default candidate		
port	same port number as in the m= line			
cand-type	"host"			

Table 5.4.3.3-5: SIP BYE from the SS-UE1 (Step 2, Table 5.4.3.2-1; Step 1, Annex A, Table A.2.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCX_SessionID_B	The URI that identifies the pre-established session		

Table 5.4.3.3-6: SIP BYE from the MCData-Server (Step 4, Table 5.4.3.2-1; Step 1, Annex A, Table A.2.17.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1, condition MO_CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCX_SessionID_B	The URI that identifies the pre-established session		

5.5 MCDATA Server - MCDATA Client / Configuration / Determination of MCDATA Service Settings / Current Active MCDATA Settings / De-subscribe

5.5.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCData client ) requests to determine the currently active MCData service
settings with SIP SUBSCRIBE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

(2)

```
with { IUT (MCData Server) having determined MCData service settings}
ensure that {
  when { the SS-UE1 (MCData client ) requests to re-subscribe to MCData service settings with SIP
SUBSCRIBE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
            }
```

(3)

```
with { IUT (MCData Server) having determined MCData service settings }
ensure that {
  when { the SS-UE1 (MCData client ) requests to de-subscribe from MCData service settings with SIP
SUBSCRIBE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
          }
```

5.5.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 7.1, 7.2.4, 7.3.3, 7.3.4, 7.3.1A, 7.3.5, 7.3.6.1, 7.3.6.2. Unless otherwise stated these are Rel-14 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.1]

"This clause describes the procedures for SIP registration and MCData service authorization for the MCData client and the MCData service. The MCData UE can use SIP REGISTER or SIP PUBLISH for MCData service settings to perform service authorization for MCData. The decision which method to use is based on implementation and on availability of an access-token received as outcome of the user authentication procedure as described in 3GPP TS 24.482 [24].

…"

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.2.4]

"In order to discover MCData service settings of another MCData client of the same MCData user or to verify the currently active MCData service settings of this MCData client, the MCData client shall generate an initial SIP SUBSCRIBE request according to 3GPP TS 24.229 [5], IETF RFC 6665 [36], and IETF RFC 4354 [35].

…"

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.3.3]

"The MCData server shall support obtaining service authorization specific information from a SIP PUBLISH request for MCData server settings.

Upon receiving a SIP PUBLISH request containing:

1) an Event header field set to the "poc-settings" value;

2) an application/poc-settings+xml MIME body; and

3) an application/vnd.3gpp.mcdata-info+xml MIME body containing an <mcdata-access-token> element and an <mcdata-client-id> element;

the MCData server:

1) shall identify the IMS public user identity from the P-Asserted-Identity header field;

2) shall perform the procedures in subclause 7.3.1A;

…

4) shall perform service authorization for the identified MCData ID as described in 3GPP TS 33.180 [26];

5) if service authorization was successful:

a) shall bind the MCData ID and MCData client ID to the IMS public user identity;

b) if the service authorization request was from an MCData user who is previously MCData service authorized on another MCData client (as determined by a comparison of the received MCData client ID with the MCData client ID of existing bindings), keep the current bindings and create a new binding between the MCData ID, MCData client ID and the IMS public user identity; and

c) if a Resource-Share header field with the value "supported" was included in the "message/sip" MIME body of the third-party REGISTER request, shall bind the MCData ID and MCData client ID to the identity of the MCData UE identified by the "+g.3gpp.registration-token" header field parameter in the Contact header field of the third-party REGISTER request that contained this IMS public user identity;

NOTE 1: The MCData server will store the binding MCData ID, MCData client ID, IMS public user identity and an identifier addressing the MCData server in an external database.

…

7) shall process the SIP PUBLISH request according to rules and procedures of IETF RFC 3903 [34] and if processing of the SIP request was not successful, do not continue with the rest of the steps;

8) shall cache the received MCData service settings until the MCData service settings expiration timer expires;

9) shall send a SIP 200 (OK) response according 3GPP TS 24.229 [5] with:

a) if more than one binding exists for the MCData ID, an application/vnd.3gpp.mcdata-info+xml MIME body as specified in annex D.1 with a <multiple-devices-ind> element set to the value "true";

10) shall download the MCData user profile from the MCData user database as defined in 3GPP TS 29.283 [37] if not already stored at the MCData server and use the <selected-user-profile-index> element of the poc-settings event package if included to identify the active MCData user profile for the MCData client;

NOTE 2: If the <selected-user-profile-index> element of the poc-settings event package is included then only that MCData user profile is needed to be downloaded from the MCData user database.

11) if there is no <selected-user-profile-index> element included in the poc-settings event package then if multiple MCData user profiles are stored at the MCData server or downloaded for the MCData user from the MCData user database, shall determine the pre-selected MCData user profile to be used as the active MCData user profile by identifying the MCData user profile (see the MCData user profile document in 3GPP TS 24.484 [12]) in the collection of MCData user profiles that contains a <Pre-selected-indication> element; and

NOTE 3: If only one MCData user profile is stored at the MCData server or only one MCData user profile is downloaded from the MCData user database, then by default this MCData user profile is the pre-selected MCData user profile.

12) if an <ImplicitAffiliations> element is contained in the <OnNetwork> element of the MCData user profile document with one or more <entry> elements containing an MCData group ID (see the MCData user profile document in 3GPP TS 24.484 [12]) for the served MCData ID, shall perform implicit affiliation as specified in subclause 8.3.2.15."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.3.4]

"Upon receiving a SIP PUBLISH request containing:

1) an Event header field set to the "poc-settings" value;

2) an application/poc-settings+xml MIME body; and

3) an application/vnd.3gpp.mcdata-info+xml MIME body containing an <mcdata-request-uri> element and an <mcdata-client-id> element;

The MCData server:

1) shall identify the IMS public user identity from the P-Asserted-Identity header field;

2) shall perform the procedures in subclause 7.3.1A;

3…

4) shall verify that a binding between the IMS public user identity in the Request-URI and the MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml exists at the MCData server;

5) if a binding exists between the IMS public user identity and the MCData ID in the request and the validity period of the binding has not expired shall download the MCData user profile from the MCData user database as defined in 3GPP TS 29.283 [37] if not already stored at the MCData server;

…

7) shall process the SIP PUBLISH request according to rules and procedures of IETF RFC 3903 [34] and if processing of the SIP request was not successful, do not continue with the rest of the steps;

8) shall cache the received MCData service settings until the MCData service settings expiration timer expires;

9) shall send a SIP 200 (OK) response according 3GPP TS 24.229 [5];

10) shall download the MCData user profile from the MCData user database as defined in 3GPP TS 29.283 [37] if not already stored at the MCData server and use the <selected-user-profile-index> element of the poc-settings event package if included to identify the active MCData user profile for the MCData client;

NOTE 1: If the <selected-user-profile-index> element of the poc-settings event package is included then only that MCData user profile is needed to be downloaded from the MCData user database.

11) if there is no <selected-user-profile-index> element included in the poc-settings event package then if multiple MCData user profiles are stored at the MCData server or downloaded for the MCData user from the MCData user database, shall determine the pre-selected MCData user profile to be used as the active MCData user profile by identifying the MCData user profile (see the MCData user profile document in 3GPP TS 24.484 [12]) in the collection of MCData user profiles that contains a <Pre-selected-indication> element; and

NOTE 2: If only one MCData user profile is stored at the MCData server or only one MCData user profile is downloaded from the MCData user database, then by default this MCData user profile is the pre-selected MCData user profile.

12) if an <ImplicitAffiliations> element is contained in the <OnNetwork> element of the MCData user profile document with one or more <entry> elements containing an MCData group ID (see the MCData user profile document in 3GPP TS 24.484 [12]) for the served MCData ID, shall perform implicit affiliation as specified in subclause 8.3.2.1."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.3.1A]

"When the MCData server receives a SIP REGISTER request sent from the MCData client contained within a message/sip MIME body of a received third-party SIP REGISTER request or a SIP PUBLISH request, it first determines whether XML MIME bodies included in the request are integrity protected. If XML MIME bodies are integrity protected the MCData server validates the signature of each of the XML MIME bodies. If the integrity protection check(s) pass or the XML MIME bodies are not integrity protected, the MCData server then determines whether the content in specific XML elements is confidentiality protected. If XML content is confidentiality protected, the MCData server decrypts the protected content.

Upon receiving:

- *a SIP REGISTER request containing an application/vnd.3gpp.mcdata-info+xml MIME body within a message/sip MIME body of the SIP REGISTER request sent from the MCData client; or*

- *a SIP PUBLISH request containing an application/vnd.3gpp.mcdata-info+xml MIME body and an application/poc-settings+xml MIME body;*

the MCData server:

1) shall determine if integrity protection has been applied to XML MIME bodies in the SIP request by following the procedures in subclause 6.5.3.4.1 for each XML MIME body;

2) if integrity protection has been applied, shall use the keying data described in subclause 6.5.3.2 and the procedures described in subclause 6.5.3.4.2 to verify the integrity of each of the XML MIME bodies; and

3) if all integrity protection checks succeed, shall continue with the remaining steps of this subclause.

Upon receiving:

- *a SIP REGISTER request containing an application/vnd.3gpp.mcdata-info+xml MIME body with an <mcdata-access-token> element and an <mcdata-client-id> element within a message/sip MIME body of the SIP REGISTER request sent from the MCData client; or*

- *a SIP PUBLISH request containing an application/vnd.3gpp.mcdata-info+xml MIME body with an <mcdata-access-token> element and an <mcdata-client-id> element, and an application/poc-settings+xml MIME body;*

the MCData server:

1) *shall determine if confidentiality protection has been applied to the <mcdata-access-token> element and the <mcdata-client-id> element in the application/vnd.3gpp.mcdata-info+xml MIME body, by following the procedures in subclause 6.5.2.4.1;*

2) *if confidentiality protection has been applied to the <mcdata-access-token> element and <mcdata-client-id> element:*

   a) *shall use the keying information received in the MIKEY-SAKKE I_MESSAGE as specified in 3GPP TS 33.180 [26], along with the procedures described in subclause 6.5.2.4.2 to:*

      i) *decrypt the received access token in the <mcdata-access-token> element in the application/vnd.3gpp.mcdata-info+xml MIME body; and*

      ii) *decrypt the received MCData client ID in the <mcdata-client-id> element in the application/vnd.3gpp.mcdata-info+xml MIME body;*

   b) *if the decryption procedure succeeds, shall identify the MCData ID and the MCData client ID from the decrypted values; and*

   c) *if the decryption procedure fails, shall determine that confidentiality protection has not been successful;*

3) *if confidentiality protection has been applied to only one of the <mcdata-access-token> element or the <mcdata-client-id> element:*

   a) *shall determine that confidentiality protection has not been successful;*

4) *if confidentiality protection has not been applied:*

   a) *shall identify the MCData ID from <mcdata-access-token> element received in the application/vnd.3gpp.mcdata-info+xml MIME body; and*

   b) *shall identify the MCData client ID from the <mcdata-client-id> element received in the application/vnd.3gpp.mcdata-info+xml MIME body.*

Upon receiving a SIP PUBLISH request containing an application/vnd.3gpp.mcdata-info+xml MIME body with an <mcdata-request-uri> element, an <mcdata-client-id> element, and an application/poc-settings+xml MIME body, the MCData server:

1) *shall determine if confidentiality protection has been applied to the <mcdata-request-uri> element and the <mcdata-client-id> element in the application/vnd.3gpp.mcdata-info+xml MIME body by following the procedures in subclause 6.5.2.4.1;*

2) *if confidentiality protection has been applied to the <mcdata-request-uri> element and the <mcdata-client-id> element:*

   a) *shall use the keying information described in subclause 6.5.2.2 along with the procedures described in subclause 6.5.2.4.2 to:*

      i) *decrypt the received encrypted MCData ID in the <mcdata-request-uri> element in the application/vnd.3gpp.mcdata-info+xml MIME body; and*

      ii) *decrypt the received encrypted MCData client ID in the <mcdata-client-id> element in the application/vnd.3gpp.mcdata-info+xml MIME body;*

b) if all decryption procedures succeed, shall identify the MCData ID and MCData client ID from the decrypted values; and

c) if the decryption procedure fails, shall determine that confidentiality protection has not been successful;

3) if confidentiality protection has been applied to only one of the <mcdata-request-uri> element or <mcdata-client-id> element:

a) shall determine that confidentiality protection has not been successful;

4) if confidentiality protection has not been applied:

a) shall identify the MCData ID from the contents of the <mcdata-request-uri> element in the application/vnd.3gpp.mcdata-info+xml MIME body; and

b) shall identify the MCData client ID from the <mcdata-client-id> element received in the application/vnd.3gpp.mcdata-info+xml MIME body."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.3.5]

"Upon receiving a SIP PUBLISH request containing:

1) an Event header field set to the "poc-settings" value; and

2) an Expires header field set to 0;

the MCData server:

1) shall identify the IMS public user identity from the P-Asserted-Identity header field;

2) shall process the SIP PUBLISH request according to rules and procedures of IETF RFC 3903 [34] and if processing of the SIP request was successful, continue with the rest of the steps;

3) shall remove the MCData service settings;

4) shall remove the binding between the MCData ID and public user identity; and

5) shall send a SIP 200 (OK) response according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.3.6.1]

"Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the participating MCData function of the served MCData user;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the<mcdata-request-uri> element which identifies an MCData ID served by the MCData server;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata,sds" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7]; and

3) the Event header field of the SIP SUBSCRIBE request contains the 'poc-settings' event type.

the MCData server:

1) shall identify the served MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

2) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the participating MCData function serving the MCData user, shall identify the originating MCData ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;

3) if the originating MCData ID is different than the served MCData ID, shall send a 403 (Forbidden) response and shall not continue with the rest of the steps; and

4) shall generate a 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [5], IETF RFC 6665 [36] and IETF RFC 4354 [35].

For the duration of the subscription, the MCData server shall notify subscriber about changes of the MCData service settings of the subscribed MCData user, as described in subclause 7.3.6.2."

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.3.6.2]

"In order to notify the subscriber about changes of the MCData service settings of the subscribed MCData client of the subscribed MCData user, the MCData server:

1) shall generate an application/poc-settings+xml MIME body as defined in 3GPP TS 24.379 [10] containing:

a) the <selected-user-profile-index> element identifying the active MCData user profile; and

2) send a SIP NOTIFY request according to 3GPP TS 24.229 [5], IETF RFC 6665 [36] and IETF RFC 4354 [35] with the constructed application/poc-settings+xml MIME body."

5.5.3 Test description

5.5.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client)
- For the underlying "transport bearer" over which the SS-UE1 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server)
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Distribution Function, the MCData User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCData Server) is connected to PLMN1
- The IUT (MCData Server) is connected to the SS-UE1 (MCData client) as defined in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Figure 4.2.4, where term "MCPTT" is replaced with "MCData".
- The SS-UE1 (MCData client) performs the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

5.5.3.2 Test procedure sequence

Table 5.5.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to determine the currently active MCData service settings of the UE (MCData client)?	-	-	1	P
2	Check: Does the MCData Server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to re-subscribe to MCData service settings and to receive later notifications?	-	-	2	P
3	Check: Does the MCData Server correctly perform procedure 'MCX Subscription and Notification' as described in Annex A, Table A.2.7.3-1 to de-subscribe from MCData service settings?	-	-	3	P

5.5.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 5.5.3.3-1: SIP SUBSCRIBE from the SS-UE1
(Step 1, Table 5.5.3.2-1; Step 1, Annex A, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition POC-SETTINGS-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.2.4	
value	"4294967295"			
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.5.3.3-2			

Table 5.5.3.3-2: MCData-Info in SIP SUBSCRIBE (Table 5.5.3.3-1, Table 5.5.3.3-4 and Table 5.5.3.3-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3, condition POC-SETTINGS-EVENT

Table 5.5.3.3-3: SIP 200 (OK) from the MCData Server
(Steps 1, 2, 3, Table 5.5.3.2-1; Step 2, Annex A, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, conditions SUBSCRIBE-RSP

**Table 5.5.3.3-4: SIP SUBSCRIBE from the SS-UE1
(Step 2, Table 5.5.3.2-1; Step 1, Annex A, Table A.2.7.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition POC-SETTINGS-EVENT, re_SUBSCRIBE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.2.4	
value	"4294967295"			
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.5.3.3-2			

**Table 5.5.3.3-5: SIP SUBSCRIBE from the SS-UE1
(Step 3, Table 5.5.3.2-1; Step 1, Annex A, Table A.2.7.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1, condition POC-SETTINGS-EVENT, re_SUBSCRIBE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.2.4	
value	"0"			
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 5.5.3.3-2			

**Table 5.5.3.3-6: SIP NOTIFY from the MCData Server
(Steps 1, 2, Table 5.5.3.2-1; Step 3, Annex A, Table A.2.7.3-1)**

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions POC-SETTINGS-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PoC-Settings**		
MIME-part-body	PoC-Settings as described in Table 5.5.3.3-7			

Table 5.5.3.3-7: PoC-Settings in SIP NOTIFY (Table 5.5.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.11.2-1

5.6 MCDATA Server - MCDATA Client / Configuration / Download CSK

5.6.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { IUT (MCData Server) requests to update the existing CSK }
    then { IUT (MCData Server) sends a SIP MESSAGE message to the SS-UE1 (MCData Client) }
           }
```

(2)

```
with { SS-UE1 (MCData Client) having sent a standalone one-to-one SDS message to SS-UE2 (MCData
Client) with disposition notification "DELIVERY" }
ensure that {
  when { IUT (MCData Server) forwards the SDS message to SS-UE2 (MCData Client) and SS-UE2 sends a
disposition notification "DELIVERED" }
    then { IUT (MCData Server) forwards the disposition notification to SS-UE1 (MCData Client) }
           }
```

5.6.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 7.3.7, ETSI TS 133 180 [12] (3GPP TS 33.180) clauses 9.2.1.1, 9.2.1.2, 9.2.1.3, 9.2.1.4, 5.4, 5.2.1, 5.2.2, 5.8.1, 5.8.2, ETSI TS 124 484 [8] (3GPP TS 24.484) clauses 4.3.1, 4.3.4, 4.4, 6.3.13.2.3, 6.3.3.2.3, 6.3.3.2.1, 4.1.4, 10.2.1, 10.2.1A, 10.3.1, 10.3.1A, 10.4.1, 6.2.3, ETSI TS 124 582 [40] (3GPP TS 24.582) clause 10.3.3. Unless otherwise stated these are Rel-16 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.3.7]

"If confidentiality protection is enabled as specified in subclause 6.5.2.3.1, and if the participating MCData function received a Client Server Key (CSK) within a SIP REGISTER request for service authorisation or SIP PUBLISH request for service authorisation, the participating MCData function may decide to update the CSK. In this case, the participating MCData function shall perform a key download procedure for the CSK. The participating MCData function:

1) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];

2) shall set the Request-URI to the URI received in the To header field in the third-party SIP REGISTER request;

3) shall include an Accept-Contact header field with the g.3gpp.icsi-ref media-feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata" along with parameters "require" and "explicit" according to IETF RFC 3841 [8];

4) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcdata";

5) shall include an application/mikey MIME body containing the algorithm to be used to protect the MCData signalling fields, the CSK-ID and the CSK encrypted within a MIKEY message to the MC client as specified in subclause 9.2.1 of 3GPP TS 33.180 [26] in the body of the SIP MESSAGE request; and

6) shall send the SIP MESSAGE request towards the MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 133 180 [12] (3GPP TS 33.180) clause 9.2.1.1]

"A Client-Server Key is required to protect unicast RTCP signalling between the MC client and the MCX Server. The use of the CSK in this context is defined in clause 9.4.

Additionally, the MC Service provider may require that MC identities, access tokens and other sensitive information transferred between clients and MC domain on the SIP-1 and SIP-2 interfaces be protected at the application layer from any viewing, including protection from viewing at the SIP signalling layer. Symmetric key based protection of SIP payload using CSK may be used to satisfy this requirement. The use of CSK in this context is defined in clause 9.3.

The uses of the CSK are shown in Figure 9.2.1-1.

…"

[ETSI TS 133 180 [12] (3GPP TS 33.180) clause 9.2.1.2]

"The 128-bit CSK is initially generated by the client and provided encrypted to the server through the SIP interface along with the CSK-ID identifying the CSK.

The key remains in use until: a new CSK is required, the SIP session is torn down, the MC user logs off, or some other indication. If during the active SIP session an update of the CSK is required, the server generates a CSK and provides it to the client using the mechanism defined in clause 5.8.

[ETSI TS 133 180 [12] (3GPP TS 33.180) clause 9.2.1.3]

"The CSK is initially distributed via the 'CSK upload' procedure as defined in clause 5.4. The 'CSK upload' procedure creates a security association between the MC client and the MCX Server and occurs during the client's initial connection with the MC Server.

The following steps describe how the client obtains the user specific key material and securely transfers the CSK to a server within the MC domain.

Prior to beginning of this procedure, the client would have obtained user-specific key material from the KMS.

1) *The client randomly generates the CSK and encapsulates the CSK as described in clause 5.4.*

2) *The client includes the encapsulated CSK in its initial SIP REGISTER or in a SIP PUBLISH message that is used to perform the MC user authorization procedure, and sends the SIP message addressed to the PSI of the server.*

An illustration is provided below as an example of how this message in included in the body of the SIP REGISTER message. The MIME media type "application/mikey" IETF RFC 3830 [22] is used in this example to insert a MIKEY I_MESSAGE in the SIP payload:

EXAMPLE:

```
REGISTER sip:MCPTT_Server_PSI SIP/2.0
Via: SIP/2.0/UDP den3.level3.com
Max-Forwards:70
From: MCPTT client IMPU
To:
Call-ID: <>
CSeq:  1 REGISTER
Contact: <URI>
Content-Type: multipart/mixed;boundary="boundary1"
Content-Length: 619

--boundary1
Content-Type: application/mikey
MIKEY I_MESSAGE
--boundary1
Content-Type: application/...
Encrypted Access token, MCPTT ID
--boundary1–
```

The following steps describe how the MCX Server retrieves the CSK from the SIP message:

1) *The server receives the SIP message and decrypts the encapsulated the CSK as described in clause 5.4.*

2) *Once the CSK has been extracted, MC user specific information (e.g. the access token) protected in the SIP message as defined in clause 9.3.4, may be decrypted."*

[ETSI TS 133 180 [12] (3GPP TS 33.180) clause 9.2.1.4]

"The MCX Server may decide to update an existing CSK at any time. This may be due to CSK revocation or expiry.

The CSK shall be updated by the MCX Server using the 'key download' procedure, defined in clause 5.8. Upon receipt of a CSK via a 'key download' procedure, the MC client shall identify the type of key as a CSK via the 4 most significant bits of the CSK-ID. The MC client shall:

- *discard any previous CSKs associated with the MC Server FQDN, and*

- *use the new CSK for uplink signaling with the MC Server."*

[ETSI TS 133 180 [12] (3GPP TS 33.180) clause 5.4]

"The key (CSK) is distributed from the MCX client to the MCX Server(s) using the 'CSK upload' procedure. The procedure shall use the common key distribution mechanism described in clause 5.2.2, transported over the SIP bearer. Identity hiding may be supported as defined in clause 5.2.6. The MCX Server may respond with a KMS Redirect Response (KRR) as described in clause 5.2.8 (e.g. if the MC client has migrated or is roaming).

The initiating entity of the CSK upload procedure shall be the MCX UE and the receiving entity shall be the MCX Server. With respect to the common key distribution procedure, the initiating entity URI shall be the MCX Service user ID of the user andthe receiving entity URI shall be the MCX Server Domain Security Identifier (MDSI). The MDSI is added to the recipient field (IDRr) of the message. The distributed key, K, shall be the CSK and the distributed identifier K-ID shall be the CSK-ID.

Clause E.4 provides MIKEY message structure for CSK distribution.

Before the CSK upload procedure can be used by the client to securely share the encryption key, the MC user shall first be authorized by KMS for key management services. Once the MC user is authorized, the KMS distributes the user's key material to the client as specified in clause 5.3.3.

The server receives the SIP message with the protected CSK and retrieves it from the message. It associates the MC User's SIP Core identity (IMPU), MC Service user ID (e.g. MCPTT ID) and the received CSK. Identity binding is used to uniquely identify the CSK used in protection of the SIP payload in subsequent SIP messages sent by both the client and the servers within a MC domain."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.2.1]

"…

The primary purpose of these procedures is to allow MCX entities to communicate with each other using end-to-end security. End-to-end security provides assurance to MCX users that no unauthorized access to communications is taking place within the MCX network. End-to-end communication security may be applied to media when operating on-network and media, floor control, transmission control, and media control when operating off-network.

A security domain is managed by a Key Management Server (KMS). The KMS is a component of the Common Services Core within the MCX system architecture. For any end-point to use or access end-to-end secure communications, it needs to be provisioned with key material associated to its identity by the KMS. Through the use of the KMS, MC administrators are able to manage the use of, and access to, secure communications within the MCX network.

…"

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.2.2]

"The security mechanism described in this clause allows a key, K, to be distributed from an initiating party to a receiving party. It provides confidentiality of the key, and integrity and authenticity of the payload. It is used within a number of different security procedures in this specification.

The key, K, is distributed encrypted specifically to the receiving entity and signed by the initiating entity. Prior to call commencement, both MCX UEs shall be provisioned by the KMS with time-limited key material associated with the MCX entity's URI. The key is distributed with a 32-bit Key Identifier (K-ID). This payload is a MIKEY-SAKKE I_MESSAGE, as defined in IETF RFC 6509 [11], which ensures the confidentiality of the key, plus integrity and authenticity of the payload.

The key is encrypted to the user identity (UID) associated to the receiving MCX entity using the security domain parameters provided in the public values in the certificate received from the KMS. The UID used to encrypt the data is derived from the receiving entity's URI (e.g. sip:user.002@mcptt.example.org) and a time-related parameter (e.g. the current year and month). The terminating entity's URI is added to the recipient field (IDRr) of the message.

The payload includes the encrypted key and the key identifier (K-ID). The key is unique within the MC domain. On creating the key, the initiator generates a 32-bit key identifier (K-ID). The 4 most significant bits of the K-ID shall indicate the purpose of the key, the other 28-bits shall be randomly generated. The key identifier (K-ID) is stored in the CSB-ID field of the MIKEY I_MESSAGE.

The payload is signed using (the KMS-provisioned key associated to) the identity of the initiating entity. The UID used to sign the data is derived from the initiating entity's URI (e.g. sip:user.001@mcptt.example.org) and a time-related parameter (e.g. the current year and month). The initiating entity's URI is added to the initiator field (IDRi) of the message.

NOTE: This solution is for the end-to-end protection of keys and does not protect the identities transmitted. Identities may be masked by transmitting the UID within the MIKEY ID fields as described in Annex E.7.

The security processes are summarized in figure 5.2.2-1."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.8.1]

"The 'key download' procedure is used to send keys from the MCX server to the MC client. It is used to distribute Multicast Signalling Keys (MuSiKs) to the MC clients, and it is used to update both the CSKs and MuSiKs.

Within the 'key download' procedure, keys (CSK or MuSiKs) are encrypted specifically to the MC user and signed using an identity representing the MC Server. Prior to group key distribution, each MC client shall be provisioned by the KMS with time-limited key material associated with the MC User as described clause 5.3. The MC Server shall also be provisioned by the MC KMS with key material for an identity which is authorised to act as an MCX Server.

The key (CSK or MuSiK) is distributed from the MCX Server to a MC client using the security mechanism described in clause 5.2.2, transported over the SIP bearer. End-point diversity is not required as end-points do not encrypt data, hence the extension in clause 5.2.3 is not applied. Additional parameters may be included as defined in clause 5.2.4. The SAKKE-to-self extension may be included as defined in clause 5.2.5. Identity hiding may be supported as defined in clause 5.2.6.

The initiating entity shall be the initiating MCX Server and the receiving entity shall be the terminating MC user. The initiating entity URI shall be the FQDN of the MCX Server (e.g. MDSI of the MC Domain) and the receiving entity URI shall be the MC Service ID of the terminating user. The distributed key, K, shall be the CSK or MuSiK and the key identifier K-ID shall be the CSK-ID or MuSiK-ID (respectively).

As a result of this 'key download' mechanism, the MC clients receive a new signalling key, CSK or MuSiK, identified by the 4 most significant bits of the key ID.

The MCData Service server may use the Key Download procedure to indicate or modify the algorithm used to protect the MCData signalling fields (i.e. MCData signaling parameters, Data signaling payload and End to end security parameters) by including a 'signalling algorithm' parameter. The 'signalling algorithm' parameter is described in clause 8.5.4.1. The available algorithms shall be as defined in clause 8.5.4.2."

[ETSI TS 133 180 [12] (3GPP TS 33.180), clause 5.8.2]

"The procedure for key download is described in figure 5.8.2-1:

…

0. The MCX UE has been provisioned by a KMS with key material associated with the MC user. The MC UE has also registered with an MCX Server. As a consequence of this registration, the MC UE is subscribed to key download notifications from the MCX Server.

1. The MCX Server sends a key download message (SIP NOTIFY or SIP MESSAGE) to the MC UE. The MC UE extracts the signalling key from the key download message.

2. Upon successful extraction of the signalling key, the MC UE returns a key download success message (200 OK response) to the MCX Server. Upon receipt of a notification of success, the MCX Server is able to begin to use the key for protection of signalling traffic."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.3.1]

"The MCS server obtains the appropriate MCS service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCS from the configuration management server.

The MCS server obtains the MCS service configuration document that contains the mission critical organisation configured parameters that defined the behaviour of the MCS from the configuration management server.

The MCS server subscribes to the MCS service configuration document for each mission critical organisation that is provisioned that is supported by the MCS server using the procedure specified in subclause 6.3.13.2.3. How the MCS server is provisioned with the identities of the mission critical organisations is out of scope of the present document.

If the MCS service configuration document has been updated since the current version stored at the MCS server, then the MCS server will receive a SIP NOTIFY request containing an HTTPS URI of the MCS service configuration document. Retrieval by the MCS server, using the notified HTTPS URI, of the MCS service configuration document is performed as specified in subclause 6.3.3.2.3.

NOTE: The MCS server can be notified of changes to the MCS service management configuration document at any time while operating the MCS."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.3.4]

"The format of the MCData service configuration document downloaded to the MCData server is defined in subclause 10.4."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.4]

"The following applies to the configuration management server used for online configuration.

The configuration management server needs to convert the MCS UE initial configuration document received from an MCS administrator into an appropriate format for configuration of the MCS UE initial configuration MO.

If the MCS UE initial configuration MO contains a <default-user-profile> element that identifies an MCS user profile configuration document, the configuration management server needs to convert the identified MCS user profile configuration document received from an MCS administrator into an appropriate format for configuration of the MCS user profile configuration MO.

Once an MCS User Profile configuration document has been created or updated by the MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to store MCS user profile configuration document as the user profile in the MC service user database.

In order to download the MCS user profile configuration document to an MCS UE or to support an MCS UE updating the MCS user profile configuration document, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to obtain the MCS user profile from the MC service user database.

In order to be notified of changes to an MCS user profile configuration document(s) that have been subscribed to by an MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to be notified of changes to the MCS user profile stored in the MC service user database.

In order to delete the MCS user profile when requested by an MCS UE, the configuration management server uses the procedures specified in 3GPP TS 29.283 [7] to delete the MCS user profile from the MC service user database.

NOTE: The configuration management server and group management server functionality for offline configuration is out of scope of the present document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.13.2.3]

"In order to subscribe to an MCS service configuration document, an MCS server shall send an initial SIP SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [22] and IETF RFC 5875 [11]. In the initial SIP SUBSCRIBE request, MCS server:

a) shall set the Request URI to a SIP URI containing:

1) the base URI being equal to the public service identity of the CMS configured in the MCS server; and

2) the "auid" parameter set to the application usage identifying th MCS service configuration document;

b) shall include a P-Asserted-Identity header field containing the public service identity of the MCS server;

c) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcptt" (coded as specified in 3GPP TS 24.229 [22]), in a P-Asserted-Service header field according to IETF RFC 6050 [23]; and

d) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field.

Upon receiving a SIP NOTIFY request associated with a subscription created as result of the sent initial SIP SUBSCRIBE request, the MCS server shall handle the SIP NOTIFY request according to IETF RFC 5875 [11].

In order to re-subscribe to notification of changes to an MCS service configuration document, an MCS server shall send a SIP re-SUBSCRIBE request to the network according to the originating AS procedures specified in 3GPP TS 24.229 [22] and IETF RFC 5875 [11]. In the SIP re-SUBSCRIBE request, MCS server:

a) shall set the Request URI to a SIP URI containing:

1) the base URI being equal to the public service identity of the CMS configured in the MCS server; and

2) the "auid" parameter set to the application usage identifying an MCS service configuration document; and

b) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcptt" in the Contact header field."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.3.2.3]

"In order to retrieve a configuration management document via the CSC-5 reference point, an MCS Server shall perform the procedures in subclause 6.3.3.2.1 specified for GC. The MCS server shall set the Request-URI of the HTTP GET request to identify the XML document based on configuration and include the "auid" as per the appropriate application usage."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.3.3.2.1]

"In order to retrieve a configuration management document, a GC shall send an HTTP GET request with the Request URI that references the document to be updated to the network according to procedures specified in IETF RFC 4825 [14] "Retrieve a Document"."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 4.1.4]

"An MCData service administrator can, using an MCData UE configure the:

- *MCData UE configuration document;*
- *MCData user profile configuration document;*
- *MCData service configuration document; and*
- *MCData related group configuration data in the MCS group document.*

The format of the MCData UE configuration document is defined in subclause 10.2.

The format of the MCData user profile configuration document is defined in subclause 10.3.

The format of the MCData service configuration document is defined in subclause 10.4."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.2.1]

"The MCData UE configuration document is specified in this subclause. The MCData UE configuration document content is based on requirements of Annex A.2 of 3GPP TS 23.282 [24], in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData UE configuration document in the MCData service is described in 3GPP TS 24.282 [25] and 3GPP TS 24.582 [26]. The schema definition is provided in subclause 10.2.2.3. An MCData UE configuration document may apply to all MCData UEs of a mission critical organization or apply to specific MCData UEs identified in the <mcdata-UE-id> element. If there is no <mcdata-UE-id> element in the MCData UE configuration document, then by default the MCData UE configuration document applies to all MCData UEs of the mission critical organization that are not specifically identified in the <mcdata-UE-id> element of another MCData UE configuration document of the mission critical organization. Each MCData UE of a mission critical organization is configured with an MCData UE configuration document that is identified by the instance ID of the MCData UE.

MCData UE configuration documents of a MCData user are contained as "XDM collections" in the user's directory of the users tree, in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. In this case, the term "user" in the XCAP sense refers to the MCData ID, as the user has been already authenticated. A MCData UE configuration document corresponding to a specific MCData UE the MCData user has used to authenticate and is authorised to use the MCData service with is placed in the user directory of the MCData user.

The MCData UE configuration document acts as a template for the CMS to generate UE configuration documents that are downloaded to specific MCData UEs. The MCData UE configuration document that acts as a template is referred to as a "master MCData UE configuration document". The master MCData UE configuration document is stored in the users tree of that MCData system administrator. The master MCData UE configuration document does not directly apply to a specific MCData UE, but instead acts as template that the CMS uses to populate the MCData UE configuration documents of MCData UEs identified by elements of the <MCData-UE-id> element. For MCData UE configuration documents that correspond to a specific MCData UE, the name of the MCData UE configuration document is created from a value defined by the corresponding element that identifies the MCData UE within the <MCData-UE-id> element. For a master MCData UE configuration documents that does not contain a <MCData-UE-id> element, the name of the MCData UE configuration document stored in the user directory is "DEFAULT-MCData-UE.xml"."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.2.1A]

"The MCData UE configuration document is accessed using the same XCAP URI, regardless of whether the MCData UE has a specific MCData UE configuration document configured or the master MCData UE configuration document applies. The CMS shall generate the UE's MCData UE configuration document from the master MCData UE configuration document if the MCData administrator did not provision a specific MCData UE configuration document. In this generated document, the <MCData-UE-id> element shall be set to the MCData client's UE ID. The UE's MCData UE configuration document shall always be stored with the filename corresponding to the UE's MCData UE ID under the user's directory of the users tree.

The XCAP URI used by the MCData client to access the UE's MCData UE configuration document shall be:

- *CMSXCAPROOTURI/org.3gpp.mcdata.ue-config/users/sip:MCDATAID/MCSUEID."*

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.3.1]

"The MCData user profile configuration document is specified in this subclause. The MCData user profile configuration document content is based on requirements of Annex A.3 of 3GPP TS 23.282 [24], and structure and procedures of OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData user profile in the MCData service is described in 3GPP TS 24.282 [25]. The schema definition is provided in subclause 10.3.2.

MCData user profile documents are "XDM collections" in the user's directory of the users tree, in accordance with OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. In this case, the term "user" in the XCAP sense refers to the MCData ID, as the user has been already authenticated.

The name of the MCData user profile document matches the value of the <ProfileName> element in the MCData user profile document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.3.1A]

"The XCAP URI used by the MCData client to access the MCData user's MCData user profile documents shall be:

CMSXCAPROOTURI/org.3gpp.mcdata.user-profile/users/sip:MCDATAID/mcdata-user-profile-INDEX.xml

Where INDEX is the index of the MCData user profile as defined in subclause 10.3.2.8."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 10.4.1]

"The MCData service configuration document is specified in this subclause. The MCData service configuration document content is based on requirements of Annex A.5 of 3GPP TS 23.282 [24], and structure and procedures of OMA OMA-TS-XDM_Core-V2_1-20120403-A [2]. The usage of an MCData service configuration in the MCData service is described in 3GPP TS 24.282[25] and 3GPP TS 24.582 [26]. The schema definition is provided in subclause 10.4.2.3. Each mission critical organization is configured with an MCData service configuration document."

[ETSI TS 124 484 [8] (3GPP TS 24.484), clause 6.2.3]

"The MCS server shall send the HTTP request as specified for the HTTP client in the network entity in annex A of 3GPP TS 24.482 [6]."

[ETSI TS 124 582 [40] (3GPP TS 24.582) clause 10.3.3]

"A participating MCData function which receives protected media information shall forward it to the next entity without any additional action related to the security framework.

A participating MCData function, when receiving a protected media control information from a MCData client shall decrypt and integrity check the protected media control using the CSK and CSK-ID negotiated with the MCData client which has sent the media control. Then, the participating MCData function shall forward the media control information to the controlling MCData function by protecting the media control information using SPK and SPK-ID, if protection is configured between the participating MCData function and the controlling MCData function.

A participating MCData function, when receiving a protected media control information from the controlling MCData function shall decrypt and integrity check the protected media control information using the SPK and SPK-ID configured between the participating MCData function and the controlling MCData function. Then, the participating MCData function shall forward the media control information to the destination MCData client by protecting the media control information using the CSK and CSK-ID if protection is negotiated between the participating MCData function and the MCData client."

5.6.3 Test description

5.6.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData Client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Distribution Function, the MCData User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCData Server) is connected to PLMN1.
- The IUT (MCData Server) is connected to the SS-UE1 (MCData client) and the SS-UE2 (MCData Client) as defined in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Figure 4.2.4, where term "MCPTT" is replaced with "MCData".
- The SS-UE1 (MCData client) and the SS-UE2 (MCData Client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

5.6.3.2 Test procedure sequence

Table 5.6.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure 'MCX Participating Originated SIP MESSAGE' as described in Annex A, Table A.2.9.3-1 to update the existing CSK between the MCData Server and SS-UE1 (MCData Client)?	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform steps 1-2 of procedure 'MCDATA Server - MCDATA Client / Short Data Service (SDS) / Standalone SDS Using Signalling Control Plane / One-to-one Standalone SDS' as described in Table 6.1.1.3.2-1 to forward a standalone one-to-one SDS message from SS-UE1 (MCData client) to SS-UE2 (MCData Client)?	-	-	2	P

5.6.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 5.6.3.3-1: SIP MESSAGE from the MCData Server (Step 1, Table 5.6.3.2-1; Step 1, Annex A, Table A.2.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, with condition MCData				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Service			ETSI TS 124 282 [13] (3GPP TS 24.282) clause 7.3.7	
Service-ID	"urn:urn-7:3gpp-service.ims.icsi.mcdata"			
Content-Type				
media-type	"application/mikey"			
Message-body				
MIKEY message	base64 encoded MIKEY message as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-1A	MIKEY message, containing the updated CSK		

5.7 MCDATA Server - MCDATA Client / Configuration / Functional Alias / Functional alias status determination / Activate functional alias / Deactivate functional alias

5.7.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCData client ) requests to subscribe to functional alias status change
notifications with SIP SUBSCRIBE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
          }
```

(2)

```
with { IUT (MCData Server) having received subscription to determine the status of a functional
alias }
ensure that {
  when { the SS-UE1 (MCData client) requests to activate a functional alias with SIP PUBLISH }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends SIP NOTIFY message to
inform the change progress status and sends SIP NOTIFY message to inform the final status }
           }
```

(3)

```
with { IUT (MCData Server) having received subscription to determine the status of a functional
alias }
ensure that {
  when { the SS-UE1 (MCData client) requests to deactivate a functional alias with SIP PUBLISH }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends SIP NOTIFY message to
inform the change progress status and sends SIP NOTIFY message to inform the final status }
           }
```

(4)

```
with { IUT (MCData Server) having received subscription to determine the status of a functional
alias }
ensure that {
  when { the SS-UE1 (MCData client ) requests to de-subscribe from functional alias status change
notifications with SIP SUBSCRIBE message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) message and sends a SIP NOTIFY message }
           }
```

5.7.2 Conformance requirements

References: The conformance requirements covered in the present TC are specified in: TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 22.2.2.2.3, 22.2.2.2.4, 22.2.2.2.5, 22.2.2.3.3, 22.2.2.3.4, 22.2.2.3.5, 22.2.2.3.6. Unless otherwise stated these are Rel-16 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 22.2.2.2.3]

"Upon receiving a SIP PUBLISH request such that:

1) *Request-URI of the SIP PUBLISH request contains either the public service identity identifying the originating participating MCData function serving the MCData user, or the public service identity identifying the terminating participating MCData function serving the MCData user;*

2) *the SIP PUBLISH request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the<mcdata-request-uri> element which identifies an MCData ID served by the MCData server;*

3) *the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7];*

4) *the Event header field of the SIP PUBLISH request contains the "presence" event type; and*

5) *SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-user functional alias information according to subclause 22.3.1;*

then the MCData server:

1) *shall identify the served MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;*

2) *if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the originating participating MCData function serving the MCData user, shall identify the originating MCData ID from public user identity in the P-Asserted-Identity header field of the SIP PUBLISH request;*

3) *if the Request-URI of the SIP PUBLISH request contains the public service identity identifying the terminating participating MCData function serving the MCData user, shall identify the originating MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;*

…

6) if the Expires header field of the SIP PUBLISH request has nonzero value, shall determine the candidate expiration interval to according to IETF RFC 3903 [34];

7) if the Expires header field of the SIP PUBLISH request has zero value, shall set the candidate expiration interval to zero;

8) shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to TS 24.229 [5], IETF RFC 3903 [34]. In the SIP 200 (OK) response, the MCData server:

a) shall set the Expires header field according to IETF RFC 3903 [34], to the candidate expiration time;

9) if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served MCData ID, shall not continue with the rest of the steps;

10) shall consider an MCData user information entry such that:

a) the MCData user information entry is in the list of MCData user information entries described in subclause 22.2.2.2.2; and

b) the MCData ID of the MCData user information entry is equal to the served MCData ID;

as the served MCData user information entry;

11) shall consider a copy of the list of the MCData functional alias entries of the served MCData user information entry as the served list of the MCData functional alias information entries;

12) if the candidate expiration interval is nonzero, shall construct the candidate list of the MCData functional alias entries as follows:

a) for each functional alias ID which has a functional alias information entry in the served list of the functional alias information entries, such that the expiration time of the functional alias information entry has not expired yet, and which is indicated in a "functionalAliasID" attribute of a <functionalAlias> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

i) shall copy the functional alias information entry into a new functional alias information entry of the candidate list of the functional alias information entries;

ii) if the functional alias status of the functional alias information entry is "deactivating" or "deactivated", shall set the functional alias status of the new functional alias information entry to the "activated" state and shall reset the activating p-id-fa of the new functional alias information entry; and

iii) shall set the expiration time of the new functional alias information entry to the current time increased with the candidate expiration interval;

b) for each functional alias ID which has a functional alias information entry in the served list of the functional alias information entries, such that the expiration time of the functional alias information entry has not expired yet, and which is not indicated in any "functionalAliasID" attribute of the <functionalAlias> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

i) shall copy the functional alias information entry into a new functional alias information entry of the candidate list of the functional alias information entries; and

ii) if the functional alias status of the functional alias information entry is "activated" or "activating":

- *shall set the functional alias status of the new functional alias entry to the "deactivating" state; and*
- *shall set the expiration time of the new functional alias information entry to the current time increased with twice the value of timer F; and*

c) for each functional alias ID:

i) which does not have a functional alias information entry in the served list of the functional alias entries; or

ii) which has a functional alias information entry in the served list of the functional alias information entries, such that the expiration time of the functional alias information entry has already expired;

and which is indicated in a "functionalAliasID" element of the <functionalAlias> element of the <status> element of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request:

i) shall add a new functional alias information entry in the candidate list of the functional alias information list for the functional alias ID;

ii) shall set the functional alias status of the new functional alias information entry to the "activating" state;

iii) shall set the expiration time of the new functional alias information entry to the current time increased with the candidate expiration interval; and

iv) shall reset the activating p-id-fa of the new functional alias information entry;

13) if the candidate expiration interval is zero, constructs the candidate list of the functional alias information entries as follows:

a) for each functional alias ID which has an entry in the served list of the functional alias information entries:

i) shall copy the functional alias entry of the served list of the functional alias information into a new functional alias information entry of the candidate list of the functional alias information entries;

ii) shall set the functional alias status of the new functional alias information entry to the "de-activating" state; and

iii) shall set the expiration time of the new functional alias information entry to the current time increased with twice the value of timer F;

14) shall replace the list of the functional alias information entries stored in the served MCData user information entry with the candidate list of the functional alias information entries;

15) shall perform the procedures specified in subclause 22.2.2.2.6 for the served MCData ID and each functional alias:

a) which does not have a functional alias information entry in the served list of the functional alias information entries and which has a functional alias information entry in the candidate list of the functional alias information entries with the functional alias status set to the "activating" state;

b) which has a functional alias information entry in the served list of the functional alias information entries with the expiration time already expired, and which has a functional alias information entry in the candidate list of the functional alias information entries with the functional alias status set to the "activating" state;

c) which has a functional alias information entry in the served list of the functional alias information entries with the functional alias status set to the "deactivating" state or the "deactivated" state and with the expiration time not expired yet, and which has an functional alias information entry in the candidate list of the functional alias information entries with the functional alias status set to the "activating" state; or

d) which has a functional alias information entry in the served list of the functional alias information entries with the functional alias status set to the "activated" state and with the expiration time not expired yet, and which has an functional alias information entry in the candidate list of the functional alias information entries with the functional alias status set to the "deactivating" state;

16) shall identify the handled p-id-fa in the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

17) shall perform the procedures specified in subclause 22.2.2.2.5 for the served MCData ID."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 22.2.2.2.4]

"Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains either the public service identity identifying the originating participating MCData function serving the MCData user, or the public service identity identifying the terminating participating MCData function serving the MCData user;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing:

a) the<mcdata-request-uri> element which identifies an MCData ID served by the MCData server; and

b) the <mcdatainfo> element with the <mcdata-Params> element with the <request-type> element set to a value of "functional-alias-status-determination";

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7]; and

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type;

the MCData server:

1) shall identify the served MCData ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

2) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the originating participating MCData function serving the MCData user, shall identify the originating MCData ID from public user identity in the P-Asserted-Identity header field of the SIP SUBSCRIBE request;

3) if the Request-URI of the SIP SUBSCRIBE request contains the public service identity identifying the terminating participating MCData function serving the MCData user, shall identify the originating MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

4) if the originating MCData ID is different than the served MCData ID and the originating MCData ID is not authorized to modify functional alias status of the served MCData ID, shall send a SIP 403 (Forbidden) response and shall not continue with the rest of the steps; and

5) shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to TS 24.229 [5], IETF RFC 6665 [36].

For the duration of the subscription, the MCData server shall notify the subscriber about changes of the information of the served MCData ID, as described in subclause 22.2.2.2.5."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 22.2.2.2.5]

"In order to notify the subscriber about changes of functional aliases of the served MCData ID, the MCData server:

1) shall consider an MCData user information entry such that:

a) the MCData user information entry is in the list of MCData user information entries described in subclause 22.2.2.2.2; and

b) the MCData ID of the MCData user information entry is equal to the served MCData ID;

as the served MCData user information entry;

2) shall generate an application/pidf+xml MIME body indicating per-user functional alias information according to clause 22.3.1 and the served list of the MCData user information entries with the following clarifications:

a) the MCData server shall not include information from functional alias information entry with the expiration time already expired;

b) the MCData server shall not include information from a functional alias information entry with the functional alias status set to the "deactivated" state;

c) if this procedures is invoked by procedure in clause 22.2.2.2.3 where the handled p-id-fa value was identified, the MCData server shall set the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id-fa value; and

3) send a SIP NOTIFY request according to 3GPP TS 24.229 [5], and IETF RFC 6665 [36] for the subscription created in clause 22.2.2.2.4. In the SIP NOTIFY request, the MCData server shall include the generated application/pidf+xml MIME body indicating per-user functional alias information."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 22.2.2.3.3]

"Upon receiving a SIP PUBLISH request such that:

1) Request-URI of the SIP PUBLISH request contains the public service identity of the controlling MCData function associated with the served functional alias;

2) the SIP PUBLISH request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the <mcdata-request-uri> element and the <mcdata-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7];

4) the Event header field of the SIP PUBLISH request contains the "presence" event type; and

5) SIP PUBLISH request contains an application/pidf+xml MIME body indicating per-functional alias information constructed according to clause 22.2.3.2;

then the MCData server:

1) shall identify the served functional alias in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;

2) shall identify the handled MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP PUBLISH request;

…

4) if the functional alias does not exist in the MCData server, shall reject the SIP PUBLISH request with SIP 403 (Forbidden) response to the SIP PUBLISH request according to 3GPP TS 24.229 [5], IETF RFC 3903 [34] and IETF RFC 3856 [39] and skip the rest of the steps;

4a) if SIP PUBLISH request is for activation of a functional alias then:

a) if handled MCPTT ID does not match with any of the entries in the <mcptt-user-list> which contains the MCPTT IDs of MCPTT users which are allowed to activate the functional alias; or

b) if no local policy exists that authorizes the request by the handled MCPTT ID;

…

5) if SIP PUBLISH request is for activation of a functional alias and the number of activations for the handled functional alias is equal <max-simultaneous-activations>, shall reject the SIP PUBLISH request with SIP 403 (Forbidden) response to the SIP PUBLISH request according to 3GPP TS 24.229 [5], IETF RFC 3903 [34] and IETF RFC 3856 [39] and skip the rest of the steps;

…

7) shall respond with SIP 200 (OK) response to the SIP PUBLISH request according to TS 24.229 [5], IETF RFC 3903 [34]. In the SIP 200 (OK) response, the MCData server:

a) shall set the Expires header field according to IETF RFC 3903 [34], to the selected expiration time;

8) if the "entity" attribute of the <presence> element of the application/pidf+xml MIME body of the SIP PUBLISH request is different than the served functional alias ID, shall not continue with the rest of the steps;

9) if the handled MCData ID is different from the MCData ID in the "id" attribute of the <tuple> element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request, shall not continue with the rest of the steps;

10) shall consider a functional alias information entry such that:

a) the functional alias information entry is in the list of functional alias information entries described in clause 22.2.2.3.2; and

b) the functional alias ID of the functional alias information entry is equal to the served functional alias ID;

as the served functional alias information entry;

11) if the selected expiration time is zero:

a) shall remove the MCData user information entry such that:

i) the MCData user information entry is in the list of the MCData user information entries of the served functional alias information entry; and

ii) the MCData user information entry has the MCData ID set to the served MCData ID;

12) if the selected expiration time is not zero:

a) shall consider an MCData user information entry such that:

i) the MCData user information entry is in the list of the MCData user information entries of the served functional alias information entry; and

ii) the MCData ID of the MCData user information entry is equal to the handled MCData ID;

as the served MCData user information entry;

b) if the MCData user information entry does not exist:

i) shall insert an MCData user information entry with the MCData ID set to the handled MCData ID into the list of the MCData user information entries of the served functional alias information entry; and

ii) shall consider the inserted MCData user information entry as the served MCData user information entry; and

c) shall set the expiration time in the served MCData user information entry according to the selected expiration time;

13) shall identify the handled p-id-fa in the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP PUBLISH request; and

14) shall perform the procedures specified in clause 22.2.2.3.5 for the served functional alias ID."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 22.2.2.3.4]

"NOTE: Usage of one SIP SUBSCRIBE request to subscribe for notification about change of functional alias state of several MCData users served by the same MCData server is not supported in this version of the specification.

Upon receiving a SIP SUBSCRIBE request such that:

1) Request-URI of the SIP SUBSCRIBE request contains the public service identity of the controlling MCData function associated with the served functional alias;

2) the SIP SUBSCRIBE request contains an application/vnd.3gpp.mcdata-info+xml MIME body containing the<mcdata-request-uri> element and the <mcdata-calling-user-id> element;

3) the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service header field according to IETF RFC 6050 [7];

4) the Event header field of the SIP SUBSCRIBE request contains the "presence" event type; and

5) the SIP SUBSCRIBE request contains an application/simple-filter+xml MIME body indicating per-user restrictions of presence event package notification information according to clause 22.3.2 indicating the same MCData ID as in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

then the MCData server:

1) shall identify the served functional alias ID in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

2) shall identify the handled MCData ID in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP SUBSCRIBE request;

…

6) shall generate a SIP 200 (OK) response to the SIP SUBSCRIBE request according to 3GPP TS 24.229 [5], IETF RFC 6665 [36].

For the duration of the subscription, the MCData server shall notify subscriber about changes of the information of the served MCData ID, as described in clause 22.2.2.3.5."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 22.2.2.3.5]

"In order to notify the subscriber identified by the handled MCData ID about changes of the functional alias status of the served functional alias ID, the MCData server:

1) shall consider a functional alias information entry such that:

a) the functional alias information entry is in the list of functional alias information entries described in clause 22.2.2.3.2; and

b) the functional alias ID of the functional alias information entry is equal to the served functional alias ID;

2) shall consider an MCData user information entry such:

a) the MCData user information entry is in the list of the MCData user information entries of the served functional alias information entry; and

b) the MCData ID of the MCData user information entry is equal to the handled MCData ID;

as the served MCData user information entry;

3) shall generate an application/pidf+xml MIME body indicating per-functional alias information according to clause 22.3.1 and the served list of the served MCData user information entry of the functional alias information entry with following clarifications:

a) the MCData server shall include the "expires" attribute in the <functionalAlias> element; and

b) if this procedures is invoked by procedure in subclause 22.2.2.3.3 where the handled p-id-fa was identified, the MCData server shall set the <p-id-fa> child element of the <presence> root element of the application/pidf+xml MIME body of the SIP NOTIFY request to the handled p-id-fa value; and

4) send a SIP NOTIFY request according to 3GPP TS 24.229 [5], and IETF RFC 6665 [36] for the subscription created in clause 22.2.2.3.4. In the SIP NOTIFY request, the MCData server shall include the generated application/pidf+xml MIME body indicating per-functional alias information."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 22.2.2.3.6]

"In order to deactivate a functional alias associated with a target MCData ID:

1) externally triggered by an MCData administrator by a mechanism outside of the scope of the standard; or

2) directly by the MCData function owning the functional alias as a result of an internal trigger like the expiration of the functional alias association;

the MCData server

1) shall consider a functional alias information entry such that:

a) the functional alias information entry is in the list of functional alias information entries described in clause 22.2.2.3.2; and

b) the functional alias ID of the functional alias information entry is equal to the served functional alias ID;

as the served functional alias information entry;

2) shall remove the MCData user information entry such that:

a) the MCData user information entry is in the list of the MCData user information entries of the served functional alias information entry; and

b) the MCData user information entry has the MCData ID set to the target MCData ID; and

3) shall perform the procedures specified in subclause 22.2.2.3.5 for the served functional alias ID."

5.7.3 Test description

5.7.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Distribution Function, the MCData User Database, the SIP AS, the HTTP Server, the HTTP Client, and the Floor Control Server.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCData Server) is connected to PLMN1.
- The IUT (MCData Server) is connected to the SS-UE1 (MCData client) as defined in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Figure 4.2.4, where term "MCPTT" is replaced with "MCData".
- The SS-UE1 (MCData client) performs the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

5.7.3.2 Test procedure sequence

Table 5.7.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server correctly perform procedure 'MCData Functional Alias Status Determination and Subscription' as described in Annex A, Table A.4.13.3-1 to determine the current status of a functional alias and to get later notification of status changes of a functional alias?	-	-	1	P
2	Check: Does the MCData Server correctly perform procedure 'MCData Functional Alias Status Change' as described in Annex A, Table A.4.14.3-1 to activate a functional alias?	-	-	2	P
3	Check: Does the MCData Server correctly perform procedure 'MCData Functional Alias Status Change' as described in Annex A, Table A.4.14.3-1 to deactivate a functional alias?	-	-	3	P
4	Check: Does the MCData Server correctly perform procedure 'MCData Functional Alias Status Determination and Subscription' as described in Annex A, Table A.4.13.3-1 to de-subscribe from functional alias status change notifications?	-	-	4	P

5.7.3.3 Specific message contents

All message contents are specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), in the subclauses referenced in each step, with the following clarifications.

Table 5.7.3.3-1: SIP PUBLISH from the SS-UE1
(Step 3, Table 5.7.3.2-1; Step 1, Annex A, Table A.4.14.3-1)

Derivation Path: Annex A, Table A.4.14.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
delta-seconds	"0"			

Table 5.7.3.3-2: SIP NOTIFY from the MCData Server
(Step 3, Table 5.7.3.2-1; Step 3, Annex A, Table A.4.14.3-1)

Derivation Path: Annex A, Table A.4.14.4-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.3.1	
MIME-part-body	PIDF as described in Table 5.7.3.3-3			

Table 5.7.3.3-3: PIDF in SIP NOTIFY (Table 5.7.3.3-2)

Derivation Path: Annex A, Table A.4.14.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"deactivating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.2.2.3.5	

Table 5.7.3.3-4: SIP NOTIFY from the MCData Server (Step 3, Table 5.7.3.2-1; Step 5, Annex A, Table A.4.14.3-1)

Derivation Path: Annex A, Table A.4.14.4-6				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.3.1	
MIME-part-body	PIDF as described in Table 5.7.3.3-5			

Table 5.7.3.3-5: PIDF in SIP NOTIFY (Table 5.7.3.3-4)

Derivation Path: Annex A, Table A.4.14.4-7				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status	not present			

Table 5.7.3.3-6: SIP NOTIFY from the MCData Server (Step 4, Table 5.7.3.2-1; Step 3, Annex A, Table A.4.13.3-1)

Derivation Path: Annex A, Table A.4.13.4-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subscription-State			IETF RFC 6665 [30]	
substate-value	"terminated"			
expires	not present			
Message-body				
MIME body part		**PIDF**		
MIME-part-body	PIDF as described in Table 5.7.3.3-5			

6 MCData Server - MCData Client operation

6.1 Short Data Service

6.1.1 MCDATA Server - MCDATA Client / Short Data Service (SDS) / Standalone SDS Using Signalling Control Plane / One-to-one Standalone SDS

6.1.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { SS-UE1 (MCData Client) requests one-to-one Standalone SDS by sending a SIP MESSAGE message
with disposition request "DELIVERY" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE2 (MCData Client) and responds with
a SIP 202 (Accepted) to UE1 (MCData Client) }
            }
```

(2)

```
with { IUT (MCData Server) having requested an one-to-one Standalone SDS }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "DELIVERED" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

(3)

```
with { IUT (MCData Server) having requested an one-to-one Standalone SDS }
ensure that {
  when { SS-UE1 (MCData Client) requests the one-to-one Standalone SDS by sending a SIP MESSAGE
message with disposition request "READ" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE2 (MCData Client) and responds with
a SIP 202 (Accepted) to UE1 (MCData Client) }
            }
```

(4)

```
with { IUT (MCData Server) having requested an one-to-one Standalone SDS }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "READ" by sending a SIP MESSAGE
message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

(5)

```
with { IUT (MCData Server) having requested an one-to-one Standalone SDS }
ensure that {
  when { SS-UE1 (MCData Client) requests the one-to-one Standalone SDS by sending a SIP MESSAGE
message with disposition request "DELIVERY AND READ" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE2 (MCData Client) and responds with
a SIP 202 (Accepted) to UE1 (MCData Client) }
            }
```

(6)

```
with { IUT (MCData Server) having requested an one-to-one Standalone SDS }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "UNDELIVERED" by sending a SIP
MESSAGE message }
    then { IUT (MCData Server) responds with a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

(7)

```
with { IUT (MCData Server) having requested an one-to-one Standalone SDS }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "DELIVERED AND READ" by sending a
SIP MESSAGE message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
          }
```

6.1.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282), clauses 9.2.2.3.1, 9.2.2.3.2, 9.2.2.4.1, 9.2.2.4.2, 9.2.2.2.1, 6.2.2.1. Unless otherwise stated, these are Rel-14 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.3.1]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for originating participating MCData function", the participating MCData function:

…

2) shall determine the MCData ID of the originating user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request, and shall authorise the calling user;

NOTE: The MCData ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in subclause 7.3.

…

4) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request is:

…

b) set to a value of "one-to-one-sds", shall determine the public service identity of the controlling MCData function hosting the one-to-one standalone SDS service for the calling user;

…

6) shall determine whether the MCData user identified by the MCData ID is authorised for MCData communications by following the procedures in subclause 11.1;

…

9) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];

10) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCData function as determined by step 4) in this subclause;

11) shall copy all MIME bodies included in the incoming SIP MESSAGE request to the outgoing SIP MESSAGE request;

12) shall include the MCData ID of the originating user in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP MESSAGE request;

12A) if the incoming SIP MESSAGE request contains an application/vnd.3gpp.mcdata-info+xml MIME body that contains a <functional-alias-URI> element, shall check if the status of the functional alias is activated for the MCData ID. If the functional alias status is activated, then the participating MCData function shall set the <functional-alias-URI> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request to the received value, otherwise shall not include a <functional-alias-URI> element;

13) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request;

14) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request; and

15) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [5].

Upon receipt of a SIP 202 (Accepted) response in response to the SIP MESSAGE request in step 15):

1) shall generate a SIP 202 (Accepted) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 202 (Accepted) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the SIP MESSAGE request in step 15):

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 200 (OK) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request in step 15) the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.3.2]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for terminating participating MCData function", the participating MCData function:

…

2) shall use the MCData ID present in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCData ID and public user identity of the terminating MCData user;

…

4) shall generate an outgoing SIP MESSAGE request as specified in subclause 6.3.2.1;

5) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request; and

6) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 200 (OK) response to the controlling MCData function according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the controlling MCData function according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.1]

"This subclause describes the procedures for sending a SIP MESSAGE from the controlling MCData function and is initiated by the controlling MCData function as a result of an action in subclause 9.2.2.4.2.

The controlling MCData function:

1) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];

2) shall include an Accept-Contact header field containing the g.3gpp.mcdata.sds media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

3) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" along with parameters "require" and "explicit" according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

4) shall copy the following MIME bodies in the received SIP MESSAGE request into the outgoing SIP MESSAGE request by following the guidelines in subclause 6.4:

a) application/vnd.3gpp.mcdata-info+xml MIME body;

b) application/vnd.3gpp.mcdata-signalling MIME body; and

c) application/vnd.3gpp.mcdata-payload MIME body

5) in the application/vnd.3gpp.mcdata-info+xml MIME body:

a) shall set the <mcdata-request-uri> element set to the MCData ID of the terminating user; and

b) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request was set to a value of "group-sds", shall set the <mcdata-calling-group-id> element to the group identity;

6) shall set the Request-URI to the public service identity of the terminating participating MCData function associated to the MCData user to be invited;

7) shall copy the public user identity of the calling MCData user from the P-Asserted-Identity header field of the incoming SIP MESSAGE request into the P-Asserted-Identity header field of the outgoing SIP MESSAGE request;

8) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds"; and

9) shall send the SIP MESSAGE request according to according to rules and procedures of 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.2]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for controlling MCData function", the controlling MCData function:

…

3) shall decode the contents of the application/vnd.3gpp.mcdata-signalling MIME body contained in the SIP MESSAGE;

4) if the application/vnd.3gpp.mcdata-signalling MIME body contains a SDS SIGNALLING PAYLOAD message with a SDS disposition request type IE, shall store the value of the Conversation ID IE and the value of the Message ID IE in the SDS SIGNALLING PAYLOAD message;

NOTE: The controlling MCData function uses the Conversation ID and Message ID for correlation with disposition notifications.

…

7) shall generate a SIP 202 (Accepted) response in response to the "SIP MESSAGE request for standalone SDS for controlling MCData function"; and

8) shall send the SIP 202 (Accepted) response towards the originating participating MCData function according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.2.1]

"The MCData client shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6] with the clarifications given below.

The MCData client:

1) shall build the SIP MESSAGE request as specified in subclause 6.2.4.1;

2) if a one-to-one standalone SDS message is to be sent, shall insert in the SIP MESSAGE request:

a) an application/resource-lists+xml MIME body with the MCData ID of the target MCData user, according to rules and procedures of IETF RFC 4826 [9];

b) an application/vnd.3gpp.mcdata-info+xml MIME body with:

i) a <request-type> element set to a value of "one-to-one-sds"; and

ii) if the MCData client is aware of active functional aliases and if an active functional alias is to be included in the SIP MESSAGE request, the <functional-alias-URI> element set to the URI of the used functional alias; and

c) if end-to-end security is required and the security context does not exist or if the existing security context has expired, an application/mikey MIME body with the MIKEY-SAKKE I_MESSAGE as specified in 3GPP TS 33.180 [26]. The MCData client:

i) if necessary, shall instruct the key management client to request keying material from the key management server as described in 3GPP TS 33.180 [26];

ii) shall use the keying material to generate a PCK as described in 3GPP TS 33.180 [26];

iii) shall use the PCK to generate a PCK-ID with the four most significant bits set to "0001" to indicate that the purpose of the PCK is to protect one-to-one communications and with the remaining twenty eight bits being randomly generated as described in 3GPP TS 33.180 [26];

iv) shall encrypt the PCK to a UID associated to the MCData client using the MCData ID of the invited user and a time related parameter as described in 3GPP TS 33.180 [26];

v) shall generate a MIKEY-SAKKE I_MESSAGE using the encapsulated PCK and PCK-ID as specified in 3GPP TS 33.180 [26]; and

vi) shall add the MCData ID of the originating MCData to the initiator field (IDRi) of the I_MESSAGE as described in 3GPP TS 33.180 [26];

vii) shall sign the MIKEY-SAKKE I_MESSAGE using the originating MCData user's signing key provided in the keying material together with a time related parameter; and

viii) shall include the MIKEY-SAKKE I_MESSAGE in an application/mikey MIME body as specified in 3GPP TS 33.180 [26];

…

4) shall generate a standalone SDS message as specified in subclause 6.2.2.1; and

5) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 6.2.2.1]

"In order to generate an SDS message, the MCData client:

1) shall generate an SDS SIGNALLING PAYLOAD message as specified in subclause 15.1.2;

2) shall generate a DATA PAYLOAD message as specified in subclause 15.1.4;

3) shall include in the SIP request, the SDS SIGNALLING PAYLOAD message in an application/vnd.3gpp.mcdata-signalling MIME body as specified in subclause E.1; and

4) shall include in the SIP request, the DATA PAYLOAD message in an application/vnd.3gpp.mcdata-payload MIME body as specified in subclause E.2.

When generating an SDS SIGNALLING PAYLOAD message as specified in subclause 15.1.2, the MCData client:

1) shall set the Date and time IE to the current time as specified in subclause 15.2.8;

2) if the SDS message starts a new conversation, shall set the Conversation ID IE to a newly generated Conversation ID value as specified in subclause 15.2.9;

3) if the SDS message continues an existing unfinished conversation, shall set the Conversation ID IE to the Conversation ID value of the existing conversation as specified in subclause 15.2.9;

4) shall set the Message ID IE to a newly generated Message ID value as specified in subclause 15.2.10;

5) if the SDS message is in reply to a previously received SDS message, shall include the InReplyTo message ID IE with the Message ID value in the previously received SDS message;

6) if the SDS message is for user consumption, shall not include an Application ID IE as specified in subclause 15.2.7and shall not include an Extended application ID IE as specified in subclause 15.2.24;

7) if the SDS message is intended for an application on the terminating MCData client, shall include:

a) an Application ID IE with a Application ID value representing the intended application as specified in subclause 15.2.7; or

b) an Extended application ID IE with an Extended application ID value representing the intended application as specified in subclause 15.2.24;

NOTE: The value chosen for the Application ID value is decided by the mission critical organisation.

8) if only a delivery disposition notification is required shall include a SDS disposition request type IE set to "DELIVERY" as specified in subclause 15.2.3;

9) if only a read disposition notification is required shall include a SDS disposition request type IE set to "READ" as specified in subclause 15.2.3;

10) if both a delivery and read disposition notification is required shall include a SDS disposition request type IE set to "DELIVERY AND READ" as specified in subclause 15.2.3; and

11) may set the User location IE to the current location of the UE as specified in subclause 15.2.25.

When generating an DATA PAYLOAD message for SDS as specified in subclause 15.1.4, the MCData client:

1) shall set the Number of payloads IE to the number of Payload IEs that needs to be encoded, as specified in subclause 15.2.12;

2) if end-to-end security is required for a one-to-one communication, shall include the Security parameters and Payload IE with security parameters as described in 3GPP TS 33.180 [26]. Otherwise, if end-to-end security is not required for a one-to-one communication, shall include the Payload IE as specified in subclause 15.1.4; and

3) for each Payload IE included:

a) if the payload is text, shall set the Payload content type as "TEXT" as specified in subclause 15.2.13;

b) if the payload is binary data, shall set the Payload content type as "BINARY" as specified in subclause 15.2.13;

c) if the payload is hyperlinks, shall set the Payload content type as "HYPERLINKS" as specified in subclause 15.2.13;

d) if the payload is location, shall set the Payload content type as "LOCATION" as specified in subclause 15.2.13;

e) if payload is enhanced status for a group, shall set the Payload content type as "ENHANCED STATUS" as specified in subclase 15.2.13; and

f) shall include the data to be sent in the Payload data."

6.1.1.3 Test description

6.1.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Storage Function, the MCData User Database, the SIP AS, the HTP Server, the HTTP Client, the MSRP TCP Server and the MSRP TCP Client.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The <max-payload-size-sds-cplane-bytes> element shall not be present in the MCData Service Configuration document so that according to ETSI TS 124 484 [8] (3GPP TS 24.484) there is no size limit imposed for the use of C-plane procedures for the SDS message.
- The IUT (MCData Server) is connected to PLMN1.
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

6.1.1.3.2 Test procedure sequence

Table 6.1.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a standalone one-to-one SDS message from SS-UE1 (MCData client) to SS-UE2 (MCData Client), with disposition request "DELIVERY"?**	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a disposition notification of "DELIVERED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	2	P
3	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a standalone one-to-one SDS message from SS-UE1 (MCData client) to SS-**	-	-	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	UE2 (MCData Client), with disposition request "READ"?				
4	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a disposition notification of "READ", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	4	P
5	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a standalone one-to-one SDS message from SS-UE1 (MCData client) to SS-UE2 (MCData Client), with disposition request "DELIVERY AND READ"?**	-	-	5	P
6	Check: Does the MCData Server (IUT) correctly perform procedure **'MCData Participating Terminated SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.2.3-1 **to receive a disposition notification of "UNDELIVERED" from SS-UE2 (MCData client)?** (Note 1)	-	-	6	P
7	Wait until timer TDP1 expires	-	-	-	-
8	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 to re-deliver the standalone one-to-one SDS message sent in step 5 to SS-UE2? (Note 2)	-	-	5	P
9	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a disposition notification of "DELIVERED AND READ", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	7	P
10	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a standalone one-to-one SDS message from SS-UE1 (MCData client) to SS-UE2 (MCData Client), with disposition request "DELIVERY AND READ"?**	-	-	5	P
11	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a disposition notification of "DELIVERED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	2	P
12	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a disposition notification of "READ", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	4	P

NOTE 1: Timer TDP1 (SDS re-delivery timer) is started upon receipt of the SIP MESSAGE message that contains an"UNDELIVERED" disposition request. clause F.2.1 Timer TDP1 (SDS re-delivery timer)=60s according to the default value defined in ETSI TS 124 282 [13] (3GPP TS 24.282).

NOTE 2: Upon TDP1 (SDS re-delivery timer) timer expiration, the MCData Participating Server shall re-deliver previously sent message, as described in ETSI TS 124 282 [13] (3GPP TS 24.282), clause F.2.1.

6.1.1.3.3 Specific message contents

Table 6.1.1.3.3-1: SIP MESSAGE from the SS-UE1 (Step 1, Table 6.1.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.1.3.3-1

Table 6.1.1.3.3-2: SIP MESSAGE from the MCData Server (IUT) to SS-UE2 (Step 1, Table 6.1.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [21] IETF RFC 5031 [22]	
Request-URI	px_MCX_SIP_PublicUserId_B			
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.1	
Message-body				
MIME body part	not present	**Resource lists**		
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.1.1.3.3-3			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing SDS SIGNALLING PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-3			
MIME body part		**MCData Data message**		
MIME-part-body	DATA PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-4			

Table 6.1.1.3.3-3: MCData-Info (Table 6.1.1.3.3-2)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A			

Table 6.1.1.3.3-4: SIP MESSAGE from the SS-UE2
(Step 2 and 11, Table 6.1.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-6				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 6.1.1.3.3-5			

Table 6.1.1.3.3-5: Resource-lists (Table 6.1.1.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.1.1.3.3-6: SIP MESSAGE from the MCData Server (IUT) to SS-UE1
(Step 2 and 11, Table 6.1.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.1.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

Table 6.1.1.3.3-7: SIP MESSAGE from the SS-UE1
(Step 3, Table 6.1.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.1.3.3-9

Table 6.1.1.3.3-8: SIP MESSAGE from the MCData Server (IUT) to SS-UE2 (Step 3, Table 6.1.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-8				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [21] IETF RFC 5031 [22]	
Request-URI	px_MCX_SIP_PublicUserId_B			
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.1	
Message-body				
MIME body part	not present	**Resource lists**		
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.1.1.3.3-9			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing SDS SIGNALLING PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-9			
MIME body part		**MCData Data message**		
MIME-part-body	DATA PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-4			

Table 6.1.1.3.3-9: MCData-Info (Table 6.1.1.3.3-8)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A			

Table 6.1.1.3.3-10: SIP MESSAGE from the SS-UE2 (Step 4 and 12, Table 6.1.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 6.1.1.3.3-11			

Table 6.1.1.3.3-11: Resource-lists (Table 6.1.1.3.3-10)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.1.1.3.3-12: SIP MESSAGE from the MCData Server (IUT) to SS-UE1 (Step 4 and 12, Table 6.1.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.1.3.3-11				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

Table 6.1.1.3.3-13: SIP MESSAGE from the SS-UE1 (Step 5 and 10, Table 6.1.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.1.3.3-13

Table 6.1.1.3.3-14: SIP MESSAGE from the MCData Server (IUT) to SS-UE2 (Steps 5, 8 and 10, Table 6.1.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-12				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [21] IETF RFC 5031 [22]	
Request-URI	px_MCX_SIP_PublicUserId_B			
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.1	
Message-body				
MIME body part	not present	**Resource lists**		
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.1.1.3.3-15			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing SDS SIGNALLING PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-13			
MIME body part		**MCData Data message**		
MIME-part-body	DATA PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-4			

Table 6.1.1.3.3-15: MCData-Info (Table 6.1.1.3.3-14)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A			

Table 6.1.1.3.3-16: SIP MESSAGE from the SS-UE2 (Step 6, Table 6.1.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-14				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 6.1.1.3.3-17			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing SDS NOTIFICATION as described in Table 6.1.1.3.3-18			

Table 6.1.1.3.3-17: Resource-lists (Table 6.1.1.3.3-16)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.1.1.3.3-18: SDS SIGNALLING PAYLOAD (Table 6.1.1.3.3-16)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.8.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
SDS disposition notification type	'00000010'B		ETSI TS 124 282 [13] (3GPP TS 24.282) clause 15.2.5	UNDELIVERED

Table 6.1.1.3.3-19: SIP MESSAGE from the SS-UE2 (Step 9, Table 6.1.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.2.3.3-14				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 6.1.1.3.3-20			

Table 6.1.1.3.3-20: Resource-lists (Table 6.1.1.3.3-20)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.1.1.3.3-21: SIP MESSAGE from the MCData Server (IUT) to SS-UE1 (Step 9, Table 6.1.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.1.3.3-15				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

6.1.2 MCDATA Server - MCDATA Client / Short Data Service (SDS) / Standalone SDS Using Media Plane / Group Standalone SDS

6.1.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCData Client) initiates establishment of a group standalone SDS }
    then { IUT (MCData Server) accepts the call from the initiator and establishes the call with SS-
UE2 (MCData Client) and responds/forwards empty MSRP messages  }
           }
```

(2)

```
with { IUT (MCData Server) having requested a group Standalone SDS }
ensure that {
  when { the SS-UE1 (MCData Client) sends an MSRP SEND message with disposition request "DELIVERY"}
    then { IUT (MCData Server) forwards an MSRP SEND message to UE2 (MCData Client) and responds
with an MSRP 200 (Ok) to UE1 (MCData Client)  }
           }
```

(3)

```
with { IUT (MCData Server) having requested a group Standalone SDS }
ensure that {
  when { the SS-UE1 (MCData client) ends the group Standalone SDS }
    then { IUT (MCData Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to SS-UE2 (MCData Client) and waits to close MSRP TCP connections }
           }
```

(4)

```
with { IUT (MCData Server) having released a group Standalone SDS }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "DELIVERED" by sending a SIP
MESSAGE message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
           }
```

6.1.2.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282), clauses 9.2.3.3.3, 9.2.3.3.4, 9.2.3.3.1, 9.2.3.3.2, 9.2.3.4.3, 9.2.3.4.4, 9.2.3.4.1, 9.2.3.4.2, ETSI TS 124 582 [40] (3GPP TS 24.582), clauses 6.2.1.4.1, 6.2.1.4.2, 6.2.1.4.3, 6.2.1.5.1, 6.2.1.5.2, 6.2.1.5.3, 6.2.1.2, 6.2.1.3, 6.3.1.2.1, 6.3.1.2.2, 6.3.1.3. Unless otherwise stated, these are Rel-14 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.3.3.3]

"Upon receipt of a "SIP INVITE request for standalone SDS over media plane for originating participating MCData function", the participating MCData function:

…

2) shall determine the MCData ID of the calling user from the public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and shall authorise the calling user;

NOTE: The MCData ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in subclause 7.3.

…

4) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP INVITE request is:

a) set to a value of "group-sds", shall determine the public service identity of the controlling MCData function associated with the MCData group identity in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the SIP INVITE request; or

…

6) shall determine whether the MCData user identified by the MCData ID

a) is authorised for MCData communications by following the procedures in subclause 11.1; and

…

8) shall generate a SIP INVITE request in accordance with 3GPP TS 24.229 [5];

9) shall include the option tag "timer" in the Supported header field;

10) should include the Session-Expires header field according to IETF RFC 4028 [38]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

11) shall set the Request-URI of the outgoing SIP INVITE request to the public service identity of the controlling MCData function as determined by step 4) in this subclause;

12) shall include the MCData ID of the originating user in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP INVITE request;

12A) if the incoming SIP INVITE request contains an application/vnd.3gpp.mcdata-info+xml MIME body that contains a <functional-alias-URI> element, shall check if the status of the functional alias is activated for the MCData ID. If the functional alias status is activated, then the participating MCData function shall set the <functional-alias-URI> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request to the received value, otherwise shall not include a <functional-alias-URI> element;

13) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

14) shall set the P-Asserted-Identity in the outgoing SIP INVITE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP INVITE request;

15) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCData client as specified in subclause 9.2.3.3.1; and

16) shall send the SIP INVITE request as specified to 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the SIP INVITE request in step 16):

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5];

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the subclause 9.2.3.3.2;

3) shall include the option tag "timer" in a Require header field;

4) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [38], "UAS Behavior". If the "refresher" parameter is not included in the received request, the "refresher" parameter in the Session-Expires header field shall be set to "uac";

5) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.sds media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds"; and

c) the isfocus media feature tag;

6) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

7) shall include an MCData session identity mapped to the MCData session identity provided in the Contact header field of the received SIP 200 (OK) response;

8) if the incoming SIP 200 (OK) response contained an application/vnd.3gpp.mcdata-info+xml MIME body, shall copy the application/vnd.3gpp.mcdata-info+xml MIME body to the outgoing SIP 200 (OK) response.

9) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response; and

10) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.2.1.4

11) shall send the SIP 200 (OK) response to the MCData client according to 3GPP TS 24.229 [5]; and

12) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [38].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP INVITE request in step 16) the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.3.3.4]

"Upon receipt of a "SIP INVITE request for standalone SDS over media plane for terminating participating MCData function", the participating MCData function:

…

3) shall use the MCData ID present in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCData ID and public user identity of the terminating MCData user;

…

5) shall generate a SIP INVITE request accordance with 3GPP TS 24.229 [5];

6) should include the Session-Expires header field according to IETF RFC 4028 [38]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

7) shall include the option tag "timer" in the Supported header field;

8) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.sds media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds";

c) the isfocus media feature tag;

d) an MCData session identity mapped to the MCData session identity provided in the Contact header field of the incoming SIP INVITE request; and

e) any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;

9) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [8] that were received (if any) in the incoming SIP INVITE request;

10) shall set the Request-URI of the outgoing SIP INVITE request to the public user identity associated to the MCData ID of the terminating MCData user;

11) shall populate the outgoing SIP INVITE request with the MIME bodies that were present in the incoming SIP INVITE request;

12) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for standalone SDS over media plane for terminating participating MCData function" as specified in subclause 9.2.3.3.1; and

14) shall send the SIP INVITE request as specified in 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the above SIP INVITE request, the participating MCData function:

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5];

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in subclause 9.2.3.3.2;

3) shall include the option tag "timer" in a Require header field;

4) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [38], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

5) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.sds media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds"; and

c) an MCData session identity mapped to the MCData session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCData function;

6) if the incoming SIP response contained an application/vnd.3gpp.mcdata-info+xml MIME body, shall copy the application/vnd.3gpp.mcdata-info+xml MIME body to the outgoing SIP 200 (OK) response.

7) shall copy the P-Asserted-Identity header field from the incoming SIP 200 (OK) response to the outgoing SIP 200 (OK) response;

8) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [38];

9) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.2.1.5; and

10) shall send the SIP 200 (OK) response to the controlling MCData function according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request, the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the controlling MCData function according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.3.3.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCData function:

1) shall contain only one SDP media-level section for SDS message as contained in the received SDP offer; and

2) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [5], the participating MCData function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCData function, if required; and

…

2) if the IP address is replaced, shall insert its MSRP URI before the MSRP URI in the "a=path" attribute in the SDP offer."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.3.3.2]

"When composing the SDP answer according to 3GPP TS 24.229 [5], the participating MCData function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCData function, for the accepted media stream in the received SDP offer, if required; and

…

2) if the IP address is replaced shall insert its MSRP URI before the MSRP URI in the "a=path" attribute in the SDP answer."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.3.4.3]

"This subclause describes the procedures for inviting an MCData user to an MCData session. The procedure is initiated by the controlling MCData function as the result of an action in subclause 9.2.3.4.4.

The controlling MCData function:

1) *shall generate a SIP INVITE request according to 3GPP TS 24.229 [5];*

2) *shall include the Supported header field set to "timer";*

3) *should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [38]. The refresher parameter shall be omitted;*

4) *shall include an Accept-Contact header field containing the g.3gpp.mcdata.sds media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8];*

5) *shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" along with parameters "require" and "explicit" according to IETF RFC 3841 [8];*

6) *shall include a Referred-By header field with the public user identity of the inviting MCData client;*

7) *shall include in the Contact header field an MCData session identity for the MCData session with the g.3gpp.mcdata.sds media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" according to IETF RFC 3840 [16];*

8) *shall include in the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request:*

   a) *the <mcdata-request-uri> element set to the MCData ID of the terminating user; and*

   b) *the <mcdata-calling-group-id> element set to the group identity;*

9) *shall set the Request-URI to the public service identity of the terminating participating MCData function associated to the MCData user to be invited;*

NOTE 1: How the controlling MCData function finds the address of the terminating participating MCData function is out of the scope of the current release.

10) *shall set the P-Asserted-Identity header field to the public service identity of the controlling MCData function;*

11) *shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [7] in the SIP INVITE request;*

12) *shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating client according to the procedures specified in subclause 9.2.3.4.1; and*

13) *shall send the SIP INVITE request towards the terminating client in accordance with 3GPP TS 24.229 [5].*

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCData function:

1) *shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.3.1.*

NOTE 2: The procedures executed by the controlling MCData function prior to sending a response to the inviting MCData client are specified in subclause 9.2.3.4.4."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.3.4.4]

"In the procedures in this subclause:

1) *MCData ID in an incoming SIP INVITE request refers to the MCData ID of the originating user from the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request;*

2) *group identity in an incoming SIP INVITE request refers to the group identity from the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request; and*

3) MCData ID in an outgoing SIP INVITE request refers to the MCData ID of the called user in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP INVITE request;

Upon receipt of a "SIP INVITE request for controlling MCData function for standalone SDS over media plane", the controlling MCData function:

…

4) shall cache SIP feature tags, if received in the Contact header field and if the specific feature tags are supported;

5) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [38];

…

7) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP INVITE request is set to a value of "group-sds":

a) shall retrieve the necessary group document(s) from the group management server for the group identity contained in the SIP INVITE request and carry out initial processing as specified in subclause 6.3.3, and shall continue with the remaining steps if the procedures in subclause 6.3.3 were successful;

…

h) shall determine targeted group members for MCData communications by following the procedures in subclause 6.3.4;

…

j) shall invite each group member determined in step h) above, to the group session, as specified in subclause 9.2.3.4.3; and

k) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.3.1.

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in subclause 9.2.3.4.3 and if the MCData ID in the SIP 200 (OK) response matches to the MCData ID in the corresponding SIP INVITE request. the controlling MCData function:

1) shall generate SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [5];

2) shall include the option tag "timer" in a Require header field;

3) shall include the Session-Expires header field and start supervising the SIP session according to rules and procedures of IETF RFC 4028 [38], "UAS Behavior". The "refresher" parameter in the Session-Expires header field shall be set to "uac";

4) shall include a P-Asserted-Identity header field with the public service identity of the controlling MCData function;

5) shall include a SIP URI for the MCData session identity in the Contact header field identifying the MCData session at the controlling MCData function;

6) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.sds media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds"; and

c) the isfocus media feature tag;

7) shall include Warning header field(s) received in incoming responses to the SIP INVITE request;

8) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the subclause 9.2.3.4.2;

9) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.3.1; and

10) shall send a SIP 200 (OK) response to the inviting MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.3.4.1]

"When composing an SDP offer according to 3GPP TS 24.229 [5], IETF RFC 4975 [17], IETF RFC 6135 [19] and IETF RFC 6714 [20] the controlling MCData function:

1) shall include an "m=message" media-level section for the MCData media stream received from the originating MCData client consisting of:

a) the port number;

b) a protocol field value of "TCP/MSRP" or "TCP/TLS/MSRP" for TLS;

c) a format list field set to '';*

d) an "a=sendonly" attribute;

e) an "a=path" attribute containing its own MSRP URI;

f) set the content type as "a=accept-types:application/vnd.3gpp.mcdata-signalling application/vnd.3gpp.mcdata-payload"; and

g) set the a=setup attribute as "actpass"."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.3.4.2]

"When composing the SDP answer according to 3GPP TS 24.229 [5], the controlling MCData function:

1) shall include an "m=message" media-level section for the accepted MCData media stream consisting of:

a) the port number;

b) a protocol field value of "TCP/MSRP" or "TCP/TLS/MSRP" for TLS according to the received SDP offer;

c) a format list field set to '';*

d) an "a=recvonly" attribute;

e) an "a=path" attribute containing its own MSRP URI;

f) set the content type as a=accept-types: application/vnd.3gpp.mcdata-signalling application/vnd.3gpp.mcdata-payload; and

g) set the a=setup attribute set to "passive" according to IETF RFC 6135 [19]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.4.1]

"The originating participating MCData function should establish the MSRP session with the originating MCData client as specified in subclause 6.2.1.2."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.4.2]

"The originating participating MCData function should establish the MSRP session with the controlling MCData function as specified in subclause 6.2.1.3."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.4.3]

"Upon receiving an MSRP SEND request from the originating MCData client, the originating participating MCData function:

1. if in a standalone one-to-one SDS using media plane, shall verify the SDS message size is less than or equal to <MacData1To1> element of the MCData user profile document for the originating MCData client (see the MCData user profile document in 3GPP TS 24.484 [7]);

2. *if the verifications in 1 above fails, shall send the MSRP response with the error code 403 to the originating MCData client and shall not continue with the rest of the procedure;*

3. *if an MSRP connection is not established with the controlling MCData function then, shall establish the MSRP connection as specified in subclause 6.2.1.4.2. Otherwise, shall use the existing MSRP connection; and*

4. *shall forward the received MSRP SEND request to the controlling MCData function according to the rules and procedures of IETF RFC 4975 [11].*

Upon receiving an MSRP 200 (OK) response from the controlling MCData function, the participating MCData function shall forward the MSRP 200 (OK) response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11].

Upon receiving an error MSRP response from the controlling MCData function, the participating MCData function shall forward the error MSRP response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.5.1]

"The terminating participating MCData function should establish the MSRP session with the controlling MCData function as specified in subclause 6.2.1.2."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.5.2]

"The terminating participating MCData function should establish MSRP session to terminating MCData client as specified in subclause 6.2.1.3."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.5.3]

"Upon receiving an MSRP SEND request from the controlling MCData function, the terminating participating MCData function:

1. *shall generate and send a MSRP 200 (OK) response for the received MSRP SEND request to the controlling MCData function, according to the rules and procedures of IETF RFC 4975 [11]; and*

2. *shall forward the received MSRP SEND request to the terminating MCData client according to the rules and procedures of IETF RFC 4975 [11].*

Upon receiving an error MSRP response from the terminating MCData client, the participating MCData function shall forward the error MSRP response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.2]

"To receive a SDS message over MSRP, the participating MCData function:

1. *shall act according to IETF RFC 6135 [12], as:*

   a. *an "passive" endpoint, if a=setup attribute in the sent SDP answer was set to "passive"; and*

   b. *an "active" endpoint, if a=setup attribute in the sent SDP answer was set to "active"; and*

2. *shall establish an MSRP connection according to the MSRP connection parameters in the sent SDP answer response as described in IETF RFC 4976 [14]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.3]

"To send a SDS message over MSRP, the participating MCData function:

1. *shall act according to IETF RFC 6135 [12], as:*

   a. *an "active" endpoint, if a=setup attribute in the received SDP answer is set to "passive"; and*

   b. *an "passive" endpoint, if a=setup attribute in the received SDP answer is set to "active"; and*

2. *shall establish the MSRP connection according to the MSRP connection parameters in the received SDP answer response as described in IETF RFC 4976 [14]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.3.1.2.1]

"To establish the MSRP connection with the originating MCData client, the controlling MCData function performs below procedures:

1. *shall act as an MSRP client and establish TLS connection with the originating participating MCData function, if exists, otherwise the originating MCData client, according to the rules and procedures as described in IETF RFC 4975 [11];*

2. *shall act as an MSRP client to send MSRP SEND requests according to the rules and procedures described in IETF RFC 6135 [12];*

3. *shall act as an "passive" endpoint according to the rules and procedures described in IETF RFC 6135 [12];*

4. *shall establish the MSRP connection with originating MCData client, according to the rules and procedures described in IETF RFC 6135 [12]; and*

5. *as a passive endpoint, shall wait for MSRP SEND request on established MSRP connection, to bind the MSRP connection to the MSRP session according to the rules and procedures of IETF RFC 4975 [11] and IETF RFC 6135 [12]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.3.1.2.2]

"To establish the MSRP connection with the terminating MCData client, the controlling MCData function performs below procedures:

1. *shall act as an MSRP client and establish TLS connection with the terminating participating MCData function, if exists, otherwise the terminating MCData client, according to the rules and procedures as described in IETF RFC 4975 [11];*

2. *shall act as an MSRP client to send MSRP SEND requests according to the rules and procedures described in IETF RFC 6135 [12];*

3. *shall act according to IETF RFC 6135 [12], as:*

   a. *an "active" endpoint, if a=setup attribute in the received SDP answer is set to "passive"; and*

   b. *an "passive" endpoint, if a=setup attribute in the received SDP answer is set to "active";*

4. *shall establish the MSRP connection with each terminating MCData client identified in the 3GPP TS 24.282 [8], according to the rules and procedures described in IETF RFC 6135 [12]; and*

5. *if acting as an "active" endpoint, shall send an empty MSRP SEND request on each established MSRP connection, to bind the MSRP connection to the MSRP session according to the rules and procedures of IETF RFC 4975 [11] and IETF RFC 6135 [12]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.3.1.3]

"Upon receiving a MSRP SEND request from the originating participating MCData function, the controlling MCData function:

1. *if in a standalone one-to-one SDS using media plane, shall verify the SDS message size is less than or equal to <max-data-size-sds-bytes> element in the MCData service configuration document as specified in 3GPP TS 24.484 [7];*

2. *if in a standalone group SDS using media plane shall verify:*

   a. *the SDS message size is less than or equal to <mcdata-max-data-in-single-request> element of the MCData group document for the (see the MCData group document in 3GPP TS 24.481 [4]); and*

   b. *the SDS message size is less than or equal to <mcdata-on-network-max-data-size-for-SDS> element of the MCData group document (see the MCData group document in 3GPP TS 24.481 [4]);*

3. *if the verifications in either 1 or 2 a or 2 b above fails, shall send the MSRP response with the error code 403 to the MCData client which sent the SDS message and shall not continue with the rest of the procedure;*

4. *shall generate and send a MSRP 200 (OK) response for the received MSRP SEND requests to the originating participating MCData function, according to the rules and procedures of IETF RFC 4975 [11]; and*

5. *shall forward the received MSRP SEND requests to each terminating MCData client with which a successful MSRP connection was established, according to the rules and procedures of IETF RFC 4975 [11]. Following clarifications apply to the generated MSRP SEND request:*

   a. *shall modify the To-Path header according to the MSRP URI received in the answer SDP from the MCData client in accordance with rules and procedures of IETF RFC 4975 [11]; and*

   b. *shall modify the From-Path header to the controlling MCData function's own MSRP URI, according to the rules and procedures of IETF RFC 4975 [11]."*

6.1.2.3 Test description

6.1.2.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Storage Function, the MCData User Database, the SIP AS, the HTP Server, the HTTP Client, the MSRP TCP Server and the MSRP TCP Client.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The <max-payload-size-sds-cplane-bytes> element of the MCData Service Configuration document shall be set to 0 to force the MCData client to send the data using the media plane.
- The IUT (MCData Server) is connected to PLMN1.
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.
- SS-UE1 (MCData client) and SS-UE2 (MCData client) are affiliated to the Prearranged group.
- The IUT (MCData Server) is the controlling server of the Prearranged group.

6.1.2.3.2 Test procedure sequence

Table 6.1.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MCData Call Establishment'** as described in Annex A, Table A.4.3.3-1 **to establish a media plane connection for a group?**	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MSRP message transfer'** as described in Annex A, Table A.4.4.3-1 **to send a group SDS message from SS-UE1 (MCData client), with disposition request "DELIVERY"?**	-	-	2	P
3	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MCData Call Release'** as described in Annex A, Table A.4.5.3-1?	-	-	3	P
4	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a disposition notification of "DELIVERED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	4	P

6.1.2.3.3 Specific message contents

Table 6.1.2.3.3-1: SIP INVITE from the SS-UE1
(Step 1, Table 6.1.2.3.2-1; Step 1, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.7.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	"1800"	The recommended initial value is 1 800 in IETF RFC 4028 [23]		
Max-Forwards				
value	"70"	The recommended initial value is 70 in IETF RFC 3261 [21]		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.3.3-2			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.7.3.3-2			

Table 6.1.2.3.3-2: SDP for SIP INVITE (Table 6.1.2.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.7.3.3-1A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line		
Connection Data	not present	c= line		
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	port number assigned by the SS-UE1	The transport port to which the media stream is sent		
Connection Data		c= line		
connection-address	IP address of the SS-UE1			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	not present			
host	IP address of the SS-UE1			
port	same value as in the media line			
session id	assigned by the SS-UE1			
transport	"tcp"			
URI-parameter	not present			

Table 6.1.2.3.3-3: SIP INVITE from the MCData Server (IUT) to the SS-UE2 (Step 1, Table 6.1.2.3.2-1; Step 2, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.8.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
To				
addr-spec				
user-info and host	px_MCX_SIP_PublicUserId_B			
Max-Forwards				
value	any allowed value			
Session-Expires				
generic-param	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.3.3-4			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.1.2.3.3-5			

Table 6.1.2.3.3-4: SDP for SIP INVITE (Table 6.1.2.3.3-4)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.8.3.3-1A				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	any allowed value	"-" indicating the concept of user IDs not being supported		
sess-id	any allowed value	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	any allowed value			
Session Name	any allowed value	s= line		
Connection Data	not required if included in all media	c= line		
nettype	"IN"			
Addrtype	"IP4" or "IP6" depending on IP address"			
connection-address	IP address of the MCData Server			
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	any allowed value	The transport port to which the media stream is sent		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCData Server			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	any value if present			
host	any allowed value			
port	same value as in the media line if present			
session id	any allowed value if present			
transport	"tcp"			
URI-parameter	not present			

Table 6.1.2.3.3-5: MCData-Info (Table 6.1.2.3.3-3)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.8.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-client-id	Encrypted <mcdata-client-id> with mcdataString set to px_MCX_Client_A_ID	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		

Table 6.1.2.3.3-6: SIP 200 (OK) from the SS-UE2 (Step 1, Table 6.1.2.3.2-1; Step 4, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.8.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.3.3-7			

Table 6.1.2.3.3-7: SDP for SIP 200 (OK) (Table 6.1.2.3.3-6)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.8.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"22222222"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"22222222"			
Session Name	" "	s= line		
Connection Data	not present	c= line		
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	port number assigned by the SS-UE2	The transport port to which the media stream is sent		
Connection Data		c= line		
connection-address	IP address of the SS-UE2			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	not present			
host	IP address of the SS-UE2			
port	same value as in the media line			
session id	assigned by the SS-UE2			
transport	"tcp"			
URI-parameter	not present			

Table 6.1.2.3.3-8: SIP 200 (OK) from the MCData Server (IUT) to the SS-UE1 (Step 1, Table 6.1.2.3.2-1; Step 5, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.7.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.2.3.3-9			

Table 6.1.2.3.3-9: SDP for SIP 200 (OK) (Table 6.1.2.3.3-8)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.7.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	any allowed value	"-" indicating the concept of user IDs not being supported		
sess-id	any allowed value	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	any allowed value			
Session Name	any allowed value	s= line		
Connection Data	not required if included in all media	c= line		
nettype	"IN"			
Addrtype	"IP4" or "IP6" depending on IP address"			
connection-address	IP address of the MCData Server			
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	any allowed value	The transport port to which the media stream is sent		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCData Server			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	any value if present			
host	any allowed value			
port	same value as in the media line if present			
session id	any allowed value if present			
transport	"tcp"			
URI-parameter	not present			

Table 6.1.2.3.3-10: MSRP SEND from the SS-UE1
(Step 2, Table 6.1.2.3.2-1; Step 1, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.7.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
data	Message or chunk of message as specified in Table 6.1.2.3.3-10A		IETF RFC 3261 [21]	

Table 6.1.2.3.3-10A: MIME Message (Table 6.1.2.3.3-10)

Derivation Path: IETF RFC 2046 [18]				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME body part		**MCData Data signalling message**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcdata-signalling"			
MIME-part-body	MCData Protected Payload Message containing SDS SIGNALLING PAYLOAD as described in Table 6.1.2.3.3-10B			
MIME body part		**MCData Data message**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcdata-payload"			
MIME-part-body	MCData Protected Payload Message containing DATA PAYLOAD as described in Table 6.1.2.3.3-10C			

Table 6.1.2.3.3-10B: SDS SIGNALLING PAYLOAD (Table 6.1.2.3.3-10A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.8.2-1, condition DELIVERED

Table 6.1.2.3.3-10C: Data Payload (Table 6.1.2.3.3-10A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.9.1-2

Table 6.1.2.3.3-11: MSRP SEND from the MCData Server (IUT) to SS-UE2 (Step 2, Table 6.1.2.3.2-1; Step 3, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.8.3.3-5

Table 6.1.2.3.3-12: SIP MESSAGE from the SS-UE2 (Step 4, Table 6.1.2.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.8.3.3-9				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 6.1.2.3.3-13			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.1.2.3.3-14			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing SDS NOTIFICATION as described in Table 6.1.2.3.3-15			

Table 6.1.2.3.3-13: Resource-lists (Table 6.1.2.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.1.2.3.3-14: MCData-Info (Table 6.1.2.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-calling-group-id	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_Group_A_ID	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-3A		
mcdata-client-id	not present			

Table 6.1.2.3.3-15: SDS NOTIFICATION (Table 6.1.2.3.3-12)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.8.3.3-11				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Date and time	The current date and time	The Date and time value is an unsigned integer containing UTC time of the time when a message was sent, in seconds since midnight UTC of January 1, 1970 (not counting leap seconds).	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 15.2.8	

Table 6.1.2.3.3-16: SIP MESSAGE from the MCData Server (IUT) to SS-UE1 (Step 4, Table 6.1.2.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.7.3.3-11				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

6.1.3 MCDATA Server - MCDATA Client / Short Data Service (SDS) / SDS Session / One-to-one SDS Session

6.1.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCData Client) initiates establishment of a one-to-one SDS session }
    then { IUT (MCData Server) accepts the call from the initiator and establishes the call with SS-
UE2 (MCData Client) and responds/forwards empty MSRP messages  }
            }
```

(2)

```
with { IUT (MCData Server) having requested an one-to-one SDS session }
ensure that {
  when { the SS-UE1 (MCData Client) sends an MSRP SEND message with disposition request "DELIVERY"}
    then { IUT (MCData Server) forwards an MSRP SEND message to SS-UE2 (MCData Client) and responds
with an MSRP 200 (Ok) to SS-UE1 (MCData Client)  }
            }
```

(3)

```
with { IUT (MCData Server) having requested an one-to-one SDS session }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "DELIVERED" by sending an MSRP SEND
message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to SS-UE1 (MCData Client) and responds
with a SIP 202 (Accepted) to SS-UE2 (MCData Client) }
            }
```

(4)

```
with { IUT (MCData Server) having requested an one-to-one SDS session }
ensure that {
  when { the SS-UE2 (MCData Client) sends an MSRP SEND message with disposition request "READ"}
    then { IUT (MCData Server) forwards an MSRP SEND message to SS-UE1 (MCData Client) and responds
with an MSRP 200 (Ok) to SS-UE2 (MCData Client)  }
            }
```

(5)

```
with { IUT (MCData Server) having requested an one-to-one SDS session }
ensure that {
  when { SS-UE1 (MCData Client) sends a disposition notification "READ" by sending an MSRP SEND
message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to SS-UE2 (MCData Client) and responds
with a SIP 202 (Accepted) to SS-UE1 (MCData Client) }
            }
```

(6)

```
with { IUT (MCData Server) having requested an one-to-one SDS session }
ensure that {
  when { the SS-UE1 (MCData client) ends the one-to-one SDS session }
    then { IUT (MCData Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to SS-UE2 (MCData Client) and waits to close MSRP TCP connections
}
            }
```

6.1.3.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282), clauses 9.2.4.3.3, 9.2.4.3.4, 9.2.4.3.1, 9.2.4.3.2, 9.2.4.4.3, 9.2.4.4.4, 9.2.4.4.1, 9.2.4.4.2, ETSI TS 124 582 [40] (3GPP TS 24.582), clauses 6.2.2.4.1, 6.2.2.4.2, 6.2.2.4.3, 6.2.2.5.1, 6.2.2.5.2, 6.2.2.5.3, 6.2.2.2, 6.2.2.3, 6.3.2.2.1, 6.3.2.2.2, 6.3.2.3. Unless otherwise stated, these are Rel-14 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.4.3.3]

"Upon receipt of a "SIP INVITE request for SDS session for originating participating MCData function", the participating MCData function:

…

2) shall determine the MCData ID of the calling user from the public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and shall authorise the calling user;

NOTE: The MCData ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in subclause 7.3.

…

4) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP INVITE request is:

…

b) set to a value of "one-to-one-sds-session", shall determine the public service identity of the controlling MCData function hosting the one-to-one SDS session service for the calling user;

…

6) shall determine whether the MCData user identified by the MCData ID is authorised for MCData communications by following the procedures in subclause 11.1;

…

8) shall generate a SIP INVITE request in accordance with 3GPP TS 24.229 [5];

9) shall include the option tag "timer" in the Supported header field;

10) should include the Session-Expires header field according to IETF RFC 4028 [38]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

11) shall set the Request-URI of the outgoing SIP INVITE request to the public service identity of the controlling MCData function as determined by step 4) in this subclause;

12) shall include the MCData ID of the originating user in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP INVITE request;

12A) if the incoming SIP MESSAGE request contains an application/vnd.3gpp.mcdata-info+xml MIME body that contains a <functional-alias-URI> element, shall check if the status of the functional alias is activated for the MCData ID. If the functional alias status is activated, then the participating MCData function shall set the <functional-alias-URI> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request to the received value, otherwise shall not include a <functional-alias-URI> element;

13) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

14) shall set the P-Asserted-Identity in the outgoing SIP INVITE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP INVITE request;

15) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCData client as specified in subclause 9.2.4.3.1; and

16) shall send the SIP INVITE request as specified to 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the SIP INVITE request in step 16):

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5];

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the subclause 9.2.4.3.2;

3) shall include the option tag "timer" in a Require header field;

4) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [38], "UAS Behavior". If the "refresher" parameter is not included in the received request, the "refresher" parameter in the Session-Expires header field shall be set to "uac";

5) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.sds media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds"; and

c) the isfocus media feature tag;

6) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

7) shall include an MCData session identity mapped to the MCData session identity provided in the Contact header field of the received SIP 200 (OK) response;

8) if the incoming SIP 200 (OK) response contained an application/vnd.3gpp.mcdata-info+xml MIME body, shall copy the application/vnd.3gpp.mcdata-info+xml MIME body to the outgoing SIP 200 (OK) response.

9) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response; and

10) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.2.2.4;

11) shall send the SIP 200 (OK) response to the MCData client according to 3GPP TS 24.229 [5]; and

12) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [38].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP INVITE request in step 16) the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.4.3.4]

"Upon receipt of a "SIP INVITE request for SDS session for terminating participating MCData function", the participating MCData function:

…

3) shall use the MCData ID present in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCData ID and public user identity of the terminating MCData user;

…

5) shall generate a SIP INVITE request accordance with 3GPP TS 24.229 [5];

6) should include the Session-Expires header field according to IETF RFC 4028 [38]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

7) shall include the option tag "timer" in the Supported header field;

8) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.sds media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds";

c) the isfocus media feature tag;

d) an MCData session identity mapped to the MCData session identity provided in the Contact header field of the incoming SIP INVITE request; and

e) any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;

9) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [8] that were received (if any) in the incoming SIP INVITE request;

10) shall set the Request-URI of the outgoing SIP INVITE request to the public user identity associated to the MCData ID of the terminating MCData user;

11) shall populate the outgoing SIP INVITE request with the MIME bodies that were present in the incoming SIP INVITE request;

12) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for SDS session for terminating participating MCData function" as specified in subclause 9.2.4.3.1; and

14) shall send the SIP INVITE request as specified in 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the above SIP INVITE request, the participating MCData function:

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5];

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in subclause 9.2.4.3.2;

3) shall include the option tag "timer" in a Require header field;

4) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [38], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

5) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.sds media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds"; and

c) an MCData session identity mapped to the MCData session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCData function;

6) if the incoming SIP response contained an application/vnd.3gpp.mcdata-info+xml MIME body, shall copy the application/vnd.3gpp.mcdata-info+xml MIME body to the outgoing SIP 200 (OK) response.

7) shall copy the P-Asserted-Identity header field from the incoming SIP 200 (OK) response to the outgoing SIP 200 (OK) response;

8) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [38].

9) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.2.2.5; and

10) shall send the SIP 200 (OK) response to the controlling MCData function according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request, the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the controlling MCData function according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.4.3.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCData function:

1) shall contain only one SDP media-level section for SDS message as contained in the received SDP offer;and

2) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [5], the participating MCData function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCData function, if required; and

…

2) if the IP address is replaced, shall insert its MSRP URI before the MSRP URI in the "a=path" attribute in the SDP offer."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.4.3.2]

"When composing the SDP answer according to 3GPP TS 24.229 [5], the participating MCData function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCData function, for the accepted media stream in the received SDP offer, if required; and

…

2) if the IP address is replaced shall insert its MSRP URI before the MSRP URI in the "a=path" attribute in the SDP answer."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.4.4.3]

"This subclause describes the procedures for inviting an MCData user to an MCData session. The procedure is initiated by the controlling MCData function as the result of:

- *an action in subclause 9.2.4.4.4; or*
- *for group SDS session, when an MCData client successfully affiliates the MCData group after the SDS session has been established.*

The controlling MCData function:

1) shall generate a SIP INVITE according to 3GPP TS 24.229 [5];

2) shall include the Supported header field set to "timer";

3) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [38]. The refresher parameter shall be omitted;

4) shall include an Accept-Contact header field containing the g.3gpp.mcdata.sds media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8];

5) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" along with parameters "require" and "explicit" according to IETF RFC 3841 [8];

6) shall include a Referred-By header field with the public user identity of the inviting MCData client;

7) shall include in the Contact header field an MCData session identity for the MCData session with the g.3gpp.mcdata.sds media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" according to IETF RFC 3840 [16];

8) shall include in the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcdata-request-uri> element set to the MCData ID of the terminating user; and

b) the <mcdata-calling-group-id> element set to the group identity if the request is for group sds;

9) shall set the Request-URI to the public service identity of the terminating participating MCData function associated to the MCData user to be invited;

NOTE 1: How the controlling MCData function finds the address of the terminating participating MCData function is out of the scope of the current release.

10) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCData function;

11) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [7] in the SIP INVITE request;

12) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating client according to the procedures specified in subclause 9.2.4.4.1; and

13) shall send the SIP INVITE request towards the terminating client in accordance with 3GPP TS 24.229 [5].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCData function:

1) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.3.2.

NOTE 2: The procedures executed by the controlling MCData function prior to sending a response to the inviting MCData client are specified in subclause 9.2.4.4.4."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.4.4.4]

"In the procedures in this subclause:

1) MCData ID in an incoming SIP INVITE request refers to the MCData ID of the originating user from the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request;

2) group identity in an incoming SIP INVITE request refers to the group identity from the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request; and

3) MCData ID in an outgoing SIP INVITE request refers to the MCData ID of the called user in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP INVITE request;

Upon receipt of a "SIP INVITE request for controlling MCData function for SDS session", the controlling MCData function:

…

4) shall cache SIP feature tags, if received in the Contact header field and if the specific feature tags are supported;

6) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [38];

7) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP INVITE request is set to a value of "one-to-one-sds-session" and the SIP INVITE request:

…

b) contains an application/resource-lists MIME body with exactly one <entry> element, shall invite the MCData user identified by the <entry> element of the MIME body, as specified in subclause 9.2.4.4.3; and

c) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.3.2;

…

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in subclause 9.2.4.4.3 and if the MCData ID in the SIP 200 (OK) response matches to the MCData ID in the corresponding SIP INVITE request the controlling MCData function:

1) shall generate SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [5];

2) shall include the option tag "timer" in a Require header field;

3) shall include the Session-Expires header field and start supervising the SIP session according to rules and procedures of IETF RFC 4028 [38], "UAS Behavior". The "refresher" parameter in the Session-Expires header field shall be set to "uac";

4) shall include a P-Asserted-Identity header field with the public service identity of the controlling MCData function;

5) shall include a SIP URI for the MCData session identity in the Contact header field identifying the MCData session at the controlling MCData function;

6) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.sds media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds"; and

c) the isfocus media feature tag;

7) shall include Warning header field(s) received in incoming responses to the SIP INVITE request;

8) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the subclause 9.2.4.4.2;

9) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 6.3.2; and

10) shall send a SIP 200 (OK) response to the inviting MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.4.4.1]

"When composing an SDP offer according to 3GPP TS 24.229 [5], IETF RFC 4975 [17], IETF RFC 6135 [19] and IETF RFC 6714 [20] the controlling MCData function:

1) shall include an "m=message" media-level section for the MCData media stream received from the originating MCData client consisting of:

a) the port number;

b) a protocol field value of "TCP/MSRP" or "TCP/TLS/MSRP" for TLS;

c) an "a=sendrecv" attribute;

d) an "a=path" attribute containing its own MSRP URI;

e) set the content type as "a=accept-types:application/vnd.3gpp.mcdata-signalling application/vnd.3gpp.mcdata-payload"; and

f) set the a=setup attribute as "actpass"."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.4.4.2]

"When composing the SDP answer according to 3GPP TS 24.229 [5], the controlling MCData function:

1) shall include an "m=message" media-level section for the accepted MCData media stream consisting of:

a) the port number;

b) a protocol field value of "TCP/MSRP" or "TCP/TLS/MSRP" for TLS according to the received SDP offer;

c) an "a=sendrecv" attribute;

d) an "a=path" attribute containing its own MSRP URI;

e) set the content type as a=accept-types: application/vnd.3gpp.mcdata-signalling application/vnd.3gpp.mcdata-payload; and

f) set the a=setup attribute set to "passive" according to IETF RFC 6135 [19]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.2.4.1]

"The originating participating MCData function should establish the MSRP session with the originating MCData client as specified in subclause 6.2.2.2."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.2.4.2]

"The originating participating MCData function should establish the MSRP session with the controlling MCData function as specified in subclause 6.2.2.3."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.2.4.3]

"Upon receiving an MSRP SEND request from the originating MCData client, the originating participating MCData function:

1. *if an MSRP connection is not established with the controlling MCData function then, shall establish the MSRP connection as specified in subclause 6.2.2.4.2. Otherwise, shall use the existing MSRP connection; and*

2. *shall forward the received MSRP SEND request to the controlling MCData function according to the rules and procedures of IETF RFC 4975 [11].*

Upon receiving an MSRP 200 (OK) response from the controlling MCData function, the participating MCData function shall forward the MSRP 200 (OK) response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11].

Upon receiving an error MSRP response from the controlling MCData function, the participating MCData function shall forward the error MSRP response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.2.5.1]

"The terminating participating MCData function should establish the MSRP session with the controlling MCData function as specified in subclause 6.2.2.2."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.2.5.2]

"The terminating participating MCData function should establish MSRP session to terminating MCData client as specified in subclause 6.2.2.3."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.2.5.3]

"Upon receiving an MSRP SEND request from the controlling MCData function, the terminating participating MCData function:

1. *shall generate and send a MSRP 200 (OK) response for the received MSRP SEND request to the controlling MCData function, according to the rules and procedures of IETF RFC 4975 [11]; and*

2. *shall forward the received MSRP SEND request to the terminating MCData client according to the rules and procedures of IETF RFC 4975 [11].*

Upon receiving an error MSRP response from the terminating MCData client, the participating MCData function shall forward the error MSRP response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.2.2]

"To receive a SDS message over MSRP, the participating MCData function:

1. *shall act according to IETF RFC 6135 [12], as:*

   a. *an "passive" endpoint, if a=setup attribute in the sent SDP answer was set to "passive"; and*

   b. *an "active" endpoint, if a=setup attribute in the sent SDP answer was set to "active"; and*

2. *shall establish an MSRP connection according to the MSRP connection parameters in the sent SDP answer response as described in IETF RFC 4976 [14]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.2.3]

"To send a SDS message over MSRP, the participating MCData function:

1. *shall act according to IETF RFC 6135 [12], as:*

   a. *an "active" endpoint, if a=setup attribute in the received SDP answer is set to "passive"; and*

   b. *an "passive" endpoint, if a=setup attribute in the received SDP answer is set to "active"; and*

2. *shall establish the MSRP connection according to the MSRP connection parameters in the received SDP answer response as described in IETF RFC 4976 [14]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.3.2.2.1]

"To establish the MSRP connection with the originating MCData client, the controlling MCData function performs below procedures:

1. *shall act as an MSRP client and establish TLS connection with the originating participating MCData function, if exists, otherwise the originating MCData client, according to the rules and procedures as described in IETF RFC 4975 [11];*

2. *shall act as an MSRP client to send MSRP SEND requests according to the rules and procedures described in IETF RFC 6135 [12];*

3. *shall act as an "passive" endpoint according to the rules and procedures described in IETF RFC 6135 [12];*

4. *shall establish the MSRP connection with originating MCData client, according to the rules and procedures described in IETF RFC 6135 [12]; and*

5. *acting as a "passive" endpoint, shall wait for MSRP SEND request on established MSRP connection, to bind the MSRP connection to the MSRP session according to the rules and procedures of IETF RFC 4975 [11] and IETF RFC 6135 [12]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.3.2.2.2]

"To establish the MSRP connection with the terminating MCData client, the controlling MCData function performs below procedures:

1. *shall act as an MSRP client and establish TLS connection with the terminating participating MCData function, if exists, otherwise the terminating MCData client, according to the rules and procedures as described in IETF RFC 4975 [11];*

2. *shall act as an MSRP client to send MSRP SEND requests according to the rules and procedures described in IETF RFC 6135 [12];*

3. *shall act according to IETF RFC 6135 [12], as:*

   a. *an "active" endpoint, if a=setup attribute in the received SDP answer is set to "passive"; and*

   b. *an "passive" endpoint, if a=setup attribute in the received SDP answer is set to "active";*

4. *shall establish the MSRP connection with each terminating MCData client identified in the 3GPP TS 24.282 [8], according to the rules and procedures described in IETF RFC 6135 [12]; and*

5. *if acting as an "active" endpoint, shall send an empty MSRP SEND request on each established MSRP connection, to bind the MSRP connection to the MSRP session according to the rules and procedures of IETF RFC 4975 [11] and IETF RFC 6135 [12]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.3.2.3]

"Upon receiving a MSRP SEND request from the originating participating MCData function, the controlling MCData function:

1. *shall generate and send a MSRP 200 (OK) response for the received MSRP SEND requests to the originating participating MCData function, according to the rules and procedures of IETF RFC 4975 [11]; and*

2. *shall forward the received MSRP SEND requests to each terminating MCData client with which a successful MSRP connection was established, according to the rules and procedures of IETF RFC 4975 [11]. Following clarifications apply to the generated MSRP SEND request:*

   a. *shall modify the To-Path header according to the MSRP URI received in the answer SDP from the MCData client in accordance with rules and procedures of IETF RFC 4975 [11]; and*

   b. *shall modify the From-Path header to the controlling MCData function's own MSRP URI, according to the rules and procedures of IETF RFC 4975 [11]."*

6.1.3.3 Test description

6.1.3.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Storage Function, the MCData User Database, the SIP AS, the HTP Server, the HTTP Client, the MSRP TCP Server and the MSRP TCP Client.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCData Server) is connected to PLMN1.
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

6.1.3.3.2 Test procedure sequence

Table 6.1.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MCData Call Establishment'** as described in Annex A, Table A.4.3.3-1 **to establish a media plane connection between SS-UE1 (MCData Client) and SS-UE2 (MCData Client)?**	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MSRP message transfer'** as described in Annex A, Table A.4.4.3-1 **to forward a SDS message from SS-UE1 (MCData client) to SS-UE2 (MCData Client), with disposition request "DELIVERY"?**	-	-	2	P
3	Check: Does the MCData Server (IUT) correctly perform procedure**'E2E MSRP message transfer'** as described in Annex A, Table A.4.4.3-1 **to forward a disposition notification of "DELIVERED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	3	P
4	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MSRP message transfer'** as described in Annex A, Table A.4.4.3-1 **to forward a SDS message from SS-UE2 (MCData client) to SS-UE1 (MCData Client), with disposition request "READ"?**	-	-	4	P
5	Check: Does the MCData Server (IUT) correctly perform procedure**'E2E MSRP message transfer'** as described in Annex A, Table A.4.4.3-1 **to forward a disposition notification of "READ", from SS-UE1 (MCData client) to SS-UE2 (MCData Client)?**	-	-	5	P
6	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MCData Call Release'** as described in Annex A, Table A.4.5.3-1?	-	-	6	P

6.1.3.3.3 Specific message contents

Table 6.1.3.3.3-1: SIP INVITE from the SS-UE1
(Step 1, Table 6.1.3.3.2-1; Step 1, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.9.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	"1800"	The recommended initial value is 1 800 in IETF RFC 4028 [23].		
Max-Forwards				
value	"70"	The recommended initial value is 70 in IETF RFC 3261 [21].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.3.3.3-2			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.9.3.3-3			

Table 6.1.3.3.3-2: SDP for SIP INVITE (Table 6.1.3.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.9.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line		
Connection Data	not present	c= line		
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	port number assigned by the SS-UE1	The transport port to which the media stream is sent		
Connection Data		c= line		
connection-address	IP address of the SS-UE1			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	not present			
host	IP address of the SS-UE1			
port	same value as in the media line			
session id	assigned by the SS-UE1			
transport	"tcp"			
URI-parameter	not present			
media attribute		a= line attribute = key-mgmt		
key-mgmt				
mikey	MIKEY-SAKKE I_MESSAGE as specified in Table 6.1.3.3.3-2A			

Table 6.1.3.3.3-2A: MIKEY-SAKKE I_MESSAGE (Table 6.1.3.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
IDRi payload				
ID data	px_MCPTT_ID_User_A	MCPTT ID associated with the initiating user		
IDRr payload				
ID data	px_MCPTT_ID_User_B	MCPTT ID associated to the receiving user		

Table 6.1.3.3.3-3: SIP INVITE from the MCData Server (IUT) to the SS-UE2 (Step 1, Table 6.1.3.3.2-1; Step 2, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
To				
addr-spec				
user-info and host	px_MCX_SIP_PublicUserId_B			
Max-Forwards				
value	any allowed value			
Session-Expires				
generic-param	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.3.3.3-4			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.1.3.3.3-5			

Table 6.1.3.3.3-4: SDP for SIP INVITE (Table 6.1.3.3.3-4)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	any allowed value	"-" indicating the concept of user IDs not being supported		
sess-id	any allowed value	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	any allowed value			
Session Name	any allowed value	s= line		
Connection Data	not required if included in all media	c= line		
nettype	"IN"			
Addrtype	"IP4" or "IP6" depending on IP address"			
connection-address	IP address of the MCData Server			
Session attribute	present only if there is no key-mgmt media attribute in the media description for data	a= line attribute = key-mgmt		
key-mgmt				
mikey	MIKEY-SAKKE I_MESSAGE as specified in Table 6.1.3.3.3-2A			
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	any allowed value	The transport port to which the media stream is sent		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCData Server			
media attribute		a= line attribute = path		

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	any value if present			
host	any allowed value			
port	same value as in the media line if present			
session id	any allowed value if present			
transport	"tcp"			
URI-parameter	not present			
media attribute	present only if there is no key-mgmt attribute at session level			
key-mgmt				
mikey	MIKEY-SAKKE I_MESSAGE as specified in Table 6.1.3.3.3-2A			

Table 6.1.3.3.3-5: MCData-Info (Table 6.1.3.3.3-3)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
request-type	"one-to-one-sds-session"			

Table 6.1.3.3.3-6: SIP 200 (OK) from the SS-UE2
(Step 1, Table 6.1.3.3.2-1; Step 4, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.3.3.3-7			

Table 6.1.3.3.3-7: SDP for SIP 200 (OK) (Table 6.1.3.3.3-6)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"22222222"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"22222222"			
Session Name	" "	s= line		
Connection Data	not present	c= line		
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	port number assigned by the SS-UE2	The transport port to which the media stream is sent		
Connection Data		c= line		
connection-address	IP address of the SS-UE2			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	not present			
host	IP address of the SS-UE2			
port	same value as in the media line			
session id	assigned by the SS-UE2			
transport	"tcp"			
URI-parameter	not present			

Table 6.1.3.3.3-8: SIP 200 (OK) from the MCData Server (IUT) to SS-UE1 (Step 1, Table 6.1.3.3.2-1; Step 5, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.9.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.1.3.3.3-9			

Table 6.1.3.3.3-9: SDP for SIP 200 (OK) (Table 6.1.3.3.3-8)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.9.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	any allowed value	"-" indicating the concept of user IDs not being supported		
sess-id	any allowed value	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	any allowed value			
Session Name	any allowed value	s= line		
Connection Data	not required if included in all media	c= line		
nettype	"IN"			
Addrtype	"IP4" or "IP6" depending on IP address"			
connection-address	IP address of the MCData Server			
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	any allowed value	The transport port to which the media stream is sent		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCData Server			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	any value if present			
host	any allowed value			
port	same value as in the media line if present			
session id	any allowed value if present			
transport	"tcp"			
URI-parameter	not present			

Table 6.1.3.3.3-10: MSRP SEND from the SS-UE1
(Step 2, Table 6.1.3.3.2-1; Step 1, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.9.3.3-6				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
data	Message or chunk of message as specified in table 6.1.3.3.3-10A		IETF RFC 3261 [21]	

Table 6.1.3.3.3-10A: MIME Message (Table 6.1.3.3.3-10)

Derivation Path: IETF RFC 2046 [18]				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME body part		**MCData Data signalling message**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcdata-signalling"			
MIME-part-body	MCData Protected Payload Message containing SDS SIGNALLING PAYLOAD as described in table 6.1.2.3.3-10B			
MIME body part		**MCData Data message**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcdata-payload"			
MIME-part-body	MCData Protected Payload Message containing DATA PAYLOAD as described in Table 6.1.2.3.3-10C			

Table 6.1.3.3.3-10B: SDS SIGNALLING PAYLOAD (Table 6.1.3.3.3-10A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.8.2-1, condition DELIVERED

Table 6.1.3.3.3-10C: Data Payload (Table 6.1.3.3.3-10A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.9.2-2

Table 6.1.3.3.3-11: MSRP SEND from the MCData Server (IUT) to SS-UE2 (Step 2, Table 6.1.3.3.2-1; Step 3, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-6

Table 6.1.3.3.3-12: MSRP SEND from the SS-UE2 (Step 3, Table 6.1.3.3.2-1; Step 1, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-8				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
data	MCData Protected Payload Message containing SDS NOTIFICATION as specified in Table 6.1.3.3.3-13			

Table 6.1.3.3.3-13: SDS SIGNALLING PAYLOAD (Table 6.1.3.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.8.4-1, condition DELIVERED

Table 6.1.3.3.3-14: MSRP SEND from the MCData Server (IUT) to SS-UE1 (Step 3, Table 6.1.3.3.2-1; Step 3, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.9.3.3-10

Table 6.1.3.3.3-15: MSRP SEND from the SS-UE2 (Step 4, Table 6.1.3.3.2-1; Step 1, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
data	Message or chunk of message as specified in Table 6.1.3.3.3-15A		IETF RFC 3261 [21]	

Table 6.1.3.3.3-15A: MIME Message (Table 6.1.3.3.3-15)

Derivation Path: IETF RFC 2046 [18]				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME body part		**MCData Data signalling message**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcdata-signalling"			
MIME-part-body	MCData Protected Payload Message containing SDS SIGNALLING PAYLOAD as described in Table 6.1.2.3.3-15B			
MIME body part		**MCData Data message**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcdata-payload"			
MIME-part-body	MCData Protected Payload Message containing DATA PAYLOAD as described in Table 6.1.2.3.3-10C			

Table 6.1.3.3.3-15B: SDS SIGNALLING PAYLOAD (Table 6.1.3.3.3-15A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.8.2-1, condition READ

Table 6.1.3.3.3-16: MSRP SEND from the MCData Server (IUT) to SS-UE1 (Step 4, Table 6.1.3.3.2-1; Step 3, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.9.3.3-12

Table 6.1.3.3.3-17: MSRP SEND from the SS-UE1 (Step 5, Table 6.1.3.3.2-1; Step 1, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.9.3.3-14				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
data	MCData Protected Payload Message containing SDS NOTIFICATION as specified in Table 6.1.3.3.3-18			

Table 6.1.3.3.3-18: SDS SIGNALLING PAYLOAD (Table 6.1.3.3.3-17)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.8.4-1, condition READ

Table 6.1.3.3.3-19: MSRP SEND from the MCData Server (IUT) to SS-UE2 (Step 5, Table 6.1.3.3.2-1; Step 3, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.10.3.3-15

6.1.4 MCDATA Server - MCDATA Client / Short Data Service (SDS) / Standalone SDS Using Media Plane / One-to-one Standalone SDS / Pre-established session

6.1.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 and having established a pre-established session with
the clients}
ensure that {
  when { the SS-UE1 (MCData Client) requests the call establishment using a pre-established session
with SIP REFER message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) and establishes the call and
responds/forwards empty MSRP messages }
            }
```

(2)

```
with { IUT (MCData Server) having established a pre-established session with the clients }
ensure that {
  when { the SS-UE1 (MCData Client) sends an MSRP SEND message with disposition request "DELIVERY"}
    then { IUT (MCData Server) forwards an MSRP SEND message to UE2 (MCData Client) and responds
with an MSRP 200 (Ok) to UE1 (MCData Client)  }
            }
```

(3)

```
with { IUT (MCData Server) having established a pre-established session with the clients }
ensure that {
  when { the SS-UE1 (MCData Client) requests the call release keeping the pre-established with SIP
REFER message }
    then { IUT (MCData Server) responds with a SIP 200 (OK) and releases the SS-UE2 (MCData Client)
}
            }
```

(4)

```
with { IUT (MCData Server) having established a pre-established session with the clients }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "DELIVERED" by sending a SIP
MESSAGE message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

6.1.4.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282), clauses 18.3.2.2, 18.3.3.1.2, 18.3.3.2.2, 9.2.5.2.2.1, 9.2.5.2.2.2, 9.2.5.4.2.1, 9.2.5.4.2.2, 9.2.5.1.1, 9.2.5.1.2, 9.2.5.1.3, ETSI TS 124 582 [40] (3GPP TS 24.582), clauses 12.2, 6.2.1.4.1, 6.2.1.4.2, 6.2.1.4.3, 6.2.1.5.1, 6.2.1.5.2, 6.2.1.5.3, 6.2.1.2, 6.2.1.3, 6.3.1.2.1, 6.3.1.2.2, 6.3.1.3. Unless otherwise stated, these are Rel-16 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 18.3.2.2]

"Upon receipt of a "SIP INVITE request for establishing a pre-established session" the participating MCData function:

…

3) shall determine whether resource sharing is supported (see subclause 18.2);

4) if resource sharing is supported by the SIP core, determine that there is a binding between the MCData ID of the MCData user establishing the pre-established session and the MCData UE identified by the "+g.3gpp.registration-token" header field parameter in the Contact header field of the third-party REGISTER request (see subclause 18.2) and that this UE identity matches the identity in the "+g.3gpp.registration-token" header field parameter in the Feature-Caps header field in the "SIP INVITE request for establishing a pre-established session";

…

8) shall allocate a URI to be used to identify the pre-established session;

9) shall generate a SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [5]; and

a) shall include a Contact header field containing the URI that identifies the pre-established session;

b) shall include the public service identity in the P-Asserted-Identity header field;

c) shall include a Supported header field containing the "norefersub" option tag;

d) shall if the SIP core supports resource sharing, include a Resource-Share header field answer as specified in 3GPP TS 24.229 [5] with:

A) the value "media-sharing";

B) an "origin" header field parameter set to "session-initiator";

C) a "timestamp" header field parameter; and

D) a "rules" header field parameter with one resource sharing rule per media stream in the same order the corresponding m-line appears in the SDP. Each resource sharing rule is constructed as follows:

- *a "new-sharing-key" part; and*

- *a "directionality" part indicating the direction of the pre-established media stream; and*

e) shall include an SDP answer as specified in 3GPP TS 24.229 [5] with the clarifications in subclause 18.3.1.2 and include ICE candidates in the SDP answer as per IETF RFC 5245 [50];

10) shall interact with the media plane as specified in 3GPP TS 24.582 [15];

11) shall send the SIP 200 (OK) response towards the MCData client according to the rules and procedures of the 3GPP TS 24.229 [5]; and

12) shall evaluate the ICE candidates according to IETF RFC 5245 [50].

NOTE: If ICE candidate evaluation results in candidate pairs other than the default candidate pair being selected a further offer answer exchange using the procedures in subclause 18.3.4 will be needed."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 18.3.3.1.2]

"Upon receiving a SIP BYE request from the participating MCData function within a pre-established session the MCData client shall check whether there are any MCData sessions using the pre-established session, and:

1) if there is an established MCData session then the MCData client shall remove the MCData client from the MCData session by performing the procedures for session release for each MCData session as specified in 3GPP TS 24.582 [15]; and

2) if there is no MCData session using the pre-established session, then the MCData client shall follow the procedure described in subclause 13.2.3.2.2."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 18.3.3.2.2]

"When a participating MCData function needs to release a pre-established session as created in subclause 8.2.2, the participating MCData function:

1) shall first release any participants of all MCData calls that are using the pre-established session. The participating MCData function shall remove the MCData client from the MCData session by performing the procedures as specified in subclause 13.2.2.2.3.1;

2) shall generate a SIP BYE request according to rules and procedures of 3GPP TS 24.229 [5];

3) shall set the Request-URI of the SIP BYE request to the URI that identifies the pre-established session;

4) shall send the SIP BYE request towards the MCData client within the SIP dialog of the pre-established session according to rules and procedures of the 3GPP TS 24.229 [5]; and

5) shall, upon receiving a SIP 200 (OK) response to the SIP BYE request interact with the media plane as specified in 3GPP TS 24.582 [15]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.5.2.2.1]

"Upon receiving a SIP REFER request, with:

1) the Request-URI set to a public service identity identifying the pre-established session on the participating MCData function;

2) the Refer-To header field containing a Content-ID ("cid") URL as specified in IETF RFC 2392 [33] that points to an application/resource-lists MIME body as specified in IETF RFC 5366 [18] containing one or more <entry> element(s) with a "uri" attribute containing a SIP URI set to the MCData ID of the called user(s);

3) an hname "body" parameter in the headers portion of the SIP URI specified above containing an application/vnd.3gpp.mcdata-info MIME body with the <request-type> element set to "one-to-one-sds" or "one-to-one-sds-session"; and

4) a Content-ID header field set to the "cid" URL;

the participating function:

…

2) shall determine the MCData ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP REFER request;

…

4) shall determine whether the MCData user identified by the MCData ID is authorised for MCData communications by following the procedures in subclause 11.1;

…

6) if the received SIP REFER request contains an application/resource-lists MIME body referenced by a "cid" URL in the Refer-To header field with more than one <entry> element each with an application/vnd.3gpp.mcdata-info MIME body with the <request-type> element set to "one-to-one-sds" or "one-to-one-sds-session", determine that the communication type is one-to-one standalone SDS or one-to-one SDS session;

7) shall determine the public service identity of the controlling MCData function associated with the originating user's MCData ID;

…

8) if the SIP REFER request contained a Refer-Sub header field containing "false" value and a Supported header field containing "norefersub" value, shall handle the SIP REFER request as specified in 3GPP TS 24.229 [5], IETF RFC 3515 [51] as updated by IETF RFC 6665 [36], and IETF RFC 4488 [53] without establishing an implicit subscription;

9) shall generate a final SIP 200 (OK) response to the SIP REFER request according to 3GPP TS 24.229 [5];

NOTE: In accordance with IETF RFC 4488 [53], the participating MCData function inserts the Refer-Sub header field containing the value "false" in the SIP 200 (OK) response to the SIP REFER request to indicate that it has not created an implicit subscription.

10) shall send the response to the SIP REFER request towards the MCData client according to 3GPP TS 24.229 [5];

11) shall generate SIP INVITE request as described in subclause 9.2.5.1.1;

12) shall set the Request-URI of the SIP INVITE request to the public service identity of the controlling MCData function serving the calling MCData user as determined above in step 7); and

13) shall forward the SIP INVITE request according to 3GPP TS 24.229 [4].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the participating MCData function:

1) shall interact with the media plane as specified in 3GPP TS 24.582 [15];

2) shall generate a SIP re-INVITE request as specified in subclause 9.2.5.1.2 with following clarifications:

i) shall set the Request-URI to a public service identity identifying the pre-established session;

3) shall send the SIP re-INVITE request towards the originating MCData client according to 3GPP TS 24.229 [5]; and

4) upon receipt of a SIP 2xx response to the SIP re-INVITE, shall interact with the media plane as specified in 3GPP TS 24.582 [15]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.5.2.2.2]

"Upon receipt of a "SIP INVITE request for standalone SDS over media plane for terminating participating MCData function" or "SIP INVITE request for SDS session for terminating participating MCData function", the participating MCData function:

…

2) shall use the MCData ID present in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCData ID and public user identity;

..

3) shall generate a SIP re-INVITE request as specified in subclause 9.2.5.1.3 with following clarifications:

i) shall set the Request-URI to a public service identity identifying the pre-established session;

ii) if the incoming SIP INVITE request contained an application/vnd.3gpp.mcdata-info+xml MIME body, shall copy the application/vnd.3gpp.mcdata-info+xml MIME body to the outgoing SIP INVITE request with following clarification:

a) shall include <mcdata-communication-state> element with a value set to "establish-request"; and

iii) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.sds media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds";

c) the isfocus media feature tag;

d) an MCData session identity mapped to the MCData session identity provided in the Contact header field of the incoming SIP INVITE request; and

e) any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;

4) shall send the SIP re-INVITE request towards the terminating MCData client according to 3GPP TS 24.229 [5]; and

5) upon receipt of a SIP 2xx response to the SIP re-INVITE, shall interact with the media plane as specified in 3GPP TS 24.582 [15]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.5.4.2.1]

"Upon receiving a SIP REFER request with the "method" SIP URI parameter set to value "BYE" in the URI in the Refer-To header field from the MCData client, the participating MCData function:

1) shall determine the MCData ID of the calling user from public user identity in the P-Asserted-Identity header field of the SIP REFER request;

…

3) *if the SIP REFER request contained a Refer-Sub header field containing "false" value and a Supported header field containing "norefersub" value, shall handle the SIP REFER request as specified in 3GPP TS 24.229 [5], IETF RFC 3515 [53] as updated by IETF RFC 6665 [36], and IETF RFC 4488 [53] without establishing an implicit subscription;*

4) *shall generate a SIP 200 (OK) response to the SIP REFER request, and in the SIP 200 (OK) response:*

   a) *shall include the Supported header field with value "norefersub" according to rules and procedures of IETF RFC 4488 [53]; and*

   b) *shall check the presence of the Refer-Sub header field of the SIP REFER request and if it is present and set to the value "false" shall include the Refer-Sub header field with value "false" according to rules and procedures of IETF RFC 4488 [53];*

5) *shall send the SIP 200 (OK) response to the SIP REFER request towards MCData client according to 3GPP TS 24.229 [5];*

6) *shall generate a SIP BYE request, and in the SIP BYE request:*

   a) *shall set the Request-URI to the MCData session identity which was included at the Refer-To header field of the received REFER request; and*

   b) *shall copy the contents of the P-Asserted-Identity header field of the received REFER request to the P-Asserted-Identity header field of the outgoing SIP BYE request; and*

7) *shall send the SIP BYE request toward the controlling MCData function according to 3GPP TS 24.229 [5].*

Upon receiving a SIP 200 (OK) response to the SIP BYE request the participating MCData function shall interact with the media plane as specified in 3GPP TS 24.582 [15] for releasing media plane resources associated with the SIP session with the controlling MCData function. The participating MCData function shall generate a SIP re-INVITE request as specified in subclause 9.2.5.1.2 with following clarifications and send the request towards the originating MCData client according to 3GPP TS 24.229 [5]:

1) *shall set the Request-URI to a public service identity identifying the pre-established session; and*

2) *shall set the <mcdata-communication-state> element with a value of "terminated"."*

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.5.4.2.2]

"Upon receiving a SIP BYE request from the controlling MCData function, the participating MCData function:

1) *shall interact with the media plane as specified in 3GPP TS 24.582 [15];*

2) *shall send a SIP 200 (OK) response to the controlling MCData function;*

3) *shall generate a SIP re-INVITE request as specified in subclause 9.2.5.1.2 with following clarifications:*

   i) *shall set the Request-URI to a public service identity identifying the pre-established session; and*

   ii) *shall set the <mcdata-communication-state> element with a value of "terminate-request";*

4) *shall send the SIP re-INVITE request towards the originating MCData client according to 3GPP TS 24.229 [5]; and*

5) *upon receipt of a SIP 2xx response to the SIP re-INVITE, shall interact with the media plane as specified in 3GPP TS 24.582 [15]."*

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.5.1.1]

"This subclause is referenced from other procedures.

When generating an initial SIP INVITE request according to 3GPP TS 24.229 [5], on receipt of an incoming SIP REFER request, the participating MCData function:

1) *shall include in the SIP INVITE request all header fields included in the headers portion of the SIP URI contained in the <entry> element of the application/resource-lists MIME body, referenced by the "cid" URL in the Refer-To header field in the incoming SIP REFER request;*

2) should include the Session-Expires header field according to IETF RFC 4028 [38].

3) shall include the option tag "timer" in the Supported header field;

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP REFER request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

5) shall include the g.3gpp.mcdata.sds media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" into the Contact header field of the outgoing SIP INVITE request;

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP INVITE request; and

7) shall include in the SIP INVITE request the option tag "tdialog" in a Supported header field according to the rules and procedures of IETF RFC 4538 [54];

8) shall include in the SIP INVITE request an SDP offer as specified in subclause 9.2.3.3.1 based upon:

a) the SDP negotiated during the pre-established session establishment and any subsequent pre-established session modification; and

b) the SDP offer (if any) included in the"body" URI parameter of the SIP URI contained in the <entry> element of the application/resource-lists MIME body, referenced by the "cid" URL in the Refer-To header field in the incoming SIP REFER request for a pre-established session;

9) shall copy the application/vnd.3gpp.mcdata-info+xml MIME body from the "body" URI parameter of the SIP URI in the application/resource-lists MIME body, referenced by the "cid" URL in the Refer-To header field of the SIP REFER request, to the outgoing SIP INVITE request;

9A) if the incoming SIP REFER request contained a <functional-alias-URI> element in an application/vnd.3gpp.mcdata-info+xml MIME body in the hname "body" parameter in the headers portion of the SIP URI in the Refer-To header field, shall check if the status of the functional alias is activated for the MCData ID. If the functional alias status is activated, then the participating MCData function shall set the <functional-alias-URI> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request to the received value, otherwise shall not include a <functional-alias-URI> element; and

10) if the incoming SIP REFER request contained an application/resource-lists MIME body in the "body" URI parameter of the SIP URI contained in the <entry> element of an application/resource-lists MIME body, referenced by the "cid" URL in the Refer-To header field, shall copy the application/resources-lists MIME body in the "body" URI parameter to the SIP INVITE request."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.5.1.2]

"This subclause is referenced from other procedures.

The participating MCData function:

1) shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [5] to be sent within the SIP dialog of the pre-established session;

2) shall include in the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing Re-INVITE request:

a) the <mcdata-communication-state> element with a value set to "establish-success", if a SIP 2xx response is received to a SIP INVITE request sent to the controlling MCData function; or

b) the <mcdata-communication-state> element with a value set to "establish-fail", if an error response is received to a SIP INVITE request sent to the controlling MCData function;"

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.5.1.3]

"This subclause is referenced from other procedures.

The participating MCData function:

1) *shall generate a SIP re-INVITE request according to 3GPP TS 24.229 [5] to be sent within the SIP dialog of the pre-established session;*

2) *should include the Session-Expires header field according to IETF RFC 4028 [38].*

3) *shall include the option tag "timer" in the Supported header field;*

4) *shall include an Accept-Contact header field containing the g.3gpp.mcdata.sds media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8];*

5) *shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" along with parameters "require" and "explicit" according to IETF RFC 3841 [8];*

6) *shall include in the Contact header field an MCData session identity for the MCData session with the g.3gpp.mcdata.sds media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" according to IETF RFC 3840 [16];"*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 12.2]

"Upon establishing one-to-one or group SDS using media plane or one-to-one or group SDS session using pre-established call, the participating MCData function shall follow the procedure as described in subclause 6.2."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.4.1]

"The originating participating MCData function should establish the MSRP session with the originating MCData client as specified in subclause 6.2.1.2."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.4.2]

"The originating participating MCData function should establish the MSRP session with the controlling MCData function as specified in subclause 6.2.1.3."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.4.3]

"Upon receiving an MSRP SEND request from the originating MCData client, the originating participating MCData function:

1. *if in a standalone one-to-one SDS using media plane, shall verify the SDS message size is less than or equal to <MacData1To1> element of the MCData user profile document for the originating MCData client (see the MCData user profile document in 3GPP TS 24.484 [7]);*

2. *if the verifications in 1 above fails, shall send the MSRP response with the error code 403 to the originating MCData client and shall not continue with the rest of the procedure;*

3. *if an MSRP connection is not established with the controlling MCData function then, shall establish the MSRP connection as specified in subclause 6.2.1.4.2. Otherwise, shall use the existing MSRP connection; and*

4. *shall forward the received MSRP SEND request to the controlling MCData function according to the rules and procedures of IETF RFC 4975 [11].*

Upon receiving an MSRP 200 (OK) response from the controlling MCData function, the participating MCData function shall forward the MSRP 200 (OK) response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11].

Upon receiving an error MSRP response from the controlling MCData function, the participating MCData function shall forward the error MSRP response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.5.1]

"The terminating participating MCData function should establish the MSRP session with the controlling MCData function as specified in subclause 6.2.1.2."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.5.2]

"The terminating participating MCData function should establish MSRP session to terminating MCData client as specified in subclause 6.2.1.3."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.5.3]

"Upon receiving an MSRP SEND request from the controlling MCData function, the terminating participating MCData function:

1. *shall generate and send a MSRP 200 (OK) response for the received MSRP SEND request to the controlling MCData function, according to the rules and procedures of IETF RFC 4975 [11]; and*
2. *shall forward the received MSRP SEND request to the terminating MCData client according to the rules and procedures of IETF RFC 4975 [11].*

Upon receiving an error MSRP response from the terminating MCData client, the participating MCData function shall forward the error MSRP response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.2]

"To receive a SDS message over MSRP, the participating MCData function:

1. *shall act according to IETF RFC 6135 [12], as:*
   a. *an "passive" endpoint, if a=setup attribute in the sent SDP answer was set to "passive"; and*
   b. *an "active" endpoint, if a=setup attribute in the sent SDP answer was set to "active"; and*
2. *shall establish an MSRP connection according to the MSRP connection parameters in the sent SDP answer response as described in IETF RFC 4976 [14]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.2.1.3]

"To send a SDS message over MSRP, the participating MCData function:

1. *shall act according to IETF RFC 6135 [12], as:*
   a. *an "active" endpoint, if a=setup attribute in the received SDP answer is set to "passive"; and*
   b. *an "passive" endpoint, if a=setup attribute in the received SDP answer is set to "active"; and*
2. *shall establish the MSRP connection according to the MSRP connection parameters in the received SDP answer response as described in IETF RFC 4976 [14]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.3.1.2.1]

"To establish the MSRP connection with the originating MCData client, the controlling MCData function performs below procedures:

1. *shall act as an MSRP client and establish TLS connection with the originating participating MCData function, if exists, otherwise the originating MCData client, according to the rules and procedures as described in IETF RFC 4975 [11];*
2. *shall act as an MSRP client to send MSRP SEND requests according to the rules and procedures described in IETF RFC 6135 [12];*
3. *shall act as an "passive" endpoint according to the rules and procedures described in IETF RFC 6135 [12];*
4. *shall establish the MSRP connection with originating MCData client, according to the rules and procedures described in IETF RFC 6135 [12]; and*
5. *as a passive endpoint, shall wait for MSRP SEND request on established MSRP connection, to bind the MSRP connection to the MSRP session according to the rules and procedures of IETF RFC 4975 [11] and IETF RFC 6135 [12]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.3.1.2.2]

"To establish the MSRP connection with the terminating MCData client, the controlling MCData function performs below procedures:

1. *shall act as an MSRP client and establish TLS connection with the terminating participating MCData function, if exists, otherwise the terminating MCData client, according to the rules and procedures as described in IETF RFC 4975 [11];*

2. *shall act as an MSRP client to send MSRP SEND requests according to the rules and procedures described in IETF RFC 6135 [12];*

3. *shall act according to IETF RFC 6135 [12], as:*

    a. *an "active" endpoint, if a=setup attribute in the received SDP answer is set to "passive"; and*

    b. *an "passive" endpoint, if a=setup attribute in the received SDP answer is set to "active";*

4. *shall establish the MSRP connection with each terminating MCData client identified in the 3GPP TS 24.282 [8], according to the rules and procedures described in IETF RFC 6135 [12]; and*

5. *if acting as an "active" endpoint, shall send an empty MSRP SEND request on each established MSRP connection, to bind the MSRP connection to the MSRP session according to the rules and procedures of IETF RFC 4975 [11] and IETF RFC 6135 [12]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 6.3.1.3]

"Upon receiving a MSRP SEND request from the originating participating MCData function, the controlling MCData function:

1. *if in a standalone one-to-one SDS using media plane, shall verify the SDS message size is less than or equal to <max-data-size-sds-bytes> element in the MCData service configuration document as specified in 3GPP TS 24.484 [7];*

2. *if in a standalone group SDS using media plane shall verify:*

    a. *the SDS message size is less than or equal to <mcdata-max-data-in-single-request> element of the MCData group document for the (see the MCData group document in 3GPP TS 24.481 [4]); and*

    b. *the SDS message size is less than or equal to <mcdata-on-network-max-data-size-for-SDS> element of the MCData group document (see the MCData group document in 3GPP TS 24.481 [4]);*

3. *if the verifications in either 1 or 2 a or 2 b above fails, shall send the MSRP response with the error code 403 to the MCData client which sent the SDS message and shall not continue with the rest of the procedure;*

4. *shall generate and send a MSRP 200 (OK) response for the received MSRP SEND requests to the originating participating MCData function, according to the rules and procedures of IETF RFC 4975 [11]; and*

5. *shall forward the received MSRP SEND requests to each terminating MCData client with which a successful MSRP connection was established, according to the rules and procedures of IETF RFC 4975 [11]. Following clarifications apply to the generated MSRP SEND request:*

    a. *shall modify the To-Path header according to the MSRP URI received in the answer SDP from the MCData client in accordance with rules and procedures of IETF RFC 4975 [11]; and*

    b. *shall modify the From-Path header to the controlling MCData function's own MSRP URI, according to the rules and procedures of IETF RFC 4975 [11]."*

6.1.4.3 Test description

6.1.4.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Storage Function, the MCData User Database, the SIP AS, the HTP Server, the HTTP Client, the MSRP TCP Server and the MSRP TCP Client.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The <max-payload-size-sds-cplane-bytes> element of the MCData Service Configuration document shall be set to 0 to force the MCData client to send the data using the media plane.
- The IUT (MCData Server) is connected to PLMN1.
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Pre-established Session Establishment as specified in step 1 on Table 5.4.3.2-1 in clause 5.4.

6.1.4.3.2 Test procedure sequence

Table 6.1.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MCData Call Establishment using pre-established session'** as described in Annex A, Table A.4.6.3-1 **to establish a media plane connection between SS-UE1 (MCData Client) and SS-UE2 (MCData Client)?**	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MSRP message transfer'** as described in Annex A, Table A.4.4.3-1 **to forward a SDS message from SS-UE1 (MCData client) to SS-UE2 (MCData Client), with disposition request "DELIVERY"?**	-	-	2	P
3	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MCData Call Release keeping the pre-established session '** as described in Annex A, Table A.4.7.3-1?	-	-	3	P
4	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a disposition notification of "DELIVERED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	4	P

6.1.4.3.3 Specific message contents

Table 6.1.4.3.3-1: SIP REFER from the SS-UE1 (Step 1, Table 6.1.4.3.2-1; Step 1, Annex A, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**Resource list**		
MIME-part-body	Resource-lists as described in Table 6.1.4.3.3-2			

Table 6.1.4.3.3-2: Resource-lists in SIP REFER (Table 6.1.4.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3, condition PRE-ESTABLISH, MCD_1to1 with the uri attribute of the entry extended with the SIP URI header fields as specified in Table 6.1.4.3.3-3

Table 6.1.4.3.3-3: SIP URI header fields extending the uri attribute of the resource-lists' single entry (Table 6.1.4.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
body				
MIME body part		**SDP Message**		
MIME-part-headers				
Content-Type	"application/sdp"			
MIME-part-body	SDP Message as described in Table 6.1.4.3.3-4			
MIME body part		**MCData-Info**		
MIME-part-body	MCdata-Info as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.13.3.3-5			

Table 6.1.4.3.3-4: SDP in SIP header fields (Table 6.1.4.3.3-3)

Derivation Path: Table 5.4.3.3-2, condition MCD_1to1 and NOT SDS_SESSION

Table 6.1.4.3.3-5: SIP re-INVITE from the MCData Server (IUT) to the SS-UE2 (Step 1, Table 6.1.4.3.2-1; Step 3, Annex A, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, condition MCDATA_SDS, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCX_SessionID_B	session identity of the pre-established session	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 9.2.5.2.2.2	
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP message as described in Table 6.1.4.3.3-6			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.1.4.3.3-7			

Table 6.1.4.3.3-6: SDP for SIP INVITE (Table 6.1.4.3.3-5)

Derivation Path: Table 5.4.3.3-4, condition SDP_OFFER, MCD_1to1, NOT SDP_ANSWER and NOT SDS_SESSION

Table 6.1.4.3.3-7: MCData-Info (Table 6.1.4.3.3-5)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.14.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		

Table 6.1.4.3.3-8: SIP 200 (OK) from the SS-UE2 (Step 1, Table 6.1.4.3.2-1; Step 5, Annex A, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1, condition INVITE-RSP, MCDATA_SDS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type				
value	"application/sdp"			
Message-body				
SDP message	As described in Table 6.1.4.3.3-9			

Table 6.1.4.3.3-9: SDP for SIP 200 (OK) (Table 6.1.4.3.3-8)

Derivation Path: Table 5.4.3.3-2, condition SDP_ANSWER, NOT OFFER and NOT SDS_SESSION

Table 6.1.4.3.3-10: SIP re-INVITE from the MCData Server (IUT) to the SS-UE1 (Step 1, Table 6.1.4.3.2-1; Step 7, Annex A, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, condition MCDATA_SDS, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCX_SessionID_B	session identity of the pre-established session	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 9.2.5.2.2.2	
Message-body				
MIME body part	not present	**SDP message**		
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.1.4.3.3-11			

Table 6.1.4.3.3-11: MCData-Info (Table 6.1.4.3.3-10)

Derivation Path: Table 5.5.3.2.2-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	not present			
mcdata-calling-user-id	not present			
anyExt				
mcdata-communication-state	"establish-success"		ETSI TS 124 282 [13] (3GPP TS 24.282) clause 9.2.5.1.2	

Table 6.1.4.3.3-12: SIP 200 (OK) from the SS-UE1 (Step 1, Table 6.1.4.3.2-1; Step 9, Annex A, Table A.4.6.3-1)

Derivation Path: Table 5.5.2.17.1.1-1, condition INVITE-RSP, MCDATA_SDS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type	not present			
Message-body	not present			

Table 6.1.4.3.3-13: MSRP SEND from the SS-UE1 (Step 2, Table 6.1.4.3.2-1; Step 1, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.13.3.3-6				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
data	Message or chunk of message as specified in table 6.1.4.3.3-13A		IETF RFC 3261 [21]	

Table 6.1.4.3.3-13A: MIME Message (Table 6.1.4.3.3-13)

Derivation Path: IETF RFC 2046 [18]				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
MIME body part		**MCData Data signalling message**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcdata-signalling"			
MIME-part-body	MCData Protected Payload Message containing SDS SIGNALLING PAYLOAD as described in Table 6.1.4.3.3-13B			
MIME body part		**MCData Data message**		
MIME-part-headers				
Content-Type	"application/vnd.3gpp.mcdata-payload"			
MIME-part-body	MCData Protected Payload Message containing DATA PAYLOAD as described in Table 6.1.4.3.3-13C			

Table 6.1.4.3.3-13B: SDS SIGNALLING PAYLOAD (Table 6.1.4.3.3-13A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.8.2-1, condition DELIVERED

Table 6.1.4.3.3-13C: Data Payload (Table 6.1.4.3.3-13A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.9.2-2

Table 6.1.4.3.3-14: MSRP SEND from the MCData Server (IUT) to SS-UE2 (Step 2, Table 6.1.4.3.2-1; Step 3, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.14.3.3-6

Table 6.1.4.3.3-15: SIP MESSAGE from the SS-UE2 (Step 4, Table 6.1.2.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.14.3.3-12				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 6.1.4.3.3-16			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing SDS NOTIFICATION as described in Table 6.1.4.3.3-17			

Table 6.1.4.3.3-16: Resource-lists (Table 6.1.4.3.3-15)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.1.4.3.3-17: SDS NOTIFICATION (Table 6.1.4.3.3-15)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.14.3.3-13				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Date and time	The current date and time	The Date and time value is an unsigned integer containing UTC time of the time when a message was sent, in seconds since midnight UTC of January 1, 1970 (not counting leap seconds).	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 15.2.8	

Table 6.1.4.3.3-18: SIP MESSAGE from the MCData Server (IUT) to SS-UE1 (Step 4, Table 6.1.2.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.1.13.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

6.2 File Distribution

6.2.1 MCDATA Server - MCDATA Client / File Distribution (FD) / FD Using HTTP / One-to-one Standalone FD / Non-Mandatory Download / FILE DOWNLOAD REQUEST ACCEPTED / FILE DOWNLOAD COMPLETED / FILE DOWNLOAD REQUEST REJECTED / FILE DOWNLOAD DEFERRED

6.2.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { SS-UE1 (MCData Client) requests discovery of the absolute URI by sending a SIP MESSAGE
message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 200 (Ok) to UE1 (MCData Client) }
           }
```

(2)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { the SS-UE1 (MCData Client) initiates FD "file 1" upload by sending a HTTP POST request }
    then { IUT (MCData Server) responds with a HTTP 201 (Created) }
            }
```

(3)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { SS-UE1 (MCData Client) requests FD transfer by sending a SIP MESSAGE message with
disposition request "FILE DOWNLOAD COMPLETED UPDATE" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE2 (MCData Client) and responds with
a SIP 202 (Accepted) to UE1 (MCData Client) }
            }
```

(4)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "FILE DOWNLOAD REQUEST REJECTED" by
sending a SIP MESSAGE message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

(5)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { the SS-UE1 (MCData Client) initiates FD "file 2" upload by sending a HTTP POST request }
    then { IUT (MCData Server) responds with a HTTP 201 (Created) }
            }
```

(6)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "FILE DOWNLOAD DEFERRED" by sending
a SIP MESSAGE message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

(7)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "FILE DOWNLOAD ACCEPTED" by sending
a SIP MESSAGE message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

(8)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { the SS-UE2 (MCData Client) initiates FD "file 2" download by sending a HTTP GET request }
    then { IUT (MCData Server) responds with a HTTP 200 (Ok) and SS-UE2 (MCData Client) contains
downloaded "file 2" with same content }
            }
```

(9)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { SS-UE2 (MCData Client) sends a disposition notification "FILE DOWNLOAD COMPLETED" by
sending a SIP MESSAGE message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

6.2.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282), clauses 10.2.4.1, 10.2.4.3.1, 10.2.4.3.2, 10.2.4.4.1, 10.2.4.4.2, 10.2.4.2.1, 10.2.1.2.3, 11.2, 6.2.3.2, 12.2.2.1, 15.2.6. Unless otherwise stated, these are Rel-14 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.1]

"The procedures in the subclauses of the parent subclause describe the SIP signalling procedures for:

- *one-to-one file distribution using HTTP; and*
- *group standalone file distribution using HTTP.*

When the MCData user wishes to perform file distribution via HTTP, the MCData client:

1) shall check that the file size is less than or equal to the:

…

b) <max-data-size-fd-bytes> element present in the service configuration document as specified in 3GPP TS 24.484 [12], if the file upload is for a one-to-one file distribution;

2) if the size of the file:

a) is acceptable for upload as determined by step 1), shall determine the value of the absolute URI associated with the media storage function of the MCData content server from the <MCDataContentServerURI> element of the MCPTT user profile document (see the MCPTT user profile document in 3GPP TS 24.484 [50]);

b) is not acceptable for upload, shall not continue with the remaining steps in this subclause;

3) shall request the media storage client to upload the file to the media storage function by following the procedures in subclause 10.2.2.1; and

4) shall initiate an FD request containing a file URL as specified in subclause 10.2.4.2.1."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.3.1]

"Upon receipt of a "SIP MESSAGE request for FD using HTTP for originating participating MCData function", the participating MCData function:

…

2) shall determine the MCData ID of the originating user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request, and shall authorise the calling user;

NOTE: The MCData ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in subclause 7.3.

…

4) if <mcdata-controller-psi> element is present in the application/vnd.3gpp.mcdata-info+xml, shall use its value as public service identity of the controlling MCData function. Otherwise, if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request is:

…

b) set to a value of "one-to-one-fd", shall determine the public service identity of the controlling MCData function hosting the one-to-one FD using HTTP service for the calling user;

…

6) shall determine whether the MCData user identified by the MCData ID is authorised for MCData communications by following the procedures in subclause 11.1;

…

8) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];

9) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCData function as determined by step 4) in this subclause;

10) shall copy all MIME bodies included in the incoming SIP MESSAGE request to the outgoing SIP MESSAGE request;

10A) if the incoming SIP MESSAGE request contains an application/vnd.3gpp.mcdata-info+xml MIME body that contains a <functional-alias-URI> element, shall check if the status of the functional alias is activated for the MCData ID. If the functional alias status is activated, then the participating MCData function shall set the <functional-alias-URI> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request to the received value, otherwise shall not include a <functional-alias-URI> element;

11) shall include the MCData ID of the originating user in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP MESSAGE request;

12) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request;

13) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request; and

14) shall send the SIP MESSAGE request as specified to 3GPP TS 24.229 [5].

Upon receipt of a SIP 202 (Accepted) response in response to the SIP MESSAGE request in step 14):

1) shall generate a SIP 202 (Accepted) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 202 (Accepted) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the SIP MESSAGE request in step 14):

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 200 (OK) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request in step 14) the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.3.2]

"Upon receipt of a:

- *"SIP MESSAGE request for FD using HTTP for terminating participating MCData function"; or*
- *"SIP MESSAGE network notification for FD using HTTP for terminating participating MCData function";*

the participating MCData function:

…

2) shall use the MCData ID present in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCData ID and public user identity of the terminating MCData user;

…

4) if the SIP MESSAGE is a "SIP MESSAGE request for FD using HTTP for terminating participating MCData function", and if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message with a FD disposition request type IE, shall store the value of the Conversation ID IE, the value of the Message ID IE and the payload IE in the FD SIGNALLING PAYLOAD message;

5) if the SIP MESSAGE is a "SIP MESSAGE network notification for FD using HTTP for terminating participating MCData function", and if FD NETWORK NOTIFICATION message within the application/vnd.3gpp.mcdata-signalling MIME body contains an FD notification type IE with value set as "FILE EXPIRED UNAVAILABLE TO DOWNLOAD" as specified in subclause 15.2.6, the file identified using Conversation ID IE shall be removed from the stored file list;

5) shall generate an outgoing SIP MESSAGE request as specified in subclause 6.3.2.1;

…

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request; and

7) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 200 (OK) response to the controlling MCData function according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the controlling MCData function according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.4.1]

"This subclause describes the procedures for sending a SIP MESSAGE from the controlling MCData function and is initiated by the controlling MCData function as a result of an action in subclause 10.2.4.4.2.

The controlling MCData function:

1) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];

2) shall include an Accept-Contact header field containing the g.3gpp.mcdata.fd media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

3) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" along with parameters "require" and "explicit" according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

4) shall copy the following MIME bodies in the received SIP MESSAGE request into the outgoing SIP MESSAGE request by following the guidelines in subclause 6.4:

a) application/vnd.3gpp.mcdata-info+xml MIME body; and

b) application/vnd.3gpp.mcdata-signalling MIME body;

5) if the application/vnd.3gpp.mcdata-signalling MIME body in the received SIP MESSAGE request contained a FD SIGNALLING PAYLOAD message without the Mandatory download IE included, then:

a) shall execute the procedures in subclause 11.2;

b) if the procedures in subclause 11.2 indicate that the mandatory download indication needs to be included,shall include the Mandatory download IE set to a value of "MANDATORY DOWNLOAD" in the FD SIGNALLING PAYLOAD message of the outgoing SIP MESSAGE request;

6) in the application/vnd.3gpp.mcdata-info+xml MIME body:

a) shall set the <mcdata-request-uri> element set to the MCData ID of the terminating user; and

b) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request was set to a value of "group-fd", shall set the <mcdata-calling-group-id> element to the group identity;

7) shall set the Request-URI to the public service identity of the terminating participating MCData function associated to the MCData user to be invited;

8) shall copy the public user identity of the calling MCData user from the P-Asserted-Identity header field of the incoming SIP MESSAGE request into the P-Asserted-Identity header field of the outgoing SIP MESSAGE request;

9) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd"; and

10) shall send the SIP MESSAGE request according to according to rules and procedures of 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.4.2]

"The procedures in this subclause are executed upon:

- *receipt of a "SIP MESSAGE request for FD using HTTP for controlling MCData function", the controlling MCData function; or*
- *a decision to now process a previously received "SIP MESSAGE request for FD using HTTP for controlling MCData function" that had been queued for later transmission;*

NOTE 1: The controlling MCData function may postpone the continuation of an FD using HTTP procedure by queuing the received "SIP MESSAGE request for FD using HTTP for controlling MCData function". The management of the queue is specified in Annex B of 3GPP TS 23.282 [2].

the controlling MCData function:

…

3) if the SIP MESSAGE does not contain:

a) an application/vnd.3gpp.mcdata-info+xml MIME body; and

b) an application/vnd.3gpp.mcdata-signalling MIME body;

…

4) shall decode the contents of the application/vnd.3gpp.mcdata-signalling MIME body contained in the SIP MESSAGE;

5) if the application/vnd.3gpp.mcdata-signalling MIME body does not contain only one FD SIGNALLING PAYLOAD message or FD HTTP TERMINATION message, shall reject the SIP MESSAGE request …, and shall not continue with the rest of the steps in this subclause;

6) if the FD SIGNALLING PAYLOAD message or FD HTTP TERMINATION messagedoes not contain only one Payload IE, shall reject the SIP MESSAGE request …, and shall not continue with the rest of the steps in this subclause;

7) if the Payload IE has Payload contents:

a) with a Payload content type set to a value other than "FILEURL" shall reject the SIP MESSAGE request …, and shall not continue with the rest of the steps in this subclause; and

b) with Payload data containing a file URL identifying a file that does not exist on the media storage function, shall reject the SIP MESSAGE request …, and shall not continue with the rest of the steps in this subclause;

8) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message with a FD disposition request type IE, shall store the value of the Conversation ID IE and the value of the Message ID IE in the FD SIGNALLING PAYLOAD message;

NOTE 2: The controlling MCData function uses the Conversation ID and Message ID for correlation with disposition notifications.

9) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message:

a) with a Metadata IE, shall derive a timer value for the file availability timer as the minimum of the file availability information in the metadata and the value contained in the <max-file-availability> element in the MCData service configuration document as specified in 3GPP TS 24.484 [12]; and

b) without a Metadata IE, shall derive a timer value for the file availability timer as the value contained in the <default-file-availability> element in the MCData service configuration document as specified in 3GPP TS 24.484 [12];

10) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request is set to a value of "one-to-one-fd" and the SIP MESSAGE request:

…

b) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message contains an application/resource-lists MIME body with exactly one <entry> element, shall send a SIP MESSAGE request to the MCData user identified in the <entry> element of the MIME body, as specified in subclause 10.2.4.4.1;

…

…

13) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message, shall start TDC2 (file availability timer) with the value derived in step 9 of this subclause;

14) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message, shall associate the running timer TDC2 (file availability timer) to the Conversation ID, Message ID, Application ID (if included), and Extended application ID (if included) contained in the FD SIGNALLING PAYLOAD message;

NOTE 3: Multiple file availability timers can be running for a file. Each file availability timer is uniquely associated to a Conversation ID and Message ID.

15) shall generate a SIP 202 (Accepted) response in response to the "SIP MESSAGE request for FD using HTTP for controlling MCData function"; and

16) shall send the SIP 202 (Accepted) response towards the originating participating MCData function according to 3GPP TS 24.229 [5].

17) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD HTTP TERMINATION message and Termination information type IE set to "TERMINATION REQUEST" then:

a) shall identify the FILE transmission with Conversation ID and Message ID and "FILE URL". If any ongoing transmission exist then execute procedure described in subclause 12.4.2.1 withfollowing clarifications:

i) shall set FD notification type IE as "FILE DELETED UNAVAILABLE TO DOWNLOAD" as specified in subclause 15.2.18;

b) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6]. In the generation of the SIP MESSAGE request, the controlling MCData function:

i) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" along with parameters "require" and "explicit" according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

ii) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd";

iii) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the participating MCData function associated to the MCData ID of the originating user mentioned in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request;

iv) shall copy the public user identity of the calling MCData user from the P-Asserted-Identity header field of the incoming SIP MESSAGE request into the P-Asserted-Identity header field of the outgoing SIP MESSAGE request;

v) shall include an application/vnd.3gpp.mcdata-info+xml MIME body in the SIP MESSAGE request, following the rules specified in subclause 6.4 for the handling of MIME bodies in a SIP message:

A) fill <mcdata-request-uri> element from <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml in received SIP MESSAGE;

vi) shall generate FD HTTP TERMINATION message as described in subclause 6.3.6.1;

vii) shall set the Termination information type IE set to "TERMINATION RESPONSE" as specified in subclause 15.2.22.

viii) if clause is successful shall set Release response type IE of FD HTTP TERMINATION MESSAGE to "RELEASE SUCCESS" else set to "RELEASE FAILED" as described in subclause 15.2.23; and

ix) shall include in the SIP request, the FD HTTP TERMINATION message in an application/vnd.3gpp.mcdata-signalling MIME body as specified in subclause E.1;

c) shall send the SIP MESSAGE request towards the originating participating MCData function as specified in 3GPP TS 24.229 [5]; and

18) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD HTTP TERMINATION message and Termination information type IE set to other than "TERMINATION REQUEST" then follow procedures described on subclause 13.2.5 and subclause 13.2.6."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.2.1]

"The MCData client shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6] with the clarifications given below.

The MCData client:

1) shall build the SIP MESSAGE request as specified in subclause 6.2.4.1;

2) if a one-to-one standalone FD message is to be sent shall insert in the SIP MESSAGE request:

a) an application/resource-lists+xml MIME body with the MCData ID of the target MCData user, according to rules and procedures of IETF RFC 4826 [9]; and

b) an application/vnd.3gpp.mcdata-info+xml MIME body with:

i) a <request-type> element set to a value of "one-to-one-fd"; and

ii) if the MCData client is aware of active functional aliases and if an active functional alias is to be included in the SIP MESSAGE request, the <functional-alias-URI> element set to the URI of the used functional alias;

…

4) shall generate a standalone FD message as specified in subclause 6.2.2.2; and

5) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.1.2.3]

"The MCData client:

1) if the FD SIGNALLING PAYLOAD message does not contain an Application ID IE and does not contain an Extended application ID IE:

a) shall determine that the payload contained in the Payload IE in the FD SIGNALLING PAYLOAD message is for user consumption;

b) shall notify the user about the incoming FD request; and

c) if the FD SIGNALLING PAYLOAD message contains a Metadata IE, shall deliver the contents of the Metadata IE to the user;

2) if the FD SIGNALLING PAYLOAD message contains an Application ID IE:

a) shall determine that the payload contained in the Payload IE in the FD SIGNALLING PAYLOAD message is not for user consumption;

b) if the Application ID value is unknown, shall discard the FD message and exit this subclause;

c) if the Application ID value is known, shall notify the application of the incoming FD request; and

NOTE 1: If FD request is addressed to a non-MCData application that is not running, the MCData client starts the local non-MCData application.

d) if the FD SIGNALLING PAYLOAD message contains a Metadata IE, shall deliver the contents of the Metadata IE to the application;

2A) if the FD SIGNALLING PAYLOAD message contains an Extended application ID IE:

a) shall determine that the payload contained in the Payload IE in the FD SIGNALLING PAYLOAD message is not for user consumption;

b) if the Extended application ID value is unknown, shall discard the FD message and exit this clause;

c) if the Extended application ID value is known, shall notify the application of the incoming FD request; and

NOTE 2: If the FD request is addressed to a non-MCData application that is not running, the MCData client starts the local non-MCData application.

d) if the FD SIGNALLING PAYLOAD message contains a Metadata IE, shall deliver the contents of the Metadata IE to the application;

3) shall start a timer TDU2 (FD non-mandatory download timer) with the timer value as specified in subclause F.2.3;

4) shall wait for the user or application to request to download the file indicated by file URL in the Payload data in the Payload IE in the FD SIGNALLING PAYLOAD message;

5) if the user or application accepts or rejects or decides to defer the FD request, shall stop timer TDU2 (FD non-mandatory download timer);

6) if the user defered the FD request while the timer TDU2 (FD non-mandatory download timer) was running, shall generate an FD NOTIFICATION indicating deferral of the FD request as specified in subclause 12.2.1.1;

NOTE 3: Once the timer TDU2 (FD non-mandatory download timer) has expired the FD request can only be accepted or rejected with an appropriate action by the MCData client.

NOTE 4: Once the timer TDU2 (FD non-mandatory download timer) has expired, no action is taken by the MCData client if the FD request is deferred.

7) if the user or application rejects the FD request, shall generate an FD NOTIFICATION indicating rejection of the FD request as specified in subclause 12.2.1.1 and shall exit this subclause; and

8) if the user accepts the FD request:

a) shall generate an FD NOTIFICATION indicating acceptance of the FD request as specified in subclause 12.2.1.1;

b) if the FD SIGNALLING PAYLOAD message contains a new Conversation ID, shall instantiate a new conversation with the Message ID in the FD SIGNALLING PAYLOAD identifying the first message in the conversation thread;

c) if the FD SIGNALLING PAYLOAD message contains an existing Conversation ID and:

i) if the FD SIGNALLING PAYLOAD message does not contain an InReplyTo message ID, shall use the Message ID in the FD SIGNALLING PAYLOAD to identify a new message in the existing conversation thread; and

ii) if the FD SIGNALLING PAYLOAD message contains an InReplyTo message ID, shall associate the message to an existing message in the conversation thread as identified by the InReplyTo message ID in the FD SIGNALLING PAYLOAD, and use the Message ID in the FD SIGNALLING PAYLOAD to identify the new message;

d) may store the Conversation ID, Message ID, InReplyTo message ID and Date and time in local storage;

e) shall attempt to download the file as identified by the file URL in the Payload IE in the FD SIGNALLING PAYLOAD message, as specified in subclause 10.2.3.1; and

f) if the received FD SIGNALLING PAYLOAD message contains an FD disposition request type IE requesting a file download completed update, then after the file download has been successfully downloaded, shall generate an FD NOTIFICATION by following the procedures in subclause 12.2.1.1."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 11.2]

"If the controlling MCData function receives a one-to-one file distribution using HTTP or a group standalone file distribution using HTTP without the mandatory download indication the controlling MCData function:

1) if the file distribution request contained metadata, shall retrieve the filesize contained in the fileselector of the Metadata IE in the FD request;

2) if the file distribution request did not contain metadata, shall determine the size of the file referenced by the file URL contained in FD request;

3) for one-to-one file distribution using HTTP, shall determine if the filesize is less than or equal to the value contained in the <max-data-size-auto-recv-bytes> element of the MCData service configuration document as specified in 3GPP TS 24.484 [12];

…

5) ifcondition 3) or 4) is true, shall determine that the mandatory download indication needs to be included in the file distribution request sent to the terminating MCData client;"

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 6.2.3.2]

"In order to generate an FD notification, the MCData client:

1) shall generate an FD NOTIFICATION message as specified in subclause 15.1.6; and

2) shall include in the SIP request, the FD NOTIFICATION message in an application/vnd.3gpp.mcdata-signalling MIME body as specified in subclause E.1.

When generating an FD NOTIFICATION message as specified in subclause 15.1.6, the MCData client:

1) *if sending a file download accept notification, shall set the FD disposition notification type IE as "FILE DOWNLOAD REQUEST ACCEPTED" as specified in subclause 15.2.6;*

2) *if sending a file download reject notification, shall set the FD disposition notification type IE as "FILE DOWNLOAD REQUEST REJECTED" as specified in subclause 15.2.6;*

3) *if sending a file download deferred notification, shall set the FD disposition notification type IE as "FILE DOWNLOAD REQUEST DEFERRED" as specified in subclause 15.2.6;*

4) *shall set the Conversation ID to the value of the Conversation ID that was received in the FD message as specified in subclause 15.2.9;*

5) *shall set the Date and time IE to the current time as specified in subclause 15.2.8; and*

6) *if sending a file download completed notification:*

   a) *shall set the FD disposition notification type IE as "FILE DOWNLOAD COMPLETED" as specified in subclause 15.2.6;*

   b) *shall set the Message ID to the value of the Message ID that was received in the FD message as specified in subclause 15.2.10;*

   c) *if the FD message was destined for the user, shall not include an Application ID IE as specified in subclause 15.2.7 and shall not include a Extended application ID IE as specified in subclause 15.2.24; and*

   d) *if the FD message was destined for an application, shall include:*

      i) *an Application ID IE set to the value of the Application ID that was included in the FD message as specified in subclause 15.2.3; or*

      ii) *an Extended application ID IE set to the value of the Extended application ID that was included in the FD message as specified in subclause 15.2.24."*

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.2.1]

"Upon receipt of a:

- *"SIP MESSAGE request for SDS disposition notification for MCData server"; or*

- *"SIP MESSAGE request for FD disposition notification for MCData server";*

the participating MCData function:

…

2) *shall determine the MCData ID of the calling user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request;*

NOTE: The MCData ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in subclause 7.3.

…

7) *shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];*

8) *shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCData function;*

NOTE: How the participating MCData function determines the controlling MCData function to forward notification message is out of scope of the present document.

9) *shall copy all MIME bodies included in the incoming SIP MESSAGE request to the outgoing SIP MESSAGE request;*

10) if not already included as part of step 8) above, shall include an application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP MESSAGE request, containing an <mcdata-calling-user-id> element set to the MCData ID of the originating user;

…

12) if the SIP MESSAGE is a "SIP MESSAGE request for FD disposition notification for MCData server ", shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request;

13) if the SIP MESSAGE is a "SIP MESSAGE request for FD disposition notification for MCData server", and the FD disposition notification type IE is set as "FILE DOWNLOAD REQUEST ACCEPTED" or "FILE DOWNLOAD REQUEST REJECTED"as specified in subclause 15.2.6, shall remove the file from the stored file list;

14) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request; and

15) shall send the SIP MESSAGE request as specified to 3GPP TS 24.229 [5].

Upon receipt of a SIP 202 (Accepted) response in response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP 202 (Accepted) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 202 (Accepted) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 200 (OK) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 15.2.6]

"The purpose of the FD disposition notification type information element is to identify the type of FD disposition notification sent from receiver to the sender.

The value part of the FD disposition notification type information element is coded as shown in Table 15.2.6-1.

The FD disposition notification type information element is a type 3 information element with a length of 1 octet.

Table 15.2.6.1: FD disposition notification type

Bits 8	7	6	5	4	3	2	1	
0	0	0	0	0	0	0	1	FILE DOWNLOAD REQUEST ACCEPTED
0	0	0	0	0	0	1	0	FILE DOWNLOAD REQUEST REJECTED
0	0	0	0	0	0	1	1	FILE DOWNLOAD COMPLETED
0	0	0	0	0	1	0	0	FILE DOWNLOAD DEFERRED
All other values are reserved.								

"

6.2.1.3 Test description

6.2.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).
- Test File 1 for CO FD as specified in annex A.2.1 and test File 2 for CO FD as specified in Annex A.2.2 are available at the SS-UE1 for upload.

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Storage Function, the MCData User Database, the SIP AS, the HTP Server, the HTTP Client, the MSRP TCP Server and the MSRP TCP Client.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- In the <on-network> element of the MCData Service Configuration document the <max-data-size-auto-recv-bytes> element of the <tx-and-rx-control> element shall be set to 0 to indicate non-mandatory download independent from the file size.
- The IUT (MCData Server) is connected to PLMN1.
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

6.2.1.3.2 Test procedure sequence

Table 6.2.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure **'Discovery of the absolute URI of the media storage function'** as described in Annex A, Table A.4.10.3-1?	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform procedure **'FD File upload using HTTP'** as described in Annex A, Table A.4.8.3-1, to send test file 1 (ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), clause A.2.1)?	-	-	2	P
3	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a one-to-one FD SIGNALLING PAYLOAD message for non mandatory download, from SS-UE1 (MCData client) to SS-UE2 (MCData Client), containing the URL of the file and with disposition request "FILE DOWNLOAD COMPLETED UPDATE"?**	-	-	3	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
4	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a FD disposition notification of "FILE DOWNLOAD REQUEST REJECTED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client), for the file sent at step 2?**	-	-	4	P
5	Check: Does the MCData Server (IUT) correctly perform procedure **'Discovery of the absolute URI of the media storage function'** as described in Annex A, Table A.4.10.3-1?	-	-	1	P
6	Check: Does the MCData Server (IUT) correctly perform procedure **'FD File upload using HTTP'** as described in Annex A, Table A.4.8.3-1, to send test file 2 (ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), clause A.2.2)?	-	-	5	P
7	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a FD SIGNALLING PAYLOAD message for non mandatory download, from SS-UE1 (MCData client) to SS-UE2 (MCData Client), containing the URL of the file and with disposition request "FILE DOWNLOAD COMPLETED UPDATE"?**	-	-	3	P
8	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a FD disposition notification of "FILE DOWNLOAD DEFERRED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client), for the file sent at step 6?**	-	-	6	P
9	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a FD disposition notification of "FILE DOWNLOAD ACCEPTED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client), for the file sent at step 6?**	-	-	7	P
10	Check: Does the MCData Server (IUT) correctly perform procedure **'FD File download using HTTP'** as described in Annex A, Table A.4.9.3-1, to download test file 2 (ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), clause A.2.2)?	-	-	8	P
11	Check: Is the content of the downloaded file the same as specified in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), clause A.2.2?	-	-	8	P
12	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a FD disposition notification of "FILE DOWNLOAD COMPLETED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client), for the file sent at step 6?**	-	-	9	P

6.2.1.3.3 Specific message contents

Table 6.2.1.3.3-1: HTTP POST from the SS-UE1 (Step 2 and 6, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.4.8.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.1.3.3-7

Table 6.2.1.3.3-2: HTTP 201 Created from the MCData Server (IUT) to the SS-UE1 (Step 2, Table 6.2.1.3.2-1; Step 3, Annex A, Table A.4.8.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.1.3.3-11

Table 6.2.1.3.3-3: SIP MESSAGE from the SS-UE1 (Step 3 and 7, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.1.3.3-12

Table 6.2.1.3.3-4: SIP MESSAGE from the MCData Server (IUT) to the SS-UE2 (Step 3 and 7, Table 6.2.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.2.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [21] IETF RFC 5031 [22]	
Request-URI	px_MCX_SIP_PublicUserId_B			
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.1	
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.2.1.3.3-5			

Table 6.2.1.3.3-5: MCData-Info (Table 6.2.1.3.3-4)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.2.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A			

Table 6.2.1.3.3-6: SIP MESSAGE from the SS-UE2 (Step 4, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.2.3.3-12				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		Resource lists		
MIME-part-body	Resource-lists as described in Table 6.2.1.3.3-7			

Table 6.2.1.3.3-7: Resource-lists (Table 6.2.1.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.2.1.3.3-8: SIP MESSAGE from the MCData Server (IUT) to the SS-UE1 (Step 4, Table 6.2.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.1.3.3-14				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

Table 6.2.1.3.3-9: HTTP 201 Created from the MCData Server (IUT) to the SS-UE1 (Step 6, Table 6.2.1.3.2-1; Step 3, Annex A, Table A.4.8.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.1.3.3-11				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Location				
uri	tsc_MCData_MSF_URI & "/file-location-2"			

Table 6.2.1.3.3-10: SIP MESSAGE from the SS-UE2 (Step 8, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.2.3.3-14				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		Resource lists		
MIME-part-body	Resource-lists as described in Table 6.2.1.3.3-7			

Table 6.2.1.3.3-11: SIP MESSAGE from the MCData Server (IUT) to the SS-UE1 (Step 8, Table 6.2.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.1.3.3-17				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

Table 6.2.1.3.3-12: SIP MESSAGE from the SS-UE2
(Step 9, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.2.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		Resource lists		
MIME-part-body	Resource-lists as described in Table 6.2.1.3.3-7			

Table 6.2.1.3.3-13: SIP MESSAGE from the MCData Server (IUT) to the SS-UE1
(Step 9, Table 6.2.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.1.3.3-19				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

Table 6.2.1.3.3-14: HTTP GET from the SS-UE2
(Step 10, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.4.9.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.2.3.3-6

Table 6.2.1.3.3-15: HTTP 200 (OK) from the MCData Server (IUT) to the SS-UE2
(Steps 10, Table 6.2.1.3.2-1; Step 2, Annex A, Table A.4.9.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.2.3.3-7

Table 6.2.1.3.3-16: SIP MESSAGE from the SS-UE2
(Step 12, Table 6.2.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.2.3.3-10				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		Resource lists		
MIME-part-body	Resource-lists as described in Table 6.2.1.3.3-7			

Table 6.2.1.3.3-17: SIP MESSAGE from the MCData Server (IUT) to the SS-UE1 (Step 12, Table 6.2.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.1.3.3-21				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

6.2.2 MCDATA Server - MCDATA Client / File Distribution (FD) / FD Using HTTP / Group Standalone FD / Mandatory Download / Without Disposition Request

6.2.2.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { SS-UE1 (MCData Client) requests discovery of the absolute URI by sending a SIP MESSAGE
message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 200 (Ok) to UE1 (MCData Client) }
            }
```

(2)

```
with { IUT (MCData Server) having requested a Group Standalone FD }
ensure that {
  when { the SS-UE1 (MCData Client) initiates FD "file 1" upload by sending a HTTP POST request }
    then { IUT (MCData Server) responds with a HTTP 201 (Created) }
            }
```

(3)

```
with { IUT (MCData Server) having requested a Group Standalone FD }
ensure that {
  when { SS-UE1 (MCData Client) requests FD upload by sending a SIP MESSAGE message without
disposition request }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE2 (MCData Client) and responds with
a SIP 202 (Accepted) to UE1 (MCData Client) }
            }
```

(4)

```
with { IUT (MCData Server) having requested a Group Standalone FD }
ensure that {
  when { SS-UE2 (MCData Client) requests FD transfer by sending a SIP MESSAGE message with
disposition request "FILE DOWNLOAD ACCEPTED" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

(5)

```
with { IUT (MCData Server) having requested a Group Standalone FD }
ensure that {
  when { the SS-UE2 (MCData Client) initiates FD "file 1" download by sending a HTTP GET request }
    then { IUT (MCData Server) responds with a HTTP 200 (Ok) and SS-UE2 (MCData Client) contains
downloaded "file 1" with same content  }
            }
```

6.2.2.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282), clauses 10.2.4.1, 10.2.4.3.1, 10.2.4.3.2, 10.2.4.4.1, 10.2.4.4.2, 10.2.4.2.1, 10.2.1.2.2, 6.2.2.2, 11.2, 6.2.3.2, 15.2.16. Unless otherwise stated, these are Rel-14 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.1]

"The procedures in the subclauses of the parent subclause describe the SIP signalling procedures for:

- *one-to-one file distribution using HTTP; and*
- *group standalone file distribution using HTTP.*

When the MCData user wishes to perform file distribution via HTTP, the MCData client:

1) shall check that the file size is less than or equal to the:

a) <mcdata-on-network-max-data-size-for-FD> element present in the group document retrieved by the group management client as specified in 3GPP TS 24.481 [11], if the file upload is for a group file distribution; or

…

2) if the size of the file:

a) is acceptable for upload as determined by step 1), shall determine the value of the absolute URI associated with the media storage function of the MCData content server from the <MCDataContentServerURI> element of the MCPTT user profile document (see the MCPTT user profile document in 3GPP TS 24.484 [50]);

b) is not acceptable for upload, shall not continue with the remaining steps in this subclause;

3) shall request the media storage client to upload the file to the media storage function by following the procedures in subclause 10.2.2.1; and

4) shall initiate an FD request containing a file URL as specified in subclause 10.2.4.2.1."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.3.1]

"Upon receipt of a "SIP MESSAGE request for FD using HTTP for originating participating MCData function", the participating MCData function:

…

2) shall determine the MCData ID of the originating user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request, and shall authorise the calling user;

NOTE: The MCData ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in subclause 7.3.

…

4) if <mcdata-controller-psi> element is present in the application/vnd.3gpp.mcdata-info+xml, shall use its value as public service identity of the controlling MCData function. Otherwise, if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request is:

a) set to a value of "group-fd", shall determine the public service identity of the controlling MCData function hosting the group standalone FD using HTTP service, associated with the MCData group identity in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the SIP MESSAGE request; or

…

…

6) shall determine whether the MCData user identified by the MCData ID is authorised for MCData communications by following the procedures in subclause 11.1;

…

8) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];

9) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCData function as determined by step 4) in this subclause;

10) shall copy all MIME bodies included in the incoming SIP MESSAGE request to the outgoing SIP MESSAGE request;

10A) if the incoming SIP MESSAGE request contains an application/vnd.3gpp.mcdata-info+xml MIME body that contains a <functional-alias-URI> element, shall check if the status of the functional alias is activated for the MCData ID. If the functional alias status is activated, then the participating MCData function shall set the <functional-alias-URI> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request to the received value, otherwise shall not include a <functional-alias-URI> element;

11) shall include the MCData ID of the originating user in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP MESSAGE request;

12) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request;

13) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request; and

14) shall send the SIP MESSAGE request as specified to 3GPP TS 24.229 [5].

Upon receipt of a SIP 202 (Accepted) response in response to the SIP MESSAGE request in step 14):

1) shall generate a SIP 202 (Accepted) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 202 (Accepted) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the SIP MESSAGE request in step 14):

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 200 (OK) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request in step 14) the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.3.2]

"Upon receipt of a:

- *"SIP MESSAGE request for FD using HTTP for terminating participating MCData function"; or*
- *"SIP MESSAGE network notification for FD using HTTP for terminating participating MCData function";*

the participating MCData function:

…

2) shall use the MCData ID present in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCData ID and public user identity of the terminating MCData user;

…

4) if the SIP MESSAGE is a "SIP MESSAGE request for FD using HTTP for terminating participating MCData function", and if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message with a FD disposition request type IE, shall store the value of the Conversation ID IE, the value of the Message ID IE and the payload IE in the FD SIGNALLING PAYLOAD message;

5) if the SIP MESSAGE is a "SIP MESSAGE network notification for FD using HTTP for terminating participating MCData function", and if FD NETWORK NOTIFICATION message within the application/vnd.3gpp.mcdata-signalling MIME body contains an FD notification type IE with value set as "FILE EXPIRED UNAVAILABLE TO DOWNLOAD" as specified in subclause 15.2.6, the file identified using Conversation ID IE shall be removed from the stored file list;

5) shall generate an outgoing SIP MESSAGE request as specified in subclause 6.3.2.1;

…

6) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request; and

7) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 200 (OK) response to the controlling MCData function according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the controlling MCData function according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.4.1]

"This subclause describes the procedures for sending a SIP MESSAGE from the controlling MCData function and is initiated by the controlling MCData function as a result of an action in subclause 10.2.4.4.2.

The controlling MCData function:

1) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];

2) shall include an Accept-Contact header field containing the g.3gpp.mcdata.fd media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

3) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" along with parameters "require" and "explicit" according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

4) shall copy the following MIME bodies in the received SIP MESSAGE request into the outgoing SIP MESSAGE request by following the guidelines in subclause 6.4:

a) application/vnd.3gpp.mcdata-info+xml MIME body; and

b) application/vnd.3gpp.mcdata-signalling MIME body;

5) if the application/vnd.3gpp.mcdata-signalling MIME body in the received SIP MESSAGE request contained a FD SIGNALLING PAYLOAD message without the Mandatory download IE included, then:

a) shall execute the procedures in subclause 11.2;

b) if the procedures in subclause 11.2 indicate that the mandatory download indication needs to be included,shall include the Mandatory download IE set to a value of "MANDATORY DOWNLOAD" in the FD SIGNALLING PAYLOAD message of the outgoing SIP MESSAGE request;

6) in the application/vnd.3gpp.mcdata-info+xml MIME body:

a) shall set the <mcdata-request-uri> element set to the MCData ID of the terminating user; and

b) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request was set to a value of "group-fd", shall set the <mcdata-calling-group-id> element to the group identity;

7) shall set the Request-URI to the public service identity of the terminating participating MCData function associated to the MCData user to be invited;

8) shall copy the public user identity of the calling MCData user from the P-Asserted-Identity header field of the incoming SIP MESSAGE request into the P-Asserted-Identity header field of the outgoing SIP MESSAGE request;

9) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd"; and

10) shall send the SIP MESSAGE request according to according to rules and procedures of 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.4.2]

"The procedures in this subclause are executed upon:

- *receipt of a "SIP MESSAGE request for FD using HTTP for controlling MCData function", the controlling MCData function; or*
- *a decision to now process a previously received "SIP MESSAGE request for FD using HTTP for controlling MCData function" that had been queued for later transmission;*

NOTE 1: The controlling MCData function may postpone the continuation of an FD using HTTP procedure by queuing the received "SIP MESSAGE request for FD using HTTP for controlling MCData function". The management of the queue is specified in Annex B of 3GPP TS 23.282 [2].

The controlling MCData function:

…

3) if the SIP MESSAGE does not contain:

a) an application/vnd.3gpp.mcdata-info+xml MIME body; and

b) an application/vnd.3gpp.mcdata-signalling MIME body;

shall reject the SIP MESSAGE request with a SIP 403 (Forbidden) response, with warning text set to "199 expected MIME bodies not in the request" in a Warning header field as specified in subclause 4.9, and shall not continue with the rest of the steps in this subclause;

4) shall decode the contents of the application/vnd.3gpp.mcdata-signalling MIME body contained in the SIP MESSAGE;

5) if the application/vnd.3gpp.mcdata-signalling MIME body does not contain only one FD SIGNALLING PAYLOAD message or FD HTTP TERMINATION message, shall reject the SIP MESSAGE …, and shall not continue with the rest of the steps in this subclause;

6) if the FD SIGNALLING PAYLOAD message or FD HTTP TERMINATION messagedoes not contain only one Payload IE, shall reject the SIP MESSAGE request …, and shall not continue with the rest of the steps in this subclause;

7) if the Payload IE has Payload contents:

a) with a Payload content type set to a value other than "FILEURL" shall reject the SIP MESSAGE request …, and shall not continue with the rest of the steps in this subclause; and

b) with Payload data containing a file URL identifying a file that does not exist on the media storage function, shall reject the SIP MESSAGE request …, and shall not continue with the rest of the steps in this subclause;

8) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message with a FD disposition request type IE, shall store the value of the Conversation ID IE and the value of the Message ID IE in the FD SIGNALLING PAYLOAD message;

NOTE 2: The controlling MCData function uses the Conversation ID and Message ID for correlation with disposition notifications.

9) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message:

a) with a Metadata IE, shall derive a timer value for the file availability timer as the minimum of the file availability information in the metadata and the value contained in the <max-file-availability> element in the MCData service configuration document as specified in 3GPP TS 24.484 [12]; and

b) without a Metadata IE, shall derive a timer value for the file availability timer as the value contained in the <default-file-availability> element in the MCData service configuration document as specified in 3GPP TS 24.484 [12];

…

11) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD HTTP TERMINATION message:

a) if the FD HTTP TERMINATION message doesn't contain Conversation Id or Message Id, shall reject the SIP MESSAGE request … and shall not continue with rest of the steps;and

…

12) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message and if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request is set to a value of "group-fd":

a) shall retrieve the group document associated with the group identity in the SIP MESSAGE request by following the procedures in subclause 6.3.3, and shall continue with the remaining steps if the procedures in subclause 6.3.3 were successful;

…

c) if the <entry> element of the <list> element of the <list-service> element in the group document does not contain an <mcdata-mcdata-id> element with a "uri" attribute matching the MCData ID of the originating user contained in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the SIP MESSAGE request, shall send a SIP 403 (Forbidden) response with the warning text set to "116 user is not part of the MCData group" in a Warning header field as specified in subclause 4.9 and shall not continue with the rest of the steps;

d) if the <list-service> element contains a <mcdata-allow-file-distribution> element in the group document set to a value of "false", shall send a SIP 403 (Forbidden) response with the warning text set to "213 file distribution not allowed for this group" in a Warning header field as specified in subclause 4.9 and shall not continue with the rest of the steps;

e) if the <supported-services> element is not present in the group document or is present and contains a <service> element containing an "enabler" attribute which is not set to the value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd", shall send a SIP 488 (Not Acceptable) response with the warning text set to "214 FD services not supported for this group" in a Warning header field as specified in subclause 4.9 and shall not continue with the rest of the steps;

…

j) shall determine targeted group members for MCData communications by following the procedures in subclause 6.3.4;

…

l) shall send SIP MESSAGE requests to the targeted group members identified in step j) above by following the procedure in subclause 10.2.4.4.1;

13) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message, shall start TDC2 (file availability timer) with the value derived in step 9 of this subclause;

14) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD SIGNALLING PAYLOAD message, shall associate the running timer TDC2 (file availability timer) to the Conversation ID, Message ID, Application ID (if included), and Extended application ID (if included) contained in the FD SIGNALLING PAYLOAD message;

NOTE 3: Multiple file availability timers can be running for a file. Each file availability timer is uniquely associated to a Conversation ID and Message ID.

15) shall generate a SIP 202 (Accepted) response in response to the "SIP MESSAGE request for FD using HTTP for controlling MCData function"; and

16) shall send the SIP 202 (Accepted) response towards the originating participating MCData function according to 3GPP TS 24.229 [5].

17) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD HTTP TERMINATION message and Termination information type IE set to "TERMINATION REQUEST" then:

a) shall identify the FILE transmission with Conversation ID and Message ID and "FILE URL". If any ongoing transmission exist then execute procedure described in subclause 12.4.2.1 withfollowing clarifications:

i) shall set FD notification type IE as "FILE DELETED UNAVAILABLE TO DOWNLOAD" as specified in subclause 15.2.18;

b) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6]. In the generation of the SIP MESSAGE request, the controlling MCData function:

i) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" along with parameters "require" and "explicit" according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

ii) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd";

iii) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the participating MCData function associated to the MCData ID of the originating user mentioned in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request;

iv) shall copy the public user identity of the calling MCData user from the P-Asserted-Identity header field of the incoming SIP MESSAGE request into the P-Asserted-Identity header field of the outgoing SIP MESSAGE request;

v) shall include an application/vnd.3gpp.mcdata-info+xml MIME body in the SIP MESSAGE request, following the rules specified in subclause 6.4 for the handling of MIME bodies in a SIP message:

A) fill <mcdata-request-uri> element from <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml in received SIP MESSAGE;

vi) shall generate FD HTTP TERMINATION message as described in subclause 6.3.6.1;

vii) shall set the Termination information type IE set to "TERMINATION RESPONSE" as specified in subclause 15.2.22.

viii) if clause is successful shall set Release response type IE of FD HTTP TERMINATION MESSAGE to "RELEASE SUCCESS" else set to "RELEASE FAILED" as described in subclause 15.2.23; and

ix) shall include in the SIP request, the FD HTTP TERMINATION message in an application/vnd.3gpp.mcdata-signalling MIME body as specified in subclause E.1;

c) shall send the SIP MESSAGE request towards the originating participating MCData function as specified in 3GPP TS 24.229 [5]; and

18) if the application/vnd.3gpp.mcdata-signalling MIME body contains an FD HTTP TERMINATION message and Termination information type IE set to other than "TERMINATION REQUEST" then follow procedures described on subclause 13.2.5 and subclause 13.2.6."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.4.2.1]

"The MCData client shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6] with the clarifications given below.

The MCData client:

5) shall build the SIP MESSAGE request as specified in subclause 6.2.4.1;

…

3) if a group standalone FD message is to be sent:

a) if the "/<x>/<x>/Common/MCData/AllowedFD" leaf node present in the group document of the requested MCData group, configured on the group management client as specified in 3GPP TS 24.483 [42] is set to "false", shall reject the request for FD and not continue with the rest of the steps in this subclause; and

b) shall insert in the SIP MESSAGE request an application/vnd.3gpp.mcdata-info+xml MIME body with:

i) the <request-type> element set to a value of "group-fd";

ii) the <mcdata-request-uri> element set to the MCData group identity;

iii) the <mcdata-client-id> element set to the MCData client ID of the originating MCData client; and

iv) if the MCData client is aware of active functional aliases and if an active functional alias is to be included in the SIP MESSAGE request, the <functional-alias-URI> element set to the URI of the used functional alias;

4) shall generate a standalone FD message as specified in subclause 6.2.2.2; and

5) shall send the SIP MESSAGE request according to rules and procedures of 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.1.2.2]

"The MCData client:

1) if the FD SIGNALLING PAYLOAD message contains a new Conversation ID, shall instantiate a new conversation with the Message ID in the FD SIGNALLING PAYLOAD identifying the first message in the conversation thread;

2) if the FD SIGNALLING PAYLOAD message contains an existing Conversation ID and:

a) if the FD SIGNALLING PAYLOAD message does not contain an InReplyTo message ID, shall use the Message ID in the FD SIGNALLING PAYLOAD to identify a new message in the existing conversation thread; and

b) if the FD SIGNALLING PAYLOAD message contains an InReplyTo message ID, shall associate the message to an existing message in the conversation thread as identified by the InReplyTo message ID in the FD SIGNALLING PAYLOAD, and use the Message ID in the FD SIGNALLING PAYLOAD to identify the new message;

3) may store the Conversation ID, Message ID, InReplyTo message ID and Date and time in local storage;

4) if the FD SIGNALLING PAYLOAD message does not contain an Application ID IE and does not contain an Extended application ID IE:

a) shall determine that the payload contained in the Payload IE in the FD SIGNALLING PAYLOAD message is for user consumption;

b) shall notify the user or application that the file identified by file URL in the Payload data in the Payload IE will be downloaded automatically; and

c) if the FD SIGNALLING PAYLOAD message contains a Metadata IE, shall deliver the contents of the Metadata IE to the user or application;

5) if the FD SIGNALLING PAYLOAD message contains an Application ID IE:

a) shall determine that the payload contained in the Payload IE in the FD SIGNALLING PAYLOAD message is not for user consumption;

b) if the Application ID value is unknown, shall discard the FD message and exit this subclause;

c) if the Application ID value is known, shall notify the application that the file identified by file URL in the Payload data in the Payload IE will be downloaded automatically; and

NOTE 1: If the FD request is addressed to a non-MCData application that is not running, the MCData client starts the local non-MCData application. Subsequent automatic download of the file is then started and the file is delivered to that application.

d) if the FD SIGNALLING PAYLOAD message contains a Metadata IE, shall deliver the contents of the Metadata IE to the application;

6) if the FD SIGNALLING PAYLOAD message contains an Extended application ID IE:

a) shall determine that the payload contained in the Payload IE in the FD SIGNALLING PAYLOAD message is not for user consumption;

b) if the Extended application ID value is unknown, shall discard the FD message and exit this clause;

c) if the Extended application ID value is known, shall notify the application that the file identified by file URL in the Payload data in the Payload IE will be downloaded automatically; and

NOTE 2: If the FD request is addressed to a non-MCData application that is not running, the MCData client starts the local non-MCData application. Subsequent automatic download of the file is then started and the file is delivered to that application.

d) if the FD SIGNALLING PAYLOAD message contains a Metadata IE, shall deliver the contents of the Metadata IE to the application;

7) shall generate an FD NOTIFICATION indicating acceptance of the FD request as specified in subclause 12.2.1.1;

8) shall attempt to download the file as identified by the file URL in the Payload IE in the FD SIGNALLING PAYLOAD message, as specified in subclause 10.2.3.1; and

9) if the received FD SIGNALLING PAYLOAD message contains an FD disposition request type IE requesting a file download completed update indication, then after the file has been successfully downloaded, shall generate an FD NOTIFICATION indicating file download completed, by following the procedures in subclause 12.2.1.1."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 6.2.2.2]

"In order to generate an FD message, the MCData client:

1) shall generate an FD SIGNALLING PAYLOAD message as specified in subclause 15.1.3; and

2) shall include in the SIP request, the FD SIGNALLING PAYLOAD message in an application/vnd.3gpp.mcdata-signalling MIME body as specified in subclause E.1.

When generating an FD SIGNALLING PAYLOAD message as specified in subclause 15.1.3, the MCData client:

1) shall set the Date and time IE to the current time as specified in subclause 15.2.8;

2) if the FD message starts a new conversation, shall set the Conversation ID IE to a newly generated Conversation ID value as specified in subclause 15.2.9;

3) if the FD message continues an existing unfinished conversation, shall set the Conversation ID IE to the Conversation ID value of the existing conversation as specified in subclause 15.2.9;

4) shall set the Message ID IE to a newly generated Message ID value as specified in subclause 15.2.10;

5) if the FD message is in reply to a previously received MCData message, shall include the InReplyTo message ID IE with the Message ID value in the previously received MCData message;

6) if the FD message is for user consumption, shall not include an Application ID IE as specified in subclause 15.2.7 and shall not include an Extended application ID IE as specified in subclause 15.2.24;

7) if the FD message is intended for an application on the terminating MCData client, shall include:

a) an Application ID IE with a Application ID value representing the intended application as specified in subclause 15.2.7; or

b) an Extended application ID IE with an Extended application ID value representing the intended application as specified in subclause 15.2.24;

NOTE: The value and field chosen for coding the identity of the application are coordinated by the mission critical organisation.

8) may include an FD disposition request type IE set to "FILE DOWNLOAD COMPLETE UPDATE" as specified in subclause 15.2.4;

9) if requiring mandatory download at the recipient side, shall include a Mandatory download IE as specified in subclause 15.2.16 set to the value of "MANDATORY DOWNLOAD";

10) shall include a Payload IE with:

a) the Payload content type set to "FILEURL" as specified in subclause 15.2.13; and

b) the URL of the file in the Payload data as as specified in subclause 15.2.13; and

11) may include a Metadata IE with the required file description information and file availability information, as specified in subclause 15.2.17."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 11.2]

"If the controlling MCData function receives a one-to-one file distribution using HTTP or a group standalone file distribution using HTTP without the mandatory download indication the controlling MCData function:

1) if the file distribution request contained metadata, shall retrieve the filesize contained in the fileselector of the Metadata IE in the FD request;

2) if the file distribution request did not contain metadata, shall determine the size of the file referenced by the file URL contained in FD request;

…

4) for group standalone file distribution using HTTP, shall determine if the filesize is less than or equal to the value contained the <mcdata-on-network-max-data-size-auto-recv> element of the MCData group document associated with the MCData group identity in the request, as specified in 3GPP TS 24.481 [11]

5) ifcondition 3) or 4) is true, shall determine that the mandatory download indication needs to be included in the file distribution request sent to the terminating MCData client;"

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 6.2.3.2]

"In order to generate an FD notification, the MCData client:

1) shall generate an FD NOTIFICATION message as specified in subclause 15.1.6; and

2) shall include in the SIP request, the FD NOTIFICATION message in an application/vnd.3gpp.mcdata-signalling MIME body as specified in subclause E.1.

When generating an FD NOTIFICATION message as specified in subclause 15.1.6, the MCData client:

1) if sending a file download accept notification, shall set the FD disposition notification type IE as "FILE DOWNLOAD REQUEST ACCEPTED" as specified in subclause 15.2.6;

2) if sending a file download reject notification, shall set the FD disposition notification type IE as "FILE DOWNLOAD REQUEST REJECTED" as specified in subclause 15.2.6;

3) if sending a file download deferred notification, shall set the FD disposition notification type IE as "FILE DOWNLOAD REQUEST DEFERRED" as specified in subclause 15.2.6;

4) shall set the Conversation ID to the value of the Conversation ID that was received in the FD message as specified in subclause 15.2.9;

5) shall set the Date and time IE to the current time as specified in subclause 15.2.8; and

6) if sending a file download completed notification:

a) shall set the FD disposition notification type IE as "FILE DOWNLOAD COMPLETED" as specified in subclause 15.2.6;

b) shall set the Message ID to the value of the Message ID that was received in the FD message as specified in subclause 15.2.10;

c) if the FD message was destined for the user, shall not include an Application ID IE as specified in subclause 15.2.7 and shall not include a Extended application ID IE as specified in subclause 15.2.24; and

d) if the FD message was destined for an application, shall include:

i) an Application ID IE set to the value of the Application ID that was included in the FD message as specified in subclause 15.2.3; or

ii) an Extended application ID IE set to the value of the Extended application ID that was included in the FD message as specified in subclause 15.2.24."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 15.2.16]

"The purpose of the Mandatory download information element is for the originating client to inform the terminating client that a file must be downloaded immediately.

The value part of the Mandatory download information element is coded as shown in Figure 15.2.16-1 and Table 15.2.16-1.

The Mandatory download information element is a type 1 information element.

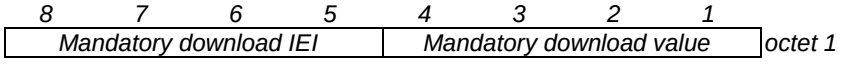


Figure 15.2.16-1: Mandatory download

Table 15.2.16-1: Mandatory download

Mandatory download value (octet 1) *Bits* *4 3 2 1* *0 0 0 1 MANDATORY DOWNLOAD* *All other values are reserved.*

"

6.2.2.3 Test description

6.2.2.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).
- Test File 1 for CO FD as specified in clause A.2.1 is available at the SS-UE1 for upload.

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Storage Function, the MCData User Database, the SIP AS, the HTP Server, the HTTP Client, the MSRP TCP Server and the MSRP TCP Client.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCData Server) is connected to PLMN1.
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.
- SS-UE1 (MCData client) and SS-UE2 (MCData client) are affiliated to the Prearranged group.
- The IUT (MCData Server) is the controlling server of the Prearranged group.

6.2.2.3.2 Test procedure sequence

Table 6.2.2.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure **'Discovery of the absolute URI of the media storage function'** as described in Annex A, Table A.4.10.3-1?	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform procedure **'FD File upload using HTTP'** as described in Annex A, Table A.4.8.3-1, to send test file 1 (ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), clause A.2.1)?	-	-	2	P
3	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a group FD SIGNALLING PAYLOAD message for mandatory download, containing the URL of the file and without disposition request?**	-	-	3	P
4	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a FD disposition notification of "FILE DOWNLOAD ACCEPTED" from SS-UE2 (MCData client), for the file sent at step 2?**	-	-	4	P
5	Check: Does the MCData Server (IUT) correctly perform procedure **'FD File download using HTTP'** as described in Annex A, Table A.4.9.3-1. to download test file 1 (ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), clause A.2.1)?	-	-	5	P
6	Check: Is the content of the downloaded file the same as specified in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), clause A.2.1?	-	-	5	P

6.2.2.3.3 Specific message contents

Table 6.2.2.3.3-1: HTTP POST from the SS-UE1 (Step 2, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.4.8.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.7.3.3-7

Table 6.2.2.3.3-2: HTTP 201 Created from the MCData Server (IUT) to the SS-UE1 (Step 2, Table 6.2.2.3.2-1; Step 3, Annex A, Table A.4.8.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.7.3.3-10

Table 6.2.2.3.3-3: SIP MESSAGE from the SS-UE1 (Step 3, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.7.3.3-11

Table 6.2.2.3.3-4: SIP MESSAGE from the MCData Server (IUT) to the SS-UE2 (Step 3, Table 6.2.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.8.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [21] IETF RFC 5031 [22]	
Request-URI	px_MCX_SIP_PublicUserId_B			
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.1	
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.2.2.3.3-5			

Table 6.2.2.3.3-5: MCData-Info (Table 6.2.2.3.3-4)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.8.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-client-id	Encrypted <mcdata-client-id> with mcdataString set to px_MCX_Client_A_ID	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		

Table 6.2.2.3.3-6: SIP MESSAGE from the SS-UE2 (Step 4, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.8.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		Resource lists		
MIME-part-body	Resource-lists as described in Table 6.2.1.3.3-7			

Table 6.2.2.3.3-7: Resource-lists (Table 6.2.2.3.3-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.2.2.3.3-8: SIP MESSAGE from the MCData Server (IUT) to the SS-UE1 (Step 4, Table 6.2.2.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.7.3.3-14				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

Table 6.2.2.3.3-9: HTTP GET from the SS-UE2 (Step 5, Table 6.2.2.3.2-1; Step 1, Annex A, Table A.4.9.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.8.3.3-8

Table 6.2.2.3.3-10: HTTP 200 (OK) from the MCData Server (IUT) to the SS-UE2 (Step 5, Table 6.2.2.3.2-1; Step 2, Annex A, Table A.4.9.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.8.3.3-9

6.2.3 MCDATA Server - MCDATA Client / File Distribution (FD) / FD Using Media Plane / One-to-one Standalone FD

6.2.3.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { the SS-UE1 (MCData Client) initiates establishment of a media plane FD session }
    then { IUT (MCData Server) accepts the call from the initiator and establishes the call with SS-
UE2 (MCData Client) and responds/forwards empty MSRP messages  }
           }
```

(2)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { the SS-UE1 (MCData Client) sends an MSRP SEND message with disposition request "FILE
DOWNLOADED COMPLETED UPDATE"}
    then { IUT (MCData Server) forwards an MSRP SEND message to UE2 (MCData Client) and responds
with an MSRP 200 (Ok) to UE1 (MCData Client)  }
           }
```

(3)

```
with { IUT (MCData Server) having requested an one-to-one Standalone FD }
ensure that {
  when { the SS-UE1 (MCData client) ends the FD session }
    then { IUT (MCData Server) responds by sending a SIP 200 (OK) message to the client ending the
call and sends a SIP BYE message to SS-UE2 (MCData Client) and waits to close MSRP TCP connections
}
            }
```

(4)

```
with { IUT (MCData Server) having received an one-to-one Standalone FD }
ensure that {
  when { SS-UE2 (MCData Client) requests the one-to-one Standalone FD by sending a SIP MESSAGE
message with disposition notification "FILE DOWNLOAD COMPLETED" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
            }
```

6.2.3.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282), clauses 10.2.5.3.3, 10.2.5.3.4, 10.2.5.3.1, 10.2.5.3.2, 10.2.5.4.3, 10.2.5.4.4, 10.2.5.4.1, 10.2.5.4.2, ETSI TS 124 582 [40] (3GPP TS 24.582), clauses 7.2.4.1, 7.2.4.2, 7.2.4.3, 7.2.5.1, 7.2.5.2, 7.2.5.3, 7.2.2., 7.2.3, 7.3.2.1, 7.3.2.2, 7.3.3, 10.1. Unless otherwise stated, these are Rel-14 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.5.3.3]

"Upon receipt of a "SIP INVITE request for file distribution for originating participating MCData function", the participating MCData function:

…

2) shall determine the MCData ID of the calling user from the public user identity in the P-Asserted-Identity header field of the SIP INVITE request, and shall authorise the calling user;

NOTE: The MCData ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in subclause 7.3.

…

4) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP INVITE request is:

…

b) set to a value of "one-to-one-fd", shall determine the public service identity of the controlling MCData function hosting the file distribution service for the calling user;

…

6) shall determine whether the MCData user identified by the MCData ID is authorised for MCData communications by following the procedures in subclause 11.1;

…

8) shall generate a SIP INVITE request in accordance with 3GPP TS 24.229 [5];

9) shall include the option tag "timer" in the Supported header field;

10) should include the Session-Expires header field according to IETF RFC 4028 [38]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

11) shall set the Request-URI of the outgoing SIP INVITE request to the public service identity of the controlling MCData function as determined by step 4) in this subclause;

12) shall include the MCData ID of the originating user in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP INVITE request;

12A) if the incoming SIP MESSAGE request contains an application/vnd.3gpp.mcdata-info+xml MIME body that contains a <functional-alias-URI> element, shall check if the status of the functional alias is activated for the MCData ID. If the functional alias status is activated, then the participating MCData function shall set the <functional-alias-URI> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request to the received value, otherwise shall not include a <functional-alias-URI> element;

13) shall include in the outgoing SIP INVITE request, the application/vnd.3gpp.mcdata-signalling MIME body that was present in the incoming SIP INVITE request;

14) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP INVITE request;

15) shall set the P-Asserted-Identity in the outgoing SIP INVITE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP INVITE request;

16) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the MCData client as specified in subclause 10.2.5.3.1; and

17) shall send the SIP INVITE request as specified to 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the SIP INVITE request in step 16):

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5];

2) shall include in the SIP 200 (OK) response an SDP answer as specified in the subclause 10.2.5.3.2;

3) shall include the option tag "timer" in a Require header field;

4) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [38], "UAS Behavior". If the "refresher" parameter is not included in the received request, the "refresher" parameter in the Session-Expires header field shall be set to "uac";

5) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.fd media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd"; and

c) the isfocus media feature tag;

6) shall include Warning header field(s) that were received in the incoming SIP 200 (OK) response;

7) shall include an MCData session identity mapped to the MCData session identity provided in the Contact header field of the received SIP 200 (OK) response;

8) if the incoming SIP 200 (OK) response contained an application/vnd.3gpp.mcdata-info+xml MIME body, shall copy the application/vnd.3gpp.mcdata-info+xml MIME body to the outgoing SIP 200 (OK) response.

9) shall include the public service identity received in the P-Asserted-Identity header field of the incoming SIP 200 (OK) response into the P-Asserted-Identity header field of the outgoing SIP 200 (OK) response; and

10) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 7.2.1;

11) shall send the SIP 200 (OK) response to the MCData client according to 3GPP TS 24.229 [5]; and

12) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [38].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP INVITE request in step 16) the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.5.3.4]

"Upon receipt of a "SIP INVITE request for file distribution for terminating participating MCData function", the participating MCData function:

…

2) shall check the presence of the isfocus media feature tag in the URI of the Contact header field and if it is not present then the participating MCData function shall reject the request .., and shall not continue with the rest of the steps;

3) shall use the MCData ID present in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request to retrieve the binding between the MCData ID and public user identity of the terminating MCData user;

…

4A) if the <IncomingOne-to-OneCommunicationList> element exists in the MCData user profile document with one or more <One-to-One-CommunicationListEntry> elements (see the MCData user profile document in 3GPP TS 24.484 [12]) and:

i) if the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request does not match with the <entry> element of any of the <One-to-One-CommunicationListEntry> elements in the <IncomingOne-to-OneCommunicationList> element of the MCData user profile document (see the MCData user profile document in 3GPP TS 24.484 [12]); and

ii) if configuration is not set in the MCData user profile document that allows the MCData user to receive one-to-one MCData communication from any user (see <allow-one-to-one-communication-from-any-user> element in MCData user profile document in 3GPP TS 24.484 [12]);

then:

i) shall reject the SIP INVITE request … and shall not continue with the rest of the steps;

5) shall generate a SIP INVITE request accordance with 3GPP TS 24.229 [5];

6) should include the Session-Expires header field according to IETF RFC 4028 [38]. It is recommended that the "refresher" header field parameter is omitted. If included, the "refresher" header field parameter shall be set to "uac";

7) shall include the option tag "timer" in the Supported header field;

8) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.fd media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd";

c) the isfocus media feature tag;

d) an MCData session identity mapped to the MCData session identity provided in the Contact header field of the incoming SIP INVITE request; and

e) any other uri-parameter provided in the Contact header field of the incoming SIP INVITE request;

9) shall include in the SIP INVITE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [8] that were received (if any) in the incoming SIP INVITE request;

10) shall set the Request-URI of the outgoing SIP INVITE request to the public user identity associated to the MCData ID of the terminating MCData user;

11) shall populate the outgoing SIP INVITE request with the MIME bodies that were present in the incoming SIP INVITE request;

12) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP INVITE request to the P-Asserted-Identity header field of the outgoing SIP INVITE request;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received "SIP INVITE request for file distribution for terminating participating MCData function" as specified in subclause 10.2.5.3.1; and

14) shall send the SIP INVITE request as specified in 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the above SIP INVITE request, the participating MCData function:

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5];

2) shall include in the SIP 200 (OK) response an SDP answer based on the SDP answer in the received SIP 200 (OK) response as specified in subclause 10.2.5.3.2;

3) shall include the option tag "timer" in a Require header field;

4) shall include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [38], "UAS Behavior". If no "refresher" parameter was included in the SIP INVITE request, the "refresher" parameter in the Session-Expires header field shall be set to "uas";

5) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.fd media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd"; and

c) an MCData session identity mapped to the MCData session identity provided in the Contact header field of the received SIP INVITE request from the controlling MCData function;

6) if the incoming SIP response contained an application/vnd.3gpp.mcdata-info+xml MIME body, shall copy the application/vnd.3gpp.mcdata-info+xml MIME body to the outgoing SIP 200 (OK) response.

7) shall copy the P-Asserted-Identity header field from the incoming SIP 200 (OK) response to the outgoing SIP 200 (OK) response;

8) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [38];

9) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 7.2.2; and

10) shall send the SIP 200 (OK) response to the controlling MCData function according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP INVITE request, the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the controlling MCData function according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.5.3.1]

"The SDP offer is generated based on the received SDP offer. The SDP offer generated by the participating MCData function:

1) shall contain only one SDP media-level section for file distribution as contained in the received SDP offer; and

2) shall contain an "a=key-mgmt" attribute field with a "mikey" attribute value, if present in the received SDP offer.

When composing the SDP offer according to 3GPP TS 24.229 [5], the participating MCData function:

1) shall replace the IP address and port number for the offered media stream in the received SDP offer with the IP address and port number of the participating MCData function, if required; and

…

2) if the IP address is replaced shall insert its MSRP URI before the MSRP URI in the "a=path" attribute in the SDP answer."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.5.3.2]

"When composing the SDP answer according to 3GPP TS 24.229 [5], the participating MCData function:

1) shall replace the IP address and port number in the received SDP answer with the IP address and port number of the participating MCData function, for the accepted media stream in the received SDP offer, if required; and

…

2) if the IP address is replaced shall insert its MSRP URI before the MSRP URI in the "a=path" attribute in the SDP answer."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.5.4.3]

"This subclause describes the procedures for inviting an MCData user to an MCData session. The procedure is initiated by the controlling MCData function as the result of an action in subclause 10.2.5.4.4.

The controlling MCData function:

1) shall generate a SIP INVITE according to 3GPP TS 24.229 [5];

2) shall include the Supported header field set to "timer";

3) should include the Session-Expires header field according to rules and procedures of IETF RFC 4028 [38]. The refresher parameter shall be omitted;

4) shall include an Accept-Contact header field containing the g.3gpp.mcdata.fd media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8];

5) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" along with parameters "require" and "explicit" according to IETF RFC 3841 [8];

6) shall include a Referred-By header field with the public user identity of the inviting MCData client;

7) shall include in the Contact header field an MCData session identity for the MCData session with the g.3gpp.mcdata.fd media feature tag, the isfocus media feature tag and the g.3gpp.icsi-ref media feature tag with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" according to IETF RFC 3840 [16];

8) shall include in the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request:

a) the <mcdata-request-uri> element set to the MCData ID of the terminating user; and

…

9) shall include in the outgoing SIP INVITE request, the application/vnd.3gpp.mcdata-signalling MIME body that was present in the incoming SIP INVITE request;

10) shall set the Request-URI to the public service identity of the terminating participating MCData function associated to the MCData user to be invited;

NOTE 1: How the controlling MCData function finds the address of the terminating participating MCData function is out of the scope of the current release.

11) shall set the P-Asserted-Identity header field to the public service identity of the controlling MCData function;

12) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" (coded as specified in 3GPP TS 24.229 [5]), in a P-Asserted-Service-Id header field according to IETF RFC 6050 [7] in the SIP INVITE request;

13) shall include in the SIP INVITE request an SDP offer based on the SDP offer in the received SIP INVITE request from the originating client according to the procedures specified in subclause 10.2.5.4.1; and

14) shall send the SIP INVITE request towards the terminating client in accordance with 3GPP TS 24.229 [5].

Upon receiving a SIP 200 (OK) response for the SIP INVITE request the controlling MCData function:

1) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 7.3.

NOTE 2: The procedures executed by the controlling MCData function prior to sending a response to the inviting MCData client are specified in subclause 10.2.5.4.4."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.5.4.4]

"In the procedures in this subclause:

1) MCData ID in an incoming SIP INVITE request refers to the MCData ID of the originating user from the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP INVITE request;

…

3) MCData ID in an outgoing SIP INVITE request refers to the MCData ID of the called user in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP INVITE request;

The procedures in this subclause are executed upon:

- *receipt of a "SIP INVITE request for controlling MCData function for file distribution"; or*
- *a decision to now process a previously received "SIP INVITE request for controlling MCData function for file distribution" that had been queued for later transmission;*

…

the controlling MCData function:

…

4) if the incoming SIP INVITE request does not contain an application/vnd.3gpp.mcdata-signalling MIME body with the FD SIGNALLING PAYLOAD as described in subclause 6.2.2.3, shall reject the SIP INVITE request with appropriate reject code;

5) shall reject the SIP request with a SIP 403 (Forbidden) response and not process the remaining steps if:

a) an Accept-Contact header field does not include the g.3gpp.mcdata.fd media feature tag; or

b) an Accept-Contact header field does not include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd";

6) shall cache SIP feature tags, if received in the Contact header field and if the specific feature tags are supported;

7) shall start the SIP Session timer according to rules and procedures of IETF RFC 4028 [38];

8) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP INVITE request is set to a value of "one-to-one-fd" and:

a) the conditions in subclause 11.1 indicate that the MCData user is not allowed to initiate FD communications due to file size exceeding allowed limits as determined by step 4) of subclause 11.1, shall reject the SIP INVITE request …, and shall not continue with the rest of the steps in this subclause; and

NOTE 2: The size of the file intended for transfer over the media plane is obtained from the 'size' selector of the file-selector attribute in the received SDP offer.

b) the SIP INVITE request:

…

ii) contains an application/resource-lists MIME body with exactly one <entry> element, shall invite the MCData user identified by the <entry> element of the MIME body, as specified in subclause 10.2.5.4.3; and

shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 7.3;

…

Upon receiving a SIP 200 (OK) response for a SIP INVITE request as specified in subclause 10.2.5.4.3 and if the MCData ID in the SIP 200 (OK) response matches to the MCData ID in the corresponding SIP INVITE request the controlling MCData function:

1) shall generate SIP 200 (OK) response to the SIP INVITE request according to 3GPP TS 24.229 [5];

2) shall include the option tag "timer" in a Require header field;

3) shall include the Session-Expires header field and start supervising the SIP session according to rules and procedures of IETF RFC 4028 [38], "UAS Behavior". The "refresher" parameter in the Session-Expires header field shall be set to "uac";

4) shall include a P-Asserted-Identity header field with the public service identity of the controlling MCData function;

5) shall include a SIP URI for the MCData session identity in the Contact header field identifying the MCData session at the controlling MCData function;

6) shall include the following in the Contact header field:

a) the g.3gpp.mcdata.fd media feature tag;

b) the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd"; and

c) the isfocus media feature tag;

7) shall include Warning header field(s) received in incoming responses to the SIP INVITE request;

8) shall include in the SIP 200 (OK) response an SDP answer to the SDP offer in the incoming SIP INVITE request as specified in the subclause 10.2.5.4.2;

9) shall interact with the media plane as specified in 3GPP TS 24.582 [15] subclause 7.3; and

10) shall send a SIP 200 (OK) response to the inviting MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.5.4.1]

"When composing an SDP offer according to 3GPP TS 24.229 [5], IETF RFC 5547 [69] IETF RFC 6135 [19] and IETF RFC 6714 [20] the MCData client:

1) shall include an "m=message" media-level section for the MCData media stream consisting of:

a) the port number;

b) a protocol field value of "TCP/MSRP" or "TCP/TLS/MSRP" for TLS;

c) an "a=sendonly" attribute;

d) an "a=path" attribute containing its own MSRP URI;

e) set the content type as "a=accept-types:application/vnd.3gpp.mcdata-signalling";

f) set the a=setup attribute as "actpass";

g) a file-selector attribute containing:

i) a 'name' selector;

ii) a 'type' selector;

iii) a 'size' selector; and

iv) a 'hash' selector; and

h) a file-date attribute;"

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 10.2.5.4.2]

"When composing the SDP answer according to 3GPP TS 24.229 [5], the controlling MCData function:

1) shall include an "m=message" media-level section for the accepted MCData media stream consisting of:

a) the port number;

b) a protocol field value of "TCP/MSRP" or "TCP/TLS/MSRP" for TLS according to the received SDP offer;

c) a format list field set to '';*

d) an "a=recvonly" attribute;

e) an "a=path" attribute containing its own MSRP URI;

f) set the content type as a=accept-types:application/vnd.3gpp.mcdata-signalling; and

g) set the a=setup attribute set to "passive", according to IETF RFC 6135 [19]; and

h) a file-selector attribute containing:

i) a 'name' selector;

ii) a 'type' selector;

iii) a 'size' selector; and

iv) a 'hash' selector."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.2.4.1]

"The originating participating MCData function should establish the MSRP session with the originating MCData client as specified in subclause 7.2.2."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.2.4.2]

"The originating participating MCData function should establish the MSRP session with the controlling MCData function as specified in subclause 7.2.3."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.2.4.3]

"Upon receiving an MSRP SEND request from the originating MCData client, the originating participating MCData function:

1. if an MSRP connection is not established with the controlling MCData function then, shall establish the MSRP connection as specified in subclause 7.2.4.2. Otherwise, shall use the existing MSRP connection; and

2. shall forward the received MSRP SEND request to the controlling MCData function according to the rules and procedures of IETF RFC 4975 [11].

Upon receiving an MSRP 200 (OK) response from the controlling MCData function, the participating MCData function shall forward the MSRP 200 (OK) response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11].

Upon receiving an error MSRP response from the controlling MCData function, the participating MCData function shall forward the error MSRP response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.2.5.1]

"The terminating participating MCData function should establish the MSRP session with the controlling MCData function as specified in subclause 7.2.2."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.2.5.2]

"The terminating participating MCData function should establish MSRP session to terminating MCData client as specified in subclause 7.2.3."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.2.5.3]

"Upon receiving an MSRP SEND request from the controlling MCData function, the terminating participating MCData function:

1. *shall generate and send a MSRP 200 (OK) response for the received MSRP SEND request to the controlling MCData function, according to the rules and procedures of IETF RFC 4975 [11]; and*
2. *shall forward the received MSRP SEND request to the terminating MCData client according to the rules and procedures of IETF RFC 4975 [11].*

Upon receiving an error MSRP response from the terminating MCData client, the participating MCData function shall forward the error MSRP response to the originating MCData client according to the rules and procedures of IETF RFC 4975 [11]."

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.2.2]

"To receive a file over MSRP, the participating MCData function:

1. *shall act according to IETF RFC 6135 [12], as:*
   a. *an "passive" endpoint, if a=setup attribute in the sent SDP answer was set to "passive"; and*
   b. *an "active" endpoint, if a=setup attribute in the sent SDP answer was set to "active"; and*
2. *shall establish an MSRP connection according to the MSRP connection parameters in the sent SDP answer response as described in IETF RFC 4976 [14]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.2.3]

"To send a file over MSRP, the participating MCData function:

1. *shall act according to IETF RFC 6135 [12], as:*
   a. *an "active" endpoint, if a=setup attribute in the received SDP answer is set to "passive"; and*
   b. *an "passive" endpoint, if a=setup attribute in the received SDP answer is set to "active"; and*
2. *shall establish the MSRP connection according to the MSRP connection parameters in the received SDP answer response as described in IETF RFC 4976 [14]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.3.2.1]

"To establish the MSRP connection with the originating MCData client, the controlling MCData function performs below procedures:

1. *shall act as an MSRP client and establish TLS connection with the originating participating MCData function, if exists, otherwise the originating MCData client, according to the rules and procedures as described in IETF RFC 4975 [11];*

2. *shall act as an MSRP client to send MSRP SEND requests according to the rules and procedures described in IETF RFC 6135 [12];*

3. *shall act as an "passive" endpoint according to the rules and procedures described in IETF RFC 6135 [12];*

4. *shall establish the MSRP connection with originating MCData client, according to the rules and procedures described in IETF RFC 6135 [12]; and*

5. *acting as a "passive" endpoint, shall wait for MSRP SEND request on established MSRP connection, to bind the MSRP connection to the MSRP session according to the rules and procedures of IETF RFC 4975 [11] and IETF RFC 6135 [12]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.3.2.2]

"To establish the MSRP connection with the terminating MCData client, the controlling MCData function performs below procedures:

1. *shall act as an MSRP client and establish TLS connection with the terminating participating MCData function, if exists, otherwise the terminating MCData client, according to the rules and procedures as described in IETF RFC 4975 [11];*

2. *shall act as an MSRP client to send MSRP SEND requests according to the rules and procedures described in IETF RFC 6135 [12];*

3. *shall act according to IETF RFC 6135 [12], as:*

    a. *an "active" endpoint, if a=setup attribute in the received SDP answer is set to "passive"; and*

    b. *an "passive" endpoint, if a=setup attribute in the received SDP answer is set to "active";*

4. *shall establish the MSRP connection with each terminating MCData client identified in the 3GPP TS 24.282 [8], according to the rules and procedures described in IETF RFC 6135 [12]; and*

5. *if acting as an "active" endpoint, shall send an empty MSRP SEND request on each established MSRP connection, to bind the MSRP connection to the MSRP session according to the rules and procedures of IETF RFC 4975 [11] and IETF RFC 6135 [12]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 7.3.3]

"Upon receiving a MSRP SEND request from the originating participating MCData function, the Controlling function:

1. *shall generate and send a MSRP 200 (OK) response for the received MSRP SEND requests to the originating participating MCData function, according to the rules and procedures of IETF RFC 4975 [11]; and*

2. *shall forward the received MSRP SEND requests to each terminating MCData client with which a successful MSRP connection was established, according to the rules and procedures of IETF RFC 4975 [11]. Following clarifications apply to the generated MSRP SEND request:*

    a. *shall modify the To-Path header according to the MSRP URI received in the answer SDP from the MCData client in accordance with rules and procedures of IETF RFC 4975 [11]; and*

    b. *shall modify the From-Path header to the controlling MCData function's own MSRP URI, according to the rules and procedures of IETF RFC 4975 [11]."*

[ETSI TS 124 582 [40] (3GPP TS 24.582), clause 10.1]

"Media plane security provides integrity and confidentiality protection for the MCData media information and media plane control information transmitted using media plane. Media plane security also provides the authentication of MCData media information.

The media plane security is based on 3GPP MCData security solution including key management as defined in 3GPP TS 33.180 [15].

NOTE: In 3GPP TS 33.180 [15] media information is denoted as MCData Data Payload and media plane control information is denoted as MCData Data signalling Payload.

Various keys and associated key identifiers protect the media information and media plane control information carried in the body of an MSRP SEND message.

The media plane control information may be:

1. *SDS SIGNALLING PAYLOAD; or*
2. *SDS NOTIFICATION.*

The media information may be:

1. *DATA PAYLOAD; or*
2. *File or file portion.*

…

In an on-network one-to-one MCData communication, if protection of media is negotiated, the PCK and the PCK-ID shall be used for protecting the media sent and received by MCData clients.

If protection of media control information sent using unicast between the MCData client and the participating MCData function serving the the MCData client is negotiated, the CSK and the CSK-ID shall be used for protecting the media control information sent and received using unicast by the MCData client and by a participating MCData function.

…

The PCK and the PCK-ID are generated by the MCData client initiating the standalone one-to-one SDS using media plane or one-to-one SDS session or one-to-one FD using media plane and provided to the MCData client receiving the SIP signalling according to 3GPP TS 24.282 [8].

The CSK and the CSK-ID are generated by the MCData client and provided to the participating MCData function serving the MCData client using SIP signalling according to 3GPP TS 24.282 [8].

…

The key material for creating and verifying the authentication signature (SSK, PVT and KPAK) is provisioned to the MCData clients by the KMS as specified in 3GPP TS 33.180 [15]."

6.2.3.3 Test description

6.2.3.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).
- Test File 1 for CO FD as specified in clause A.2.1 is available at the SS-UE1 for upload.

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Storage Function, the MCData User Database, the SIP AS, the HTP Server, the HTTP Client, the MSRP TCP Server and the MSRP TCP Client.

- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The IUT (MCData Server) is connected to PLMN1.
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

6.2.3.3.2 Test procedure sequence

Table 6.2.3.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MCData Call Establishment'** as described in Annex A, Table A.4.3.3-1 **to establish a media plane connection between SS-UE1 (MCData Client) and SS-UE2 (MCData Client)?**	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MSRP message transfer'** as described in Annex A, Table A.4.4.3-1 **to send a FD message containing test file 1 from SS-UE1 (MCData client) and with disposition request "FILE DOWNLOAD COMPLETED UPDATE"?**	-	-	2	P
3	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E MCData Call Release'** as described in Annex A, Table A.4.5.3-1?	-	-	3	P
4	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a disposition notification of "FILE DOWNLOAD COMPLETED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	4	P

6.2.3.3.3 Specific message contents

**Table 6.2.3.3.3-1: SIP INVITE from the SS-UE1
(Step 1, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.4.3.3-1)**

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.9.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [23].		
Max-Forwards				
value	"70"	The recommended initial value is 70 in IETF RFC 3261 [21].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.3.3.3-2			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.9.3.3-3			
MIME body part		**MCData Data signalling message**		
MIME-part-headers				
MIME-Content-Type	"application/vnd.3gpp.mcdata-signalling"			
MIME-part-body	MCData Protected Payload Message containing FD SIGNALLING PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.9.3.3-3A			

Table 6.2.3.3.3-2: SDP for SIP INVITE (Table 6.2.3.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.9.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line		
Connection Data	not present	c= line		
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	port number assigned by the SS-UE1	The transport port to which the media stream is sent		
Connection Data		c= line		
connection-address	IP address of the SS-UE1			

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.9.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	not present			
host	IP address of the SS-UE1			
port	same value as in the media line			
session id	assigned by the SS-UE1			
transport	"tcp"			
URI-parameter	not present			
media attribute		a= line attribute = key-mgmt		
key-mgmt				
mikey	MIKEY-SAKKE I_MESSAGE as specified in Table 6.2.3.3.3-2A			
media attribute		a= line attribute = file-transfer-id	IETF RFC 5547 [25]	
file-transfer-id	value assigned by the SS-UE1	randomly chosen globally unique identification (IETF RFC 5547 [25])		
media attribute		a= line attribute = file-selector	IETF RFC 5547 [25]	
file-selector				
selector[1]				
filename	name of the file to be transferred	e.g. "TestFile.txt"		
filesize	size of the file to be transferred			
filetype	type of the file to be transferred	e.g. "text/plain"		
hash				
algorithm	"sha-1"			
value	hash value of the file to be transferred			
media attribute		a= line attribute = file-date	IETF RFC 5547 [25]	
file-date				
date-param[1]				
type	"creation"			
date-time	date and time when the file has been created	e.g. "Mon, 20 Dec 2021 15:01:31 +0100"	IETF RFC 5322 [26]	

Table 6.2.3.3.3-2A: MIKEY-SAKKE I_MESSAGE (Table 6.2.3.3.3-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
IDRi payload				
ID data	px_MCPTT_ID_User_A	MCPTT ID associated with the initiating user		
IDRr payload				
ID data	px_MCPTT_ID_User_B	MCPTT ID associated to the receiving user		

Table 6.2.3.3.3-3: SIP INVITE from the MCData Server (IUT) to the SS-UE2 (Step 1, Table 6.2.3.3.2-1; Step 2, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
To				
addr-spec				
user-info and host	px_MCX_SIP_PublicUserId_B			
Max-Forwards				
value	any allowed value			
Session-Expires				
generic-param	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.3.3.3-4			
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.2.3.3.3-5			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing FD SIGNALLING PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-3A			

Table 6.2.3.3.3-4: SDP for SIP INVITE (Table 6.2.3.3.3-4)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	any allowed value	"-" indicating the concept of user IDs not being supported		
sess-id	any allowed value	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	any allowed value			
Session Name	any allowed value	s= line		
Connection Data	not required if included in all media	c= line		
nettype	"IN"			
Addrtype	"IP4" or "IP6" depending on IP address"			
connection-address	IP address of the MCData Server			
Session attribute	present only if there is no key-mgmt media attribute in the media description for data	a= line attribute = key-mgmt		
key-mgmt				
mikey	MIKEY-SAKKE I_MESSAGE as specified in Table 6.2.3.3.3-2A			
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	any allowed value	The transport port to which the media stream is sent		

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCData Server			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	any value if present			
host	any allowed value			
port	same value as in the media line if present			
session id	any allowed value if present			
transport	"tcp"			
URI-parameter	not present			
media attribute	present only if there is no key-mgmt attribute at session level			
key-mgmt				
mikey	MIKEY-SAKKE I_MESSAGE as specified in Table 6.2.3.3.3-2A			

Table 6.2.3.3.3-5: MCData-Info (Table 6.2.3.3.3-3)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
request-type	"one-to-one-fd"			

Table 6.2.3.3.3-6: SIP 200 (OK) from the SS-UE2 (Step 1, Table 6.2.3.3.2-1; Step 4, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.3.3.3-7			

Table 6.2.3.3.3-7: SDP for SIP 200 (OK) (Table 6.2.3.3.3-6)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"22222222"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"22222222"			
Session Name	" "	s= line		
Connection Data	not present	c= line		
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	port number assigned by the SS-UE2	The transport port to which the media stream is sent		
Connection Data		c= line		
connection-address	IP address of the SS-UE2			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	not present			
host	IP address of the SS-UE2			
port	same value as in the media line			
session id	assigned by the SS-UE2			
transport	"tcp"			
URI-parameter	not present			

Table 6.2.3.3.3-8: SIP 200 (OK) from the MCData Server (IUT) to SS-UE1 (Step 1, Table 6.2.3.3.2-1; Step 5, Annex A, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.9.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table 6.2.3.3.3-9			

Table 6.2.3.3.3-9: SDP for SIP 200 (OK) (Table 6.2.3.3.3-8)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.9.3.3-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Origin		o= line		
username	any allowed value	"-" indicating the concept of user IDs not being supported		
sess-id	any allowed value	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	any allowed value			
Session Name	any allowed value	s= line		
Connection Data	not required if included in all media	c= line		
nettype	"IN"			
Addrtype	"IP4" or "IP6" depending on IP address"			
connection-address	IP address of the MCData Server			
Media description[1]		**Media description for data**		
media description		m= line media = message		
port	any allowed value	The transport port to which the media stream is sent		
Connection Data	present if session description does not contain a c=line; optional otherwise	c= line		
connection-address	IP address of the MCData Server			
media attribute		a= line attribute = path		
path	MSRP URI according to IETF RFC 4975 [24] clauses 6 and 9			
scheme	"msrp"			
authority				
userinfo	any value if present			
host	any allowed value			
port	same value as in the media line if present			
session id	any allowed value if present			
transport	"tcp"			
URI-parameter	not present			

Table 6.2.3.3.3-10: MSRP SEND from the SS-UE1
(Step 2, Table 6.2.3.3.2-1; Step 1, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.9.3.3-6

Table 6.2.3.3.3-11: MSRP SEND from the MCData Server (IUT) to SS-UE2
(Step 2, Table 6.2.3.3.2-1; Step 3, Annex A, Table A.4.4.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-6

Table 6.2.3.3.3-12: SIP MESSAGE from the SS-UE2 (Step 4, Table 6.1.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-11				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 6.2.3.3.3-5			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing FD NOTIFICATION as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.10.3.3-12			

Table 6.2.3.3.3-13: Resource-lists (Table 6.2.3.3.3-12)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.2.3.3.3-14: SIP MESSAGE from the MCData Server (IUT) to SS-UE1 (Step 4, Table 6.1.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.9.3.3-13				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

6.2.4 MCDATA Server - MCDATA Client / File Distribution (FD) / Accessing list of deferred data group communications

6.2.4.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { SS-UE1 (MCData Client) requests discovery of the absolute URI by sending a SIP MESSAGE
message }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 200 (Ok) to UE1 (MCData Client) }
            }
```

(2)

```
with { IUT (MCData Server) having requested an FD }
ensure that {
  when { the SS-UE1 (MCData Client) initiates FD "file 1" upload by sending a HTTP POST request }
    then { IUT (MCData Server) responds with a HTTP 201 (Created) }
           }
```

(3)

```
with { IUT (MCData Server) having requested an FD }
ensure that {
  when { SS-UE1 (MCData Client) requests FD upload by sending a SIP MESSAGE message with disposition
request "FILE DOWNLOAD COMPLETED" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE2 (MCData Client) and responds with
a SIP 202 (Accepted) to UE1 (MCData Client) }
           }
```

(4)

```
with { IUT (MCData Server) having requested an FD }
ensure that {
  when { SS-UE2 (MCData Client) requests FD upload by sending a SIP MESSAGE message with disposition
notification "FILE DOWNLOAD DEFERRED" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
           }
```

(5)

```
with { IUT (MCData Server) having requested an FD }
ensure that {
  when { SS-UE1 (MCData Client) requests the list of deferred data group communications by sending a
SIP MESSAGE message }
    then { IUT (MCData Server) responds with a SIP 200 (Ok) to UE1 (MCData Client) and sends a SIP
MESSAGE message to UE1 (MCData Client) containing the list of deferred data group communications }
           }
```

6.2.4.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282), clauses 11.3.1, 11.3.3.1, 11.3.3.2, 6.3.2.1, 15.1.12.1, 11.3.2.1, 6.2.4.1, 15.1.11.1. Unless otherwise stated, these are Rel-14 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 11.3.1]

"Accessing list of deferred data group communication allows a MCData user to request for the list of files that have been deferred for future download. The procedures are applicable for FD using HTTP."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 11.3.3.1]

"Upon receipt of a "SIP MESSAGE request for the list of deferred group communications", the participating MCData function:

1) shall generate a SIP 200 (OK) response according to 3GPP TS 24.229 [5];

2) shall send SIP 200 (OK) response towards MCData server according to 3GPP TS 24.229 [5]; and

3) shall follow the procedure described in subclause 11.3.3.2 to send response."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 11.3.3.2]

"To send the list of deferred group communications, the participating MCData function:

1) shall build the SIP MESSAGE request as specified in subclause 6.3.2.1;

2) shall generate DEFERRED DATA RESPONSE message as specified in subclause 15.1.12.1;

3) shall include in the SIP request, the DEFERRED DATA RESPONSE message in an application/vnd.3gpp.mcdata-signalling MIME body as specified in subclause E.1; and

4) shall send the SIP MESSAGE request towards the participating MCData function according to rules and procedures of 3GPP TS 24.229 [5].

When generating a DEFERRED DATA RESPONSE message as specified in subclause 15.1.12, the MCData client:

1) shall set the number of payloads IE to the number of FD using HTTP communication which are deferred as per the stored file list:

a) for each deferred file from the list, shall copy the payload IE value from the stored list to the payload IE value of the outgoing message being generated;"

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 6.3.2.1]

"This subclause is referenced from other procedures.

The participating MCData function shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6] and:

1) shall include in the SIP MESSAGE request all Accept-Contact header fields and all Reject-Contact header fields, with their feature tags and their corresponding values along with parameters according to rules and procedures of IETF RFC 3841 [8] that were received (if any) in the incoming SIP MESSAGE request;

2) shall set the Request-URI of the outgoing SIP MESSAGE request to the public user identity associated to the MCData ID of the terminating MCData user;

3) shall populate the outgoing SIP MESSAGE request MIME bodies as specified in subclause 6.4 and

4) shall copy the contents of the P-Asserted-Identity header field of the incoming SIP MESSAGE request to the P-Asserted-Identity header field of the outgoing SIP MESSAGE request."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 15.1.12.1]

"This message is sent by the MCData server to the MCData UE as response to the list of deferred group communications request from the MCData UE.

Message type: DEFERRED DATA RESPONSE

Direction: Server to UE

Table 15.1.12.1-1: DEFERRED DATA RESPONSE message content

IEI	Information Element	Type/Reference	Presence	Format	Length
	Deferred data response message identity	Message type 15.2.2	M	V	1
	Number of payloads	Number of payloads 15.2.12	M	V	1
7A	Security parameters and Payload	MCData Protected Payload message 3GPP TS 33.180 [26]	O	TLV-E	32-x
78	Payload	Payload 15.2.13	O	TLV-E	3-x

"

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 11.3.2.1]

"Upon receiving a request from the MCData user to access the list of deferred data group communications, the MCData client:

1) shall build the SIP MESSAGE request as specified in subclause 6.2.4.1;

2) shall generate DEFERRED DATA REQUEST message as specified in subclause 15.1.11.1;

3) shall include in the SIP request, the DEFERRED DATA GROUP COMM message in an application/vnd.3gpp.mcdata-signalling MIME body as specified in subclause E.1; and

4) shall send the SIP MESSAGE request towards the participating MCData function according to rules and procedures of 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 6.2.4.1]

"This subclause is referenced from other procedures.

In a SIP MESSAGE request, the MCData client:

…

2) when sending FD messages, FD disposition notifications or FD media storage function discovery messages:

a) shall include an Accept-Contact header field containing the g.3gpp.mcdata.fd media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8];

b) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref containing the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8]; and

c) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.fd" (coded as specified in 3GPP TS 24.229 [5]), in a P-Preferred-Service header field according to IETF RFC 6050 [7] in the SIP MESSAGE request;

3) may include a P-Preferred-Identity header field in the SIP MESSAGE request containing a public user identity as specified in 3GPP TS 24.229 [5]; and

4) shall set the Request-URI to the public service identity identifying the participating MCData function serving the MCData user."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 15.1.11.1]

"This message is sent by the MCData UE to MCData server to request the list of group communications which was deferred by the MCData user.

Message type: DEFERRED DATA REQUEST

Direction: UE to server

Table 15.1.11.1-1: DEFERRED DATA REQUEST message content

IEI	*Information Element*	*Type/Reference*	*Presence*	*Format*	*Length*
	Deferred data request message identity	*Message type 15.2.2*	*M*	*V*	*1*

"

6.2.4.3 Test description

6.2.4.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).
- Test File 1 for CO FD as specified in clause A.2.1 is available at the SS-UE1 for upload.

IUT:

- IUT (MCData Server).

- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Storage Function, the MCData User Database, the SIP AS, the HTP Server, the HTTP Client, the MSRP TCP Server and the MSRP TCP Client.

- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- In the MCData Group Configuration document the <mcdata-on-network-max-data-size-auto-recv> shall be set to 0 to indicate non-mandatory download independent from the file size.

- The IUT (MCData Server) is connected to PLMN1.

- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

- In the MCData Group Configuration document the <mcdata-on-network-max-data-size-auto-recv> shall be set to 0 to indicate non-mandatory download independent from the file size.

- SS-UE1 (MCData client) and SS-UE2 (MCData client) are affiliated to the Prearranged group.

- The IUT (MCData Server) is the controlling server of the Prearranged group.

6.2.4.3.2 Test procedure sequence

Table 6.2.4.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure **'Discovery of the absolute URI of the media storage function'** as described in Annex A, Table A.4.10.3-1?	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform procedure **'FD File upload using HTTP'** as described in Annex A, Table A.4.8.3-1, to send test file 1 (ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), clause A.2.1)?	-	-	2	P
3	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a group FD SIGNALLING PAYLOAD message for non mandatory download, containing the URL of the file and with disposition request "FILE DOWNLOAD COMPLETED UPDATE"?**	-	-	3	P
4	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a FD disposition notification of "FILE DOWNLOAD DEFERRED" from SS-UE2 (MCData client), for the file sent at step 2?**	-	-	4	P
5	The SS-UE1 (MCData Client) sends a SIP MESSAGE **to request the list of deferred data group communications**.	-->	SIP MESSAGE	-	-
6	Check: Does the MCData Server (IUT) respond with a SIP 200 (OK) message?	<--	SIP 200 (OK)	5	P
7	Check: Does the MCData Server (IUT) send a SIP MESSAGE **containing the list of deferred data group communications?**	<--	SIP MESSAGE	5	P
8	The SS-UE1 (MCData Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-

6.2.4.3.3 Specific message contents

Table 6.2.4.3.3-1: HTTP POST from the SS-UE1
(Step 2, Table 6.2.4.3.2-1; Step 1, Annex A, Table A.4.8.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.13.3.3-7

Table 6.2.4.3.3-2: HTTP 201 Created from the MCData Server (IUT) to the SS-UE1
(Step 2, Table 6.2.4.3.2-1; Step 3, Annex A, Table A.4.8.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.13.3.3-10

Table 6.2.4.3.3-3: SIP MESSAGE from the SS-UE1
(Step 3, Table 6.2.4.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.13.3.3-11

Table 6.2.4.3.3-4: SIP MESSAGE from the MCData Server (IUT) to the SS-UE2
(Step 3, Table 6.2.4.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.2-1, condition MCDATA_FD, MCDATA_SIGNALLING				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [21] IETF RFC 5031 [22]	
Request-URI	px_MCX_SIP_PublicUserId_B			
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.1	
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.2.4.3.3-5			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing FD SIGNALLING PAYLOAD as described in Table 6.2.4.3.3-6			

Table 6.2.4.3.3-5: MCData-Info (Table 6.2.4.3.3-4)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 5.5.3.2.2-3, condition MCD_grp				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
request-type	"group-fd"			
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		
mcdata-client-id	Encrypted <mcdata-client-id> with mcdataString set to px_MCX_Client_A_ID	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.2-3A		

Table 6.2.4.3.3-6: FD SIGNALLING PAYLOAD (Table 6.2.4.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.8.5-1, condition FD_HTTP

Table 6.2.4.3.3-7: SIP MESSAGE from the SS-UE2 (Step 4, Table 6.2.4.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.7.1-1, condition MCDATA_FD, RESOURCE_LISTS, MCDATA_SIGNALLING				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 6.2.4.3.3-8			
MIME body part		**MCData-Info**		
MIME-part-headers	MCData-Info as described in Table 6.2.4.3.3-9			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing FD NOTIFICATION as described in Table 6.2.4.3.3-10			

Table 6.2.4.3.3-8: Resource-lists (Table 6.2.4.3.3-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.2.4.3.3-9: MCData-Info (Table 6.2.4.3.3-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-calling-group-id	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_Group_A_ID	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1) Table 5.5.3.2.1-3A		

Table 6.2.4.3.3-10: FD NOTIFICATION (Table 6.2.4.3.3-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.8.7-1, condition FD_DEFERRED

Table 6.2.4.3.3-11: SIP MESSAGE from the MCData Server (IUT) to the SS-UE1 (Step 4, Table 6.2.4.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.13.3.3-13				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

Table 6.2.4.3.3-12: SIP MESSAGE from the SS-UE1 (Step 5, Table 6.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.13.3.3-16

Table 6.2.4.3.3-13: SIP MESSAGE from the MCData Server (IUT) to the SS-UE1 (Step 7, Table 6.2.4.3.2-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.2.13.3.3-18				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

6.3 Enhanced Status (ES)

6.3.1 MCDATA Server - MCDATA Client / Enhanced Status (ES)

6.3.1.1 Test Purpose (TP)

(1)

```
with { IUT (MCData Server) connected to PLMN1 }
ensure that {
  when { SS-UE1 (MCData Client) requests one-to-one Standalone SDS by sending a SIP MESSAGE message
with disposition request "DELIVERY" and an enhanced status payload with id = 0}
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE2 (MCData Client) and responds with
a SIP 202 (Accepted) to UE1 (MCData Client) }
            }
```

(2)

```
with { IUT (MCData Server) having requested Enhanced Status }
ensure that {
  when { SS-UE2 (MCData Client) requests one-to-one Standalone SDS by sending a SIP MESSAGE message
with disposition notification "DELIVERED" }
    then { IUT (MCData Server) sends a SIP MESSAGE message to UE1 (MCData Client) and responds with
a SIP 202 (Accepted) to UE2 (MCData Client) }
          }
```

6.3.1.2 Conformance requirements

References: The conformance requirements covered in the current TC are specified in: ETSI TS 124 282 [13] (3GPP TS 24.282), clauses 14.2.2.1, 14.2.2.2, 14.2.3.1, 14.2.3.2, 9.2.2.3.1, 9.2.2.3.2, 9.2.2.4.1, 9.2.2.4.2. Unless otherwise stated, these are Rel-15 requirements.

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 14.2.2.1]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for originating participating MCData function", the participating MCData function should follow the procedure described in subclause 9.2.2.3.1."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 14.2.2.2]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for terminating participating MCData function", the participating MCData function should follow the procedure described in subclause 9.2.2.3.2."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 14.2.3.1]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for controlling MCData function", the controlling MCData function should follow the procedure described in subclause 9.2.2.4.1."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 14.2.3.2]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for controlling MCData function", the controlling MCData function should follow the procedure described in subclause 9.2.2.4.2."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.3.1]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for originating participating MCData function", the participating MCData function:

…

2) shall determine the MCData ID of the originating user from the public user identity in the P-Asserted-Identity header field of the SIP MESSAGE request, and shall authorise the calling user;

NOTE: The MCData ID of the calling user is bound to the public user identity at the time of service authorisation, as documented in subclause 7.3.

…

4) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request is:

a) set to a value of "group-sds", shall determine the public service identity of the controlling MCData function associated with the MCData group identity in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the SIP MESSAGE request; or

b) set to a value of "one-to-one-sds", shall determine the public service identity of the controlling MCData function hosting the one-to-one standalone SDS service for the calling user;

…

6) shall determine whether the MCData user identified by the MCData ID is authorised for MCData communications by following the procedures in subclause 11.1;

…

9) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];

10) shall set the Request-URI of the outgoing SIP MESSAGE request to the public service identity of the controlling MCData function as determined by step 4) in this subclause;

11) shall copy all MIME bodies included in the incoming SIP MESSAGE request to the outgoing SIP MESSAGE request;

12) shall include the MCData ID of the originating user in the <mcdata-calling-user-id> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the outgoing SIP MESSAGE request;

12A) if the incoming SIP MESSAGE request contains an application/vnd.3gpp.mcdata-info+xml MIME body that contains a <functional-alias-URI> element, shall check if the status of the functional alias is activated for the MCData ID. If the functional alias status is activated, then the participating MCData function shall set the <functional-alias-URI> element of the application/vnd.3gpp.mcdata-info+xml MIME body in the outgoing SIP INVITE request to the received value, otherwise shall not include a <functional-alias-URI> element;

13) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request;

14) shall set the P-Asserted-Identity in the outgoing SIP MESSAGE request to the public user identity in the P-Asserted-Identity header field contained in the received SIP MESSAGE request; and

15) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [5].

Upon receipt of a SIP 202 (Accepted) response in response to the SIP MESSAGE request in step 15):

1) shall generate a SIP 202 (Accepted) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 202 (Accepted) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the SIP MESSAGE request in step 15):

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 200 (OK) response to the MCData client according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the SIP MESSAGE request in step 15) the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the MCData client according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.3.2]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for terminating participating MCData function", the participating MCData function:

…

2) shall use the MCData ID present in the <mcdata-request-uri> element of the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request to retrieve the binding between the MCData ID and public user identity of the terminating MCData user;

…

4) shall generate an outgoing SIP MESSAGE request as specified in subclause 6.3.2.1;

5) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" (coded as specified in 3GPP TS 24.229 [5]), into the P-Asserted-Service header field of the outgoing SIP MESSAGE request; and

6) shall send the SIP MESSAGE request as specified in 3GPP TS 24.229 [5].

Upon receipt of a SIP 200 (OK) response in response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP 200 (OK) response as specified in 3GPP TS 24.229 [5]; and

2) shall send the SIP 200 (OK) response to the controlling MCData function according to 3GPP TS 24.229 [5].

Upon receipt of a SIP 4xx, 5xx or 6xx response to the above SIP MESSAGE request, the participating MCData function:

1) shall generate a SIP response according to 3GPP TS 24.229 [5];

2) shall include Warning header field(s) that were received in the incoming SIP response; and

3) shall forward the SIP response to the controlling MCData function according to 3GPP TS 24.229 [5]."

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.1]

"This subclause describes the procedures for sending a SIP MESSAGE from the controlling MCData function and is initiated by the controlling MCData function as a result of an action in subclause 9.2.2.4.2.

The controlling MCData function:

1) shall generate a SIP MESSAGE request in accordance with 3GPP TS 24.229 [5] and IETF RFC 3428 [6];

2) shall include an Accept-Contact header field containing the g.3gpp.mcdata.sds media feature tag along with the "require" and "explicit" header field parameters according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

3) shall include an Accept-Contact header field with the media feature tag g.3gpp.icsi-ref with the value of "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds" along with parameters "require" and "explicit" according to IETF RFC 3841 [8] in the outgoing SIP MESSAGE request;

4) shall copy the following MIME bodies in the received SIP MESSAGE request into the outgoing SIP MESSAGE request by following the guidelines in subclause 6.4:

   a) application/vnd.3gpp.mcdata-info+xml MIME body;

   b) application/vnd.3gpp.mcdata-signalling MIME body; and

   c) application/vnd.3gpp.mcdata-payload MIME body

5) in the application/vnd.3gpp.mcdata-info+xml MIME body:

   a) shall set the <mcdata-request-uri> element set to the MCData ID of the terminating user; and

   b) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the incoming SIP MESSAGE request was set to a value of "group-sds", shall set the <mcdata-calling-group-id> element to the group identity;

6) shall set the Request-URI to the public service identity of the terminating participating MCData function associated to the MCData user to be invited;

7) shall copy the public user identity of the calling MCData user from the P-Asserted-Identity header field of the incoming SIP MESSAGE request into the P-Asserted-Identity header field of the outgoing SIP MESSAGE request;

8) shall include a P-Asserted-Service header field with the value "urn:urn-7:3gpp-service.ims.icsi.mcdata.sds"; and

9) shall send the SIP MESSAGE request according to according to rules and procedures of 3GPP TS 24.229 [5].

[ETSI TS 124 282 [13] (3GPP TS 24.282), clause 9.2.2.4.2]

"Upon receipt of a "SIP MESSAGE request for standalone SDS for controlling MCData function", the controlling MCData function:

…

2) if the SIP MESSAGE does not contain:

a) an application/vnd.3gpp.mcdata-info+xml MIME body;

b) an application/vnd.3gpp.mcdata-signalling MIME body; and

c) an application/vnd.3gpp.mcdata-payload MIME body;

shall reject the SIP MESSAGE request …, and shall not continue with the rest of the steps in this subclause;

3) shall decode the contents of the application/vnd.3gpp.mcdata-signalling MIME body contained in the SIP MESSAGE;

4) if the application/vnd.3gpp.mcdata-signalling MIME body contains a SDS SIGNALLING PAYLOAD message with a SDS disposition request type IE, shall store the value of the Conversation ID IE and the value of the Message ID IE in the SDS SIGNALLING PAYLOAD message;

NOTE: The controlling MCData function uses the Conversation ID and Message ID for correlation with disposition notifications.

5) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request is set to a value of "one-to-one-sds" and:

…

b) the SIP MESSAGE request:

…

ii) contains an application/resource-lists MIME body with exactly one <entry> element, shall send a SIP MESSAGE request to the MCData user identified in the <entry> element of the MIME body, as specified in subclause 9.2.2.4.1;

6) if the <request-type> element in the application/vnd.3gpp.mcdata-info+xml MIME body of the SIP MESSAGE request is set to a value of "group-sds":

a) shall retrieve the group document associated with the group identity in the SIP MESSAGE request by following the procedures in subclause 6.3.3, and shall continue with the remaining steps if the procedures in subclause 6.3.3 were successful;

…

h) shall determine targeted group members for MCData communications by following the procedures in subclause 6.3.4;

j) if the procedures in subclause 6.3.4 result in no affiliated members found in the selected MCData group, shall return a SIP 403 (Forbidden) response with the warning text set to "198 no users are affiliated to this group" in a Warning header field as specified in subclause 4.9, and skip the rest of the steps below; and

k) shall send SIP MESSAGE requests to the targeted group members identified in step h) above by following the procedure in subclause 9.2.2.4.1;

7) shall generate a SIP 202 (Accepted) response in response to the "SIP MESSAGE request for standalone SDS for controlling MCData function"; and

8) shall send the SIP 202 (Accepted) response towards the originating participating MCData function according to 3GPP TS 24.229 [5]."

6.3.1.3 Test description

6.3.1.3.1 Pre-test conditions

System Simulator:

- SS-UE1 (MCData client) and SS-UE2 (MCData Client).
- For the underlying "transport bearer" over which the SS-UE1 (MCData client), the SS-UE2 (MCData client) and the MCData Server will communicate, Parameters are set to the default parameters for the basic E-UTRA Single cell network scenarios, as defined in ETSI TS 136 508 [10] (3GPP TS 36.508) clause 4.4. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCData operation in the MCData configuration document).

IUT:

- IUT (MCData Server).
- The IUT (MCData Server) consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCData Server) also consists of all sub-systems of the MCData Server, including the Media Storage Function, the MCData User Database, the SIP AS, the HTP Server, the HTTP Client, the MSRP TCP Server and the MSRP TCP Client.
- The IUT (MCData Server) is the acting Participating Server and Controlling Server.

Preamble:

- The <max-payload-size-sds-cplane-bytes> element in the MCData service configuration document as specified in 3GPP ETSI TS 124 484 [8] (3GPP TS 24.484), shall be large enough to allow the sending of the standalone SDS message using the signalling plane.
- The IUT (MCData Server) is connected to PLMN1.
- The SS-UE1 (MCData client) and the SS-UE2 (MCData client) perform the Procedure for MCData Authorization, Configuration and Key Generation as specified in clause 5.1.

6.3.1.3.2 Test procedure sequence

Table 6.3.1.3.2-1: Main Behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward an Enhanced Status from SS-UE1 (MCData client) to SS-UE2 (MCData Client), with disposition request "DELIVERY" and an enhanced status payload with id=0 ("going to the operation site")?**	-	-	1	P
2	Check: Does the MCData Server (IUT) correctly perform procedure **'E2E SDS or FD message transfer using signalling plane'** as described in Annex A, Table A.4.1.3-1 **to forward a disposition notification of "DELIVERED", from SS-UE2 (MCData client) to SS-UE1 (MCData Client)?**	-	-	2	P

6.3.1.3.3 Specific message contents

Table 6.3.1.3.3-1: SIP MESSAGE from the SS-UE1 (Step 1, Table 6.3.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.3.1.3.3-1

Table 6.3.1.3.3-2: SIP MESSAGE from the MCData Server (IUT) to SS-UE2 (Step 1, Table 6.3.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.3.2.3.3-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line			IETF RFC 3261 [21] IETF RFC 5031 [22]	
Request-URI	px_MCX_SIP_PublicUserId_B			
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.		
Message-body				
MIME body part	not present	**Resource lists**		
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table 6.3.1.3.3-3			
MIME body part		**MCData Data signalling message**		
MIME-part-body	MCData Protected Payload Message containing SDS SIGNALLING PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.3.2.3.3-3			
MIME body part		**MCData Data message**		
MIME-part-body	DATA PAYLOAD as described in ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.3.2.3.3-4			

Table 6.3.1.3.3-3: MCData-Info (Table 6.3.1.3.3-2)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.3.2.3.3-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	Encrypted <mcdata-request-uri> with mcdataURI set to px_MCData_ID_User_B			
mcdata-calling-user-id	Encrypted <mcdata-calling-user-id> with mcdataURI set to px_MCData_ID_User_A			

Table 6.3.1.3.3-4: SIP MESSAGE from the SS-UE2 (Step 2, Table 6.3.1.3.2-1; Step 1, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.3.2.3.3-6				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
From			IETF RFC 3261 [21]	
addr-spec				
user-info and host	Default public user id (px_MCX_SIP_PublicUserId_B)	The URI of the SS-UE2		
Message-body				
MIME body part		**Resource lists**		
MIME-part-body	Resource-lists as described in Table 6.3.1.3.3-5			

Table 6.3.1.3.3-5: Resource-lists (Table 6.3.1.3.3-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
resource-lists				
list				
name attribute				
display-name				
entry[1]				
uri attribute	px_MCData_ID_User_A	The MCData ID of the target MCData user		

Table 6.3.1.3.3-6: SIP MESSAGE from the MCData Server (IUT) to SS-UE1 (Step 2, Table 6.1.1.3.2-1; Step 3, Annex A, Table A.4.1.3-1)

Derivation Path: ETSI TS 136 579-7 [4] (3GPP TS 36.579-7), Table 6.3.1.3.3-4				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
P-Asserted-Identity			IETF RFC 3261 [21]	
name-addr	tsc_MCData_PublicServiceId_A	The public service identity identifying the controlling MCData function.	ETSI TS 124 282 [13] (3GPP TS 24.282), clause 12.2.3	

Annex A (normative): Common Test Environment

A.1 Introduction

This annex specifies additional generic test procedures to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), to be used by the test cases wherever applicable.

A.2 Generic test procedures for Server-Client MCS operation

A.2.1 General

The purpose of the procedures specified in the following clauses is to facilitate test description by providing procedure sequences which can be referred from the relevant server-to-client TCs specified in ETSI TS 136 579-3 [2] (3GPP TS 36.579-3), 3GPP TS 36.579-8 [31] and the present document.

The procedures specified are required to ensure that any MC service can take place or specific MC relevant pre-conditions are met before a test case can be executed.

A.2.2 Void

A.2.3 MCX Server Originated notifications regarding temporary group creation or tear down

A.2.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.2.3.2 Definition of system information messages

N/A.

A.2.3.3 Procedure

Table A.2.3.3-1: MCX Server originated notifications regarding temporary group creation or tear down

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP NOTIFY to the SS-UE informing about change of group A's configuration document?	<--	SIP NOTIFY	-	P
2	The SS-UE (MCX Client) sends a SIP 200 (OK) message to the MCX Server.	-->	SIP 200 (OK)	-	-
3	The SS-UE (MCX Client) sends an HTTP GET Request message to the MCX Server that contains the access token and the XCAP-URI of the Group Configuration document.	-->	HTTP GET	-	-
4	Check: Does the MCX Server send a HTTP 200 (OK) message including the updated Group Document?	<--	HTTP 200 (OK)	-	P
5	Check: Does the MCX Server send a SIP NOTIFY message to the SS-UE (MCX Client) containing the group key transport payloads (GKTP) document including the group keys?	<-	SIP NOTIFY	-	P
6	The SS-UE (MCX Client) sends a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.3.4 Specific message contents

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).
- Clarifications specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.3.22.4 shall also be followed:

Table A.2.3.4-1: SIP NOTIFY (Step 1, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-1

Table A.2.3.4-2: SIP 200 (OK) (Steps 2, 6, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-2

Table A.2.3.4-3: HTTP GET (Step 3, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-3

7.3.4-4: HTTP 200 (OK) (Step 4, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-4

Table A.2.3.4-5: SIP NOTIFY (Step 5, Table A.2.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.22.4-6

A.2.4 MCX Server Terminated Group Creation

A.2.4.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.4.2 Definition of system information messages

N/A.

A.2.4.3 Procedure

Table A.2.4.3-1: MCX Server Terminated Group Creation procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP PUT to the Group Management Server for requesting the creation of the new group. (Note)	-->	HTTP PUT	-	-
2	Check: Does the MCX Server respond with a HTTP 201 (Created) response message?	<--	HTTP 201 (Created)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.4.4 Specific message contents

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.4.4-1: HTTP PUT (Step 1, Table A.2.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.4-1, condition GROUPCREATE

A.2.5 MCX Server Terminated Temporary Group Creation

A.2.5.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.5.2 Definition of system information messages

N/A.

A.2.5.3 Procedure

A.2.5.3-1: MCX Participating Terminated Temporary Group Creation procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP POST to the MCX Server to request for creation of a temporary group. (Note)	-->	HTTP POST	-	-
2	Check: Does the MCX Server send a HTTP 200 (OK) containing the GMOP group-regroup-creation-response?	<--	HTTP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.5.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.5.4-1: HTTP POST (Step 1, Table A.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.3-1, condition TEMPGROUP

Table A.2.5.4-2: HTTP 200 (OK) (Step 2, Table A.2.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.6-1, condition TEMPGROUP

A.2.6 MCX Server Terminated Temporary Group Tear Down

A.2.6.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.6.2 Definition of system information messages

N/A.

A.2.6.3 Procedure

Table A.2.6.3-1: MCX Server Terminated Temporary Group Tear Down procedure

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a HTTP DELETE to the MCX Server to request for tear down of a temporary group. (Note)	-->	HTTP DELETE	-	-
2	Check: Does the MCX Server send a HTTP 200 (OK) response message?	<--	HTTP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.6.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.6.4-1: HTTP DELETE (Step 1, Table A.2.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.4.5-1, condition TEMPGROUP

A.2.7 MCX Subscription and Notification

A.2.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.7.2 Definition of system information messages

N/A.

A.2.7.3 Procedure

Table A.2.7.3-1: MCX Subscription and Notification

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP SUBSCRIBE message request.	-->	SIP SUBSCRIBE	-	-
2	Check: Does the MCX Server respond to the SIP SUBSCRIBE message with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCX Server send a SIP NOTIFY message?	<--	SIP NOTIFY	-	P
4	The SS-UE (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)		
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.7.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.7.4-1: SIP SUBSCRIBE (Step 1, Table A.2.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.14-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires				
value	"4294967295"	to receive the current status and later notification	ETSI TS 124 379 [5] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3; ETSI TS 124 281 [32] (3GPP TS 24.281) clause 8.2.1.3; ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
Message-body				
MIME body part		**MCPTT Info**	ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9.2.1.3 and 9A.2.1.3	MCPTT
MIME-part-body	MCPTT-Info as described in Table A.2.7.4-2			
MIME body part		**MCVIDEO Info**	ETSI TS 124 281 [32] (3GPP TS 24.281) clause 8.2.1.3	MCVIDEO
MIME-part-body	MCVIDEO-Info as described in Table A.2.7.4-3			
MIME body part		**MCDATA Info**	ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	MCDATA
MIME-part-body	MCDATA-Info as described in Table A.2.7.4-4			

Table A.2.7.4-2: MCPTT-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [5] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table A.2.7.4-3: MCVideo-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	encrypted (note) <mcvideo-request-uri> with mcvideoURI set to px_MCVIDEO_ID_User_A		ETSI TS 124 281 [32] (3GPP TS 24.281) clause 8.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

Table A.2.7.4-4: MCData-Info in SIP SUBSCRIBE (Table A.2.7.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
mcdata-request-uri	encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCDATA_ID_User_A		ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3A.				

Table A.2.7.4-5: SIP 200 (OK) (Step 2, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition SUBSCRIBE-RSP

A.2.8 MCX Participating Terminated SIP MESSAGE

A.2.8.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData

A.2.8.2 Definition of system information messages

N/A.

A.2.8.3 Procedure

Table A.2.8.3-1: MCX Participating Terminated SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) send a SIP MESSAGE message (Note)	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server responds with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.8.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.9 MCX Participating Originated SIP MESSAGE

A.2.9.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData

A.2.9.2 Definition of system information messages

N/A.

A.2.9.3 Procedure

Table A.2.9.3-1: MCX Participating Originated SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP MESSAGE message?	<--	SIP MESSAGE	-	P
2	The SS-UE (MCX Client) responds with a SIP 200 (OK) message (Note)	→	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.9.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.10 MCX E2E SIP MESSAGE

A.2.10.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo

A.2.10.2 Definition of system information messages

N/A.

A.2.10.3 Procedure

Table A.2.10.3-1: MCX E2E SIP MESSAGE

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP MESSAGE message	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server send a SIP MESSAGE to SS-UE2?	<--	SIP MESSAGE	-	P
3	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX Server respond with a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	-	P

A.2.10.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.11 MCX E2E SIP MESSAGE Request - Accept

A.2.11.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo

A.2.11.2 Definition of system information messages

N/A.

A.2.11.3 Procedure

Table A.2.11.3-1: MCX E2E SIP MESSAGE Request - Accept Participating Originated

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP MESSAGE message.	-->	SIP MESSAGE	-	-
2	Check: Does the MCX Server respond with a SIP 200 (OK) message	<--	SIP 200 (OK)	-	P
3	Check: Does the MCX server send a SIP MESSAGE accepting the request to SS-UE1?	<--	SIP MESSAGE	-	P
4	The SS-UE1 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
-	EXCEPTION: Depending on the MCX Server implementation, the following steps may be performed in parallel to steps 3-4.	-	-	-	-
5	Check: Does the MCX Server send a SIP MESSAGE message to SS-UE2?	<--	SIP MESSAGE	-	P
6	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-

A.2.11.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.12 MCX Presence Event Status Change

A.2.12.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData

A.2.12.2 Definition of system information messages

N/A.

A.2.12.3 Procedure

Table A.2.12.3-1: MCX Presence Event Status Change

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP PUBLISH message.	-->	SIP PUBLISH	-	P
2	Check: Does the MCX Server respond to the SIP PUBLISH message with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	-
3	Check: Does the MCX Server send a SIP NOTIFY message informing about the status change progress?	<--	SIP NOTIFY	-	P
4	The SS-UE (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
5	Check: Does the MCX Server send a SIP NOTIFY informing about the final status?	<--	SIP NOTIFY	-	P
6	The SS-UE responds with a SIP 200 (OK).	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.12.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.12.4-1: SIP PUBLISH (Step 1, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.11-1, condition PRESENCE-EVENT

Table A.2.12.4-2: MCPTT-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_A		ETSI TS 124 379 [5] (3GPP TS 24.379) clauses 9.2.1.3 and 9A.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1A.				

Table A.2.12.4-3: MCVideo-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcvideoinfo				
mcvideo-Params				
mcvideo-request-uri	encrypted (note) <mcvideo-request-uri> with mcvideoURI set to px_MCVIDEO_ID_User_A		ETSI TS 124 281 [32] (3GPP TS 24.281) clause 8.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-2A.				

Table A.2.12.4-4: MCData-Info in SIP PUBLISH (Table A.2.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-request-uri	encrypted (note) <mcdata-request-uri> with mcdataURI set to px_MCDATA_ID_User_A		ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 8.2.3 and 22.2.1.3	
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3A.				

Table A.2.12.4-5: SIP 200 (OK) (Step 2, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, condition PUBLISH-RSP

Table A.2.12.4-6: SIP NOTIFY (Steps 3 and 5, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, condition PRESENCE-EVENT

Table A.2.12.4-7: SIP 200 (OK) (Steps 4 and 6, Table A.2.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1

A.2.13 MCX Participating Terminated pre-established session establishment

A.2.13.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo

A.2.13.2 Definition of system information messages

N/A.

A.2.13.3 Procedure

Table A.2.13.3-1: MCX Participating Terminated pre-established session establishment

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP INVITE message in order to create a pre-established session	-->	SIP INVITE	-	-
-	EXCEPTION: Step 2a1 describes behaviour that depends on the MCX Server implementation; the "lower case letter" identifies a step sequence that take place if the MCX Server responds to a SIP INVITE with a SIP 100 (Trying)	-	-	-	-
2a1	Check: Does the MCX Server respond with a SIP 100 (Trying) message?	<--	SIP 100 Trying	-	P
3	Check: Does the MCX Server respond with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
4	The SS-UE (MCX Client) responds with a SIP ACK message.	-->	SIP ACK	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.13.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.13.4-1: SIP INVITE (Step 1, Table A.2.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.3.4-1

Table A.2.13.4-2: SIP 200 (OK) (Step 3, Table A.2.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3.3.4-2

A.2.14 MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying

A.2.14.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.14.2 Definition of system information messages

N/A.

A.2.14.3 Procedure

Table A.2.14.3-1: MCX Participating Terminated session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call (note)	-->	SIP INVITE	-	-
-	EXCEPTION: Step 2a1 describes behaviour that depends on the MCX Server implementation; the "lower case letter" identifies a step sequence that take place if the MCX Server responds to a SIP INVITE with a SIP 100 (Trying)	-	-	-	-
2a1	Check: Does the MCX Server respond with a SIP 100 Trying?	<--	SIP 100 (Trying)	-	P
3	Check: Does the MCX Server respond with a SIP 200 (OK)?	<--	SIP 200 (OK)	-	P
4	The SS-UE (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification (note)	-->	SIP ACK	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.14.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.14.4-1: SIP INVITE (Step 1, Table A.2.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (Note)
NOTE: With condition re_INVITE for **session modifications**.

Table A.2.14.4-1: SIP 200 (OK) (Step 3, Table A.2.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.15 MCX Participating Terminated call release

A.2.15.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.15.2 Definition of system information messages

N/A.

A.2.15.3 Procedure

Table A.2.15.3-1: MCX Participating Terminated call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCX Client) sends a SIP BYE request to terminate the MCX session note)	-->	SIP BYE	-	-
2	Check: Does the MCX Server respond with a SIP 200 (OK) message?	<--	SIP 200 (OK)	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.15.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.16 MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying

A.2.16.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.16.2 Definition of system information messages

N/A.

A.2.16.3 Procedure

Table A.2.16.3-1: MCX Participating Originated session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP INVITE requesting the establishment/modification of a call?	<--	SIP INVITE	-	P
2	The SS-UE (MCX Client) sends a SIP 100 Trying (note)	-->	SIP 100 (Trying)	-	-
3	The SS-UE (MCX Client) responds with a SIP 200 (OK) (note)	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification?	<--	SIP ACK	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.16.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.16.4-1: SIP INVITE (Step 1, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note)
NOTE: With condition re_INVITE for session modifications.

Table A.2.16.4-2: SIP 200 (OK) (Step 3, Table A.2.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

A.2.17 MCX Participating Originated call release

A.2.17.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.17.2 Definition of system information messages

N/A.

A.2.17.3 Procedure

Table A.2.17.3-1: MCX Participating Originated call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCX Server send a SIP BYE request to terminate the MCX session?	<--	SIP BYE	-	P
2	The SS-UE (MCX Client) responds with a SIP 200 (OK) message (note).	-->	SIP 200 (OK)	-	-
NOTE: The SS-UE being used is specified in the test case calling the procedure.					

A.2.17.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5. and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

A.2.18 MCX E2E session establishment/modification without provisional responses other than 100 Trying

A.2.18.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT or MCVideo.

A.2.18.2 Definition of system information messages

N/A.

A.2.18.3 Procedure

Table A.2.18.3-1: MCX E2E session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
6	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
7	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
8	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.18.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.18.4-1: SIP INVITE (Step 1, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (note)
NOTE: With condition re_INVITE **for session modifications**.

Table A.2.18.4-2: SIP INVITE (Step 2, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [**1**] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note)
NOTE: With condition re_INVITE **for session modifications**.

Table A.2.18.4-1: SIP 200 (OK) (Step 5, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [**1**] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.18.4-1: SIP 200 (OK) (Step 6, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.19 MCX E2E call release

A.2.19.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.19.2 Definition of system information messages

N/A.

A.2.19.3 Procedure

Table A.2.19.3-1: MCX E2E call release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP BYE request to terminate the MCX session	-->	SIP BYE	-	-
2	Check: Does the MCX Server respond a SIP BYE request to terminate the MCX session to SS-UE2 (MCX Client)?	<--	SIP BYE	-	P
3	The SS-UE2 (MCX Client) responds with a SIP 200 (OK) message.	-->	SIP 200 (OK)	-	-
4	Check: Does the MCX Server respond a SIP 200 (OK) message to SS-UE1?	<--	SIP 200 (OK)	-	P

A.2.19.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.19.4-1: SIP BYE from the SS-UE1 (Step 1, Table A.2.19.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1

Table A.2.19.4-2: SIP BYE from the MCPTT Server (Step 2, Table A.2.19.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1

A.2.20 MCX E2E group call establishment, manual commencement

A.2.20.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.20.2 Definition of system information messages

N/A.

A.2.20.3 Procedure

Table A.2.20.3-1: MCX E2E group call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) sends a SIP 183 (Session Progress) reliably.	-->	SIP 183 (Session Progress)	-	-
6	Check: Does the MCX Server acknowledge the receipt of SIP 183 (Session Progress) from SS-UE2?	<--	PRACK	-	P
7	The SS-UE2 (MCX Client) responds a SIP 200 (OK) to PRACK message	-->	SIP 200 (OK)	-	-
8	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
9	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
10	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
11	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.20.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with conditions GROUP-CALL and MANUAL; and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

For chat group calls, CHAT-GROUP-CALL condition applies instead of GROUP-CALL.

Table A.2.20.4-1: SIP 200 (OK) (Step 8, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.20.4-2: SIP 200 (OK) (Step 9, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.2.21 MCX E2E private call establishment, manual commencement

A.2.21.1 Initial conditions

As specified in the test case which calls the procedure.

Within the context of this procedure, MCX refers to MCPTT, MCVideo or MCData.

A.2.21.2 Definition of system information messages

N/A.

A.2.21.3 Procedure

Table A.2.21.3-1: MCX E2E private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCX Client) sends a SIP INVITE requesting the establishment/modification of a call	-->	SIP INVITE	-	-
2	Check: Does the MCX Server forward the SIP INVITE requesting the establishment/modification of a call to SS-UE2 (MCX Client)?	<--	SIP INVITE	-	P
3	The SS-UE2 (MCX Client) sends a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
4	Check: Does the MCX Server respond with a SIP 100 Trying to SS-UE1 (MCX Client)?	<--	SIP 100 (Trying)	-	P
5	The SS-UE2 (MCX Client) sends a SIP 180 (Ringing) unreliably.	-->	SIP 180 (Ringing)	-	-
6	Check: Does the MCX Server respond with a unreliable SIP 180 (Ringing) to SS-UE1 (MCX Client)?	<--	SIP 180 (Ringing)	-	P
7	The SS-UE2 (MCX Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
8	Check: Does the MCX Server respond with a SIP 200 (OK) to SS-UE1 (MCX Client)?	<--	SIP 200 (OK)	-	P
9	The SS-UE1 (MCX Client) sends a SIP ACK to acknowledge the session establishment/modification	-->	SIP ACK	-	-
10	Check: Does the MCX server send a SIP ACK to acknowledge the session establishment/modification to SS-UE2?	<--	SIP ACK	-	P

A.2.21.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with conditions PRIVATE-CALL and MANUAL; and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.2.21.4-1: SIP 200 (OK) (Step 7, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition INVITE-RSP

Table A.2.21.4-1: SIP 200 (OK) (Step 8, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition INVITE-RSP

A.3 Generic test procedures for MCPTT Server operation

A.3.1 MCPTT E2E session establishment/modification without provisional responses other than 100 Trying

A.3.1.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.1.2 Definition of system information messages

N/A.

A.3.1.3 Procedure

Table A.3.1.3-1: MCPTT E2E session establishment/modification without provisional responses other than 100 Trying

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure MCX E2E session establishment/modification without provisional responses other than 100 Trying' for as described in Table A.2.18.3-1 with MCPTT as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit floor control in the SIP INVITE request but it is not implicitly granted (NOTE1)	-	-	-	-
2a1	Check: Does the MCPTT server send a Floor Granted message to SS-UE1?	<--	Floor Granted	-	P
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is granted to SS-UE1, either implicit or explicitly. (Note)	-	-	-	-
3a1	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<--	Floor Taken	-	P
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is not granted to SS-UE1. (Note)	-	-	-	-
4a1	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<--	Floor Idle	-	P
4a2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<--	Floor Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:
a. SS-UE1 sends SDP offer with media description for floor control but without implicit floor request.
b. SS-UE1 sends SDP offer with media description for floor control and with implicit floor request:
i. SDP answer from MCPTT Server contains "mc_implicit_request" and "mc_granted" (Floor is implicitly granted).
ii. SDP answer from MCPTT Server contains "mc_implicit request" and but no "mc_granted" (Floor needs to be explicitly granted at step 2a1).
iii. SDP answer from MCPTT Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the floor).
c. SS-UE1 sends SDP offer without media description for floor control.

A.3.1.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCPTT and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.1.4-1: SIP INVITE
(Step 1, Table A.3.1.3-1; Step 1, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1 (note), with condition MCPTT NOTE: With condition re_INVITE for session modifications.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [23]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [23].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-2			
MIME body part		MCPTT-Info		
MIME-part-body	MCPTT-Info as described in Table A.3.1.4-3			

Table A.3.1.4-2: SDP in SIP INVITE (Table A.3.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with condition INITIAL_SDP_OFFER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [35] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [36]	
"RR"	2000		IETF RFC 3556 [36]	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with condition INITIAL_SDP_OFFER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_FLOORCONTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1) Note 3	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with condition INITIAL_SDP_OFFER.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
NOTE 1: If the SS-UE requests a call with media floor control and implicit floor (Option b of note on Table A.3.1.3-1), apply condition IMPLICIT_GRANT_REQUESTED. NOTE 2: If the SS-UE requests a call without media floor (Option c of NOTE1 on Table A.3.1.3-1), apply condition WITHOUT_FLOORCONTROL. NOTE 3: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.1.4-3: MCPTT-Info in SIP INVITE (Table A.3.1.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition INVITE_REFER

Table A.3.1.4-4: SIP INVITE from the MCPTT-Server (Step 1, Table A.3.1.3-1; Step 2 Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note), with condition MCPTT NOTE: With condition re_INVITE for session modifications.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [38] IETF RFC 7134 [20] IETF RFC 8101 [39]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 (note), with condition MCPTT NOTE: With condition re_INVITE for session modifications.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
Session-Expires			IETF RFC 4028 [23]	
delta-seconds	any allowed value			
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.1.4-6			

Table A.3.1.4-5: SDP in SIP INVITE (Table A.3.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1) Note 2	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	Any value lower or equal to the priority specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE 1: If the SS-UE1 requests a call without media floor control (Option c of note on Table A.3.1.3-1), apply condition WITHOUT_FLOORCONTROL. NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.1.4-6: MCPTT-Info in SIP INVITE (Table A.3.1.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A				

Table A.3.1.4-7: SIP 200 (OK) (Step 1, Table A.3.1.3-1; Step 5, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [19]	
value	"application/sdp"			
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-8			

Table A.3.1.4-8: SDP in SIP 200 (OK) (Table A.3.1.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [35] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [36]	
"RR"	2000		IETF RFC 3556 [36]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_FLOORCONTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_FLOORCONTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.1.4-9: SIP 200 (OK) (Step 1, Table A.3.1.3-1; Step 6, Table A.2.18.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.1.4-10			

Table A.3.1.4-10: SDP in SIP 200 (OK) (Table A.3.1.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_ FLOORCO NTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_granted	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_F LOOR_GR ANTED
mc_implicit_request	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_ GRANT_RE QUESTED

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mc_ssrc	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_ GRANT_RE QUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
NOTE 1: If the SS-UE requests an implicit floor (Option b of NOTE 1 on Table A.3.1.3-1), more conditions may apply: i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied. ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies. NOTE 2: If the SS-UE requests a call without media floor (Option c of NOTE 1 on Table A.3.1.3-1) apply condition WITHOUT_FLOORCONTROL NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.				

A.3.2 MCPTT E2E group call establishment, manual commencement

A.3.2.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.2.2 Definition of system information messages

N/A.

A.3.2.3 Procedure

Table A.3.2.3-1: MCPTT E2E group call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure MCX E2E group call establishment, manual commencement' as described in Table A.2.20.3-1 with MCPTT as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit floor control in the SIP INVITE request but it is not implicitly granted (note)	-	-	-	-
2a1	Check: Does the MCPTT server send a Floor Granted message to SS-UE1?	<--	Floor Granted	-	P
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is granted to SS-UE1, either implicit or explicitly. (NOTE1)	-	-	-	-
3a1	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<--	Floor Taken	-	P
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is not granted to SS-UE1. (note)	-	-	-	-
4a1	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<--	Floor Idle	-	P
4a2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<--	Floor Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:

a. SS-UE1 sends SDP offer with media description for floor control but without implicit floor request.
b. SS-UE1 sends SDP offer with media description for floor control and with implicit floor request:
   i. SDP answer from MCPTT Server contains "mc_implicit_request" and "mc_granted" (Floor is implicitly granted).
   ii. SDP answer from MCPTT Server contains "mc_implicit request" and but no "mc_granted" (Floor needs to be explicitly granted at step 2a1).
   iii. SDP answer from MCPTT Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the floor).

A.3.2.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCPTT and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.2.4-1: SIP INVITE from the SS-UE1
(Step 1, Table A.3.2.3-1; Step 1, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [23]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [23]		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-2			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.2.4-3			

Table A.3.2.4-2: SDP in SIP INVITE (Table A.3.2.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and IMPLICIT_GRANT_REQUESTED if the SS-UE1 requests implicit floor.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [35] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [36]	
"RR"	2000		IETF RFC 3556 [36]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and IMPLICIT_GRANT_REQUESTED if the SS-UE1 requests implicit floor.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1) Note	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
NOTE: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.2.4-3: MCPTT-Info in SIP INVITE (Table A.3.2.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition GROUP-CALL, INVITE_REFER

Table A.3.2.4-4: SIP INVITE from the MCPTT-Server (Step 1, Table A.3.2.3-1; Step 2, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [38] IETF RFC 7134 [20] IETF RFC 8101 [39]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
Session-Expires			IETF RFC 4028 [23]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.2.4-6			

Table A.3.2.4-5: SDP in SIP INVITE (Table A.3.2.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1) Note	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.2.4-6: MCPTT-Info in SIP INVITE (Table A.3.2.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1, condition GROUP-CALL (note 1)				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (note 2) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (note 2) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE 1: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL. NOTE 2: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.2.4-7: SIP 200 (OK) from the SS-UE2 (Step 1, Table A.3.2.3-1; Step 8, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1, with condition MCPTT, INVITE-RSP and GROUP-CALL (note)				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [19]	
value	"application/sdp"			
Message-body				
MIME body part		SDP message		
MIME-part-body	SDP Message as described in Table A.3.2.4-8			
NOTE: For chat group calls CHAT-GROUP-CALL condition shall be used instead of GROUP-CALL.				

Table A.3.2.4-8: SDP in SIP 200 (OK) (Table A.3.2.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [35] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [36]	
"RR"	2000		IETF RFC 3556 [36]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE1			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.2.4-9: SIP 200 (OK) from the MCPTT Server (Step 1, Table A.3.2.3-1; Step 9, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.2.4-10			

Table A.3.2.4-10: SDP in SIP 200 (OK) (Table A.3.2.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_ssrc	any value		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
NOTE 1: If the SS-UE requests an implicit floor (Option b of NOTE 1 on Table A.3.2.3-1), more conditions may apply: i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied. ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies. NOTE 2: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.				

Table A.3.2.4-11: SIP 183 (Session Progress) from the SS-UE2 (Step 1, Table A.3.2.3-1; Step 5, Table A.2.20.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.3.2-1, with condition MCPTT and 100rel

A.3.3 MCPTT E2E private call establishment, manual commencement

A.3.3.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.3.2 Definition of system information messages

N/A.

A.3.3.3 Procedure

Table A.3.3.3-1: MCPTT E2E private call establishment, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCPTT server correctly perform test procedure MCX E2E private call establishment, manual commencement' as described in Table A.2.21.3-1 with MCPTT as MCX?	-	-	-	P
-	EXCEPTION: Step 2a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place if the SS-UE1 requests implicit floor control in the SIP INVITE request but it is not implicitly granted (Note)	-	-	-	-
2a1	Check: Does the MCPTT server send a Floor Granted message to SS-UE1?	<--	Floor Granted	-	P
-	EXCEPTION: Step 3a1 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is granted to SS-UE1, either implicit or explicitly. (NOTE1)	-	-	-	-
3a1	Check: Does the MCPTT server send a Floor Taken message to SS-UE2?	<--	Floor Taken	-	P
-	EXCEPTION: Steps 4a1-4a2 describes behaviour that depends on the test case requirements; the "lower case letter" identifies a step sequence that takes place when the floor is not granted to SS-UE1. (Note)	-	-	-	-
4a1	Check: Does the MCPTT server send a Floor Idle message to SS-UE1?	<--	Floor Idle	-	P
4a2	Check: Does the MCPTT server send a Floor Idle message to SS-UE2?	<--	Floor Idle	-	P

NOTE: Possibilities in SDP-offer/answer depend on the test case requirements:

a. SS-UE1 sends SDP offer with media description for floor control but without implicit floor request.
b. SS-UE1 sends SDP offer with media description for floor control and with implicit floor request:
   i. SDP answer from MCPTT Server contains "mc_implicit_request" and "mc_granted" (Floor is implicitly granted).
   ii. SDP answer from MCPTT Server contains "mc_implicit request" and but no "mc_granted" (Floor needs to be explicitly granted at step 2a1).
   iii. SDP answer from MCPTT Server contains no "mc_implicit_request"and no "mc_granted" (the UE needs to explicitly request the floor).
c. SS-UE1 sends SDP offer without media description for floor control.

A.3.3.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 with condition MCPTT and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.3.4-1: SIP INVITE from the SS-UE1
(Step 1, Table A.3.3.3-1; Step 1, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.1-1, with condition MCPTT, PRIVATE-CALL and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires			IETF RFC 4028 [23]	
generic-param	"1800"	The recommended initial value is 1800 in IETF RFC 4028 [23].		
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-2			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.3.4-3			

Table A.3.3.4-2: SDP in SIP INVITE (Table A.3.3.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [35] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [36]	
"RR"	2000		IETF RFC 3556 [36]	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, with conditions INITIAL_SDP_OFFER and PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_FLOORCONTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	
NOTE 1: If the SS-UE requests a call with media floor control and implicit floor (Option b of note on Table A.3.3.3-1), apply condition IMPLICIT_GRANT_REQUESTED. NOTE 2: If the SS-UE requests a call without media floor (Option c of NOTE1 on Table A.3.3.3-1), apply condition WITHOUT_FLOORCONTROL				

Table A.3.3.4-3: MCPTT-Info in SIP INVITE (Table A.3.3.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition PRIVATE-CALL, INVITE_REFER

Table A.3.3.4-4: SIP INVITE from the MCPTT-Server to SS-UE2 (Step 1, Table A.3.3.3-1; Step 2, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, with condition MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [38] IETF RFC 7134 [20] IETF RFC 8101 [39]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
Session-Expires			IETF RFC 4028 [23]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.3.4-6			

Table A.3.3.4-5: SDP in SIP INVITE (Table A.3.3.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER NOTE: If the SS-UE1 requests a call without media floor control (Option c of note on Table A.3.3.3-1), apply condition WITHOUT_FLOORCONTROL.				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing

Table A.3.3.4-6: MCPTT-Info in SIP INVITE (Table A.3.3.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1, condition PRIVATE-CALL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.3.4-7: SIP 200 (OK) from the SS-UE2 to MCPTT Server (Step 1, Table A.3.3.3-1; Step 7, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1, with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [19]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-8			

Table A.3.3.4-8: SDP in SIP 200 (OK) (Table A.3.3.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [35] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [36]	
"RR"	2000		IETF RFC 3556 [36]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_FLOORCONTROL

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.3.4-9: SIP 200 (OK) from the MCPTT Server to SS-UE1 (Step 1, Table A.3.3.3-1; Step 8, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1, with condition MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session-Expires				
generic-param	Same value as session expires header in SIP INVITE or lower			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.3.4-10			

Table A.3.3.4-10: SDP in SIP 200 (OK) (Table A.3.3.4-9)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER.
NOTE 1: If the SS-UE1 request an implicit floor (Option b of note on Table A.3.3.3-1), more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.
NOTE 2: If the SS-UE1 request a call without media floor (Option c of note on Table A.3.3.3-1) apply condition WITHOUT_FLOORCONTROL.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Information Element	Value/remark	Comment	Reference	Condition
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_FLOOR_GRANTED
mc_implicit_request	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.3.4-11: SIP 180 (Ringing) response (Step 1, Table A.3.3.3-1; Steps 5, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.1-1, with condition MCPTT

Table A.3.3.4-12: SIP 180 (Ringing) response (Step 1, Table A.3.3.3-1; Steps 6, Table A.2.21.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.16.2.2-1, with condition MCPTT

A.3.4 MCPTT E2E call establishment using a pre-established session, manual commencement

A.3.4.1 Initial conditions

As specified in the test case which calls the procedure.

A.3.4.2 Definition of system information messages

N/A.

A.3.4.3 Procedure

Table A.3.4.3-1: MCPTT E2E call establishment using a pre-established session, manual commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a SIP REFER message to request the establishment of an MCPTT call using a pre-established session	-->	SIP REFER	-	-
2	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1 indicating that the MCPTT call has been properly requested?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCPTT Server send a SIP re-INVITE requesting the establishment of a call to SS-UE2 (MCPTT Client)?	<--	SIP re-INVITE	-	P
4	The SS-UE2 (MCPTT Client) responds a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
5	The SS-UE2 (MCPTT Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
6	Check: Does the MCPTT server send a SIP ACK to acknowledge the session establishment to SS-UE2?	<--	SIP ACK	-	P
-	EXCEPTION: Depending on the MCPTT Server implementation, the following steps might be performed in parallel.	-	-	-	-
7	Check: Does the MCPTT server send a Connect message to SS-UE1 (MCPTT Client)?	<--	Connect	-	P
8	The SS-UE1 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-
9	Check: Does the MCPTT server send a Connect message to SS-UE2 (MCPTT Client)?	<--	Connect	-	P
10	The SS-UE2 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-

A.3.4.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.4.4-1: SIP REFER (Step 1, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCPTT and MANUAL				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body			IETF RFC 3261 [21]	
MIME body part		Resource list	IETF RFC 5366 [17]	
MIME-part-body	Resource-lists as described in Table A.3.4.4-2			

Table A.3.4.4-2: Resource-lists in SIP REFER (Table A.3.4.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1, condition PRE-ESTABLISH with the uri attribute of the entry extended with the SIP URI header fields as specified in Table A.3.4.4-3A

Table A.3.4.4-2A: SIP header fields extending the uri attribute of the resource-lists' single entry (Table A.3.4.4-2)

Derivation Path:ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-2, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
body				
MIME body part		**SDP Message**		
MIME-part-headers				
Content-Type	"application/sdp"			
MIME-part-body	SDP Message as described in Table A.3.4.4-2B			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.4.4-2C			

Table A.3.4.4-2B: SDP in SIP header fields (Table A.3.4.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [35] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [36]	
"RR"	2000		IETF RFC 3556 [36]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for RTP		
foundation	1234	arbitrarily selected		
priority	2130706431	IETF RFC 5245 [29] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for RTCP		
foundation	1234	Same as for RTP		
priority	2130706430	IETF RFC 5245 [29] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]		Media description for media control		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1) Note 2	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_implicit_request	not present			
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	any value if present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
mc_priority	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for Media Control messages		
foundation	4321	arbitrarily selected; different than for RTP/RTCP		
priority	2130706430	IETF RFC 5245 [29] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		

NOTE 1: If the SS-UE requests an implicit floor condition IMPLCIT_GRAN_REQUESTED also applies.
NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).

Table A.3.4.4-2C: MCPTT-Info in SIP header fields (Table A.3.4.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition INVITE-REFER

Table A.3.4.4-3: SIP 200 (OK) (Step 2, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCPTT and REFER-RSP

Information Element	Value/remark	Comment	Reference	Condition
Content-Type				
media-type	"application/sdp"			
Message-body				
SDP message	SDP message as described in Table A.3.4.4-3A			

Table A.3.4.4-3A: SDP in SIP 200 (OK) (Table A.3.4.4-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
candidate		candidate for RTP		
foundation	any value			
priority	any value	IETF RFC 5245 [29] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for RTCP		
foundation	any value			
priority	any value	IETF RFC 5245 [29] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_granted	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_F LOOR_GR ANTED
mc_implicit_request	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_ GRANT_RE QUESTED

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
mc_ssrc	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for Media Control messages		
foundation	any value			
priority	any value	IETF RFC 5245 [29] clause 4.2: $2^{24} * 126 + 2^{8} * 65535 +$ 256 - component id		

NOTE 1: If the SS-UE requests an implicit floor more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.
NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Table A.3.4.4-4: SIP INVITE (Step 3, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1 with conditions MCPTT, MANUAL and re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Resource-Priority			IETF RFC 4412 [38] IETF RFC 7134 [20] IETF RFC 8101 [39]	EMERGENCY-CALL or IMMPERIL-CALL
r-value				EMERGENCY-CALL
namespace	value of the <resource-priority-namespace> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-priority	value of the <resource-priority-priority> element contained in the <emergency-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-value				IMMPERIL-CALL
namespace	value of the <resource-priority-namespace> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration documents	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
r-priority	value of the <resource-priority-priority> element contained in the <imminent-peril-resource-priority> element contained in the <OnNetwork> element of the MCX service configuration document	As configured in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1 for MCPTT and in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.8-1 for MCVIdeo		
Session-Expires			IETF RFC 4028 [23]	
delta-seconds	any allowed value			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.4.4-5			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.4.4-6			

Table A.3.4.4-5: SDP in SIP INVITE (Table A.3.4.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, with condition INITIAL_SDP_OFFER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Media description[2]				NOT WITHOUT_FLOORCONTROL
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				
mc_priority	Any value lower or equal to the priority specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1) Note 2	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	Any value lower or equal to the priority specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
NOTE 1: If the SS-UE requests a call without media floor control condition WITHOUT_FLOORCONTROL shall be applied. NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).				

Table A.3.4.4-6: MCPTT-Info in SIP INVITE (Table A.3.4.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
mcptt-request-uri	Encrypted (note) <mcptt-request-uri> with mcpttURI set to px_MCPTT_ID_User_B	The URI of the called user		
mcptt-calling-user-id	Encrypted (note) <mcptt-calling-user-id> with mcpttURI set to px_MCPTT_ID_User_A	The URI of the calling user		
NOTE: Encrypted element as described in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-1A.				

Table A.3.4.4-7: SIP 200 (OK) (Step 4, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.1-1 with conditions MCPTT and INVITE-RSP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Content-Type			IETF RFC 5621 [19]	
value	"application/sdp"			
Message-body				
MIME body part		**SDP message**		
MIME-part-body	SDP Message as described in Table A.3.4.4-8			

Table A.3.4.4-8: SDP in SIP 200 (OK) (Table A.3.4.4-7)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [35] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [36]	
"RR"	2000		IETF RFC 3556 [36]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
Media description[2]		Media description for media control		NOT WITHOUT_ FLOORCO NTROL
media description		m= line media = application SDP media-level section for a media control entity		
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	

Table A.3.4.4-9: Connect (Step 7, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Session Identity field			
Session Type	"00000000"	No session type	
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	
Inviting MCPTT User Identity field	not present		

Table A.3.4.4-10: Connect (Step 9, Table A.3.4.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	

A.3.5 MCPTT E2E call establishment using a pre-established session, automatic commencement

A.3.5.1 Initial conditions

As specified in the test case which calls the procedure.

A.3.5.2 Definition of system information messages

N/A.

A.3.5.3 Procedure

Table A.3.5.3-1: MCPTT E2E call establishment using a pre-established session, automatic commencement

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a SIP REFER message to request the establishment of an MCPTT call using a pre-established session	-->	SIP REFER	-	-
2	Check: Does the MCPTT Server respond with a SIP 200 (OK) message to SS-UE1 indicating that the MCPTT call has been properly requested?	<--	SIP 200 (OK)	-	P
-	EXCEPTION: Depending on the MCPTT Server implementation, the following steps might be performed in parallel.	-	-	-	-
3	Check: Does the MCPTT server send a Connect message to SS-UE1 (MCPTT Client)?	<--	Connect	-	P
4	The SS-UE1 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-
5	Check: Does the MCPTT server send a Connect message to SS-UE2 (MCPTT Client)?	<--	Connect	-	P
6	The SS-UE2 (MCPTT Client) sends an Acknowledgement in response to the Connect message.	-->	Acknowledge	-	-

A.3.5.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.5.4-1: SIP REFER (Step 1, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body			IETF RFC 3261 [21]	
MIME body part		Resource list	IETF RFC 5366 [17]	
MIME-part-body	Resource-lists as described in Table A.3.5.4-2			

Table A.3.5.4-2: Resource-lists in SIP REFER (Table A.3.5.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.3.1-1, condition PRE-ESTABLISH with the uri attribute of the entry extended with the SIP URI header fields as specified in Table A.3.5.4-3A

Table A.3.5.4-2A: SIP header fields extending the uri attribute of the resource-lists' single entry (Table A.3.5.4-2)

Derivation Path:ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-2, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
body				
MIME body part		**SDP Message**		
MIME-part-headers				
Content-Type	"application/sdp"			
MIME-part-body	SDP Message as described in Table A.3.5.4-2B			
MIME body part		**MCPTT-Info**		
MIME-part-body	MCPTT-Info as described in Table A.3.5.4-2C			

Table A.3.5.4-2B: SDP in SIP header fields (Table A.3.5.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Session description:				
Origin		o= line		
username	"-"	"-" indicating the concept of user IDs not being supported		
sess-id	"11111111"	A numeric string such that the tuple of <username>, <sess-id>, <nettype>, <addrtype>, and <unicast-address> forms a globally unique identifier for the session.		
sess-version	"11111111"			
Session Name	" "	s= line single empty space indicating no session name		
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
Media description[1]		**Media description for audio**		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
media description		m= line media = audio	IETF RFC 4867 [34]	
port	port number assigned by the SS-UE (even integer)	The transport port to which the media stream is sent	IETF RFC 6335 [35] clause 6	
fmt	"99"	RTP/SAVP payload type for AMR-WB is dynamic		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
Bandwidth		b= line		
"AS"	38		ETSI TS 126 114 [33] (3GPP TS 26.114) Table K.6	
"RS"	0		IETF RFC 3556 [36]	
"RR"	2000		IETF RFC 3556 [36]	
media attribute		a= line attribute = rtpmap		
rtpmap	"rtpmap"			
payload type	"99"			
encoding parameter	"1"	Channel number		
media attribute		a= line attribute = fmtp		
fmtp	"fmtp"			
format	"99"			
media attribute		a= line attribute =ptime		
ptime	"20"	packet time		
media attribute		a= line attribute =maxptime		
maxptime	"240"	maximum packet time		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for RTP		
foundation	1234	arbitrarily selected		
priority	2130706431	IETF RFC 5245 [29] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for RTCP		
foundation	1234	Same as for RTP		
priority	2130706430	IETF RFC 5245 [29] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]		Media description for media control		
media description		m= line media = application SDP media-level section for a media control entity		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
port	port number assigned by the SS-UE being different than the port number of the audio channel (RTP) and its associated control channel (RTCP)"	The port for the media control entity		
Connection Data		c= line		
connection-address	IP address of the SS-UE			
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	Present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	pc_MCPTT_FloorRequestQueueing
mc_priority	The priority value specified in the MCPTT Group configuration	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1) Note 2	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	The max priority value specified in the MCPTT Service configuration	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_implicit_request	not present			
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_GRANT_REQUESTED
mc_ssrc	any value if present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_FLOORCONTROL
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.1-1, condition SDP_OFFER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
mc_granted	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for Media Control messages		
foundation	4321	arbitrarily selected; different than for RTP/RTCP		
priority	2130706430	IETF RFC 5245 [29] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		

NOTE 1: If the SS-UE requests an implicit floor condition IMPLCIT_GRAN_REQUESTED also applies.
NOTE 2: The <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration has higher value than any <user-priority> element. (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1).

Table A.3.5.4-2C: MCPTT-Info in SIP header fields (Table A.3.5.4-2A)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1, condition INVITE-REFER

Table A.3.5.4-3: SIP 200 (OK) (Step 2, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCPTT and REFER-RSP

Information Element	Value/remark	Comment	Reference	Condition
Content-Type				
media-type	"application/sdp"			
Message-body				
SDP message	SDP message as described in Table A.3.5.4-3A			

Table A.3.5.4-3A: SDP in SIP 200 (OK) (Table A.3.5.4-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
Media description[1]		**Media description for audio**		
media description		m= line media = audio	IETF RFC 4867 [34]	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for RTP		
foundation	any value			
priority	any value	IETF RFC 5245 [29] clause 4.2:		

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
		2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for RTCP		
foundation	any value			
priority	any value	IETF RFC 5245 [29] clause 4.2: 2^{24} * 126 + 2^{8} * 65535 + 256 - component id		
Media description[2]				
media attribute		a= line attribute = fmtp		
fmtp				
format	"MCPTT"			
format specific parameters				NOT WITHOUT_ FLOORCO NTROL
mc_queueing	present if included in the offer	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Group Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <user-priority> element for the user in the MCPTT Group Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.A.2.1-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	GROUP-CALL OR CHAT-GROUP-CALL
mc_priority	if a value is provided in the offer: the value defined in the MCPTT Service Configuration or the value provided in the offer, whichever is the lower value; otherwise not present	The value of the <num-levels-priority-hierarchy> element for on-network in the MCPTT Service Configuration (ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.8.4-1)	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 4.1.1.4 and cl. 14.3.3	PRIVATE-CALL
mc_granted	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_F LOOR_GR ANTED
mc_implicit_request	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_ GRANT_RE QUESTED
mc_ssrc	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-1, condition SDP_ANSWER and PRE_ESTABLISHED_SESSION

Information Element	Value/remark	Comment	Reference	Condition
	same value as in the offer if provided in the offer and there is no collision with the value used by the SS; otherwise value assigned by the SS		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	IMPLICIT_ GRANT_RE QUESTED
mc_no_floor_ctrl	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
format specific parameters				WITHOUT_ FLOORCO NTROL
mc_queueing	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_priority	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3 and cl. 14.3.3	
mc_granted	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_implicit_request	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_ssrc	not present		ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
mc_no_floor_ctrl	present	Parameter has no value	ETSI TS 124 380 [37] (3GPP TS 24.380) cl. 12.1.2.3	
media attribute		a=line attribute="candidate"	IETF RFC 5245 [29]	
candidate		candidate for Media Control messages		
foundation	any value			
priority	any value	IETF RFC 5245 [29] clause 4.2: $2^{24} * 126 + 2^{8} * 65535 + 256$ - component id		

NOTE 1: If the SS-UE requests an implicit floor more conditions may apply:
i. If the server implicitly grants the floor, conditions IMPLICIT_GRANT_REQUESTED and IMPLICIT_FLOOR_GRANTED are also applied.
ii. If the floor needs to be explicitly granted, condition IMPLICIT_GRANT_REQUESTED applies.
NOTE 2: If the SS-UE requests a call without media floor condition WITHOUT_FLOORCONTROL shall be applied.
NOTE 3: Elements whose values mention the SS shall be understood as if they were referring to the MCPTT Server.

Table A.3.5.4-4: Connect (Step 3, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Session Identity field			
Session Type	"00000000"	No session type	
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	
Inviting MCPTT User Identity field	not present		

Table A.3.5.4-5: Connect (Step 5, Table A.3.5.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.12-1 with condition ACK			
Information Element	**Value/remark**	**Comment**	**Condition**
RTCP header			
SSRC	The SSRC of the MCPTT Participating Server		
MCPTT Group Identity field			GROUP-CALL OR CHAT-GROUP-CALL
MCPTT Group Identity	px_MCPTT_Group_A_ID	a URI, which identifies the MCPTT group	
PCK I_MESSAGE field	MIKEY-SAKKE I_MESSAGE as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.9.1-2A		PRIVATE-CALL

A.3.6 MCPTT E2E call release keeping the pre-established session

A.3.6.1 Initial conditions

As specified in the test case which calls the procedure.

A.3.6.2 Definition of system information messages

N/A.

A.3.6.3 Procedure

Table A.3.6.3-1: MCPTT E2E call release keeping the pre-established session

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCPTT Client) sends a SIP REFER message with method "BYE" to release the MCPTT session and keep the pre-established session.	-->	SIP REFER	-	-
2	Check: Does the MCPTT Server respond with a SIP 200 (OK) to SS-UE1?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCPTT Server release the call by sending a Disconnect message to SS-UE2 (MCPTT Client)?	<--	Disconnect	-	P
4	The SS-UE2 (MCPTT Client) sends an Acknowledgement to accept the release of the call	-->	Acknowledge	-	-

A.3.6.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.6.4-1: SIP REFER (Step 1, Table A.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCPTT and METHOD-BYE

Table A.3.6.4-2: SIP 200 (OK) (Step 2, Table A.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCPTT and REFER-RSP

Table A.3.6.4-3: Disconnect (Steps 3, Table A.3.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.13-1 with condition ACK

A.3.7 MCPTT Group Affiliation Status Determination and Subscription

A.3.7.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.7.2 Definition of system information messages

N/A.

A.3.7.3 Procedure

Table A.3.7.3-1: MCPTT Group affiliation status determination and subscription

Same procedure as defined in: Table A.2.7.3-1

A.3.7.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.7.4-1: SIP SUBSCRIBE (Step 1, Table A.3.7.3-1)

Derivation Path: Table A.2.7.4-1, conditions MCPTT and PRESENCE-EVENT

Table A.3.7.4-2: SIP NOTIFY (Step 3, Table A.3.7.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.7.4-3			

Table A.3.7.4-3: PIDF in SIP NOTIFY (Table A.3.7.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1 (note)
NOTE: PIDF document contains tuple with empty <status> element

A.3.8 MCPTT Group Affiliation Status Change

A.3.8.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.8.2 Definition of system information messages

N/A.

A.3.8.3 Procedure

Table A.3.8.3-1: MCPTT Group Affiliation Status Change

Same procedure as defined in Table A.2.12.3-1

A.3.8.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.8.4-1: SIP PUBLISH (Step 1, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-1, condition MCPTT

Table A.3.8.4-2: PIDF in SIP PUBLISH (Table A.3.8.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.1-1, condition AFFILIATION

Table A.3.8.4-3: SIP 200 (OK) (Step 2, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			IETF RFC 3261 [21] IETF RFC 3903 [16]	
value	same value as in the request or lower			

Table A.3.8.4-4: SIP NOTIFY (Step 3, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-6, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.8.4-5			

Table A.3.8.4-5: PIDF in SIP NOTIFY (Table A.3.8.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9.2.2.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

Table A.3.8.4-6: SIP NOTIFY (Step 5, Table A.3.8.3-1)

Derivation Path: Table A.2.12.4-6, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.8.4-7			

Table A.3.8.4-7: PIDF in SIP NOTIFY (Table A.3.8.4-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliated"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9.2.2.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

A.3.9 MCPTT Functional Alias Status Determination and Subscription

A.3.9.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.9.2 Definition of system information messages

N/A.

A.3.9.3 Procedure

Table A.3.9.3-1: MCPTT Functional Alias Status Determination and Subscription

Same procedure as defined in Table A.2.7.3-1

A.3.9.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.9.4-1: SIP SUBSCRIBE (Step 1, Table A.3.9.3-1)

Derivation Path: Table A.2.7.4-1, conditions MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCPTT Info**	ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9A.2.1.3	
MIME-part-body	MCPTT-Info as described in Table A.2.7.4-2			

Table A.3.9.4-2: MCPTT-Info in SIP SUBSCRIBE (Table A.3.9.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcpttinfo				
mcptt-Params				
request-type	functional-alias-status-determination		ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9A.2.1.3	

Table A.3.9.4-3: SIP NOTIFY (Step 3, Table A.3.9.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1 with conditions MCPTT and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.9.4-4			

Table A.3.9.4-4: PIDF in SIP NOTIFY (Table A.3.9.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1 (note)
NOTE: PIDF document contains tuple with empty <status> element

A.3.10 MCPTT Functional Alias Status Change

A.3.10.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.10.2 Definition of system information messages

N/A.

A.3.10.3 Procedure

Table A.3.10.3-1: MCPTT Functional Alias Status Change

Same procedure as defined in Table A.2.12.3-1

A.3.10.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.10.4-1: SIP PUBLISH (Step 1, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-1, condition MCPTT

Table A.3.10.4-2: PIDF in SIP PUBLISH (Table A.3.10.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.1-1, condition FUNCTIONAL_ALIAS_STATUS_CHANGE

Table A.3.10.4-3: SIP 200 (OK) (Step 2, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			IETF RFC 3261 [21] IETF RFC 3903 [16]	
value	same value as in the request or lower			

Table A.3.10.4-4: SIP NOTIFY (Step 3, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-6, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.10.4-5			

Table A.3.10.4-5: PIDF in SIP NOTIFY (Table A.3.10.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, conditions FUNCTIONAL_ALIAS_ACTIVATED and NOTIFY_FOR_PUBLISH				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"activating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9A.2.2.3.5	

Table A.3.10.4-6: SIP NOTIFY (Step 5, Table A.3.10.3-1)

Derivation Path: Table A.2.12.4-6, condition MCPTT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9A.2.2.2.5	
MIME-part-body	PIDF as described in Table A.3.10.4-7			

Table A.3.10.4-7: PIDF in SIP NOTIFY (Table A.3.10.4-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-1, conditions FUNCTIONAL_ALIAS_ACTIVATED and NOTIFY_FOR_PUBLISH				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"activated"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 379 [5] (3GPP TS 24.379) clause 9A.2.2.3.5	

A.3.11 MCPTT Floor Request - Floor Granted

A.3.11.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.11.2 Definition of system information messages

N/A.

A.3.11.3 Procedure

Table A.3.11.3-1: MCPTT Floor Request - Floor Granted

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Request message. (Note)	-->	Floor Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Granted message with acknowledgement required?	<--	Floor Granted	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Granted message. (Note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Granted message with no acknowledgement required?	<--	Floor Granted	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.11.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.11.4-1: Floor Granted (Step 2a1, Table A.3.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.3-1 condition ACK

A.3.12 MCPTT Floor Request - Floor Queue Position Info

A.3.12.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.12.2 Definition of system information messages

N/A.

A.3.12.3 Procedure

Table A.3.12.3-1: MCPTT Floor Request - Floor Queue Position Info

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Request message. (Note)	-->	Floor Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with acknowledgement required?	<--	Floor Queue Position Info	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Queue Position Info message. (Note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with no acknowledgement required?	<--	Floor Queue Position Info	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.12.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.12.4-1: Floor Queue Position Info (Step 2a1, Table A.3.12.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1 condition ACK

A.3.13 MCPTT Queuing Position Request

A.3.13.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.13.2 Definition of system information messages

N/A.

A.3.13.3 Procedure

Table A.3.13.3-1: MCPTT Queuing Position Request

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Queue Position Request message. (Note)	-->	Floor Queue Position Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with acknowledgement required?	<--	Floor Queue Position Info	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Queue Position Info message. (Note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Queue Position Info message indicating that the Floor Request message was queued, with no acknowledgement required?	<--	Floor Queue Position Info	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.13.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.13.4-1: Floor Queue Position Info (Step 2a1, Table A.3.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.10-1 condition ACK

A.3.14 MCPTT Floor Request - Floor Deny

A.3.14.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.14.2 Definition of system information messages

N/A.

A.3.14.3 Procedure

Table A.3.14.3-1: MCPTT Floor Request - Floor Deny

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Request message. (Note)	-->	Floor Request	-	-
-	EXCEPTION: Steps 2a1-2b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
2a1	Check: Does the MCPTT Server send a Floor Deny message indicating that the Floor Request message has been rejected, with acknowledgement required?	<--	Floor Deny	-	P
2a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Deny message. (Note)	-->	Floor Ack	-	-
2b1	Check: Does the MCPTT Server send a Floor Deny message indicating that the Floor Request message has been rejected, with no acknowledgement required?	<--	Floor Deny	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.14.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.14.4-1: Floor Deny (Step 2a1, Table A.3.14.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.4-1 condition ACK

A.3.15 MCPTT Floor Release - Floor Idle

A.3.15.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.15.2 Definition of system information messages

N/A.

A.3.15.3 Procedure

Table A.3.15.3-1: MCPTT Floor Release - Floor Idle

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Release message with acknowledgment required. (Note)	-->	Floor Release	-	-
2	Check: Does the MCPTT Server send a Floor Ack message in response to the Floor Release message?	<--	Floor Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
3a1	Check: Does the MCPTT Server send a Floor Idle message with acknowledgement required?	<--	Floor Idle	-	P
3a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Idle message. (Note)	-->	Floor Ack	-	-
3b1	Check: Does the MCPTT Server send a Floor Idle message with no acknowledgement required?	<--	Floor Idle	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.15.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.15.4-1: Floor Release (Step 1, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
RTCP header				
Subtype	10100			

Table A.3.15.4-2: Floor Idle (Step 3a1, Table A.3.15.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.6-1 condition ACK

A.3.16 MCPTT Floor Release - Floor Taken

A.3.16.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.3.16.2 Definition of system information messages

N/A.

A.3.16.3 Procedure

Table A.3.16.3-1: MCPTT Floor Release - Floor Taken

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCPTT Client) sends a Floor Release message with acknowledgment required. (Note)	-->	Floor Release	-	-
2	Check: Does the MCPTT Server send a Floor Ack message in response to the Floor Release message?	<--	Floor Ack	-	P
-	EXCEPTION: Steps 3a1-3b1 describe behaviour that depends on the MCPTT Server implementation; the "lower case letter" identifies a step sequence that takes place if the MCPTT Server requests acknowledgement of sent floor control messages.	-	-	-	-
3a1	Check: Does the MCPTT Server send a Floor Taken message with acknowledgement required?	<--	Floor Taken	-	P
3a2	The SS-UE (MCPTT Client) sends a Floor Ack message in response to the Floor Taken message. (Note)	-->	Floor Ack	-	-
3b1	Check: Does the MCPTT Server send a Floor Taken message with no acknowledgement required?	<--	Floor Taken	-	P
NOTE: The SS-UE being used is specified in the test case calling the procedure. If not stated, SS-UE1 (MCPTT Client) will be assumed.					

A.3.16.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCX Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCX Client) or SS-UE2 (MCX Client).

Table A.3.16.4-1: Floor Release (Step 1, Table A.3.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.5-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
RTCP header				
Subtype	10100			

Table A.3.16.4-2: Floor Taken (Step 3a1, Table A.3.16.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.6.7-1 condition ACK

A.4 Generic test procedures for MCData Server operation

A.4.1 E2E SDS or FD message transfer using signalling plane

A.4.1.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.1.2 Definition of system information messages

N/A.

A.4.1.3 Procedure

Table A.4.1.3-1: E2E SDS or FD message transfer using signalling plane

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCData Client) sends a SIP MESSAGE request.	-->	SIP MESSAGE	-	-
-	EXCEPTION: Depending on the MCData Server implementation, steps 2 and 3 might be performed in parallel.	-	-	-	-
2	Check: Does the MCData Server (IUT) send a SIP 202 (Accepted) response?	<--	SIP 202 (Accepted)	-	P
3	Check: Does the MCData Server (IUT) forward the SIP Message request to SS-UE2?	<--	SIP MESSAGE	-	P
4	The SS-UE2 (MCData Client) sends a SIP 200 (OK) response.	-->	SIP 200 (OK)	-	-
NOTE: The test case calling this procedure may switch the role of SS-UE1 and SS-UE2 clients.					

A.4.1.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

A.4.2 MCData Participating Terminated SDS or FD message transfer using signalling plane

A.4.2.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.2.2 Definition of system information messages

N/A.

A.4.2.3 Procedure

Table A.4.2.3-1: Participating Terminated SDS or FD message transfer using signalling plane

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE (MCData Client) sends a SIP MESSAGE request.	-->	SIP MESSAGE	-	-
2	Check: Does the MCData Server (IUT) send a SIP 202 (Accepted) response?	<--	SIP 202 (Accepted)	-	P

A.4.2.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to SS-UE (MCData Client).

A.4.3 E2E MCData Call Establishment

A.4.3.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.3.2 Definition of system information messages

N/A.

A.4.3.3 Procedure

Table A.4.3.3-1: E2E MCData Call Establishment

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCData client) sends a SIP INVITE requesting the establishment of an MCData call.	-->	SIP INVITE	-	-
2	Check: Does the MCData Server (IUT) send a SIP INVITE requesting the establishment of an MCData call to SS-UE2 (MCData Client)?	<--	SIP INVITE		P
-	EXCEPTION: Step 3 describes behaviour that depends on the MCData server implementation; it identifies a step sequence that take place if the MCDATA Server responds to a SIP INVITE with a SIP 100 (Trying).	-	-	-	-
3	The MCData server (IUT) sends a SIP 100 Trying	<--	SIP 100 (Trying)	-	-
4	The SS-UE2 (MCData Client) responds with a SIP 200 (OK).	-->	SIP 200 (OK)		-
5	Check: Does the MCData Server (IUT) respond with a SIP 200 (OK) to SS-UE1 (MCData Client)?	<--	SIP 200 (OK)	-	P
6	The SS-UE1 (MCData Client) sends a SIP ACK to acknowledge the session establishment.	-->	SIP ACK	-	-
7	Check: Does the MCData server (IUT) send a SIP ACK to acknowledge the session establishment to SS-UE2 (MCData client)?	<--	SIP ACK	-	P
8	The SS-UE1 (MCData client) connects to the TCP server at the MCData server side to establish an MSRP connection. (Note 1)	-	-	-	-
9	The SS-UE1 (MCData client) sends an empty MSRP SEND request to bind the TCP connection to the MSRP session.	-->	MSRP SEND	-	-
10	Check: Does the MCData server (IUT) send an MSRP 200 (OK) response?	<--	MSRP 200 (OK)	-	P
-	EXCEPTION: Steps 11a1 - 11b3 describe behaviour that depends on which role of an endpoint the SS-UE2 (MCData client) has chosen in its SDP answer sent at step 4.	-	-	-	-
11a1	IF the SS-UE2 (MCData client) acts as passive endpoint (Note 2) THEN the MCData Server connects to the TCP server at the SS-UE2 side to establish an MSRP connection. Depending on the MCData Server implementation, steps 11a1-11a3 might be performed in parallel to steps 8-10.	-	-	-	-
11a2	Check: Does the MCData Server (IUT) send an empty MSRP SEND request to bind the TCP connection to the MSRP session?	<--	MSRP SEND	-	P
11a3	The SS-UE2 (MCData client) sends an MSRP 200 (OK) response.	-->	MSRP 200 (OK)	-	-
11b1	ELSE (Note 3) the SS-UE2 (MCData client) connects to the TCP server at the MCData Server side to establish an MSRP connection.	-	-	-	-
11b2	The SS-UE2 (MCData client) sends an empty MSRP SEND request to bind the TCP connection to the MSRP session.	-->	MSRP SEND	-	-
11b3	Check: Does the MCData server (IUT) send an MSRP 200 (OK) response?	<--	MSRP 200 (OK)	-	P

NOTE 1: According to ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 9.2.3.4.2, 9.2.4.4.2 and 10.2.5.4.2 the MCData Server sets the a=setup attribute set to "passive" (see ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-3) ⇒ The SS-UE's MCData client has the role of the active endpoint.

NOTE 2: The MCData client indicates to act as passive endpoint by setting the a=setup attribute of the SDP answer at step 4 to "passive" (according to IETF RFC 4145 [15]).

NOTE 3: The MCData client indicates to act as active endpoint by setting the a=setup attribute of the SDP answer at step 4 to "active" (according to IETF RFC 4145 [15]).

A.4.3.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

Table A.4.3.4-1: MSRP SEND (Steps 9 and 11b2, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.12.1.1-1, condition EMPTY_SEND_REQ

Table A.4.3.4-2: MSRP SEND (Step 11a2, Table A.4.3.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.12.1.2-1, condition EMPTY_SEND_REQ

A.4.4 E2E MSRP message transfer

A.4.4.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.4.2 Definition of system information messages

N/A.

A.4.4.3 Procedure

Table A.4.4.3-1: E2E MSRP message transfer

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Steps 1-2 are repeated until the SS-UE1 (MCData client) indicates the end of the message by setting the continuation-flag to "$" in the End-line of the MSRP SEND request at step 1. (Note 1)	-	-	-	-
1	The SS-UE1 (MCData client) sends an MSRP SEND request.	-->	MSRP SEND	-	-
2	Check: Does the MCData server (IUT) send an MSRP 200 (OK) response?	<--	MSRP 200 (OK)	-	P
3	Check: Does the MCData Server (IUT) send an MSRP SEND request to SS-UE2 (MCData Client), containing the message received from the SS-UE1 (MCData Client)? (Note 2)	<--	MSRP SEND	-	P
4	The SS-UE2 (MCData client) sends an MSRP 200 (OK) response.	-->	MSRP 200 (OK)	-	-

NOTE 1: In case of no chunking there is only one MSRP SEND request which contains the entire data. In case of chunking there are more than one MSRP SEND requests containing the chunks of data and the content type shall be the same for all MSRP SEND requests.
NOTE 2: It is assumed that in general there is no chunking in DL for MCData test cases and the MCData Server reassembles the data contained in the bodies of the MSRP SEND requests of step 1.
NOTE 3: The test case calling this procedure may switch the role of SS-UE1 and SS-UE2 clients.

A.4.4.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

A.4.5 E2E MCData Call Release

A.4.5.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.5.2 Definition of system information messages

N/A.

A.4.5.3 Procedure

Table A.4.5.3-1: E2E MCData Call Release

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: Does the MCData Server (IUT) correctly perform procedure 'MCX E2E call release' as described in Table A.2.19.3-1 to release the call?	-	-	-	P
2	The SS-UE1 (MCData Client) closes the MSRP TCP connection. (Note 1)	-	-	-	-
3	Check: Does the MCData Server (IUT) wait for the SS-UE1 (MCData Client) to close the MSRP TCP connection?	-	-	-	P
-	EXCEPTION: Steps 4a1 - 4b2 describe behaviour that depends on the endpoint role the SS-UE2 (MCData client) has chosen at call establishment. (Note 2)	-	-	-	-
4a1	IF the SS-UE2 (MCData Client) is the active endpoint THEN	-	-	-	-
4a2	Check: Does the MCData Server (IUT) wait for the SS-UE2 (MCData Client) to close the MSRP TCP connection?	-	-	-	P
4b1	ELSE the MCData Server closes the MSRP TCP connection.	-	-	-	-
4b2	Check: Does the MCDATA Server (IUT) close the MSRP TCP connection?	-	-	-	P
NOTE 1: The MCData Originating Participating Server chooses the role of the passive endpoint during call establishment (Table A.4.3.3-1). NOTE 2: The endpoint role is negotiated in the SDP signalling at call establishment (Table A.4.3.3-1).					

A.4.5.4 Specific message contents

All message contents are as specified in, ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).

- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

Table A.4.5.4-1: SIP BYE from the SS-UE1
(Step 1, Table A.4.5.3-1; Step 1, Table A.2.19.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.1-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Reason			IETF RFC 3326 [14]	
reason-value	"SIP"			
protocol-cause	"cause="200""			
reason-text	"text="transmission succeeded""			

Table A.4.5.4-2: SIP BYE from the MCPTT Server
(Step 1, Table A.4.5.3-1; Step 2, Table A.2.19.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.2.2-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Reason			IETF RFC 3326 [14]	
reason-value	"SIP"			
protocol-cause	"cause="200""			
reason-text	"text="transmission succeeded""			

A.4.6 E2E MCData Call Establishment using pre-established session

A.4.6.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.6.2 Definition of system information messages

N/A.

A.4.6.3 Procedure

Table A.4.6.3-1: E2E MCData Call Establishment using a pre-established session

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCData Client) sends a SIP REFER message to request the establishment of an MCData call using a pre-established session.	-->	SIP REFER	-	-
2	Check: Does the MCData Server (IUT) respond with a SIP 200 (OK) message to SS-UE1 (MCData Client) indicating that the MCData call has been properly requested?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCData Server (IUT) send a SIP re-INVITE requesting the establishment of a call to SS-UE2 (MCData Client)?	<--	SIP re-INVITE	-	P
4	The SS-UE2 (MCData Client) responds a SIP 100 Trying.	-->	SIP 100 (Trying)	-	-
5	The SS-UE2 (MCData Client) responds with a SIP 200 (OK).	-->	SIP 200 (OK)	-	-
6	Check: Does the MCData server (IUT) send a SIP ACK to acknowledge the session establishment to SS-UE2?	<--	SIP ACK	-	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
7	Check: Does the MCData Server (IUT) send a SIP re-INVITE to SS-UE1, to confirm that the MCData call has been successfully established?	<--	SIP re-INVITE	-	P
8	The SS-UE1 (MCData Client) responds a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
9	The SS-UE1 (MCData Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
10	Check: Does the MCData Server (IUT) sends a SIP ACK in response to the SIP 200 (OK) message.	<--	SIP ACK	-	P
11	The SS-UE1 (MCData client) connects to the TCP server at the MCData server side to establish an MSRP connection. (Note 1)	-	-	-	-
12	The SS-UE1 (MCData client) sends an empty MSRP SEND request to bind the TCP connection to the MSRP session.	-->	MSRP SEND	-	-
13	Check: Does the MCData server (IUT) send an MSRP 200 (OK) response?	<--	MSRP 200 (OK)	-	P
-	EXCEPTION: Steps 14a1 - 14b3 describe behaviour that depends on which role of an endpoint the SS-UE2 (MCData client) has chosen in its SDP answer sent at step 5.	-	-	-	-
14a1	IF the SS-UE2 (MCData client) acts as passive endpoint (Note 2) THEN the MCData Server connects to the TCP server at the SS-UE2 side to establish an MSRP connection. Depending on the MCData Server implementation, steps 14a1-14a3 might be performed in parallel to steps 11-13.	-	-	-	-
14a2	Check: Does the MCData Server (IUT) send an empty MSRP SEND request to bind the TCP connection to the MSRP session?	<--	MSRP SEND	-	P
14a3	The SS-UE2 (MCData client) sends an MSRP 200 (OK) response.	-->	MSRP 200 (OK)	-	-
14b1	ELSE (Note 3) the SS-UE2 (MCData client) connects to the TCP server at the MCData Server side to establish an MSRP connection.	-	-	-	-
14b2	The SS-UE2 (MCData client) sends an empty MSRP SEND request to bind the TCP connection to the MSRP session.	-->	MSRP SEND	-	-
14b3	Check: Does the MCData server (IUT) send an MSRP 200 (OK) response?	<--	MSRP 200 (OK)	-	P
NOTE 1: According to ETSI TS 124 282 [13] (3GPP TS 24.282) clauses 9.2.3.4.2, 9.2.4.4.2 and 10.2.5.4.2 the MCData Server sets the a=setup attribute set to "passive" (see ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.1.2-3) ⇒ The SS-UE's MCData client has the role of the active endpoint NOTE 2: The MCData client indicates to act as passive endpoint by setting the a=setup attribute of the SDP answer at step 4 to "passive" (according to IETF RFC 4145 [15]) NOTE 3: The MCData client indicates to act as active endpoint by setting the a=setup attribute of the SDP answer at step 4 to "active" (according to IETF RFC 4145 [15])					

A.4.6.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

Table A.4.6.4-1: MSRP SEND (Steps 12 and 14b2, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.12.1.1-1, condition EMPTY_SEND_REQ

Table A.4.6.4-2: MSRP SEND (Step 14a2, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.12.1.2-1, condition EMPTY_SEND_REQ

A.4.7 E2E MCData Call Release keeping the pre-established session

A.4.7.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.7.2 Definition of system information messages

N/A.

A.4.7.3 Procedure

Table A.4.7.3-1: E2E MCData Call Release keeping the pre-established session

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCData Client) sends a SIP REFER message with method "BYE" to release the MCData session and keep the pre-established session.	-->	SIP REFER	-	-
2	Check: Does the MCData Server (IUT) respond with a SIP 200 (OK) message to SS-UE1 (MCData Client)?	<--	SIP 200 (OK)	-	P
3	Check: Does the MCData Server (IUT) send a SIP re-INVITE informing about the call release to SS-UE2 (MCData Client)?	<--	SIP re-INVITE	-	P
4	The SS-UE2 (MCData Client) responds a SIP 100 Trying.	-->	SIP 100 (Trying)	-	-
5	The SS-UE2 (MCData Client) responds with a SIP 200 (OK).	-->	SIP 200 (OK)	-	-
6	Check: Does the MCData server (IUT) send a SIP ACK to acknowledge the session establishment to SS-UE2?	<--	SIP ACK	-	P
7	Check: Does the MCData Server (IUT) send a SIP re-INVITE to SS-UE1, to confirm that the MCData call has been successfully released?	<--	SIP re-INVITE	-	P
8	The SS-UE1 (MCData Client) responds a SIP 100 Trying	-->	SIP 100 (Trying)	-	-
9	The SS-UE1 (MCData Client) responds with a SIP 200 (OK)	-->	SIP 200 (OK)	-	-
10	Check: Does the MCData Server (IUT) sends a SIP ACK in response to the SIP 200 (OK) message.	<--	SIP ACK	-	P
11	The SS-UE1 (MCData Client) closes the MSRP TCP connection. (Note 1)	-	-	-	-
12	Check: Does the MCData Server (IUT) wait for the SS-UE1 (MCData Client) to close the MSRP TCP connection?	-	-	-	P
-	EXCEPTION: Steps 13a1 - 13b2 describe behaviour that depends on the endpoint role the SS-UE2 (MCData client) has chosen at call establishment. (Note 2)	-	-	-	-
13a1	IF the SS-UE2 (MCData Client) is the active endpoint THEN	-	-	-	-

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
13a2	Check: Does the MCData Server (IUT) wait for the SS-UE2 (MCData Client) to close the MSRP TCP connection?	-	-	-	P
13b1	ELSE the MCData Server closes the MSRP TCP connection.	-	-	-	-
13b2	Check: Does the MCDATA Server (IUT) close the MSRP TCP connection?	-	-	-	P
NOTE 1: The MCData Originating Participating Server chooses the role of the passive endpoint during call establishment (Table A.4.6.3-1). NOTE 2: The endpoint role is negotiated in the SDP signalling at call establishment (Table A.4.6.3-1).					

A.4.7.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

Table A.4.7.4-1: SIP REFER (Step 1, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.12-1 with conditions MCDATA and METHOD-BYE

Table A.4.7.4-2: SIP 200 (OK) (Step 2, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.17.1.2-1 with conditions MCDATA and REFER-RSP

Table A.4.7.4-3: SIP re-INVITE from the MCData Server (IUT) to SS-UE2 (Step 3, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.5.2-1, condition MCDATA_SDS, re_INVITE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Request-Line				
Request-URI	tsc_MCX_SessionID_B	session identity of the pre-established session	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 9.2.5.4.2.1	
Message-body				
MIME body part	not present	**SDP message**		
MIME body part		**MCData Info**		
MIME-part-body	MCData-Info message as described in Table A.4.7.4-4			

Table A.4.7.4-4: MCData-Info (Table A.4.7.4-3)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.2-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
mcdata-request-uri	not present			
mcdata-calling-user-id	not present			
anyExt				
mcdata-communication-state	"terminate request"		ETSI TS 124 282 [13] (3GPP TS 24.282) clause 9.2.5.4.2.2	

Table A.4.7.4-5: SIP re-INVITE from the MCData Server (IUT) to SS-UE1 (Step 7, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3C.13.4-3

Table A.4.7.4-6: SIP 200 (OK) from the SS-UE1 and SS-UE2 (Steps 5 and 9, Table A.4.6.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3C.13.4-5

A.4.8 FD file upload using HTTP

A.4.8.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.8.2 Definition of system information messages

N/A.

A.4.8.3 Procedure

Table A.4.8.3-1: FD file upload using HTTP

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE1 (MCData client) sends an HTTP POST request to upload a file to the media storage function?	-->	HTTP POST	-	-
2	Check: Does the MCData server (IUT) send an HTTP 201 Created response containing a Location header field with a URL identifying the location of the resource where the file has been stored at the media storage function?	<--	HTTP 201 Created	-	P

A.4.8.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to SS-UE1 (MCData Client).

A.4.9 FD file download using HTTP

A.4.9.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.9.2 Definition of system information messages

N/A.

A.4.9.3 Procedure

Table A.4.9.3-1: FD file download using HTTP

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	The SS-UE2 (MCData client) sends an HTTP GET request to download the file.	-->	HTTP GET	-	-
2	Check: Does the MCData server (IUT) send an HTTP 200 OK response containing the requested file?	<--	HTTP 200 OK	-	P

A.4.9.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to SS-UE2 (MCData Client).

A.4.10 Discovery of the absolute URI of the media storage function

A.4.10.1 Initial conditions

As specified in the test case which calls the procedure.

A.4.10.2 Definition of system information messages

N/A.

A.4.10.3 Procedure

Table A.4.10.3-1: Discovery of the absolute URI of the media storage function

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
-	EXCEPTION: Steps 1a1 - 1b1 describe behaviour that depends on the MCData Server implementation.	-	-	-	-
1a1	IF pc_MCData_MSFDiscoverySignalling THEN The SS-UE1 (MCData client) sends a SIP MESSAGE request to discover the absolute URI of the media storage function.	-->	SIP MESSAGE	-	-
1a2	Check: Does the MCData server (IUT) send an SIP 200 OK response?	<--	SIP 200 (OK)	-	P
1a3	Check: Does the MCData server (IUT) send a SIP MESSAGE containing the absolute URI of the media storage function in the <mcdata-controller-psi> element of the mcdata-info?	<--	SIP MESSAGE	-	P
1a4	The SS-UE sends SIP 200 OK response.	-->	SIP 200 (OK)	-	-
1b1	ELSE the SS-UE determines the value of the absolute URI associated with the media storage function of the MCData content server from the <MCDataContentServerURI> element of the MCData user profile document	-	-	-	-

A.4.10.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to SS-UE1 (MCData Client).

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure, with the following clarifications.

Table A.4.10.4-1: SIP MESSAGE from the SS-UE (Step 1a1, Table A.4.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3C.8.4-1				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCData-Info**		
MIME-part-body	MCData-Info as described in Table A.4.10.4-2			

Table A.4.10.4-2: MCDATA-Info from the SS-UE (Table A.4.10.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3C.8.4-2				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdata-info				
mcdata-Params				
request-type	"msf-disc-req"			
mcdata-calling-group-id	Encrypted <mcdata-calling-group-id> with mcdataURI set to px_MCData_Group_A_ID	Encrypted according to ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3A		MCD_grp

Table A.4.10.4-3: SIP MESSAGE from the MCData Server (IUT) (Step 1a3, Table A.4.10.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.3C.8.4-3

A.4.11 MCData Group Affiliation Status Determination and Subscription

A.4.11.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.11.2 Definition of system information messages

N/A.

A.4.11.3 Procedure

Table A.4.11.3-1: MCData Group affiliation status determination and subscription

Same procedure as defined in: Table A.2.7.3-1

A.4.11.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

Table A.4.11.4-1: SIP SUBSCRIBE (Step 1, Table A.4.11.3-1)

Derivation Path: Table A.2.7.4-1, conditions MCDATA and PRESENCE-EVENT

Table A.4.11.4-2: SIP NOTIFY (Step 3, Table A.4.11.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1, conditions MCDATA and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.4.1	
MIME-part-body	PIDF as described in Table A.4.11.4-3			

Table A.4.11.4-3: PIDF in SIP NOTIFY (Table A.4.11.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3 (note)
NOTE: PIDF document contains tuple with empty <status> element.

A.4.12 MCData Group Affiliation Status Change

A.4.12.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.12.2 Definition of system information messages

N/A.

A.4.12.3 Procedure

Table A.4.12.3-1: MCData Group Affiliation Status Change

Same procedure as defined in Table A.2.12.3-1

A.4.12.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

Table A.4.12.4-1: SIP PUBLISH (Step 1, Table A.4.12.3-1)

Derivation Path: Table A.2.12.4-1, condition MCDATA

Table A.4.12.4-2: PIDF in SIP PUBLISH (Table A.4.12.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.1-3, condition AFFILIATION

Table A.4.12.4-3: SIP 200 (OK) (Step 2, Table A.4.12.3-1)

Derivation Path: Table A.2.12.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			IETF RFC 3261 [21] IETF RFC 3903 [16]	
value	same value as in the request or lower			

Table A.4.12.4-4: SIP NOTIFY (Step 3, Table A.4.12.3-1)

Derivation Path: Table A.2.12.4-6, condition MCDATA				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.4.1	
MIME-part-body	PIDF as described in Table A.4.12.4-5			

Table A.4.12.4-5: PIDF in SIP NOTIFY (Table A.4.12.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

Table A.4.12.4-6: SIP NOTIFY (Step 5, Table A.4.12.3-1)

Derivation Path: Table A.2.12.4-6, condition MCDATA				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.4.1	
MIME-part-body	PIDF as described in Table A.4.12.4-7			

Table A.4.12.4-7: PIDF in SIP NOTIFY (Table A.4.12.4-5)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, condition AFFILIATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
affiliation				
status	"affiliated"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 282 [13] (3GPP TS 24.282) clause 8.3.3.5	
..p-id	same value as received in the SIP PUBLISH message.			

A.4.13 MCData Functional Alias Status Determination and Subscription

A.4.13.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.13.2 Definition of system information messages

N/A.

A.4.13.3 Procedure

Table A.4.13.3-1: MCData Functional Alias Status Determination and Subscription

Same procedure as defined in Table A.2.7.3-1

A.4.13.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

Table A.4.13.4-1: SIP SUBSCRIBE (Step 1, Table A.4.13.3-1)

Derivation Path: Table A.2.7.4-1, condition MCDATA				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**MCData Info**	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.2.1.3	
MIME-part-body	MCData-Info as described in Table A.2.7.4-2			

Table A.4.13.4-2: MCData-Info in SIP SUBSCRIBE (Table A.4.13.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.2.1-3				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
mcdatainfo				
mcdata-Params				
request-type	functional-alias-status-determination		ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.2.1.3	

Table A.4.13.4-3: SIP NOTIFY (Step 3, Table A.4.13.3-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.2.8-1 with conditions MCDATA and PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.3.1	
MIME-part-body	PIDF as described in Table A.4.13.4-4			

Table A.4.13.4-4: PIDF in SIP NOTIFY (Table A.4.13.4-2)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3 (note)
NOTE: PIDF document contains tuple with empty <status> element

A.4.14 MCData Functional Alias Status Change

A.4.14.1 Initial conditions

As specified in the test case which calls the procedure in its entirety or refers to parts of it.

A.4.14.2 Definition of system information messages

N/A.

A.4.14.3 Procedure

Table A.4.14.3-1: MCData Functional Alias Status Change

Same procedure as defined in Table A.2.12.3-1

A.4.14.4 Specific message contents

All message contents are as specified in ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), clause 5.5 and in the test case calling the procedure with the following clarifications:

- Subclauses referring to messages from SS relate to the IUT (MCData Server).
- Subclauses referring to messages from UE relate to either SS-UE1 (MCData Client) or SS-UE2 (MCData Client).

Table A.4.14.4-1: SIP PUBLISH (Step 1, Table A.4.14.3-1)

Derivation Path: Table A.2.12.4-1, condition MCDATA

Table A.4.14.4-2: PIDF in SIP PUBLISH (Table A.4.14.4-1)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.1-3, condition FUNCTIONAL_ALIAS_STATUS_CHANGE

Table A.4.14.4-3: SIP 200 (OK) (Step 2, Table A.4.14.3-1)

Derivation Path: Table A.2.12.4-5				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Expires			IETF RFC 3261 [21] IETF RFC 3903 [16]	
value	same value as in the request or lower			

Table A.4.14.4-4: SIP NOTIFY (Step 3, Table A.4.14.3-1)

Derivation Path: Table A.2.12.4-6, condition MCDATA				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.3.1	
MIME-part-body	PIDF as described in Table A.4.14.4-5			

Table A.4.14.4-5: PIDF in SIP NOTIFY (Table A.4.14.4-4)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, conditions FUNCTIONAL_ALIAS_ACTIVATED and NOTIFY_FOR_PUBLISH				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"activating"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.2.2.3.5	

Table A.4.14.4-6: SIP NOTIFY (Step 5, Table A.4.14.3-1)

Derivation Path: Table A.2.12.4-6, condition MCDATA				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Message-body				
MIME body part		**PIDF**	ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.3.1	
MIME-part-body	PIDF as described in Table A.4.14.4-7			

Table A.4.14.4-7: PIDF in SIP NOTIFY (Table A.4.14.4-6)

Derivation Path: ETSI TS 136 579-1 [1] (3GPP TS 36.579-1), Table 5.5.3.5.2-3, conditions FUNCTIONAL_ALIAS_ACTIVATED and NOTIFY_FOR_PUBLISH				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
presence				
tuple				
status				
functionalAlias				
status	"activated"			
expires	Current time plus the value set in the Expires header of the SIP PUBLISH response.		ETSI TS 124 282 [13] (3GPP TS 24.282) clause 22.2.2.3.5	

Annex B (informative): Bibliography

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